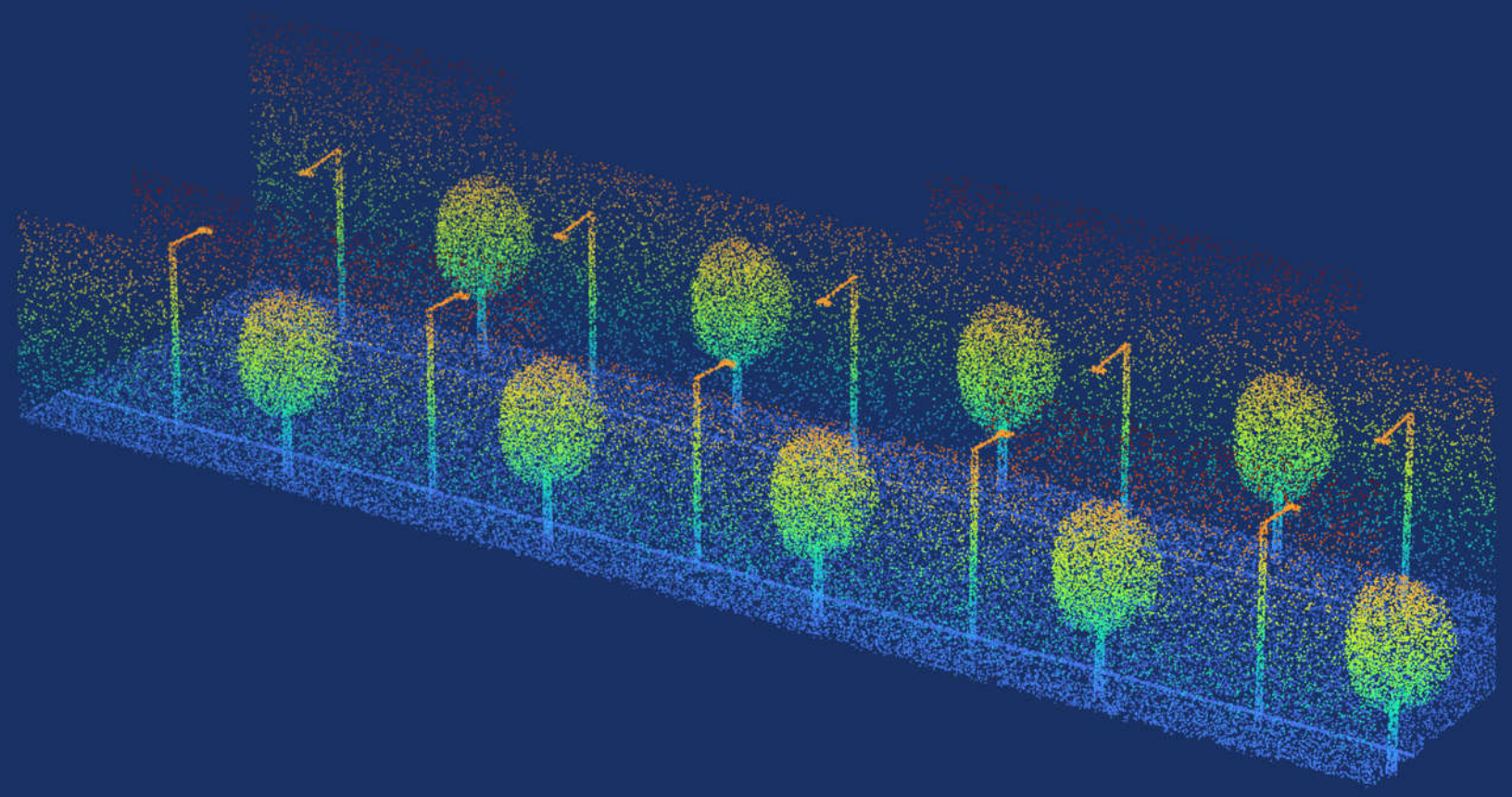
LiDAR from the Ground Up

Mobile Laser Scanning, IMUs, SLAM and Survey-Grade Deliverables

D.S. Banks, PSM

A two-track course: a simple approach anyone can follow,
and a rigorous engineering treatment — on every topic.



21 modules • 85+ illustrations • 80+ fully worked examples • every equation explained

Course Text • October 2026

How to Use This Course

This course teaches mobile LiDAR — the technology that lets a car, a backpack or a drone measure millions of 3-D points while it moves — from the very first idea ("light bounces back") all the way to the factor-graph mathematics inside modern SLAM software such as LIO-SAM, and the accuracy statistics a surveyor signs off on.

It was built by pulling together three source documents — a four-module course outline, a 59-page *Mobile LiDAR Basics* seminar, and the LIO-SAM research paper (Shan *et al.*, 2020) — then reorganizing, correcting and greatly expanding them. Wherever we fixed or sharpened something from those sources you will see a red *Improved from the source material* box, so you can see exactly what changed and why.

Two readers, one book

Every module is written on two levels at the same time. You can read only one track, or both.

SIMPLE APPROACH

Green boxes like this one take the simple approach: they explain the idea with everyday objects and familiar experiences. No algebra is needed. If you want the picture in your head before the math, read these first.

ENGINEER'S VIEW

Blue boxes give the precise statement: the governing equations, the assumptions behind them, the conventions, and how the idea connects to surveying practice, standards and software.

The other boxes

KEY EQUATION — Name of the equation

Every important equation is framed like this, and is *always* followed by a table called **What every symbol means** — each letter, what it stands for, and its units. No symbol is ever left unexplained.

WORKED EXAMPLE — a fully solved problem

Orange boxes solve a realistic problem step by step: *Given*, then numbered steps with every bit of arithmetic shown, then the *Result*. All numbers in this book were checked by computer.

TRY IT YOURSELF

Purple boxes give you a problem to solve on your own. Answers are at the back of the book (Appendix A).

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SURVEYOR'S FIELD TIP

Teal boxes connect the idea to real field and office practice for surveyors and civil engineers.

FOR SURVEY SEMINAR PARTICIPANTS

If you are taking this course as a **survey seminar** (for example, for continuing-education credit), here is exactly what you are responsible for:

- **You are responsible for the graphs and the concepts.** Read the Simple Approach boxes, the Engineer's View text, the figures and their captions, the Surveyor's Field Tips, the Big Ideas, and the Improved From the Source Material boxes. You should be able to explain what each graph shows and why it matters in the field.
- **You are not responsible for the equations.** The Key Equation boxes, the arithmetic inside the Worked Examples, the Try It Yourself problems, and Appendices A and C are there for engineers and for reference. Skim them for the story they tell; you will not be asked to reproduce or solve them.
- **All exam questions are conceptual and standards related.** The final exam is **not printed in this book**: the instructor administers it from a separate instructor packet (two versions). It has no calculations and needs no calculator. The **pass mark is 70%**. It tests the ideas behind LiDAR, IMUs, SLAM and error propagation, and the professional standards discussed in the course: the ASPRS Positional Accuracy Standards (Edition 2), the Florida Standards of Practice (Chapter 5J-17, F.A.C.), FDOT TML requirements, the NSRS modernization, FAA Part 107, laser eye safety, control versus checkpoints, datums and documentation of the work.

Suggested path: follow the six-hour agenda below; give extra attention to Modules 13–14 (accuracy standards and error budgets), 19–20 (Florida standards and professional practice) and 21 (capstone and sample dataset); study the sample SOP and client accuracy agreement in Appendix E.

Six-hour seminar agenda

Time	Topic	Modules
0:00–0:15	Welcome, objectives, how the course works	front matter
0:15–1:00	What LiDAR measures; light, beam, reflectivity, what it can see through	1, 3, 4, 5
1:00–1:45	Pose, the right-hand rule, IMUs, timing, calibration	6, 7, 8, 9
1:45–2:00	Break	
2:00–2:45	SLAM, scan matching, factor graphs; how the trajectory is built	10, 11, 12
2:45–3:30	Accuracy standards (ASPRS Edition 2) and error budgets	13, 14
3:30–4:00	Lunch	
4:00–4:45	Case studies: OS0, GX5/GQ7, RIEGL at 500 ft	15, 16, 17
4:45–5:30	Processing, Florida standards, professional practice	18, 19, 20
5:30–5:45	Capstone and sample dataset walk-through	21, App. E
5:45–6:15	Final exam (conceptual, 50 questions, 70% to pass) and evaluation	instructor packet

Six hours of instruction plus breaks; adjust to the approving board's contact-hour rules. Module 2 (math toolkit) is optional reading for seminar participants.

Learning objectives

After the seminar, participants will be able to:

1. explain how a LiDAR measures range and position and what it can and cannot see (Modules 1–5);
2. describe pose, the right-hand rule and how IMU, LiDAR scans and GNSS build a trajectory (Modules 6–12);
3. interpret accuracy statements and checkpoint tests under ASPRS Edition 2 (Module 13);
4. read an error budget and identify the dominant error source for a given system and range (Modules 14–17);
5. apply Florida 5J-17, FDOT TML and NSRS-modernization requirements to a LiDAR deliverable (Module 19);
6. identify professional-practice obligations: responsible charge, sealing digital data, scope, Part 107, safety and records (Module 20);
7. evaluate a sample project's trajectory, checkpoint table and accuracy report (Module 21).

What you need

For seminar participants and the simple track: curiosity — no calculator is needed. For the engineer's track: algebra, a little trigonometry (Module 2 teaches all of it from scratch), and a scientific calculator. Matrices appear from Module 6 onward and are explained as they arrive.

Course map

Part	Modules	Source outline module
A	1 What is LiDAR? • 2 Math toolkit	— (foundation)
B	3 Time of flight • 4 The beam • 5 Angles to X,Y,Z	Module 1: Sensor architecture
C	6 Rotation & pose • 7 IMU • 8 Timing & motion compensation	Module 1: IMUs, time sync
D	9 Lever arm & boresight calibration	Module 3: System calibration
E	10 SLAM • 11 Scan matching • 12 Factor graphs & LIO-SAM	Module 2: SLAM & trajectory
F	13 Control & accuracy standards • 14 Error propagation & all error budgets	Module 3: Control & error propagation
G	Short case studies: 15 Ouster OS0 • 16 GX5-25 / GX5-45 / GQ7 • 17 RIEGL VUX-240[24]	Module 1: Sensors & IMUs (applied)
H	18 Processing & deliverables • 19 Florida standards, FDOT & NSRS • 20 Professional practice • 21 Capstone & sample dataset	Module 4: Workflows & QA/QC; Florida practice

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MODULE 1

What Is LiDAR?

SIMPLE APPROACH

Close your eyes and clap your hands in a big gym. You hear an echo. If the wall is far away, the echo takes longer to come back. If it is close, the echo comes back fast. A bat flies in the dark this way!
A LiDAR does the same thing — but instead of a clap it uses a tiny flash of laser light, and instead of ears it has a super-fast light catcher. It does this hundreds of thousands of times every second, in every direction, so it can "see" the shape of everything around it: roads, trees, buildings, people. Each echo becomes one dot. Millions of dots together make a 3-D picture called a **point cloud**.

1.1 The name

LiDAR stands for **Li**ght **D**etection **A**nd **R**anging. "Ranging" is the surveyor's word for measuring distance. A LiDAR is, at heart, an electronic distance meter (EDM) — the same physics as the EDM inside a total station — that fires extremely fast and sweeps its beam in many directions.

1.2 The four families of LiDAR

Platform	How it moves	Where the position comes from	Typical use
Airborne	aircraft / drone	GNSS + IMU (very open sky)	terrain, floodplains
Terrestrial (TLS)	stands still on a tripod	surveyed setups, targets	buildings, plants, forensics
Mobile (MLS)	car, train, boat	GNSS + IMU + LiDAR SLAM	corridors, roads, rail
Backpack / handheld	a walking person	IMU + LiDAR SLAM (+ control)	indoors, tunnels, under canopy

This course concentrates on the two moving families at the bottom of the table: **mobile** and **handheld** systems. They are the hardest, because the instrument never holds still — and the most productive, because you can capture a kilometer of street in the time it takes to set up a total station a few times.

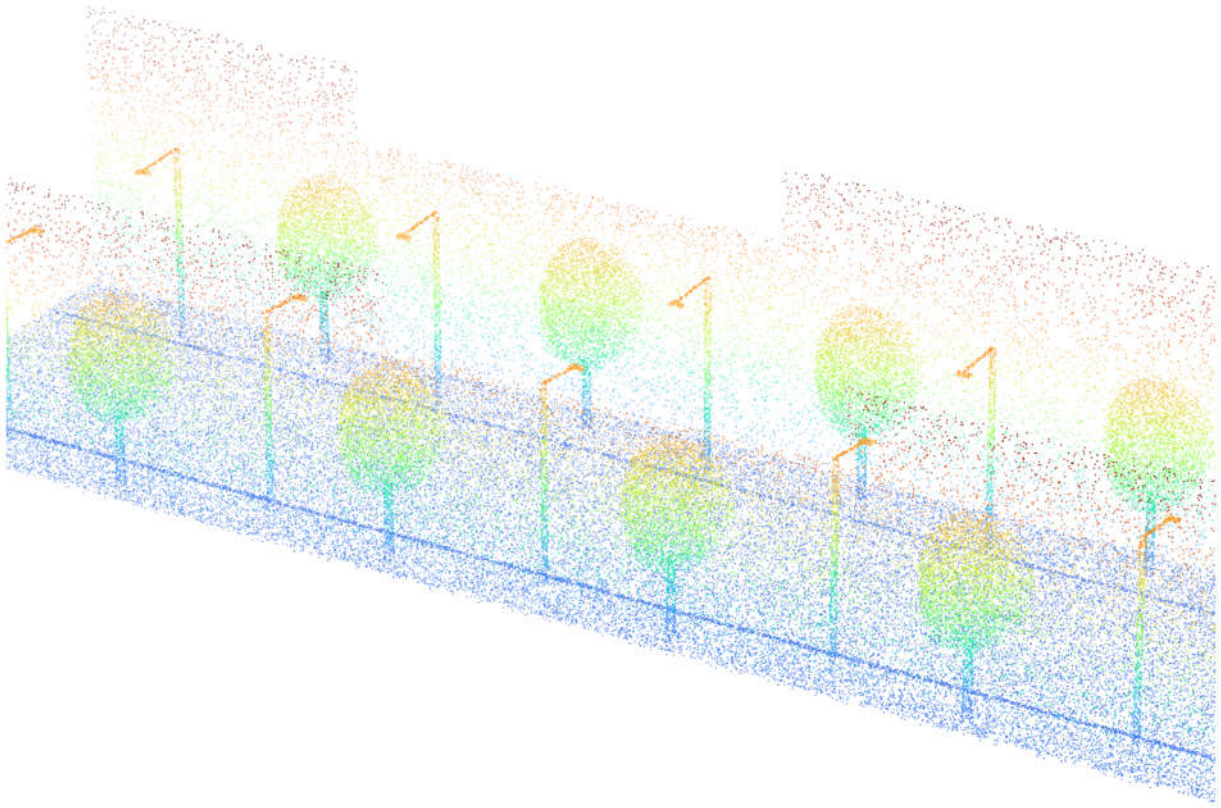


Figure 1.1: A synthetic mobile-LiDAR point cloud of a street, colored by height: road, curbs, poles, trees and building faces. Every dot is one laser echo.

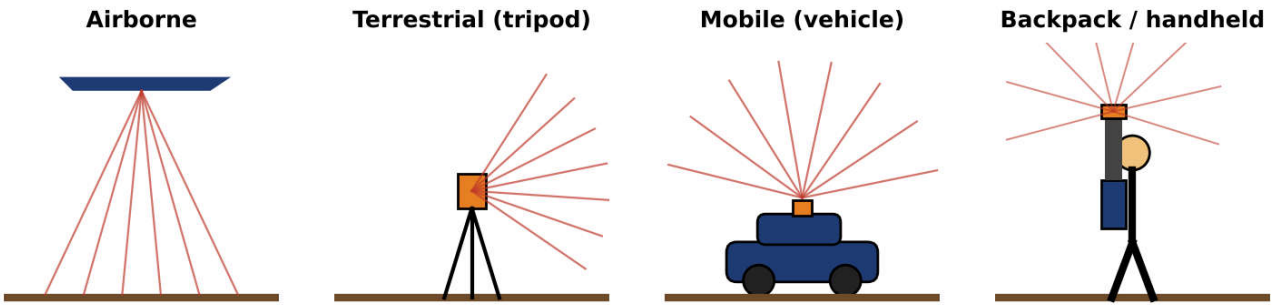


Figure 1.2: The same measuring idea mounted four ways.

1.3 The one big problem: the instrument is moving

BIG IDEA

A laser can only tell you *"that wall is 25.000 m away in that direction from me."* If you do not know exactly where "me" is — and which way "me" was pointing — at the *exact instant* of each pulse, the measurement is useless. Most of this course is about knowing "me": the sensor's **pose**.

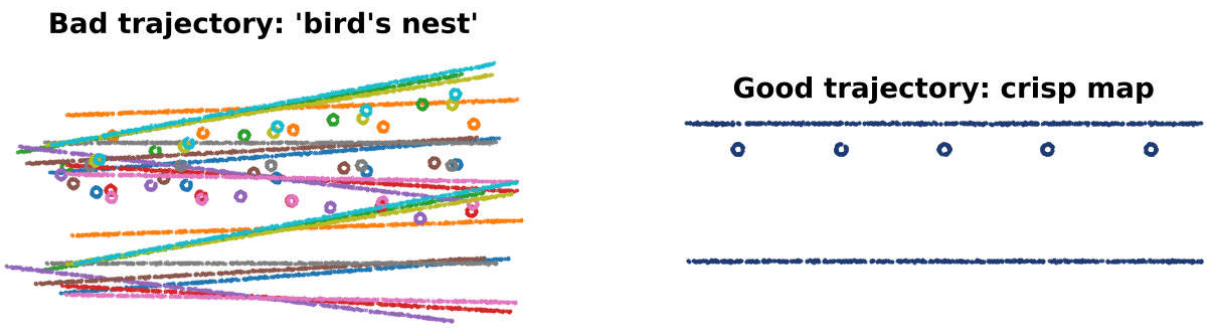


Figure 1.3: Left: the same scans placed with a bad trajectory form the famous "bird's nest." Right: with a correct high-rate trajectory every scan stacks into a crisp map.

ENGINEER'S VIEW

For a moving scanner each laser return must be placed in the project frame using the full six-degree-of-freedom pose (three positions X, Y, Z and three angles roll, pitch, yaw) of the sensor at the time that particular pulse was fired. A spinning scanner at 10 Hz fires up to a few million pulses per second, so the pose must be known at very high rate. That is achieved by combining:

1. an **inertial measurement unit (IMU)** for high-rate, short-term motion (Module 7);
2. **precise time synchronization** so every measurement shares one clock (Module 8);
3. measured **lever arm and boresight** between the sensors (Module 9);
4. a **trajectory estimator** — GNSS/INS fusion and/or LiDAR SLAM — that keeps drift bounded (Modules 10–12);
5. **ground control and independent checkpoints** that tie the result to a datum and prove its accuracy (Module 13).

1.4 Why surveyors already understand most of this

Mobile-LiDAR idea	Classical surveying equivalent
Time-of-flight range	EDM distance
Polar-to-Cartesian conversion	Total-station angle + distance → coordinates
Pose (position + orientation)	Instrument station + backsight orientation
IMU dead reckoning	An open traverse — error accumulates
Loop closure	Closing a traverse
Factor-graph optimization	Least-squares network adjustment
Control points vs checkpoints	Constrained points vs independent check shots
Error budget (root-sum-square)	Traverse / level-network design

1.5 What you will be able to do at the end

- Compute a range from a time of flight and explain why we divide by two.
- Turn a range and two angles into X, Y, Z — and then into project coordinates using lever arm, boresight, attitude and trajectory.

- Predict how fast an IMU of a given grade drifts, and how far a timing or boresight error moves a point.
- Explain, with numbers, what an extended Kalman filter, ICP, and a factor graph actually do.
- Read the LIO-SAM paper's equations and know what every symbol means.
- Build an error budget, compute RMSE from checkpoints, and report accuracy in the language of current standards.

TRY IT YOURSELF

1.1 (Simple) If you clap and hear the echo after 1 second, and sound travels about 343 meters each second, how far away is the wall? (Hint: the sound went there *and* back.)
1.2 (Engineer) List the five ingredients (besides the laser) needed to turn a moving scanner's ranges into a survey-grade map.

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Shan, T. *et al.* (2020), LIO-SAM, IEEE/RSJ IROS, arXiv:2007.00258; *Mobile LiDAR Basics for Surveying and Mapping* seminar (2026); *LiDAR Mobile Mapping with SLAM* course outline (2026).

MODULE 2

The Math Toolkit

This module teaches every piece of math used later in the course. If you already know it, skim the worked examples and move on.

2.1 Big and small numbers (scientific notation)

SIMPLE APPROACH

Light is *so* fast that it could go around the Earth about seven times in one second. To write such big numbers — and tiny ones like "one billionth of a second" — scientists use a shortcut: count the zeros and write that count as a little number up high. 10^3 means 1 followed by 3 zeros = 1000. 10^{-3} means "1 divided into 1000 pieces" = 0.001.

KEY EQUATION — Scientific notation

$$N = m \times 10^k \qquad \text{and} \qquad 10^a \times 10^b = 10^{a+b}$$

What every symbol means

Symbol	Meaning	Units
N	the number being written	any
m	the *mantissa*, a number between 1 and 10	—
k	the power of ten (how many places to move the decimal point; negative = move left)	—
a, b	any two exponents; multiplying powers of ten adds the exponents	—

Prefix	Symbol	Factor	Light travels about…
milli	m	10^{-3}	300 km in 1 ms
micro	μ	10^{-6}	300 m in 1 μs
nano	n	10^{-9}	30 cm in 1 ns
pico	p	10^{-12}	0.3 mm in 1 ps

WORKED EXAMPLE — multiplying with powers of ten

Given: speed $c = 2.997 \times 10^8$ m/s, time $t = 4.00 \times 10^{-7}$ s. Find $c \times t$.

Step 1 — multiply the mantissas

$2.997 \times 4.00 = 11.988$

Step 2 — add the exponents

$10^8 \times 10^{-7} = 10^{8+(-7)} = 10^1$

Step 3 — combine

$11.988 \times 10^1 = 119.88$ m

Result: $c\,t = 119.88$ m.

2.2 Angles: degrees and radians

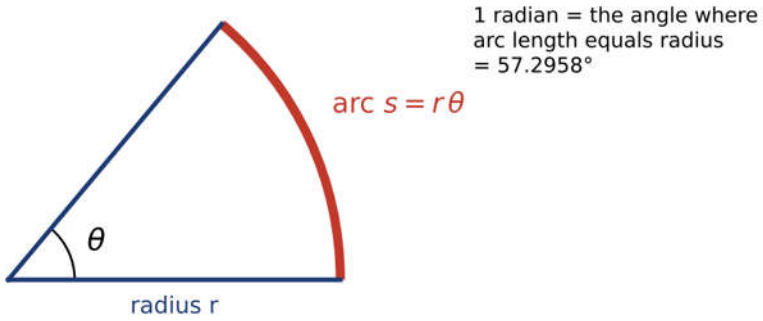


Figure 2.1: A radian is the angle whose arc is exactly one radius long.

SIMPLE APPROACH

Imagine a pizza. If you cut a slice whose crust is exactly as long as the line from the center to the crust, that slice's angle is **one radian**. A whole pizza has a bit more than 6 of those slices (6.28, which is 2π).

KEY EQUATION — Degree–radian conversion and arc length

$$\theta_{\text{rad}} = \theta_{\text{deg}} \times \frac{\pi}{180} \qquad s = r\,\theta_{\text{rad}}$$

What every symbol means

Symbol	Meaning	Units
θ_{deg}	an angle in degrees (a full circle = 360°)	degrees
θ_{rad}	the same angle in radians (a full circle = 2π)	radians
π	3.14159…, ratio of a circle's circumference to its diameter	—
r	radius (in LiDAR: the range to the target)	m
s	arc length: how far you move along the circle	m

Why care? Because **sideways error = range × angle error (in radians)**. This "small-angle rule" will appear in nearly every module. It works because for small angles the arc and the straight sideways distance are almost identical ($\sin\theta \approx \tan\theta \approx \theta$ for θ in radians).

WORKED EXAMPLE — a tiny angle at a long range

Given: an aiming error of 20 arc-seconds at 100 m.

Step 1 — seconds to degrees

$20'' = 20/3600 = 0.005556°$

Step 2 — degrees to radians

$0.005556 \times \pi/180 = 9.70 \times 10^{-5}$ rad

Step 3 — arc length
$s = 100 \times 9.70 \times 10^{-5} = 0.0097$ m

Result: about 1 cm sideways — the classic surveyor's rule "20 seconds at 100 m is a centimeter."

2.3 Sine and cosine

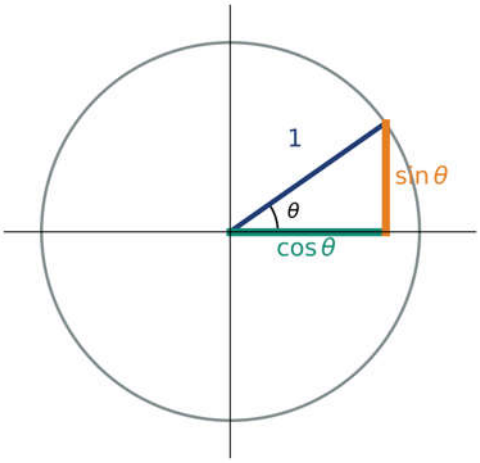


Figure 2.2: On a circle of radius 1, cosine is the "across" part and sine is the "up" part.

SIMPLE APPROACH

Lean a 1-meter stick against a wall. The bottom of the stick is some distance from the wall (that is **cosine**) and the top is some height up the wall (that is **sine**). If the stick leans a lot, cosine is big and sine is small. If it stands almost straight up, sine is big and cosine is small.

KEY EQUATION — Right-triangle trigonometry

$$\text{across} = L\cos\theta, \qquad \text{up} = L\sin\theta, \qquad L = \sqrt{\text{across}^2 + \text{up}^2}, \qquad \theta = \text{atan2}(\text{up}, \text{across})$$

What every symbol means

Symbol	Meaning	Units
L	length of the slanted side (hypotenuse)	m
θ	angle measured from the "across" direction	deg or rad
cos, sin	functions that turn an angle into a number between -1 and $+1$	—
atan2	"two-argument arctangent": returns the angle and gets the quadrant right	rad

WORKED EXAMPLE — a 3-4-5 triangle

Given: a slope distance of 5.000 m at an angle of 36.87° above horizontal.

Step 1
$\text{across} = 5.000\cos 36.87^\circ = 5.000 \times 0.8000 = 4.000$ m

Step 2

$\text{up} = 5.000 \sin 36.87^\circ = 5.000 \times 0.6000 = 3.000$ m

Step 3 (check)

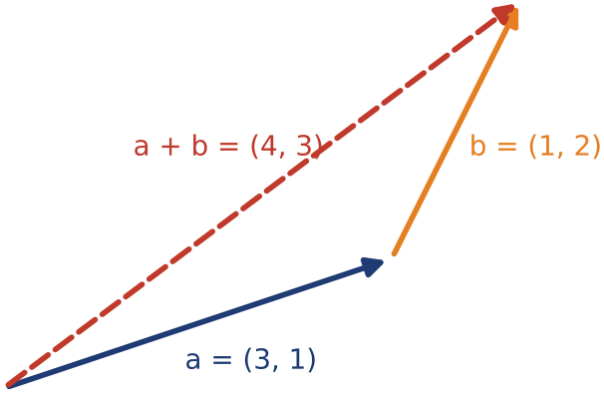
$\sqrt{4^2 + 3^2} = \sqrt{25} = 5$ ✓

Result: 4.000 m across and 3.000 m up.

2.4 Vectors

SIMPLE APPROACH

A vector is an arrow: it says *how far* and *which way*. "Walk 3 steps east and 1 step north" is a vector. Two walks one after the other add up to one bigger walk.



ENGINEER'S VIEW

A 3-D vector is written $\mathbf{v} = (v_x, v_y, v_z)^\mathsf{T}$. Vectors add component by component. Its length (norm) is $\|\mathbf{v}\| = \sqrt{v_x^2 + v_y^2 + v_z^2}$. Two products appear later:

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z \quad \text{(dot: a number)}, \qquad \mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{pmatrix} \quad \text{(cross: a vector)}.$$

The dot product measures how much two arrows point the same way. The cross product is perpendicular to both, and its length equals the area of the parallelogram they span — which is why LIO-SAM uses it to measure distances to lines and planes (Module 11).

WORKED EXAMPLE — dot and cross products

Given: $\mathbf{a} = (1, 2, 0)$, $\mathbf{b} = (3, 0, 0)$.

Dot

$1 \cdot 3 + 2 \cdot 0 + 0 \cdot 0 = 3$

Cross

$(2 \cdot 0 - 0 \cdot 0,\ 0 \cdot 3 - 1 \cdot 0,\ 1 \cdot 0 - 2 \cdot 3) = (0, 0, -6)$

Meaning

The parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$ has area $\|(0, 0, -6)\| = 6$; with base $\|\mathbf{b}\| = 3$ its height is $6/3 = 2$ — exactly the distance from the tip of $\mathbf{a}$ to the line along $\mathbf{b}$.

Result: dot = 3, cross = (0, 0, −6), point-to-line distance = 2.

2.5 Matrices in one page

ENGINEER'S VIEW

A matrix is a table of numbers that transforms a vector. Multiplying a 3×3 matrix by a vector gives a new vector whose each entry is a row of the matrix "dotted" with the vector:

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} ax + by + cz \\ dx + ey + fz \\ gx + hy + iz \end{pmatrix}.$$

Key facts used later: the **transpose** $\mathbf{A}^{\mathsf{T}}$ swaps rows and columns; the **identity** $\mathbf{I}$ changes nothing; the **inverse** $\mathbf{A}^{-1}$ undoes $\mathbf{A}$; for a *rotation* matrix the inverse is simply the transpose, $\mathbf{R}^{-1} = \mathbf{R}^{\mathsf{T}}$.

WORKED EXAMPLE — matrix times vector

$$\begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 2\cdot 1 + 0\cdot 4 + 1\cdot 2 \\ 0\cdot 1 + 1\cdot 4 + 0\cdot 2 \\ 1\cdot 1 + 0\cdot 4 + 3\cdot 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 7 \end{pmatrix}$$

2.6 Statistics: mean, standard deviation, RMSE



Figure 2.3: Random measurement errors usually follow a bell curve. About 68% fall within one standard deviation σ and 95% within 1.96σ.

SIMPLE APPROACH

Throw ten paper balls at a target. The **mean** tells you where they landed on average (are you aiming too far left?). The **spread** (standard deviation) tells you how scattered they are. A great thrower has a mean right on the bullseye *and* a small spread.

KEY EQUATION — Mean, standard deviation, RMSE

$$\bar{e} = \frac{1}{n}\sum_{i=1}^{n} e_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(e_i - \bar{e})^2}, \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2}$$

What every symbol means

Symbol	Meaning	Units
e_i	the i-th error (measured minus true)	m
n	how many errors we have	—
$\sum$	"add them all up," from $i = 1$ to n	—
$\bar{e}$	mean error = *bias* (systematic part)	m
s	sample standard deviation = *precision* (random part); $n-1$ because one degree of freedom was spent computing $\bar{e}$	m
RMSE	root-mean-square error = total *accuracy*; combines bias and spread: $\text{RMSE}^2 \approx \bar{e}^2 + s^2$	m

WORKED EXAMPLE — five elevation errors

Given: errors $+2, -1, +3, 0, +1$ cm.

Mean

$(2 - 1 + 3 + 0 + 1)/5 = 5/5 = 1.0$ cm (a 1 cm high bias)

Deviations from mean

$1, -2, 2, -1, 0$; squares $1, 4, 4, 1, 0$; sum $= 10$

Std. dev.

$s = \sqrt{10/4} = \sqrt{2.5} = 1.58$ cm

RMSE

squares of errors $4, 1, 9, 0, 1$; sum $= 15$; $\text{RMSE} = \sqrt{15/5} = \sqrt{3} = 1.73$ cm

Check

$\bar{e}^2 + s^2 \cdot \frac{n-1}{n} = 1 + 2.5 \times 0.8 = 3.0 = \text{RMSE}^2$ ✓

Result: bias 1.0 cm, precision 1.58 cm, accuracy (RMSE) 1.73 cm.

2.7 Adding independent errors: root-sum-square

KEY EQUATION — Propagation of independent errors

$$\sigma_{\text{total}} = \sqrt{\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_k^2}$$

What every symbol means

Symbol	Meaning	Units
σ_j	standard deviation of the j-th independent error source	m
σ_{total}	standard deviation of their sum	m

Errors do *not* add like ordinary numbers because random errors partly cancel. Two independent 3 cm and 4 cm errors combine to $\sqrt{9+16} = 5$ cm, not 7 cm. The rule only holds when the sources

are *independent*; a systematic error that pushes every point the same way must be treated separately (see Module 13).

TRY IT YOURSELF

2.1 Convert $0.05°$ to radians, then find the sideways error it causes at 80 m.
2.2 Compute $\sqrt{2^2 + 1.5^2 + 3^2}$.
2.3 A wall is 12 m across and 5 m up from you along the slope. How long is the slope distance?
2.4 (Simple) You need 3 big steps forward and then 4 big steps to the right. If you could walk straight there instead, how many steps would it be?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Ghilani, C. D. (2017), *Adjustment Computations: Spatial Data Analysis*, 6th ed., Wiley (statistics, RMSE, propagation); NIST SP 811, *Guide for the Use of the International System of Units*.

MODULE 3

Measuring Distance with Light

3.1 Time of flight

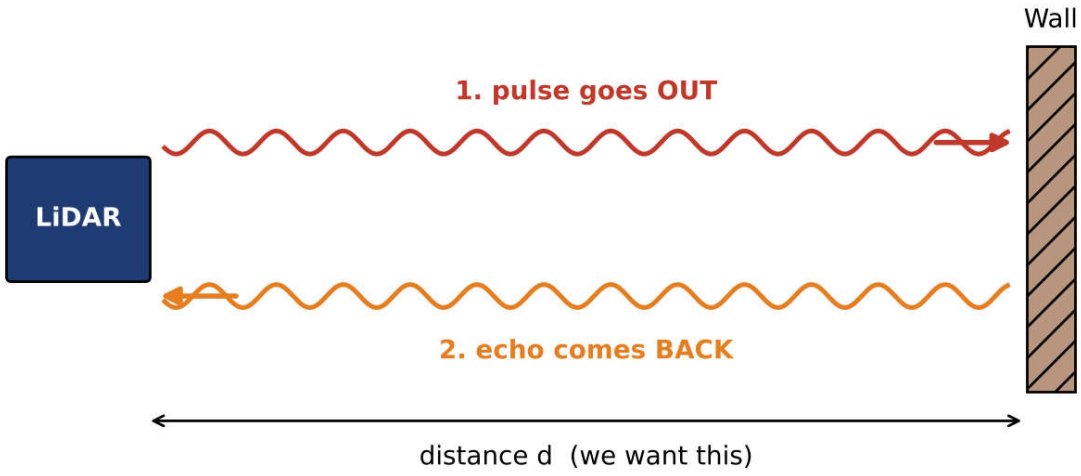


Figure 3.1: A pulse travels out to the wall and back. The clock measures the whole round trip.

SIMPLE APPROACH

Light is the fastest thing there is. In the time it takes you to blink, light could go around the world more than 2 times! The LiDAR starts a stopwatch when it flashes, and stops it when the flash bounces back. Because the light went *to* the wall and *back*, the wall is only half as far as the light traveled. So we cut the answer in half.

KEY EQUATION — Time-of-flight range

$$\rho = \frac{c\,\Delta t}{2}, \qquad c = \frac{c_0}{n}$$

What every symbol means

Symbol	Meaning	Units
ρ	range (one-way distance from sensor to surface). Greek "rho."	m
Δt	measured round-trip time of flight	s
c	speed of light *in air*	m/s
c_0	speed of light in vacuum, exactly 299 792 458	m/s
n	refractive index of air, about 1.0003 at sea level (depends on temperature, pressure, humidity)	—
2	the light travels the distance twice (out and back)	—

WORKED EXAMPLE — range from a 400 ns echo

Given: $\Delta t = 400$ ns; standard air with $n = 1.000300$.

Step 1 — speed of light in air

$c = 299\,792\,458 / 1.000300 = 299\,702\,547$ m/s

Step 2 — time in seconds

$400 \text{ ns} = 400 \times 10^{-9} = 4.00 \times 10^{-7}$ s

Step 3 — round-trip path

$c\,\Delta t = 299\,702\,547 \times 4.00 \times 10^{-7} = 119.881$ m

Step 4 — halve it

$\rho = 119.881 / 2 = 59.941$ m

Result: $\rho \approx 59.94$ m.

IMPROVED FROM THE SOURCE MATERIAL

The seminar used $c \approx 2.997 \times 10^8$ m/s and got 59.94 m. That value is right, but the reason matters: it is the *vacuum* speed (2.9979×10^8) divided by the refractive index of air. Using the vacuum value instead would give 59.958 m — an 18 mm difference at 60 m, i.e. about 300 parts per million. Survey-grade processing applies an atmospheric correction for this, exactly as you set temperature and pressure on a total station's EDM.

3.2 How precise must the clock be?

Rearranging the key equation for a small timing error δt:

KEY EQUATION — Range error from timing error

$$\delta\rho = \frac{c\,\delta t}{2} \approx 0.15 \text{ m per nanosecond}$$

What every symbol means

Symbol	Meaning	Units
$\delta\rho$	error in the computed range	m
δt	error in the measured time of flight	s

So a 1 ns timing error ruins the range by 15 cm, and *millimeter* ranging needs timing to about 7 picoseconds. Real sensors do not count clock ticks that finely; they digitize the returning pulse shape and interpolate its center, which is why waveform quality and signal strength matter so much.

TRY IT YOURSELF

3.1 An echo returns after 166.8 ns. What is the range (use $c = 2.997 \times 10^8$ m/s)?
3.2 What timing precision is needed for 1 cm range precision?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

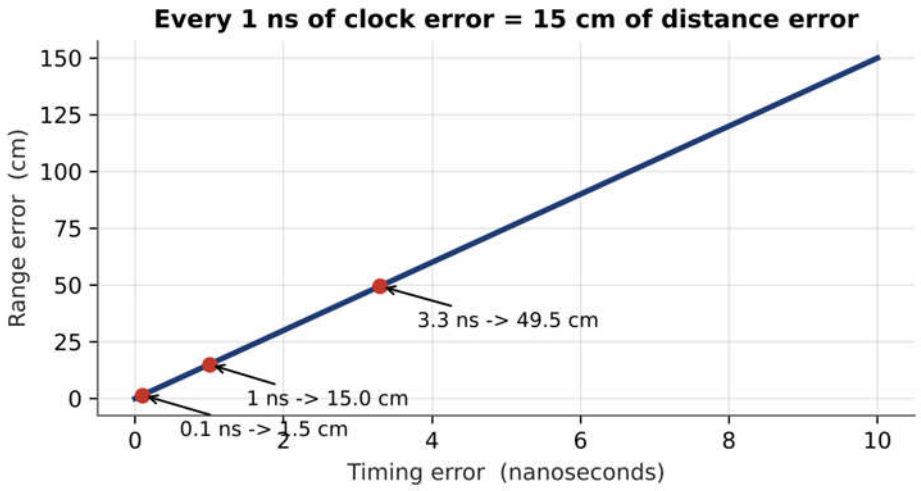


Figure 3.2: Range error grows in a straight line with timing error.

3.3 Pulse rate and range ambiguity

SIMPLE APPROACH

Imagine shouting "Hello!" every second in a giant canyon. If an echo takes 3 seconds, you won't know which "Hello" it belongs to. A LiDAR has the same problem if it fires the next flash before the last echo has come home.

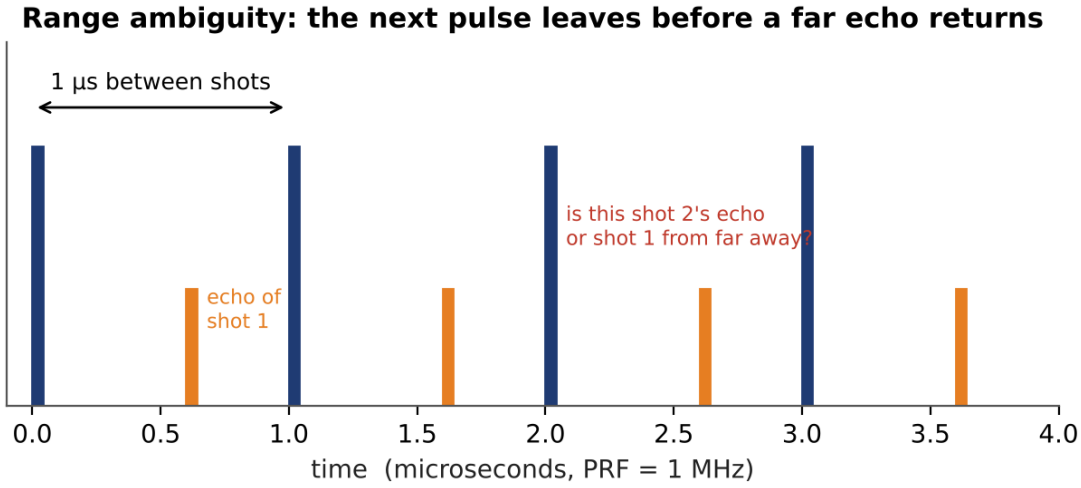


Figure 3.3: If the pulse repetition interval is shorter than the round-trip time, echoes get mixed up.

KEY EQUATION — Maximum unambiguous range

$$\rho_{\max} = \frac{c}{2\, f_{\mathrm{PRF}}}$$

What every symbol means

Symbol	Meaning	Units
$\rho_{\max}$	farthest range that can be measured without confusion	m
f_{PRF}	pulse repetition frequency: pulses fired per second	Hz (1/s)

WORKED EXAMPLE — how fast can we fire?

Given: $f_{\mathrm{PRF}} = 1$ MHz $= 10^6$ pulses/s, $c = 2.997 \times 10^8$ m/s.

Step 1

time between pulses $= 1/10^6 = 1\ \mu$s

Step 2

$\rho_{\max} = 2.997 \times 10^8 / (2 \times 10^6) = 149.9$ m

Step 3 — double the rate

at 2 MHz, $\rho_{\max} = 74.9$ m

Result: faster firing = denser points but a shorter safe range. Modern sensors use "multiple pulses in the air" (MPiA) coding to beat this limit.

3.4 Phase-shift (continuous-wave) ranging

Some scanners send a continuous light wave whose brightness is modulated, and measure how far the returning wave has slid relative to the outgoing one.

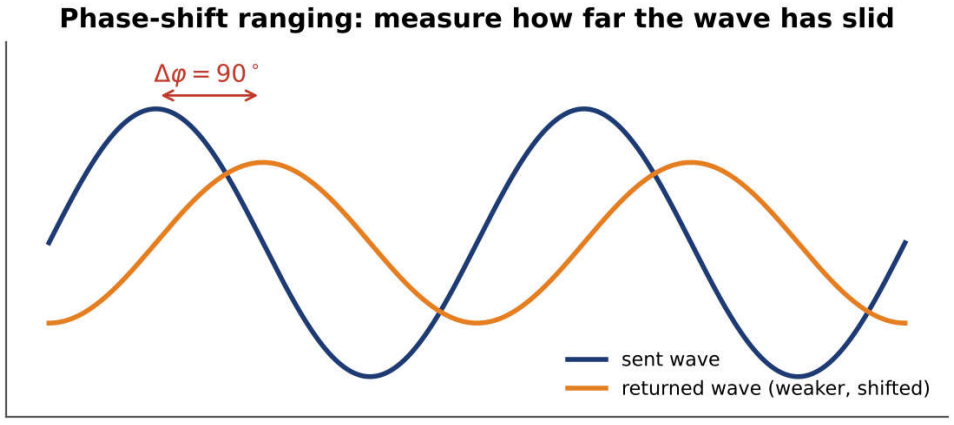


Figure 3.4: The returned wave is shifted by a fraction of a wavelength.

KEY EQUATION — Phase-shift range

$$\rho = \frac{\lambda_m}{2}\left(N + \frac{\Delta\varphi}{2\pi}\right), \qquad \lambda_m = \frac{c}{f_m}$$

What every symbol means

Symbol	Meaning	Units
λ_m	modulation wavelength (not the laser color!)	m
f_m	modulation frequency	Hz
$\Delta\varphi$	measured phase difference	rad
N	whole number of half-wavelengths (the "ambiguity"); solved by using several frequencies	—

WORKED EXAMPLE — phase measurement

Given: $f_m = 10$ MHz, $\Delta\varphi = 90°$, $c = 2.997 \times 10^8$ m/s, $N = 0$.

Step 1

$\lambda_m = 2.997 \times 10^8 / 10^7 = 29.97$ m; half-wavelength $= 14.985$ m

Step 2

$\Delta\varphi/2\pi = 90/360 = 0.25$

Step 3

$\rho = 14.985 \times 0.25 = 3.746$ m

Result: 3.746 m — or 18.731 m, or 33.716 m… (add multiples of 14.985 m). A second, lower frequency decides which.

This is the same principle as most total-station EDMs. Phase scanners are very precise at short range (terrestrial scanners), while pulsed time-of-flight dominates long-range and mobile systems.

3.5 Multiple returns

ENGINEER'S VIEW

Because the beam has a finite width (Module 4), one pulse can partially hit several surfaces at different ranges. A *discrete-return* sensor reports the first, intermediate and last peaks; a *full-waveform* sensor records the whole curve. Each return gets its own range from its own time, $\rho_k = c\,t_k/2$. Last returns are the ones most likely to reach bare ground under vegetation — the basis of terrain models in forests. Two surfaces closer together than about $c\,\tau/2$ (with τ the pulse width, e.g. 4 ns → 0.6 m) merge into a single peak.

SURVEYOR'S FIELD TIP

Edges of poles, curbs and building corners produce "mixed pixels": part of the beam hits the near object and part the far background, and the reported range can land in empty air between them. These show up as ghost points trailing behind edges and are removed in cleaning (Module 18).

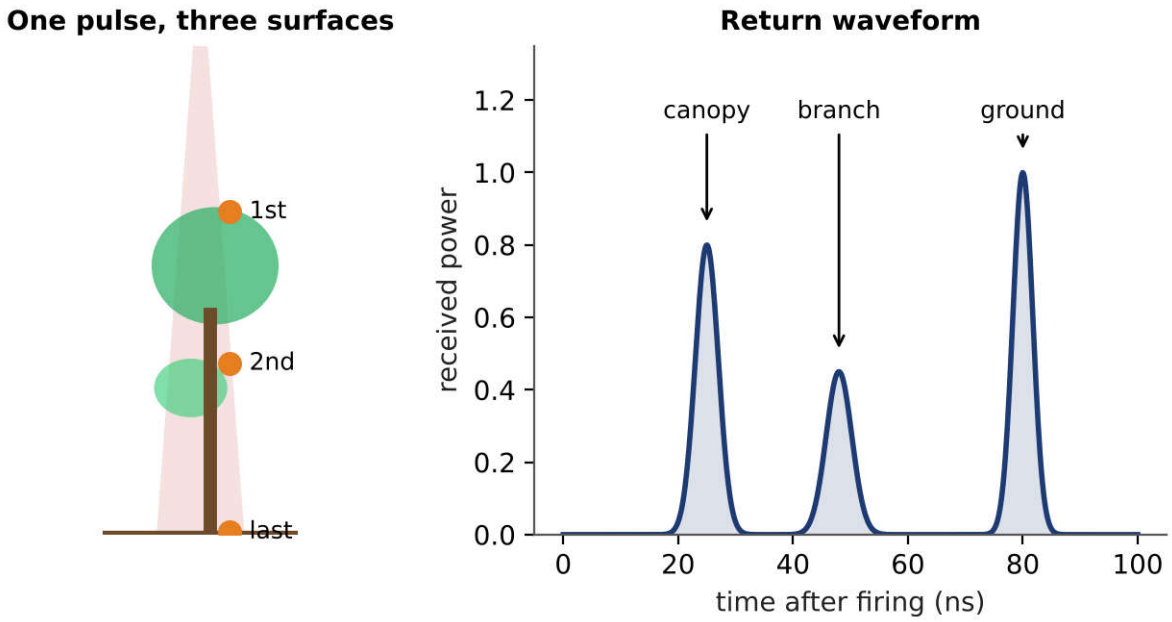


Figure 3.5: One wide pulse can hit leaves, a branch and the ground. The receiver sees several peaks.

SOURCES FOR THIS MODULE

BIPM, *The International System of Units* (speed of light $c_0 = 299\,792\,458$ m/s, exact); Ciddor, P. E. (1996), refractive index of air, *Applied Optics* 35(9); Baltsavias, E. P. (1999), Airborne laser scanning: basic relations and formulas, *ISPRS J. Photogramm. Remote Sens.* 54; RIEGL VUX-240[24] datasheet (MTA processing).

MODULE 4

The Laser Beam and the Echo

4.1 The beam spreads out

SIMPLE APPROACH

Shine a flashlight at your hand: you see a small circle. Now shine it at a wall across the room: the circle is much bigger. A laser beam spreads too — just very, very slowly. Far away, the laser "dot" can be as big as a coin, so it can touch the edge of a pole and the wall behind it at the same time.

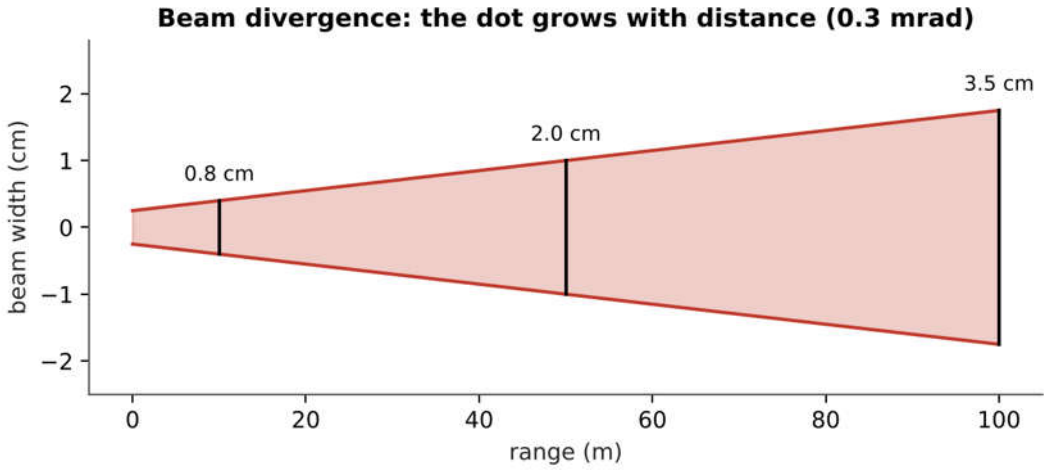


Figure 4.1: The footprint of a beam with 5 mm exit diameter and 0.3 mrad divergence.

KEY EQUATION — Beam footprint

$$D(\rho) \approx d_0 + \rho\,\gamma$$

What every symbol means

Symbol	Meaning	Units
D	diameter of the laser spot on the target	m
d_0	diameter of the beam as it leaves the sensor	m
ρ	range	m
γ	full-angle beam divergence (small-angle rule again!)	rad (often given in mrad = 10^{-3} rad)

WORKED EXAMPLE — spot size at 50 m and 100 m

Given: $d_0 = 5$ mm $= 0.005$ m, $\gamma = 0.3$ mrad $= 0.0003$ rad.

At 50 m

$D = 0.005 + 50 \times 0.0003 = 0.005 + 0.015 = 0.020$ m $= 2.0$ cm

At 100 m

$D = 0.005 + 100 \times 0.0003 = 0.035$ m $= 3.5$ cm

Result: a 2–3.5 cm "blur" at mapping ranges. You cannot locate an edge more sharply than about half the spot size.

4.2 How strong is the echo? The LiDAR range equation

SIMPLE APPROACH

The farther away something is, the dimmer its echo. It gets dim fast: twice as far means only one quarter as bright. Shiny white things (a road sign) send back lots of light; black asphalt or wet surfaces send back very little. And if the light hits a surface at a slant, most of it bounces away like a ball skipping off water.

KEY EQUATION — Received power for an extended diffuse target

$$P_r = P_t \frac{\rho_{\text{refl}} \cos\theta_i}{\pi R^2} A_r \eta_{\text{sys}} \eta_{\text{atm}}^2$$

What every symbol means

Symbol	Meaning	Units
P_r	power arriving at the detector	W
P_t	power transmitted in the pulse	W
ρ_{refl}	target reflectance (0 = black, 1 = perfect white)	—
θ_i	incidence angle between beam and surface normal	rad
R	range	m
A_r	area of the receiving lens	m^2
η_{sys}	optical efficiency of the sensor	—
η_{atm}	one-way atmospheric transmission (squared: out and back)	—
π	comes from a diffuse (Lambertian) surface scattering into a half-sphere	—

WORKED EXAMPLE — comparing two targets

Given: Target A: white concrete ($\rho_{\text{refl}} = 0.6$) at 10 m, head-on. Target B: asphalt ($\rho_{\text{refl}} = 0.1$) at 50 m, hit at 60°. Everything else equal.

Step 1 — ratio

$$\frac{P_B}{P_A} = \frac{0.1}{0.6} \times \frac{\cos 60^\circ}{\cos 0^\circ} \times \frac{10^2}{50^2}$$

Step 2

$= 0.1667 \times 0.5 \times 0.04 = 0.00333$

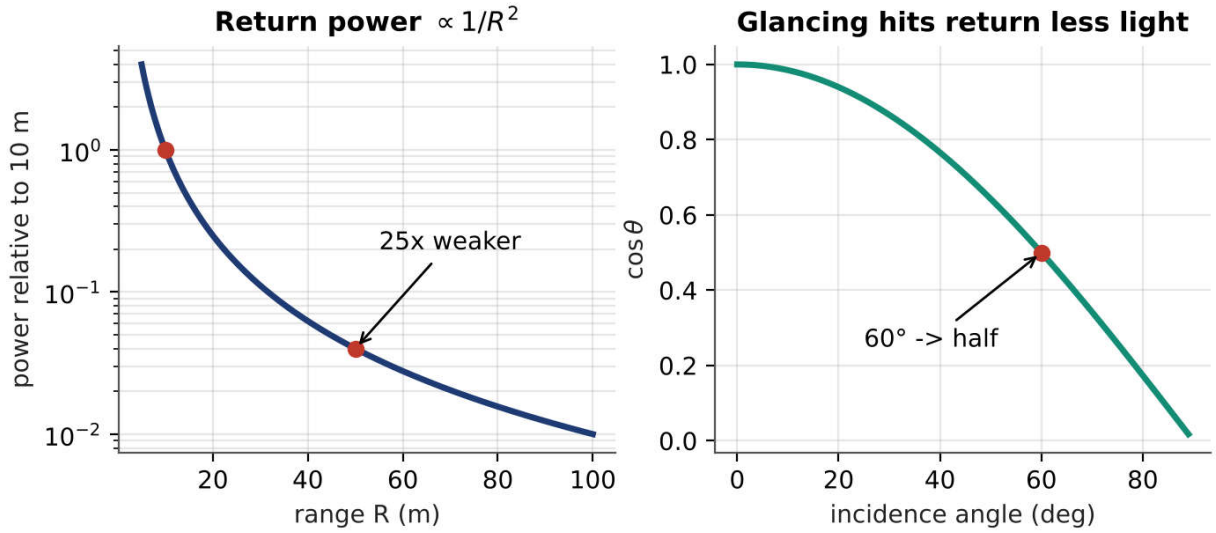


Figure 4.2: Left: the $1/R^2$ law. Right: the $\cos\theta_i$ incidence-angle law.

Result: B returns about 1/300 of A's power. Weak echoes have noisier ranges, which is why range precision gets worse with distance and grazing angle — and why road surfaces far from the vehicle are the noisiest part of a mobile scan.

4.3 Intensity

Every point carries an **intensity** value, a number related to P_r. Because of the R^2 and $\cos\theta_i$ effects, raw intensity of the same paint changes with range and angle; software can *normalize* it. Intensity makes road markings, signs and retro-reflective targets jump out of the cloud, which is handy for extracting line work and for finding control targets.

4.4 Wavelength and eye safety

ENGINEER'S VIEW

Most automotive and mobile scanners use **905 nm** (near-infrared, cheap silicon detectors) or **1550 nm** (absorbed by the eye's cornea and lens before reaching the retina, so far more power may be used eye-safely — giving longer range). All survey scanners used in public must be **Class 1** eye-safe under IEC 60825-1. Water strongly absorbs 1550 nm, so wet surfaces and rain reduce returns more at that wavelength; green (532 nm) lasers are used for bathymetric LiDAR because green light penetrates water.

4.5 What can LiDAR see through?

SIMPLE APPROACH

A laser is light, so it acts like a flashlight beam, not like X-ray vision. It cannot see *through* a wall, a tree trunk, or the ground. It *can* slip through holes — the gaps between leaves, the openings in a fence — and reach whatever is behind them. It passes through clear glass the way sunlight does. And it is stopped almost right away by water.

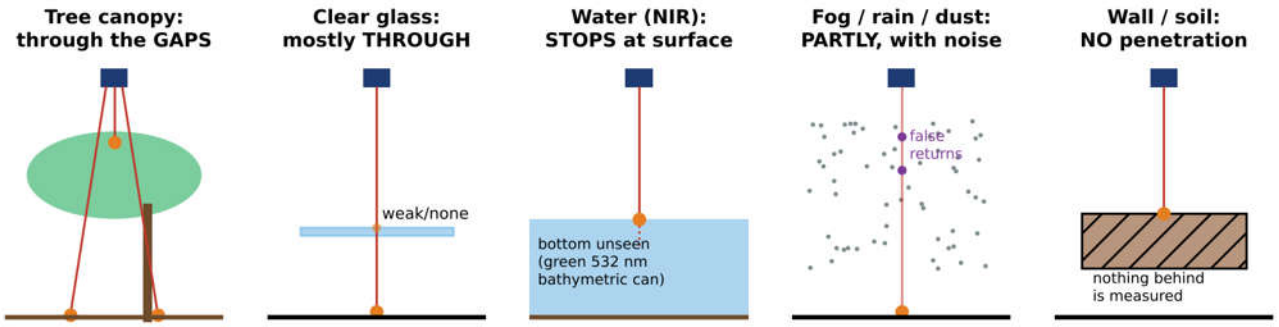


Figure 4.3: The five situations every LiDAR user meets. Orange dots are recorded returns.

Material or condition	Does the laser get through?	What you get and what to do
Tree canopy, shrubs	**Through the gaps only**	First returns on leaves, last returns on the ground where there are openings. Dense evergreen or summer canopy may leave no ground hits: collect in leaf-off season, slow down, add passes.
Tall grass, crops, mangroves	**Rarely**	Points sit on top of the vegetation, not the soil. Heights are biased high; take ground shots by conventional methods.
Chain-link fence, mesh, railings	**Partly** (gaps)	Some points on the fence, some behind it; the fence appears as a sparse ghost surface.
Clear glass windows	**Mostly through**	Few returns from the glass itself (except head-on); you see the room behind. Reflections can create "mirror-world" ghost points.
Tinted, frosted or coated glass	**Partly / varies**	Unpredictable; check the cloud.
Mirrors, polished metal, chrome	**No — reflects away**	Dropouts or ghost points placed at false ranges behind the surface.
Water (865–1550 nm)	**No**	Absorbed within centimeters; calm water often returns nothing except straight down. Only green (532 nm) bathymetric LiDAR maps the bottom, in clear water.
Wet pavement, puddles	**No**	Weak or missing returns; specular reflection; schedule collection after surfaces dry.
Rain, fog, dust, smoke, steam	**Partly**	Range shortens; "floating" false points near the sensor; heavy fog can block completely. Filter them out (Module 18).
Falling snow	**Partly**	Noise points in the air; snow cover hides the ground surface.
Walls, soil, pavement, people, vehicles	**No**	Opaque: anything behind is in shadow. Plan extra passes or positions to see behind objects.
Underground utilities	**No**	LiDAR maps only visible surfaces (manhole lids, valve covers). Use ground-penetrating radar or locating for what is buried.

BIG IDEA

LiDAR never "penetrates" solid material. It reaches hidden ground only through openings, which is why multiple returns (Module 3) and many viewing angles matter more than laser power.

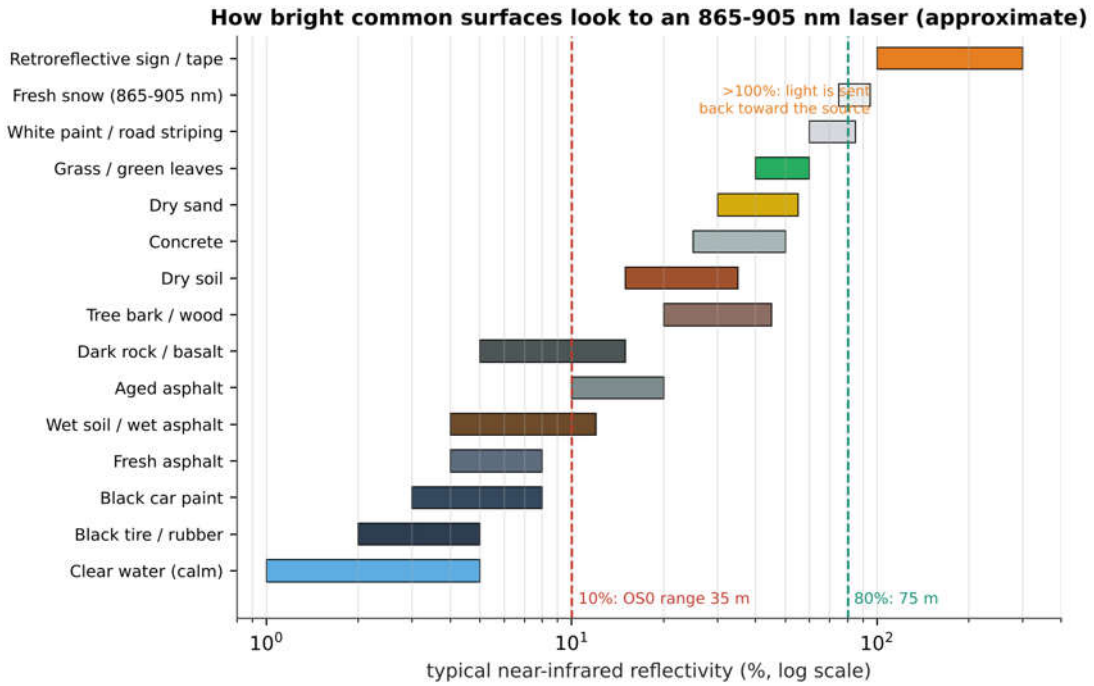


Figure 4.4: Typical reflectivity of common surfaces at the near-infrared wavelengths used by most mobile scanners (865–905 nm). Values vary with moisture, age, roughness, color and incidence angle.

4.6 Reflectivity of common surfaces

Surface	Typical reflectivity	What it means in the field
Clear, calm water	1–5%	often no return at all except straight below the sensor
Black tire, rubber	2–5%	among the darkest targets; shortest detection range
Black or dark car paint	3–8%	cars may appear with holes; glossy paint can also mirror
Fresh asphalt	4–8%	dark: road returns thin out first at long range
Wet soil, wet asphalt	4–12%	water darkens surfaces in the near infrared
Dark rock (basalt)	5–15%	dark
Aged, gray asphalt	10–20%	brighter as it weathers
Dry soil	15–35%	medium
Tree bark, wood	20–45%	medium
Concrete	25–50%	good, consistent target
Dry sand	30–55%	bright
Grass, green leaves	40–60%	bright in the near infrared (healthy plants reflect strongly there)
White paint, road striping	60–85%	very bright: striping stands out in intensity images
Fresh snow	75–95% at 865–905 nm	very bright at 905 nm — but much darker at 1550 nm
Retroreflective signs and tape	“over 100%”	sends light straight back; can saturate the receiver and bias the range (Module 15)

Values are typical ranges compiled for teaching; they are not specifications. Sensors that report *calibrated reflectivity* (such

as the Ouster OS0) let you measure your own surfaces.

WORKED EXAMPLE — how far can I see dark asphalt?

Given: the OS0 detects a 10% target to 35 m (Module 15). Received power falls as $1/R^2$ (Section 4.2), so for the same detection threshold the maximum range grows as the *square root* of reflectivity.

Fresh asphalt, 5%
$35 \times \sqrt{5/10} = 35 \times 0.71 = 25$ m

Black car, 4%
$35 \times \sqrt{0.4} = 22$ m

Concrete, 40%
$35 \times \sqrt{4} = 70$ m (the datasheet's 75 m at 80% shows the rule flattening out in bright sunlight)

Result: plan the usable swath on the *darkest* surface you must map: on new asphalt, expect road coverage to end near 25 m, long before the brochure range.

4.7 Range precision in practice

Situation	Typical range noise (1σ)
Good mobile unit, bright hard target, under 30 m (100 ft)	1–3 cm (0.03–0.10 ft)
Same unit, dark or grazing target, 50–100 m (165–330 ft)	5–15 cm (0.16–0.49 ft)
Survey-grade mobile mapping head	3–8 mm (0.01–0.03 ft)
Terrestrial phase scanner, short range	1–3 mm (0.003–0.010 ft)

These are order-of-magnitude illustrations; always read the specification sheet and, more importantly, *test on your own control*.

TRY IT YOURSELF

4.1 A beam has $d_0 = 8$ mm and $\gamma = 0.5$ mrad. How wide is the spot at 40 m?
4.2 By what factor does echo power drop when a target moves from 20 m to 60 m?
4.3 (Simple) Which gives a brighter echo: a white sign or a black tire? Why?
4.4 (Simple) Which of these can a mobile LiDAR measure the ground behind: a leafy tree, a brick wall, a glass storefront, a pond?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Baltsavias (1999), *ISPRS J.* 54 (range equation, footprint); IEC 60825-1:2014 (laser classes); Ouster OS0 Rev8 datasheet (calibrated reflectivity, detection ranges at 10% and 80%); reflectivity values are typical teaching ranges — measured spectra are available in the USGS Spectral Library Version 7 (Kokaly *et al.*, 2017).

MODULE 5

From Angles and Distance to X, Y, Z

5.1 The idea

SIMPLE APPROACH

To tell a friend where a treasure is hidden, you could say: "Face the big tree. Turn a little to the left. Look a little up. Now walk 25 steps that way." That is a **turn**, a **tilt** and a **distance**. A LiDAR records the same three things for every dot. Then it does some triangle math to change them into "how far forward, how far sideways, how far up" — which we call X, Y and Z.

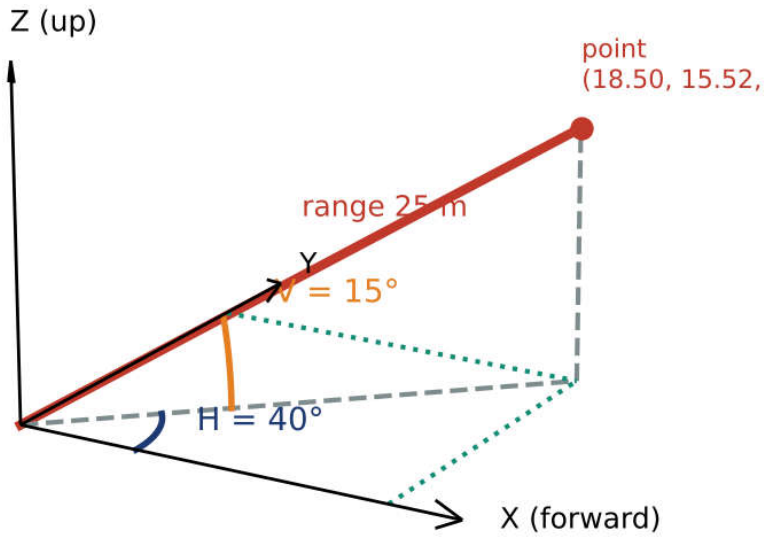


Figure 5.1: A range of 25 m with horizontal angle 40° and vertical angle 15° becomes a point at (18.50, 15.52, 6.47) m.

KEY EQUATION — Polar (spherical) to Cartesian

$$x_s = \rho \cos V \cos H, \qquad y_s = \rho \cos V \sin H, \qquad z_s = \rho \sin V$$

What every symbol means

Symbol	Meaning	Units
x_s, y_s, z_s	point coordinates in the *sensor* frame (subscript s)	m
ρ	measured range	m
H	horizontal angle (azimuth in the sensor), measured from the sensor's x-axis toward its y-axis	deg/rad
V	vertical angle above the sensor's horizontal plane (elevation). Note: a total station's *zenith* angle Z_a is $90° - V$	deg/rad
$\rho \cos V$	the horizontal part of the range (its shadow on the floor)	m

ENGINEER'S VIEW

Read the equation in two moves. **First tilt:** the vertical angle splits the range into a horizontal part $\rho \cos V$ and a height $\rho \sin V$. **Then turn:** the horizontal angle splits the horizontal part into x and y. That is why $\cos V$ appears in both x and y. In a spinning scanner, H comes from the motor's angle encoder at the instant of firing, and V is the fixed elevation angle of the laser channel ("ring") that fired. Conventions differ by manufacturer (some measure H clockwise, some use zenith angles), so always check the sensor's documentation.

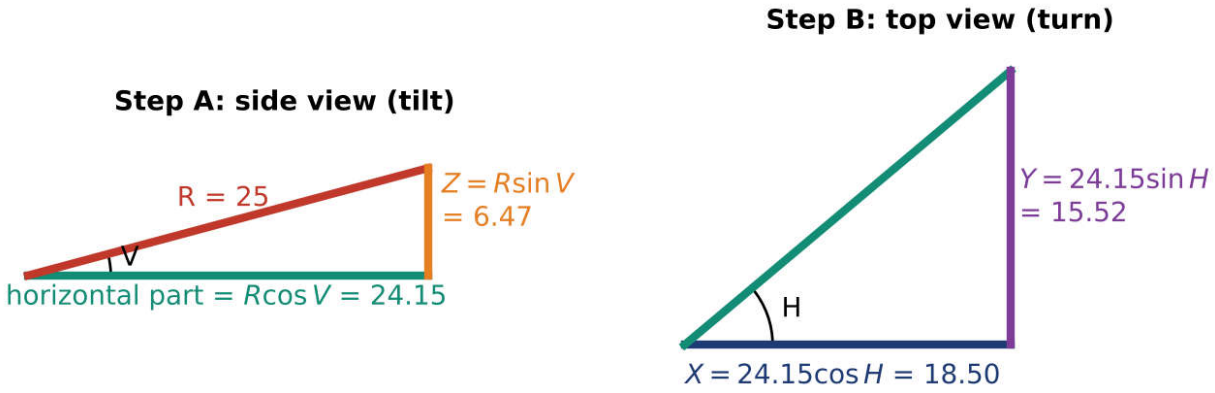


Figure 5.2: The two moves: tilt (side view), then turn (top view).

WORKED EXAMPLE — one laser return to sensor coordinates

Given: $\rho = 25.000$ m, $H = 40.0°$, $V = 15.0°$. Sensor frame: x forward, y left, z up.

Step 1 — trig values

$\cos 15° = 0.96593$, $\sin 15° = 0.25882$, $\cos 40° = 0.76604$, $\sin 40° = 0.64279$

Step 2 — horizontal part

$\rho \cos V = 25.000 \times 0.96593 = 24.148$ m

Step 3 — split it

$x_s = 24.148 \times 0.76604 = 18.499$ m; $\quad y_s = 24.148 \times 0.64279 = 15.522$ m

Step 4 — height

$z_s = 25.000 \times 0.25882 = 6.470$ m

Step 5 — check

$\sqrt{18.499^2 + 15.522^2 + 6.470^2} = \sqrt{342.21 + 240.93 + 41.86} = \sqrt{625.00} = 25.000$ ✓

Result: $(x_s, y_s, z_s) = (18.50,\ 15.52,\ 6.47)$ m.

IMPROVED FROM THE SOURCE MATERIAL

The seminar used "Y right" with a counter-clockwise H. Those two choices are inconsistent for a right-handed frame with Z up: with X forward and Z up, a right-handed Y points *left*. We use the robotics standard (ROS REP-103: x forward, y left, z up) throughout this course, and we added the length check in Step 5 — a habit worth keeping: the computed coordinates must reproduce the measured range.

5.2 Going backward: Cartesian to polar

KEY EQUATION — Inverse conversion

$$\rho = \sqrt{x^2 + y^2 + z^2}, \qquad H = \operatorname{atan2}(y, x), \qquad V = \operatorname{atan2}\left(z, \sqrt{x^2 + y^2}\right)$$

What every symbol means

Symbol	Meaning	Units
$\operatorname{atan2}(a, b)$	the angle whose tangent is a/b, placed in the correct quadrant using the signs of a and b	rad

This is what the SLAM software uses to *predict* what range and bearing the sensor should see to a known object (Module 10).

5.3 Mechanical versus solid-state scanners

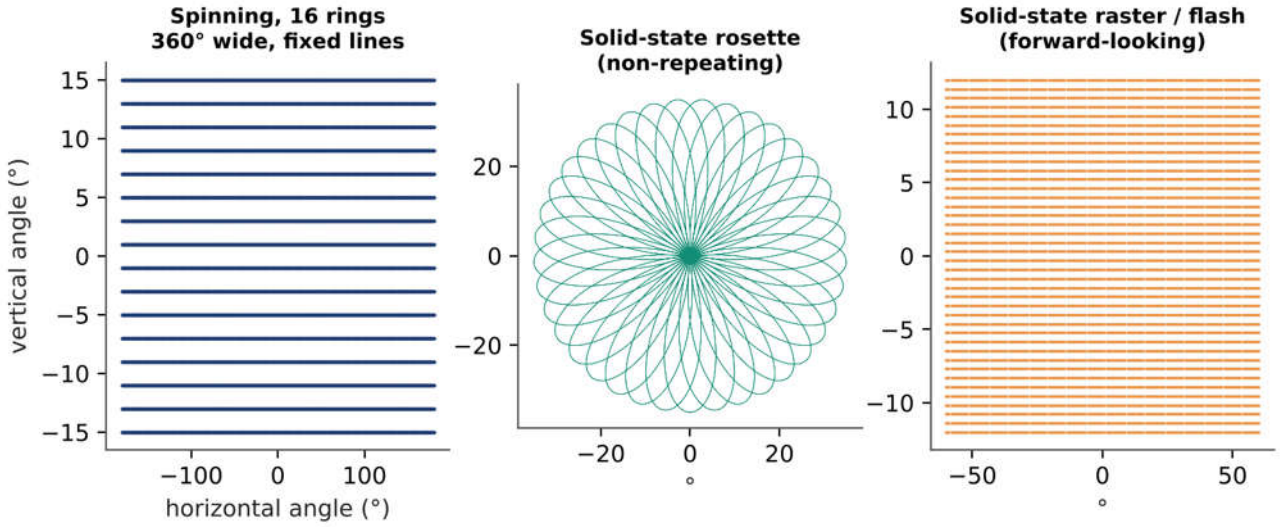


Figure 5.3: How three kinds of scanner sample the world (angles in degrees).

	Mechanical (spinning)	Solid-state
How the beam moves	a motor spins a stack of 16–128 lasers	MEMS mirror, prisms (rosette), optical phased array, or flash (no motion)
Field of view	360° horizontal, 30–90° vertical	typically 60–120° wide, forward-looking
Point pattern	regular rings	raster or non-repeating; coverage fills in over time
Strengths	full surround view, mature, well-modeled	small, light, rugged, cheap, no large moving parts
Watch-outs	motor wear, size, cost	narrower view; need several units or motion for full coverage; pattern-specific calibration

5.4 How dense will my points be?

SIMPLE APPROACH

Imagine painting a road with a paint-roller that only touches the road once every big step. If you walk faster, the stripes get further apart. If you roll faster, they get closer.

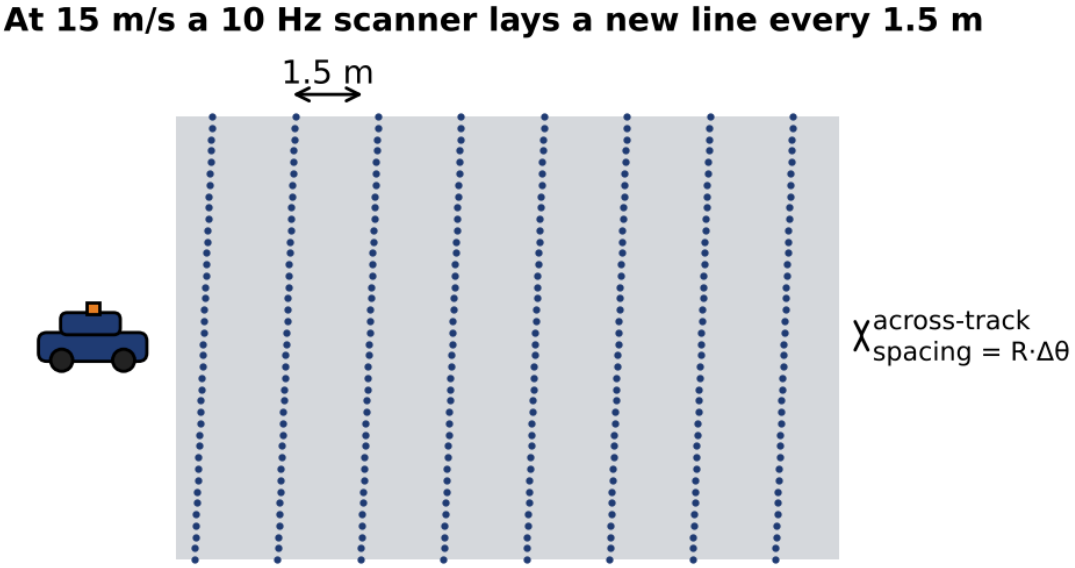


Figure 5.4: Each sweep of the scanner lays a line of points across the road.

KEY EQUATION — Point spacing and point rate

$$\Delta_{\text{along}} = \frac{v}{f_{\text{rot}}}, \qquad \Delta_{\text{across}} \approx \rho\,\Delta\theta, \qquad N_{\text{pts/s}} = N_{\text{ch}} \times N_{\text{az}} \times f_{\text{rot}}$$

What every symbol means

Symbol	Meaning	Units
Δ_{along}	spacing between scan lines in the direction of travel	m
v	platform speed	m/s
f_{rot}	rotation (scan-line) rate	Hz
Δ_{across}	spacing between neighboring points along one scan line	m
$\Delta\theta$	angular step between pulses	rad
N_{ch}	number of laser channels	—
N_{az}	azimuth steps per revolution ($360^\circ / \Delta\theta$)	—

WORKED EXAMPLE — planning a corridor survey

Given: 16-channel scanner, 0.2° horizontal step, 10 Hz, driving 15 m/s (54 km/h). Target 20 m away.

Step 1 — along-track

$\Delta_{\text{along}} = 15/10 = 1.5$ m between sweeps (for each channel)

Step 2 — across-track
$0.2^\circ = 0.003491$ rad; $\Delta_{\text{across}} = 20 \times 0.003491 = 0.070$ m

Step 3 — point rate
$N_{\text{az}} = 360/0.2 = 1800$; $N = 16 \times 1800 \times 10 = 288\,000$ points/s

Step 4 — slow down
at 5 m/s the along-track spacing becomes 0.5 m

Result: 1.5 m between lines is fine for a pavement surface but too coarse for curb lines or small utilities — drive slower, tilt the scanner, or use more channels.

SURVEYOR'S FIELD TIP

Mobile mapping heads for survey work are often mounted *tilted* so the spinning plane slices the road at an angle; then several channels hit the same strip of road at different times and along-track gaps close up. Point-density specifications should be checked on the delivered data (points per m^2 on a test patch), not assumed from the brochure.

TRY IT YOURSELF

5.1 Convert $\rho = 40.000$ m, $H = -30^\circ$, $V = 5^\circ$ into (x_s, y_s, z_s). Check the length.
5.2 A point is at $(3.0, 4.0, 12.0)$ m in the sensor frame. Find ρ, H and V.
5.3 A 32-channel scanner spins at 20 Hz with 0.4° steps. How many points per second? What is the along-track spacing at 10 m/s?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

ROS REP-103, *Standard Units of Measure and Coordinate Conventions*; Ouster OS0 Rev8 datasheet (channels, columns, field of view).

MODULE 6

Rotation, Pose and Frames

6.1 Pose = where you are + which way you face

SIMPLE APPROACH

Stand in the middle of a room. "Where you are" is your spot on the floor (and how tall you are). "Which way you face" is whether you look at the board or the window, whether you tilt your head, whether you lean. A LiDAR on a car needs to know both of these things, all the time, because the car bumps, turns and leans. Together they are called its **pose**.

ENGINEER'S VIEW

Pose has six degrees of freedom: position $\mathbf{p} = (X, Y, Z)$ and orientation, usually written as three angles — **roll** ϕ, **pitch** θ, **yaw** ψ — or as a 3×3 rotation matrix $\mathbf{R}$, or a unit quaternion. In LIO-SAM's notation the pose is a rigid-body transformation $\mathbf{T} = [\mathbf{R} \mid \mathbf{p}] \in SE(3)$, where $SE(3)$ ("special Euclidean group") is simply the set of all rotations-plus-translations, and $\mathbf{R} \in SO(3)$ ("special orthogonal group") the set of all proper rotation matrices.

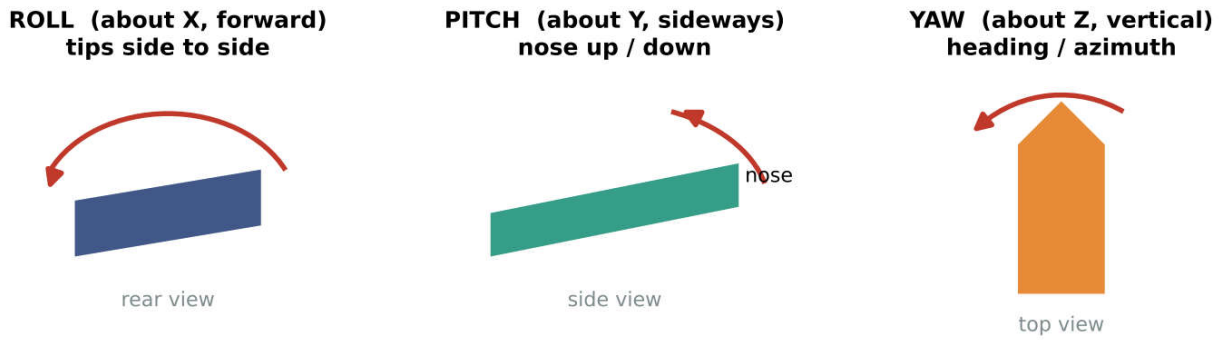


Figure 6.1: Roll, pitch and yaw on a mapping vehicle.

Angle	About axis	Everyday example	Effect of a residual error on the map
Roll ϕ	x (forward)	banked curve	tilts points left/right of the drive line; heights wrong on one side
Pitch θ	y (sideways)	climbing a grade	far points too high/low ahead of and behind the vehicle
Yaw ψ	z (vertical)	turning a corner	rotates the whole scan; lateral error grows with range

6.2 The right-hand rule: naming X, Y, Z and roll, pitch, yaw

SIMPLE APPROACH

Hold up your **right** hand. Point your **first (index) finger** straight ahead — that is **X**. Point your **middle finger** to the left — that is **Y**. Stick your **thumb** straight up — that is **Z**. **The thumb is always Z.** Another way to say it: curl your fingers from X toward Y, and your thumb points to Z. Now make a thumbs-up: the way your curled fingers wrap around your thumb is the "plus" direction for turning around that thumb. Point your thumb along X and your fingers show plus **roll**; along Y, plus **pitch**; along Z (straight up), plus **yaw**.

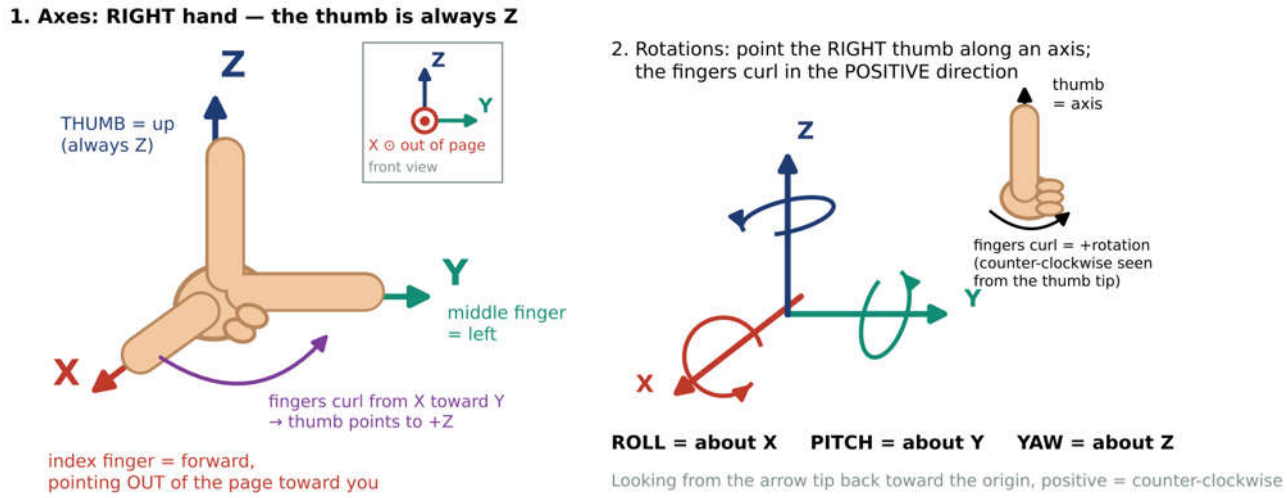


Figure 6.2: The right-hand rule. Left: index finger = X, middle finger = Y, and the thumb is always Z; equivalently, the fingers curl from X toward Y and the thumb points to Z. Right: with the thumb along an axis, the curled fingers show the positive rotation about it — roll about X, pitch about Y, yaw about Z.

ENGINEER'S VIEW

Rule 1 — the axes form a right-handed set. With the right hand: index finger $= +X$, middle finger $= +Y$, **thumb** $= +Z$ — the thumb is always Z. Equivalently, curl the fingers from $+X$ toward $+Y$ and the thumb points along $+Z$; mathematically $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$ (Module 2). (Some textbooks label the fingers in a different order, e.g. thumb $= X$; any cyclic relabeling gives the same right-handed set, but this course keeps the thumb on Z throughout so that axis naming and the rotation rule below use the hand the same way.) Every coordinate frame in this course — sensor, body, IMU, world (ENU) and map projection with height — is right-handed. A left-handed frame (one axis flipped) is a mirror image, and no rotation can turn one into the other; mixing them is a classic cause of a "mirrored" point cloud.

Rule 2 — positive rotations follow the curl. Point the right thumb along the positive axis; the fingers curl in the direction of a *positive* angle. Seen from the arrow tip looking back toward the origin, positive is **counter-clockwise**.

- **Roll** ϕ = rotation about $+X$
- **Pitch** θ = rotation about $+Y$
- **Yaw** ψ = rotation about $+Z$

Because the *direction* of each axis decides the sign, the same physical motion can be $+$ in one convention and $-$ in another. The two conventions you will meet most:

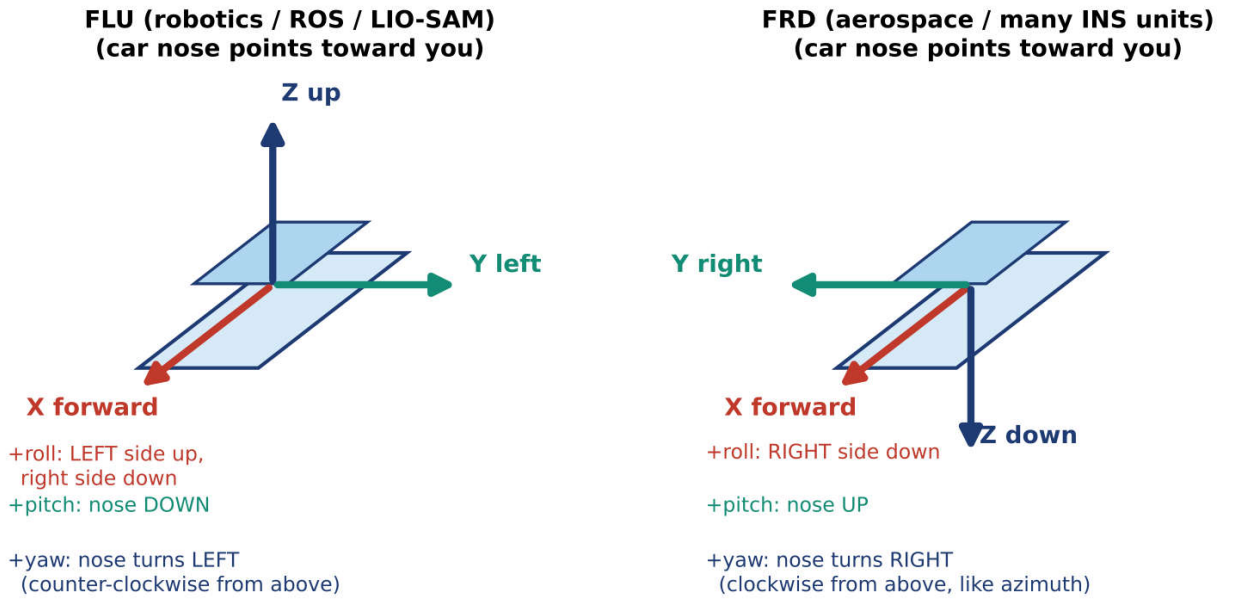


Figure 6.3: The same vehicle in the two common right-handed body frames.

	FLU: X forward, Y left, Z up	FRD: X forward, Y right, Z down
Used by	ROS (REP-103), LIO-SAM, most LiDAR drivers, this course	aircraft, many INS/IMU products (NED world frame)
+ roll (about X)	left side *up*, right side down	right side *down*
+ pitch (about Y)	nose *down*	nose *up*
+ yaw (about Z)	nose turns *left* (counter-clockwise from above)	nose turns *right* (clockwise from above, like azimuth)
World frame	ENU: East, North, Up	NED: North, East, Down

WORKED EXAMPLE — checking a frame with your hand (and with the cross product)

Given: a datasheet says a sensor's axes are "X forward, Y right, Z up."

Hand test

right index finger forward (X), middle finger right (Y) — the thumb (Z) now points *down*, not up.

Cross-product test

forward $\times$ right $= (1,0,0) \times (0,-1,0) = (0,0,-1)$ = down in a frame where $(0,1,0)$ is left.

Result: "X forward, Y right, Z up" is **left-handed**. Either the datasheet means Z down (FRD) or one axis is mislabeled. Resolve it with the manufacturer before computing a single lever arm or boresight; a sign error here mirrors or tilts the entire cloud.

SURVEYOR'S FIELD TIP

When setting up a rig, stand behind the vehicle, hold up your right hand in the way the manual defines the axes, and *physically* roll, pitch and yaw the IMU a few degrees while watching the live output. Each angle should increase in the direction your fingers curl. Write the verified convention in the calibration record (Appendix E).

6.3 Rotating a point in 2-D

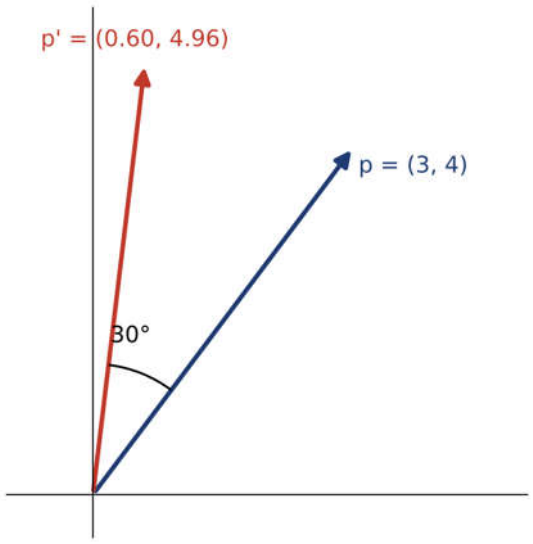


Figure 6.4: Rotating $\mathbf{p} = (3, 4)$ by 30° counter-clockwise.

KEY EQUATION — 2-D rotation

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \underbrace{\begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix}}_{\mathbf{R}(\psi)} \begin{pmatrix} x \\ y \end{pmatrix}$$

What every symbol means

Symbol	Meaning	Units
(x, y)	original point	m
(x', y')	point after rotation	m
ψ	rotation angle, positive counter-clockwise ("psi")	rad
$\mathbf{R}(\psi)$	the rotation matrix; its columns are where the x- and y-axes end up	—

WORKED EXAMPLE — rotate (3, 4) by 30°

Step 1

$\cos 30^\circ = 0.86603$, $\sin 30^\circ = 0.50000$

Step 2

$x' = 0.86603 \times 3 - 0.5 \times 4 = 2.59808 - 2.00000 = 0.598$

Step 3

$y' = 0.5 \times 3 + 0.86603 \times 4 = 1.50000 + 3.46410 = 4.964$

Step 4 — check

length before $= \sqrt{9+16} = 5$; after $= \sqrt{0.598^2 + 4.964^2} = \sqrt{0.358 + 24.642} = 5.000$ ✓

Result: $(0.598,\ 4.964)$. A rotation never changes lengths — a great error check.

6.4 Rotations in 3-D

KEY EQUATION — The three elementary rotations

$$\mathbf{R}_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{pmatrix},\ \mathbf{R}_y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix},\ \mathbf{R}_z(\psi) = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{R} = \mathbf{R}_z(\psi)\,\mathbf{R}_y(\theta)\,\mathbf{R}_x(\phi) \qquad (\text{“ZYX” or yaw–pitch–roll order})$$

What every symbol means

Symbol	Meaning	Units
ϕ, θ, ψ	roll, pitch, yaw (“phi, theta, psi”)	rad
$\mathbf{R}_x$ etc.	rotation about one axis only; the axis’s own row and column hold 1 and 0s	—
$\mathbf{R}$	the full body-to-world rotation; applied right-to-left: roll first, then pitch, then yaw	—

ENGINEER’S VIEW

Order matters. Rotating 90° about x then 90° about z ends somewhere different than the reverse order — try it with a book. Every software package documents its order and sign convention; a large share of real-world georeferencing bugs are convention mismatches. Two other important facts:

- $\mathbf{R}^\mathsf{T}\mathbf{R} = \mathbf{I}$ and $\det \mathbf{R} = +1$. So the inverse rotation (world-to-body) is just $\mathbf{R}^\mathsf{T}$. LIO-SAM writes $\mathbf{R}_t = \mathbf{R}_t^{\mathsf{WB}} = \mathbf{R}_t^{\mathsf{BW}\mathsf{T}}$ for exactly this reason.
- In the FLU frame used here (x forward, y left, z up), a *positive* rotation about y tips the nose *down*. Many INS products use FRD (x forward, y right, z down) where positive pitch is nose up. Always confirm.

SURVEYOR’S FIELD TIP

Yaw is not azimuth. Surveyors measure azimuth clockwise from grid north. ROS/ENU systems measure yaw counter-clockwise from east. Convert with $\psi = 90^\circ - \alpha_{\text{az}}$. A vehicle heading due north has $\alpha_{\text{az}} = 0^\circ$ but $\psi = 90^\circ$. Grid north also differs from true north by the *convergence angle* of your map projection — another place a heading can quietly go wrong.

6.5 Homogeneous transforms: rotation and shift in one matrix

KEY EQUATION — Rigid-body transformation

$$\mathbf{r}_W = \mathbf{R}\,\mathbf{r}_B + \mathbf{p} \quad \Longleftrightarrow \quad \begin{pmatrix} \mathbf{r}_W \\ 1 \end{pmatrix} = \underbrace{\begin{pmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{pmatrix}}_{\mathbf{T}} \begin{pmatrix} \mathbf{r}_B \\ 1 \end{pmatrix}$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{r}_B$	a point expressed in the body (vehicle/IMU) frame B	m
$\mathbf{r}_W$	the same point expressed in the world frame W	m
$\mathbf{R}$	body-to-world rotation	—
$\mathbf{p}$	position of the body origin in W	m
$\mathbf{T}$	4×4 transform; chaining frames = multiplying $\mathbf{T}$'s	—

The relative motion between two poses is $\Delta\mathbf{T}_{i,j} = \mathbf{T}_i^{-1}\mathbf{T}_j$. That is LIO-SAM's equation (12), which defines the *lidar odometry factor* (Module 12). (The paper prints it as $\mathbf{T}_i^{\mathsf{T}}\mathbf{T}_{i+1}$; for the full 4×4 matrix the correct operation is the inverse, which equals the transpose only for the rotation block.)

6.6 Rotation from a turning rate: the exponential map

Gyroscopes give angular *rates*. Over a short time Δt the body turns through a rotation vector $\boldsymbol{\omega}\Delta t$ (direction = axis, length = angle). LIO-SAM's equation (6) converts that small turn into a rotation matrix with the exponential map:

KEY EQUATION — Rotation update (Rodrigues' formula)

$$\mathbf{R}_{t+\Delta t} = \mathbf{R}_t \exp\big([\boldsymbol{\omega}\Delta t]_\times\big), \qquad \exp([\mathbf{a}]_\times) = \mathbf{I} + \frac{\sin\alpha}{\alpha}[\mathbf{a}]_\times + \frac{1-\cos\alpha}{\alpha^2}[\mathbf{a}]_\times^2$$

What every symbol means

Symbol	Meaning	Units
$\boldsymbol{\omega}$	angular-rate vector from the gyros (bias-corrected)	rad/s
$\mathbf{a}$	rotation vector $\boldsymbol{\omega}\Delta t$; $\alpha = \|\mathbf{a}\|$ is the turn angle	rad
$[\mathbf{a}]_\times$	skew-symmetric matrix $\begin{pmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{pmatrix}$; $[\mathbf{a}]_\times\mathbf{b} = \mathbf{a} \times \mathbf{b}$	rad

WORKED EXAMPLE — one gyro step

Given: yaw rate $\omega_z = 0.0842$ rad/s (from a real IMU message in the seminar), $\Delta t = 0.005$ s (200 Hz). Start level, heading $\psi = 0$.

Step 1

$\alpha = 0.0842 \times 0.005 = 0.000421$ rad $= 0.0241°$, all about z

Step 2

For rotation purely about z, Rodrigues reduces to $\mathbf{R}_z(\alpha)$: $\cos\alpha = 0.99999991$, $\sin\alpha = 0.000421$

Step 3

New heading $\psi = 0.0241°$. If the rate stays constant, 200 such steps (1 s) add up to $200 \times 0.000421 = 0.0842$ rad $= 4.82°$

Result: the strapdown computer repeats this tiny multiplication 200 times per second to track attitude.

6.7 Why tiny angle errors matter



Figure 6.5: Lateral error versus range for three residual angle errors.

WORKED EXAMPLE — 0.05° heading error at 80 m

Step 1 — radians

$0.05 \times \pi/180 = 0.0008727$ rad

Step 2 — small-angle rule

$80 \times 0.0008727 = 0.0698$ m

Result: 7.0 cm. At 100 m: 8.7 cm. A 0.1° error at 100 m: 17.5 cm — bigger than most engineering tolerances.

TRY IT YOURSELF

6.1 Rotate $(10, 0)$ by 90°; then rotate $(0, 10)$ by $-90°$. What do you notice?
6.2 A vehicle drives with azimuth 135° (southeast). What is its ENU yaw?
6.3 What heading error produces 2 cm lateral error at 40 m?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

ROS REP-103 (FLU body frame, ENU world frame); Groves, P. D. (2013), *Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems*, 2nd ed. (rotations, FRD/NED conventions); Shan *et al.* (2020), eq. (6), (12).

MODULE 7

The IMU: Feeling Motion

7.1 What is inside an IMU?

SIMPLE APPROACH

When a car speeds up, you get pushed back into your seat. When it turns, you lean to the side. Your body *feels* the motion even with your eyes closed. An IMU is a little box that feels motion the same way. It has three "push-feelers" (accelerometers) — forward, sideways, up — and three "spin-feelers" (gyroscopes). By adding up all those feelings many times a second, the computer can guess where the car went, even inside a tunnel. But like walking with your eyes closed, the guess slowly gets worse.

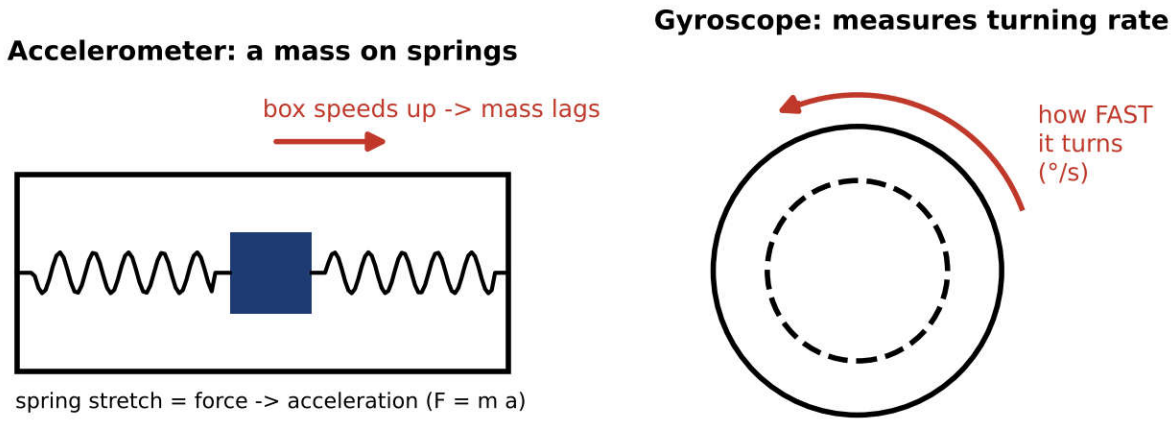


Figure 7.1: An accelerometer is a mass on springs; a gyroscope senses turning rate.

ENGINEER'S VIEW

An **inertial measurement unit** contains a triad of accelerometers measuring *specific force* $\mathbf{f}$ (acceleration minus gravity) and a triad of gyroscopes measuring *angular rate* $\boldsymbol{\omega}$, both in the body frame, typically at 100–1000 Hz. The sensors are rigidly fixed ("strapped down") to the platform; the computer — not a gimbal — keeps track of orientation. An at-rest, level accelerometer reads $+9.81$ m/s^2 upward (it feels the table pushing up), which is why an IMU message at rest shows $a_z \approx 9.78$–9.81.

7.2 The IMU measurement model (LIO-SAM eqs. 2–3)

KEY EQUATION — What the IMU actually reports

$$\hat{\omega}_t = \omega_t + \mathbf{b}_t^{\omega} + \mathbf{n}_t^{\omega} \qquad \hat{\mathbf{a}}_t = \mathbf{R}_t^{\mathrm{BW}}(\mathbf{a}_t - \mathbf{g}) + \mathbf{b}_t^{a} + \mathbf{n}_t^{a}$$

What every symbol means

Symbol	Meaning	Units
$\hat{\omega}_t$, $\hat{\mathbf{a}}_t$	raw gyro and accelerometer readings at time t (the "hat" means *measured*)	rad/s, m/s^2
ω_t	the true angular rate of the body	rad/s
$\mathbf{a}_t$	the true acceleration of the body, in the world frame W	m/s^2
$\mathbf{g}$	gravity vector in W, about $(0, 0, -9.81)$	m/s^2
$\mathbf{R}_t^{\mathrm{BW}}$	rotation from world to body: expresses the world vector in sensor axes	—
$\mathbf{b}_t^{\omega}, \mathbf{b}_t^{a}$	**biases**: slowly drifting offsets (the big villains)	rad/s, m/s^2
$\mathbf{n}_t^{\omega}, \mathbf{n}_t^{a}$	white noise: fast random jitter	rad/s, m/s^2

In words: *reading = truth + slow offset + fast jitter*. Note the accelerometer equation: subtracting $\mathbf{g}$ (which points down) *adds* 9.81 upward — that is the "table pushing up" reading at rest.

7.3 Strapdown integration (LIO-SAM eqs. 4–6)

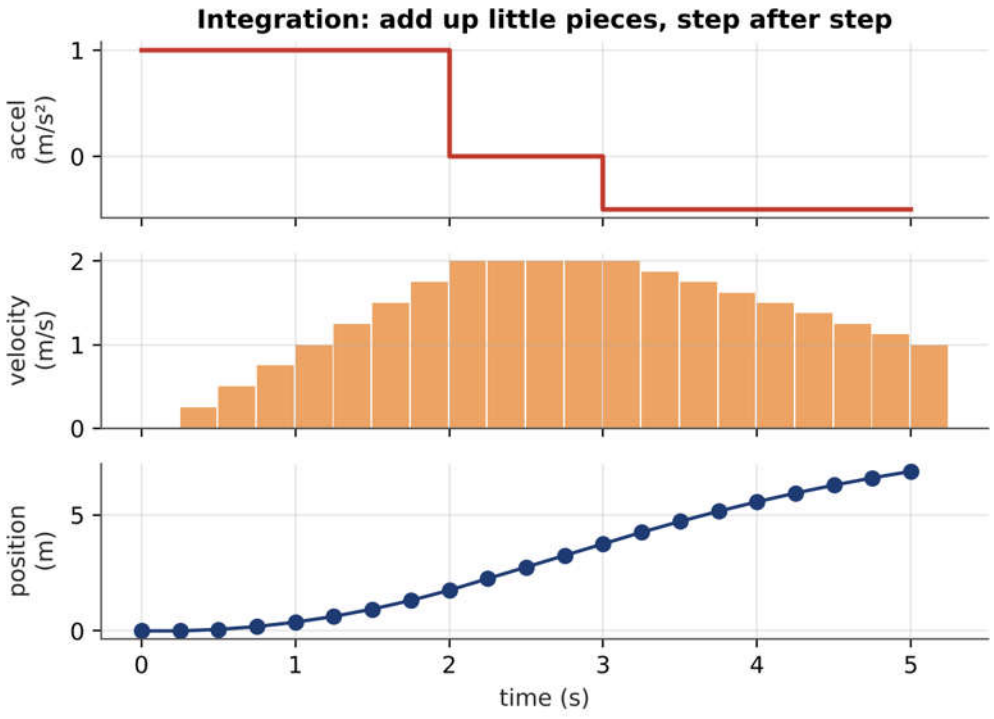


Figure 7.2: Acceleration is summed into velocity; velocity is summed into position.

KEY EQUATION — Dead-reckoning update over one IMU interval

$$\mathbf{v}_{t+\Delta t} = \mathbf{v}_t + \mathbf{g}\Delta t + \mathbf{R}_t(\hat{\mathbf{a}}_t - \mathbf{b}_t^a - \mathbf{n}_t^a)\Delta t$$
$$\mathbf{p}_{t+\Delta t} = \mathbf{p}_t + \mathbf{v}_t\Delta t + \tfrac{1}{2}\mathbf{g}\Delta t^2 + \tfrac{1}{2}\mathbf{R}_t(\hat{\mathbf{a}}_t - \mathbf{b}_t^a - \mathbf{n}_t^a)\Delta t^2$$
$$\mathbf{R}_{t+\Delta t} = \mathbf{R}_t \exp\big((\hat{\boldsymbol{\omega}}_t - \mathbf{b}_t^\omega - \mathbf{n}_t^\omega)\Delta t\big)$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{v}_t, \mathbf{p}_t$	velocity and position of the body in W	m/s, m
$\mathbf{R}_t$	body-to-world rotation (attitude) at time t	—
Δt	IMU sample interval (0.01 s at 100 Hz)	s
$\hat{\mathbf{a}}_t - \mathbf{b}_t^a$	the bias-corrected accelerometer reading (noise unknown, treated as zero mean)	m/s^2
$\mathbf{R}_t(\cdot)$	rotates the body-frame reading into world axes	—
$\mathbf{g}\Delta t$	puts gravity back so a sensor at rest does not "fly upward"	m/s
$\frac{1}{2}(\cdot)\Delta t^2$	the familiar $s = ut + \frac{1}{2}at^2$ from school physics	m

WORKED EXAMPLE — one strapdown step along a road

Given: 100 Hz ($\Delta t = 0.01$ s); level, heading straight along x ($\mathbf{R} = \mathbf{I}$); $v = 15.000$ m/s; position $x = 100.000$ m; accelerometer reads $\hat{a}_x = 0.80$ m/s^2, $\hat{a}_z = 9.81$ m/s^2; biases already removed.

Step 1 — vertical check

v_z: $\mathbf{g}\Delta t + \hat{a}_z\Delta t = (-9.81 + 9.81) \times 0.01 = 0$ ✓ (gravity cancels, the car stays on the road)

Step 2 — forward velocity

$v_x = 15.000 + 0.80 \times 0.01 = 15.008$ m/s

Step 3 — forward position

$x = 100.000 + 15.000 \times 0.01 + \frac{1}{2} \times 0.80 \times 0.01^2 = 100.000 + 0.15000 + 0.00004 = 100.15004$ m

Result: the vehicle moved 0.150 m and sped up by 8 mm/s. Repeat 100 times a second, in 3-D, with rotation.

7.4 Why IMUs drift: biases integrate

SIMPLE APPROACH

Imagine a scale that always reads 1 gram too heavy. Weigh one candy and the mistake is tiny. Weigh a thousand candies one by one and add them up — now you are off by a whole kilogram! An IMU adds up its readings thousands of times, so even a tiny error grows and grows.

KEY EQUATION — Error growth from constant biases

$$\delta\psi(t) = b_g\, t, \qquad \delta y(t) \approx \tfrac{1}{2}\, v\, b_g\, t^2; \qquad \delta v(t) = b_a\, t, \qquad \delta x(t) = \tfrac{1}{2}\, b_a\, t^2$$

What every symbol means

Symbol	Meaning	Units
b_g	constant gyro bias	rad/s
$\delta\psi$	heading error (grows linearly)	rad
v	vehicle speed	m/s
δy	sideways position error from heading error (grows with t^2)	m
b_a	constant accelerometer bias	m/s^2
$\delta v,\ \delta x$	velocity and along-track position errors	m/s, m

ENGINEER'S VIEW

Derivation of the sideways term. Heading error at time τ is $b_g\tau$. Driving at speed v, the sideways velocity error is $v\sin(b_g\tau) \approx v\, b_g\tau$. Integrating from 0 to t: $\delta y = \int_0^t v\, b_g\tau\, d\tau = \frac{1}{2} v\, b_g t^2$.

WORKED EXAMPLE — tactical IMU coasting through a tunnel

Given: gyro bias 1°/h, accel bias 0.5 mg, speed 15 m/s, 60 s with no GNSS or LiDAR aiding.

Step 1 — gyro bias in rad/s

$1^\circ/\text{h} = \frac{1}{3600}\,^\circ/\text{s} \times \frac{\pi}{180} = 4.848 \times 10^{-6}$ rad/s

Step 2 — heading error

$4.848 \times 10^{-6} \times 60 = 2.909 \times 10^{-4}$ rad $= 0.0167^\circ$

Step 3 — sideways error

$\frac{1}{2} \times 15 \times 4.848 \times 10^{-6} \times 60^2 = 0.131$ m

Step 4 — accel bias

$0.5\,\text{m}g = 0.5 \times 10^{-3} \times 9.80665 = 0.004903$ m/s^2

Step 5 — along-track error

$\frac{1}{2} \times 0.004903 \times 60^2 = 8.83$ m

Result: 13 cm sideways from the gyro, but almost 9 m along-track from the accelerometer — after only one minute! (After 30 s: 2.2 m.) This is why no survey system coasts on its IMU for long.

IMPROVED FROM THE SOURCE MATERIAL

The seminar computed the gyro-induced lateral error as $v \times t \times \delta\psi$ using the *final* heading error for the whole interval (and an extra 60 s), giving 0.26 m. Because the heading error grows from zero, the correct integral is $\frac{1}{2} v\, b_g t^2 = 0.13$ m for 60 s. The two agree only if the heading error were already fully present at the start.

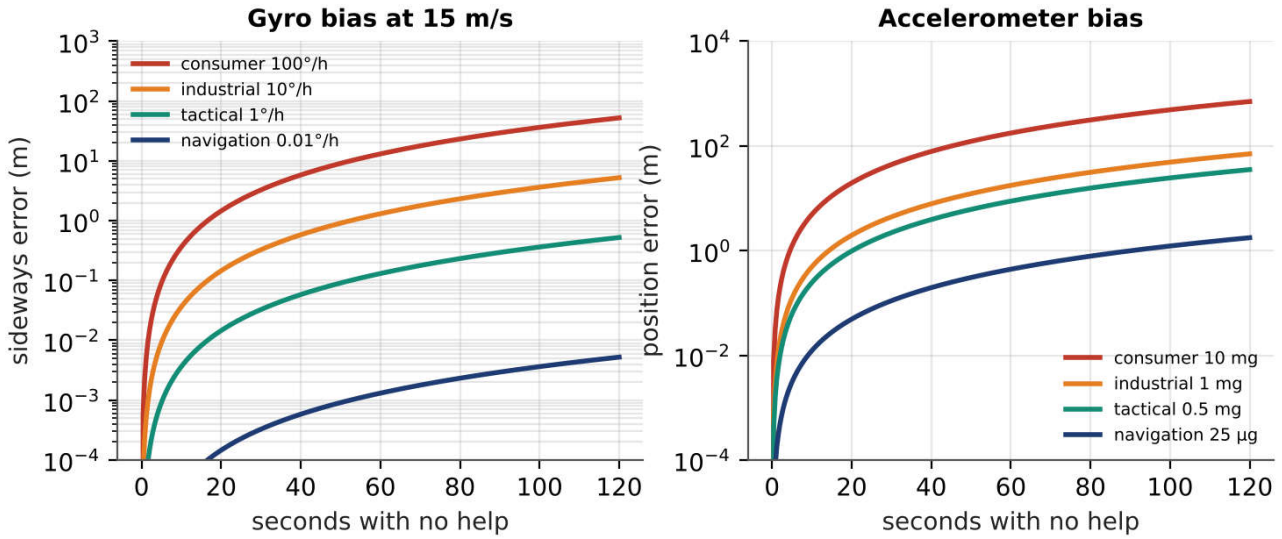


Figure 7.3: Position error versus unaided time for four IMU grades (log scale). Each grade is about ten times better than the one before.

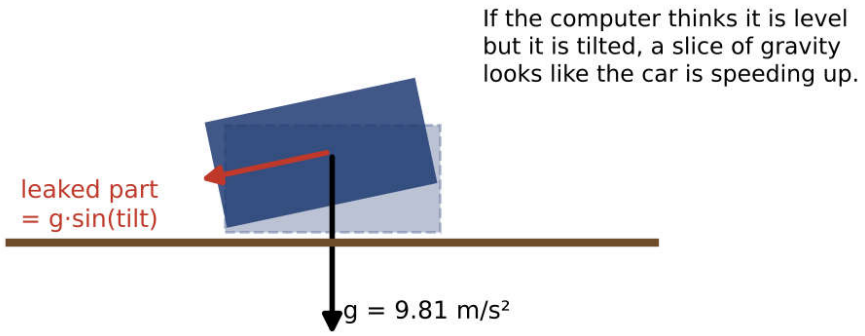


Figure 7.4: A small attitude error makes part of gravity look like acceleration.

7.5 The hidden killer: tilt error leaks gravity

WORKED EXAMPLE — 0.1 ° tilt error

Step 1

leaked acceleration $= g \sin 0.1^\circ = 9.80665 \times 0.001745 = 0.01712$ m/s^2 (that is 1.7 mg — worse than the accel bias above!)

Step 2

after 10 s: $\frac{1}{2} \times 0.01712 \times 10^2 = 0.86$ m

Result: attitude must be known very well, or gravity itself becomes the biggest error. Good INS filters estimate roll and pitch continuously from the gravity direction while the vehicle is not accelerating.

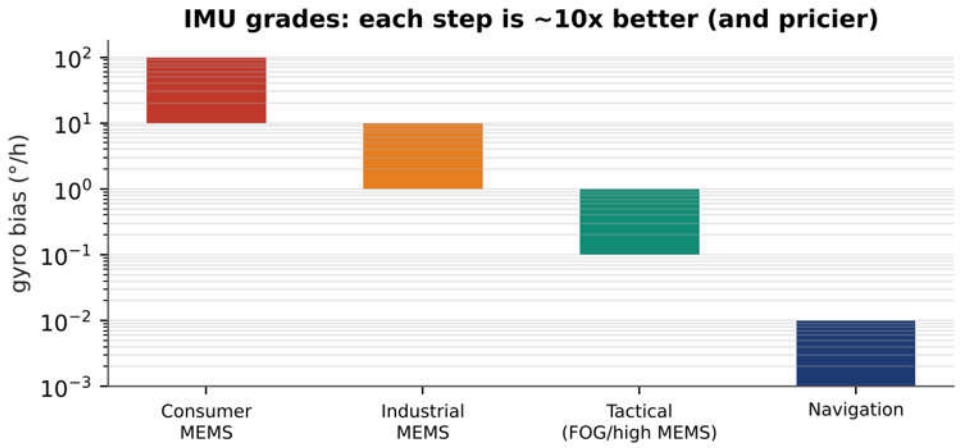


Figure 7.5: Typical gyro bias-instability ranges (order of magnitude).

7.6 IMU grades

Grade	Gyro bias	Accel bias	Unaided drift and use
Consumer MEMS	10–100+ °/h	several–tens mg	meters in seconds; phones, games
Industrial MEMS	1–10 °/h	0.1–1 mg	dm–m in 30–60 s; many handheld SLAM units
Tactical (FOG / high MEMS)	0.1–1 °/h	0.05–0.5 mg	bridges short GNSS outages; survey vehicles
Navigation	$\leq$0.01 °/h	$\leq$25 μg	aircraft, ships; rarely needed for street mapping

7.7 IMU preintegration (LIO-SAM eqs. 7–9)

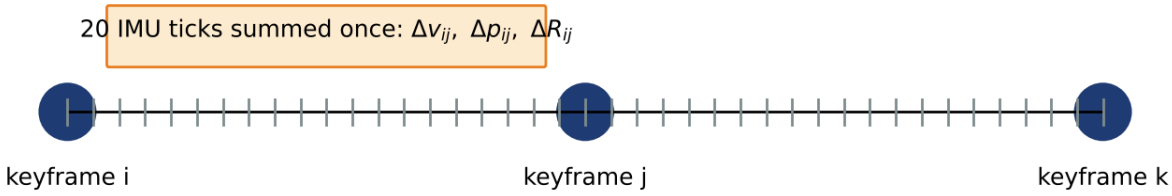


Figure 7.6: Between two keyframes, hundreds of IMU samples are summarized once into a relative motion.

SIMPLE APPROACH

Imagine writing down every step you take on a walk: “step, step, turn a little, step…” Instead of keeping that whole long list, you could write one note: “from the door to the swing: 20 steps forward and a quarter turn.” Preintegration is that short note.

KEY EQUATION — Preintegrated measurements between keyframes i and j

$$\Delta\mathbf{v}_{ij} = \mathbf{R}_i^\mathsf{T}(\mathbf{v}_j - \mathbf{v}_i - \mathbf{g}\Delta t_{ij}), \quad \Delta\mathbf{p}_{ij} = \mathbf{R}_i^\mathsf{T}\left(\mathbf{p}_j - \mathbf{p}_i - \mathbf{v}_i\Delta t_{ij} - \tfrac{1}{2}\mathbf{g}\Delta t_{ij}^2\right), \quad \Delta\mathbf{R}_{ij} = \mathbf{R}_i^\mathsf{T}\mathbf{R}_j$$

What every symbol means

Symbol	Meaning	Units
i, j	two consecutive keyframe times	—
Δt_{ij}	time between them	s
$\mathbf{R}_i^\mathsf{T}$	rotates world quantities into the body frame *at time i*, so the result does not depend on where the vehicle was in the world	—
$\Delta\mathbf{v}_{ij}$, $\Delta\mathbf{p}_{ij}$, $\Delta\mathbf{R}_{ij}$	change in velocity, position and attitude, with gravity and initial velocity removed — computable from IMU readings alone	m/s, m, —

ENGINEER'S VIEW

Why this is clever. The right-hand sides contain the unknown states (what the optimizer is solving for); the left-hand sides can be computed by summing raw IMU samples *once*, in the body frame at i, without knowing $\mathbf{R}_i$ or $\mathbf{v}_i$. When the optimizer later adjusts the poses, it does not have to re-integrate the hundreds of samples — it just compares the stored Δ's with the new state differences. If the bias estimate changes slightly, a first-order (Jacobian) correction is applied instead of re-integrating (Forster *et al.*, 2017). The mismatch between stored and predicted Δ's is the **IMU preintegration factor** in the factor graph, and the IMU biases are estimated jointly with everything else.

WORKED EXAMPLE — preintegration in one dimension

Given: level, straight road; between keyframes 10 IMU samples at 100 Hz ($\Delta t_{ij} = 0.1$ s), each reading $\hat{a} = 0.50$ m/s^2 forward (bias-corrected).

Step 1 — sum velocity

$\Delta v_{ij} = \sum \hat{a}\,\delta t = 10 \times 0.50 \times 0.01 = 0.050$ m/s

Step 2 — sum position

$\Delta p_{ij} = \frac{1}{2}\hat{a}\,\Delta t_{ij}^2 = \frac{1}{2} \times 0.50 \times 0.1^2 = 0.0025$ m

Step 3 — use it

if the optimizer says $v_i = 12.000$ m/s, then it must find $v_j = 12.050$ m/s and $p_j - p_i = 12.000 \times 0.1 + 0.0025 = 1.2025$ m; any difference is a residual weighted by the IMU noise.

Result: $\Delta v = 0.050$ m/s, $\Delta p = 2.5$ mm (beyond $v_i\Delta t$). Ten numbers became two.

TRY IT YOURSELF

7.1 A consumer IMU has a 10 mg accel bias. How far off is it after 10 s unaided?

7.2 An industrial gyro has a 10°/h bias. Find the sideways error after 30 s at 20 m/s.

7.3 An IMU at rest reads $\hat{a}_z = 9.83$ m/s^2 while local gravity is 9.80 m/s^2. Estimate the z accelerometer bias.

Answers: worked solutions are in Appendix A, “Answers to Try It Yourself,” starting on page 158.

SOURCES FOR THIS MODULE

Shan *et al.* (2020), eqs. (2)–(9); Forster, C. *et al.* (2017), On-manifold preintegration, *IEEE Trans. Robotics* 33(1); Groves (2013), IMU error models and grades; MicroStrain 3DM-GX5 and 3DM-GQ7 datasheets (example bias values).

MODULE 8

Timing, GNSS and Motion Compensation

8.1 One clock to rule them all

SIMPLE APPROACH

Imagine you and a friend take photos of a race. If your camera's clock is one second slow, your photo of the runner crossing the line will be labeled at the wrong time — and the runner will seem to be in the wrong place. Every sensor on a mapping car must use the *same* clock, set perfectly, so every laser dot gets matched to exactly where the car was at that instant.

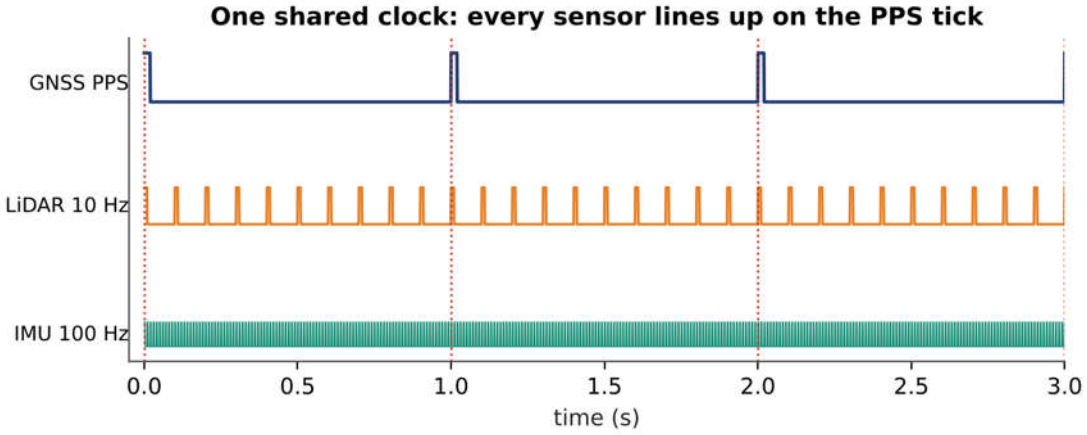


Figure 8.1: GNSS PPS, LiDAR and IMU events all share one time base.

ENGINEER'S VIEW

A GNSS receiver outputs a **PPS** (pulse-per-second) electrical edge aligned to GNSS time to within tens of nanoseconds, plus a serial/NMEA message saying *which* second that edge was. The LiDAR and IMU discipline their internal clocks to this edge (hardware synchronization), or the system uses the IEEE 1588 Precision Time Protocol (PTP) over Ethernet. Every point then carries a timestamp in one common time base. If the LiDAR's driver instead stamps packets when the computer *receives* them (software time), latency and jitter of milliseconds are common — unacceptable for survey work.

8.2 How far does a timing error move a point?

KEY EQUATION — Timing-error displacement

$$\delta_{\text{along}} = v\,\delta t \qquad \delta_{\text{lat}} \approx \rho\,\dot{\psi}\,\delta t$$

What every symbol means

Symbol	Meaning	Units
δt	time offset between LiDAR and trajectory clocks	s
v	platform speed	m/s
δ_{along}	resulting shift along the direction of travel	m
$\dot{\psi}$	turning (yaw) rate — the dot means "rate of change"	rad/s
ρ	range to the point	m
δ_{lat}	sideways shift caused by the rotation that happened during δt	m

WORKED EXAMPLE — a 1 ms clock offset

Given: $\delta t = 1$ ms $= 0.001$ s; driving 20 m/s (72 km/h); turning a corner at 30°/s; point 50 m away.

Step 1 — along-track

$20 \times 0.001 = 0.020$ m $= 2.0$ cm

Step 2 — rotation during the offset

$30°/\text{s} \times 0.001$ s $= 0.03° = 0.000524$ rad

Step 3 — sideways at 50 m

$50 \times 0.000524 = 0.026$ m $= 2.6$ cm

Result: 2 cm along-track on straights, plus 2.6 cm sideways in turns. Because the second term depends on turning, the error shows up as "corners look smeared but straights look fine" — a classic timing-error signature.

IMPROVED FROM THE SOURCE MATERIAL

The seminar computed only the along-track term. In practice the rotational term is often larger (scanner heads on vehicles turn fast and points are far away), and it is the one that makes buildings "double" at intersections. Both terms are included here.

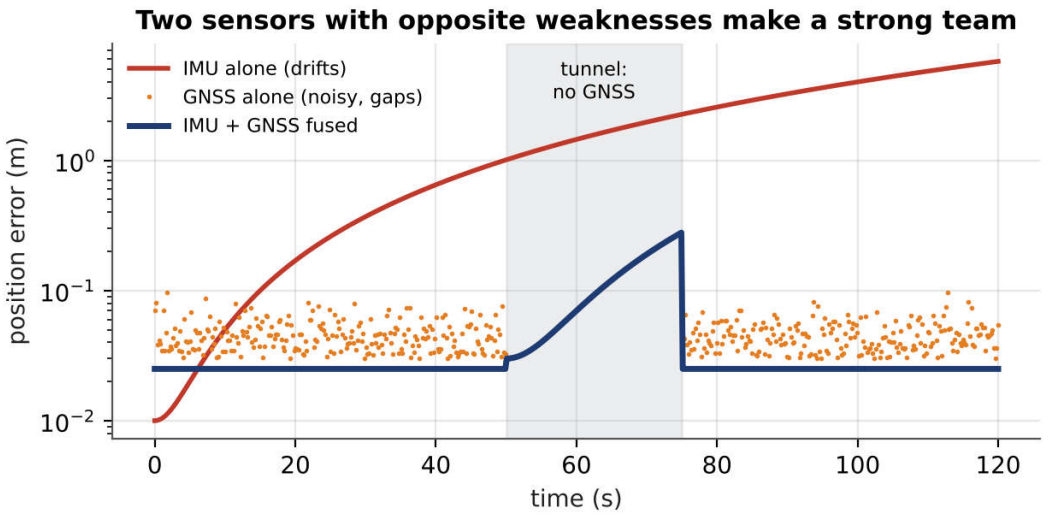


Figure 8.2: The IMU drifts but never stops; GNSS never drifts but can disappear. Together they are far better than either.

8.3 GNSS and IMU: a perfect partnership

	GNSS	IMU
Measures	absolute position (cm-level with RTK/PPK), velocity	angular rate, specific force
Rate	1–20 Hz	100–1000 Hz
Long-term	no drift (bounded error)	unbounded drift
Short-term	noisy, multipath, outages	smooth, precise
Fails in	tunnels, canopy, urban canyons, indoors	never "fails" — just drifts
Attitude	weak (needs two antennas for heading)	full roll, pitch, yaw

ENGINEER'S VIEW

Loosely coupled fusion feeds GNSS position solutions into a Kalman filter that corrects the INS. **Tightly coupled** fusion feeds raw pseudoranges/carrier phases, so even 2–3 visible satellites still help in an urban canyon. Survey trajectories are usually post-processed with a forward-and-backward *smoother* (PPK), which uses future data to improve past estimates — the same "smoothing" idea that LIO-SAM's name refers to.

8.4 Motion compensation (de-skewing)

SIMPLE APPROACH

Take a photo of a fast car with a slow camera and the car looks stretched and blurry. A spinning LiDAR takes 1/10 of a second to sweep all the way around, and the vehicle moves during that time. If the computer pretends the whole sweep happened at one instant, straight walls come out bent. De-skewing fixes the blur by moving each dot back to where it really was.

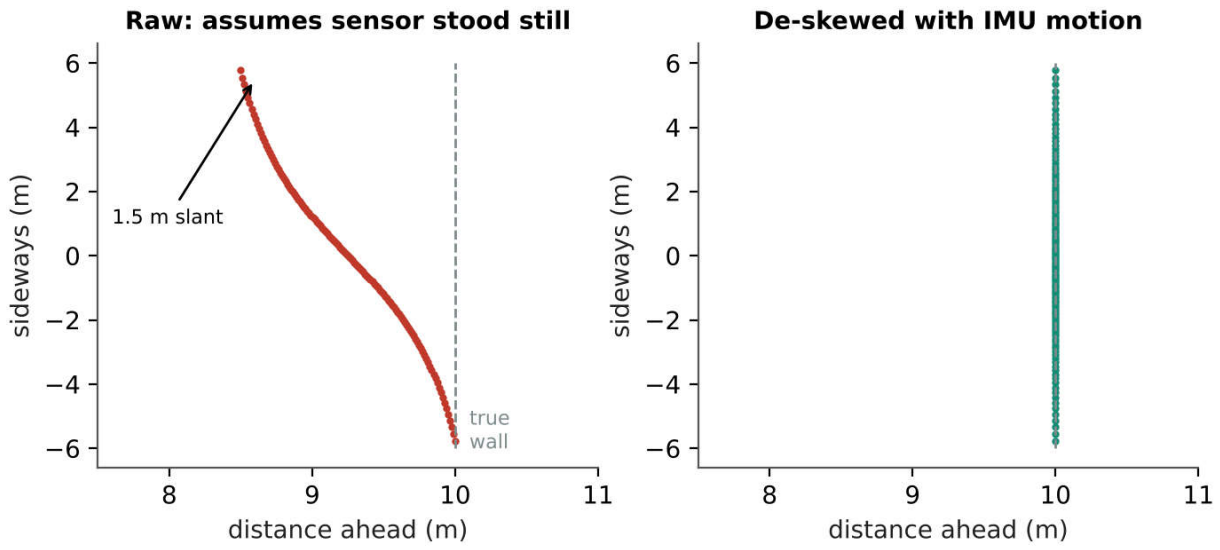


Figure 8.3: A wall scanned while driving toward it at 15 m/s. Left: raw sweep. Right: after de-skewing each point with its own pose.

KEY EQUATION — Per-point de-skew

$$\mathbf{r}_W^{(k)} = \mathbf{T}(t_0 + \tau_k)\,\mathbf{r}_s^{(k)} \qquad \text{or, in the scan's start frame:} \qquad \mathbf{r}_{s_0}^{(k)} = \mathbf{T}(t_0)^{-1}\,\mathbf{T}(t_0 + \tau_k)\,\mathbf{r}_s^{(k)}$$

What every symbol means

Symbol	Meaning	Units
k	index of the point inside one sweep	—
τ_k	the point's time offset within the sweep (the per-point "time" field in the LiDAR message)	s
t_0	the sweep's reference time (start or middle)	s
$\mathbf{T}(t)$	sensor pose at time t, interpolated from the high-rate IMU trajectory	—
$\mathbf{r}_s^{(k)}$	the point as measured, in the sensor frame	m

WORKED EXAMPLE — how big is the skew?

Given: 10 Hz sweep (0.1 s), vehicle at 15 m/s; a point fired at $\tau = 0.075$ s into the sweep; reference t_0 = sweep start.

Step 1 — distance moved

$15 \times 0.075 = 1.125$ m

Step 2 — whole sweep

first-to-last difference $= 15 \times 0.1 = 1.5$ m

Step 3 — correction

shift this point by 1.125 m in the direction of travel (expressed in the start frame)

Result: without de-skewing, points within one sweep are up to 1.5 m inconsistent — orders of magnitude beyond tolerance. LIO-SAM uses its IMU-predicted motion for exactly this step before any matching is done.

SURVEYOR'S FIELD TIP

A practical QA check before processing: confirm that (1) the IMU stream is present at its nominal rate with no gaps, (2) every LiDAR point has a per-point time field, (3) LiDAR and IMU timestamps overlap and are in the same time base, and (4) the static lever-arm/boresight transform is loaded. If any of these is missing, stop — no amount of processing will fix it later.

TRY IT YOURSELF

8.1 At highway speed (30 m/s), what along-track error does a 0.5 ms offset cause?
8.2 A backpack operator turns their body at 90°/s. With a 2 ms offset, what is the sideways error at 15 m?
8.3 (Simple) Why does a photo of a fast-moving car look stretched?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Ouster OS0 Rev8 datasheet (PTP/PPS sync < 1 ms, timestamp < 1 μs); IEEE 1588 Precision Time Protocol; Groves (2013), GNSS/INS integration; Shan *et al.* (2020), de-skewing.

MODULE 9

Lever Arm and Boresight Calibration

9.1 Two boxes on one plate

SIMPLE APPROACH

Put a toy camera on your shoulder and close your eyes. A friend asks, "Where is the camera?" You know where *you* are, but the camera is a little to the side and a little higher — and it might be turned slightly. If you forget that, you will point to the wrong place. The car's "body" (the IMU) and its laser "eye" are in different spots too. The **lever arm** is how far apart they are. The **boresight** is how much one is twisted compared to the other.

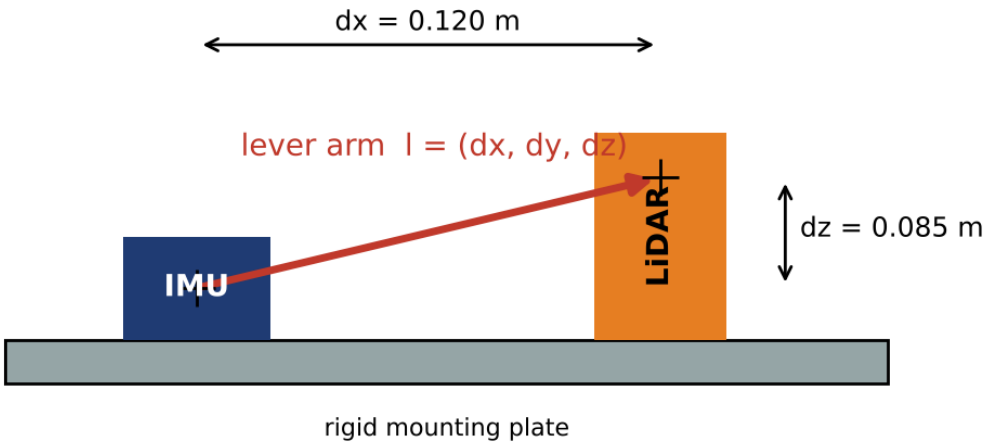


Figure 9.1: The lever arm is the vector from the IMU origin to the LiDAR origin.

ENGINEER'S VIEW

The trajectory solution describes the motion of the IMU (body frame B). The LiDAR measures in its own frame S. Two fixed calibration quantities connect them:

- **Lever arm** $\mathbf{l}^B = (d_x, d_y, d_z)$: position of the LiDAR's optical center in B. Measured with a total station, a calibration bay or the manufacturer's CAD; mm-level is needed.
- **Boresight** $\mathbf{R}_S^B$: the small rotation from S to B, three angles $(\delta\phi, \delta\theta, \delta\psi)$. These cannot be measured with a tape — they are estimated from data (see below).

A GNSS antenna has its own lever arm too; an error there shifts the whole trajectory.

9.2 The direct georeferencing equation

This is the single most important equation in mobile mapping. Every point in every deliverable went through it.

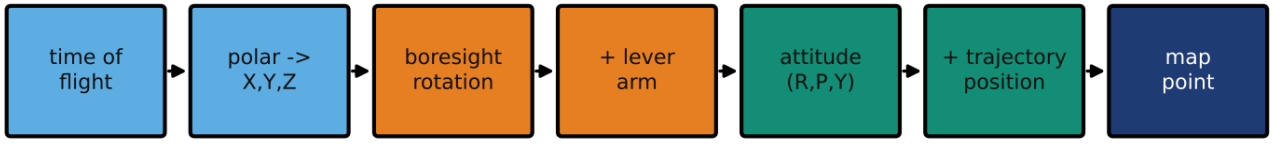


Figure 9.2: The chain of operations from a laser echo to a map coordinate.

KEY EQUATION — Direct georeferencing

$$\mathbf{r}_W = \mathbf{p}_W(t) + \mathbf{R}_B^W(t)\left(\mathbf{R}_S^B\,\mathbf{r}_S + \mathbf{l}^B\right)$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{r}_W$	the final point in the world/project frame	m
$\mathbf{p}_W(t)$	IMU position at the pulse time t (from the trajectory)	m
$\mathbf{R}_B^W(t)$	attitude: body-to-world rotation from roll, pitch, yaw at t	—
$\mathbf{R}_S^B$	boresight: sensor-to-body rotation (fixed)	—
$\mathbf{r}_S$	the point in the LiDAR frame (from polar-to-Cartesian, Module 5)	m
$\mathbf{l}^B$	lever arm: LiDAR origin in the body frame (fixed)	m

Read it from the inside out: (1) turn the LiDAR's point into body axes (boresight), (2) shift it to the IMU origin (lever arm), (3) rotate into world axes (attitude), (4) add where the IMU was (position).

SURVEYOR'S FIELD TIP

In ROS-based systems the lever arm and boresight live in the static transform `/tf_static` between `imu_link` and the LiDAR frame, e.g. translation $[0.120,\ 0.000,\ 0.085]$ m and roll-pitch-yaw $[0.11^\circ,\ -0.23^\circ,\ 0.01^\circ]$. If that transform is missing or wrong, you get the bird's nest of Module 1.

9.3 Effect of calibration errors

KEY EQUATION — Calibration-error rules of thumb

$$\delta r_{\text{lever}} \approx \|\delta\mathbf{l}\| \qquad \delta r_{\text{bore}} \approx \rho\,\delta\alpha$$

What every symbol means

Symbol	Meaning	Units
$\delta\mathbf{l}$	error in the lever arm	m
δr_{lever}	resulting point error: the *same size at every range*, but its direction rotates with the vehicle's heading	m
$\delta\alpha$	residual boresight angle	rad
δr_{bore}	resulting point error: *grows with range*	m

WORKED EXAMPLE — boresight residual at mapping range

Given: residual boresight 0.05°; target 80 m away.

Step 1

$0.05 \times \pi/180 = 8.727 \times 10^{-4}$ rad

Step 2

$80 \times 8.727 \times 10^{-4} = 0.0698$ m

Result: 7.0 cm. Compare: a 7 mm lever-arm error causes 7 mm everywhere. Boresight dominates at long range; lever arm dominates close in.

9.4 How boresight is calibrated

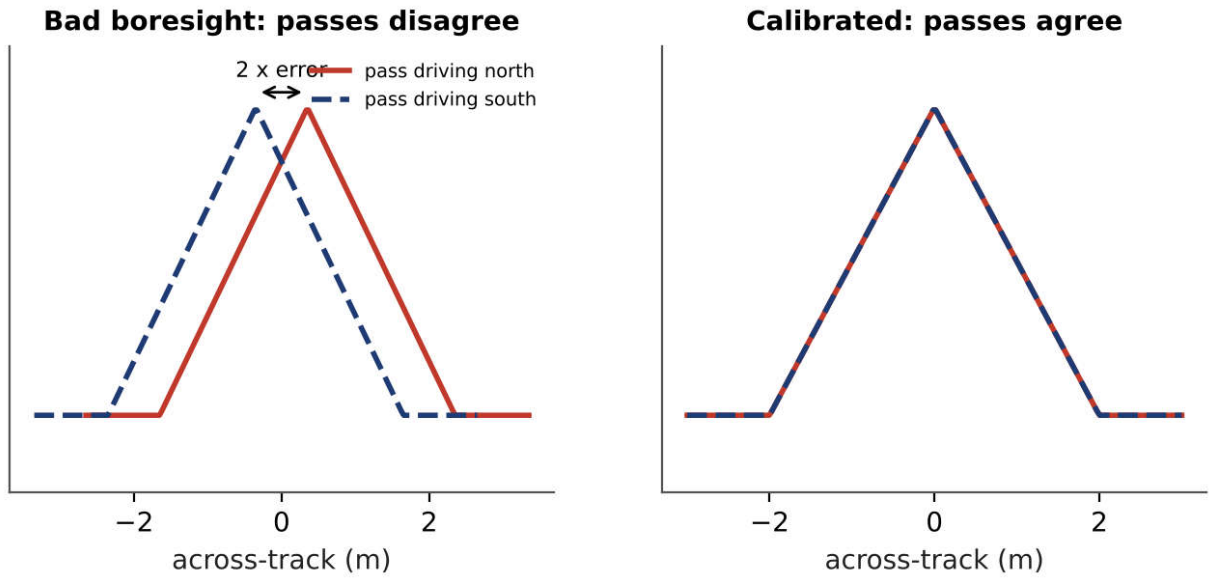


Figure 9.3: Driving past the same building in opposite directions doubles the visible effect of a boresight error.

ENGINEER'S VIEW

Because a boresight error rotates with the vehicle, scanning the same features from **opposite directions** and **different ranges** makes it visible: a roll error tilts the road one way on the northbound pass and the other way on the southbound pass; a yaw error shifts facades forward on one pass and backward on the other. Calibration software extracts planes (road, walls, roofs) or targets from overlapping passes and solves, by least squares, for the three angles (and optionally lever-arm refinements) that minimize the plane-to-plane disagreement. A good calibration site has:

- planar surfaces in many orientations (walls facing different ways, flat road, sloped roofs);
- passes in opposite directions, at two or more offsets/ranges, plus a turn;
- surveyed targets or control for an absolute check.

SURVEYOR'S FIELD TIP

Treat calibration like any instrument adjustment: re-check after any mount change, impact, or long storage, and at the start of every project with a short loop over known features. Document the values used for every dataset.

9.5 Full worked example: one pulse to project coordinates

WORKED EXAMPLE — direct georeferencing, every step

Given (FLU body frame, ENU world, ZYX angles): IMU position $\mathbf{p}_W = (1000.000,\ 2000.000,\ 10.000)$ m; roll $1.0°$, pitch $-0.5°$, yaw $30.0°$. Lever arm $\mathbf{l} = (0.120,\ 0.000,\ 0.085)$ m. Boresight $(0.05°,\ -0.03°,\ 0.02°)$. Laser: $\rho = 25.000$ m, $H = 10.0°$, $V = 2.0°$.

Step 1 — polar to Cartesian (Module 5)

$x_S = 25\cos 2°\cos 10° = 24.6052$; $y_S = 25\cos 2°\sin 10° = 4.3386$; $z_S = 25\sin 2° = 0.8725$ m

Step 2 — boresight rotation $\mathbf{R}_S^B = \mathbf{R}_z(0.02°)\mathbf{R}_y(-0.03°)\mathbf{R}_x(0.05°)$

For tiny angles $\mathbf{R} \approx \begin{pmatrix} 1 & -\delta\psi & \delta\theta \\ \delta\psi & 1 & -\delta\phi \\ -\delta\theta & \delta\phi & 1 \end{pmatrix}$ with $\delta\phi = 0.000873$, $\delta\theta = -0.000524$, $\delta\psi = 0.000349$ rad:

$x = 24.6052 - 0.000349(4.3386) - 0.000524(0.8725) = 24.6032$

$y = 0.000349(24.6052) + 4.3386 - 0.000873(0.8725) = 4.3464$

$z = 0.000524(24.6052) + 0.000873(4.3386) + 0.8725 = 0.8892$ m

Step 3 — add the lever arm

$\mathbf{r}_B = (24.6032 + 0.120,\ 4.3464 + 0,\ 0.8892 + 0.085) = (24.7232,\ 4.3464,\ 0.9742)$ m

Step 4 — attitude $\mathbf{R}_B^W = \mathbf{R}_z(30°)\mathbf{R}_y(-0.5°)\mathbf{R}_x(1.0°)$

Multiplying out gives $\mathbf{R}_B^W = \begin{pmatrix} 0.86599 & -0.50006 & 0.00117 \\ 0.49998 & 0.86582 & -0.01948 \\ 0.00873 & 0.01745 & 0.99981 \end{pmatrix}$.

Row by row: $X = 0.86599(24.7232) - 0.50006(4.3464) + 0.00117(0.9742) = 19.2378$

$Y = 0.49998(24.7232) + 0.86582(4.3464) - 0.01948(0.9742) = 16.1053$

$Z = 0.00873(24.7232) + 0.01745(4.3464) + 0.99981(0.9742) = 1.2656$ m

Step 5 — add the trajectory position

$\mathbf{r}_W = (1000 + 19.2378,\ 2000 + 16.1053,\ 10 + 1.2656)$

Result: $\mathbf{r}_W = (1019.238,\ 2016.105,\ 11.266)$ m.

IMPROVED FROM THE SOURCE MATERIAL

The seminar's version of this example applied only the yaw rotation and stated that roll and pitch "would adjust this by centimeters." Doing the full rotation shows the height is actually **31 cm** higher (11.266 m versus 10.957 m): the $0.5°$ nose-up pitch lifts a point 24.7 m ahead by $24.7 \times \sin 0.5° = 0.216$ m, and the $1°$ roll adds $4.35 \times \sin 1° = 0.076$ m. Roll and pitch must never be ignored — this is exactly why a leveled total station and an unleveled moving vehicle are different problems. (In this FLU convention a negative rotation about y is nose up.) The boresight alone moved the point by about 1.9 cm, mostly vertically.

TRY IT YOURSELF

9.1 A lever-arm d_z was entered as 0.850 m instead of 0.085 m. How big is the resulting error, and does it depend on range?
9.2 What residual boresight keeps errors under 2 cm at 60 m?
9.3 Repeat Step 4 of the worked example with roll $=$ pitch $= 0$. What is Z?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Glennie, C. (2007), Rigorous 3D error analysis of kinematic scanning LIDAR systems, *J. Applied Geodesy* 1; Skaloud, J. & Lichti, D. (2006), Rigorous approach to bore-sight self-calibration in airborne laser scanning, *ISPRS J.* 61; *Mobile LiDAR Basics* seminar (2026), worked example (corrected here).

MODULE 10

SLAM: Mapping While Finding Yourself

10.1 The chicken-and-egg problem

SIMPLE APPROACH

You are blindfolded in a new house and must draw a map of it. To draw a wall in the right place, you need to know where you are standing. But to know where you are standing, you need a map! SLAM is the clever trick of doing both a little at a time: take a step, feel a wall, guess where you are, add the wall to your map, take another step… and when you bump into a wall you already drew, you fix up your whole map.

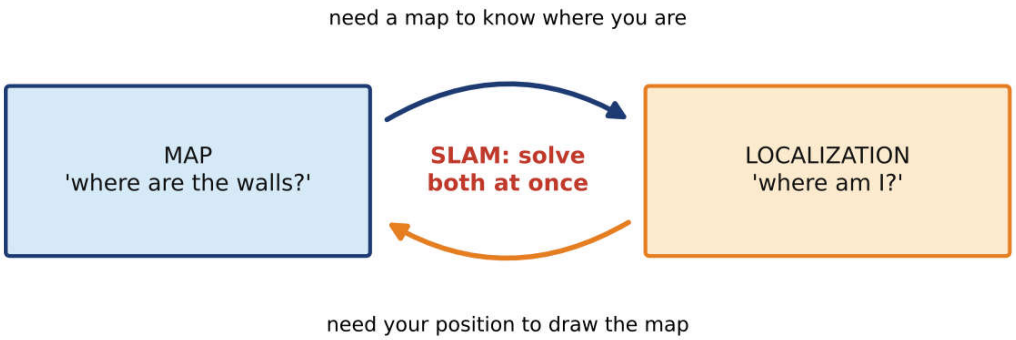


Figure 10.1: Simultaneous Localization And Mapping solves both questions together.

ENGINEER'S VIEW

SLAM estimates the trajectory $\mathbf{x}_{0:t}$ and the map $\mathbf{m}$ jointly from controls/odometry $\mathbf{u}_{1:t}$ and observations $\mathbf{z}_{1:t}$:

$$p(\mathbf{x}_{0:t}, \mathbf{m} \mid \mathbf{z}_{1:t}, \mathbf{u}_{1:t}).$$

Surveyors know this problem as *a traverse run through new terrain while simultaneously locating the detail points*; each new shot both extends the traverse and adds map points, and the whole network is adjusted at the end. This module builds the classical tools — motion model, measurement model, Bayes filter, Kalman filters, particle filters — following Claus Brenner's well-known SLAM lecture series. Modules 11 and 12 add scan matching and the modern factor-graph form used by LIO-SAM.

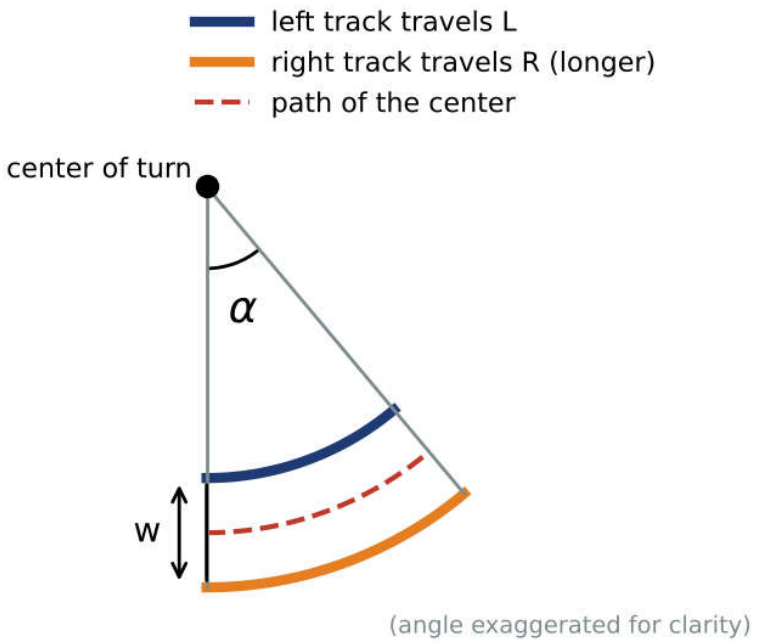


Figure 10.2: A two-track (differential drive) platform: when one side travels farther, the platform turns.

10.2 The motion model

KEY EQUATION — Differential-drive motion model

$$\alpha = \frac{R - L}{w}, \qquad r = \frac{L}{\alpha}, \qquad \begin{aligned} x' &= x + \left(r + \tfrac{w}{2}\right)\left(\sin(\theta + \alpha) - \sin\theta\right) \\ y' &= y - \left(r + \tfrac{w}{2}\right)\left(\cos(\theta + \alpha) - \cos\theta\right) \\ \theta' &= \theta + \alpha \end{aligned}$$

What every symbol means

Symbol	Meaning	Units
x, y, θ	pose before the move (position and heading)	m, m, rad
L, R	distance traveled by the left and right tracks	m
w	track width (distance between the tracks)	m
α	how much the platform turned	rad
r	turning radius of the *left* track; $r + w/2$ is the radius of the center	m
x', y', θ'	predicted pose after the move	m, m, rad

If $L = R$ the platform drives straight: $x' = x + L\cos\theta$, $y' = y + L\sin\theta$, $\theta' = \theta$.

WORKED EXAMPLE — predicting the next pose

Given: pose $(10.00,\ 5.00,\ 0°)$; $L = 0.40$ m, $R = 0.45$ m, $w = 0.30$ m.

Step 1 — turn angle

$\alpha = (0.45 - 0.40)/0.30 = 0.16667$ rad $= 9.549°$

Step 2 — radius

$r = 0.40/0.16667 = 2.400$ m; center radius $r + w/2 = 2.550$ m

Step 3 — x'
$10 + 2.550(\sin 9.549^\circ - \sin 0^\circ) = 10 + 2.550(0.16590) = 10.4230$ m

Step 4 — y'
$5 - 2.550(\cos 9.549^\circ - 1) = 5 - 2.550(-0.01386) = 5.0353$ m

Step 5 — θ'
$0 + 9.549^\circ = 9.549^\circ$

Result: new pose $(10.423,\ 5.035,\ 9.55^\circ)$: about 42 cm forward while turning 9.55° left.

On a mapping vehicle the same "prediction" role is played by IMU strapdown integration (Module 7) or wheel odometry.

10.3 Uncertainty grows when you move and shrinks when you look

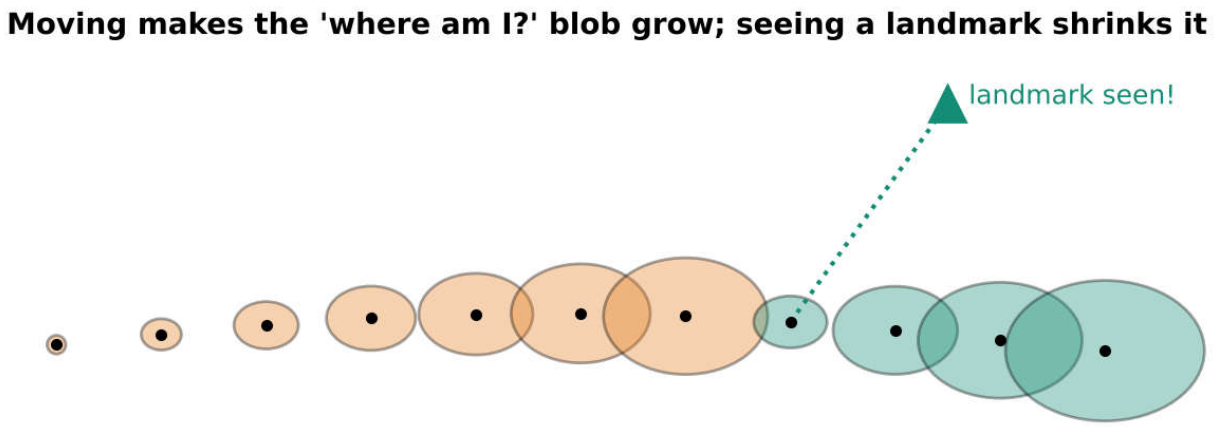


Figure 10.3: Each prediction inflates the 2σ uncertainty ellipse; one good landmark observation collapses it.

KEY EQUATION — The Bayes filter — predict, then correct

$$\underbrace{\overline{bel}(\mathbf{x}_t) = \int p(\mathbf{x}_t \mid \mathbf{u}_t, \mathbf{x}_{t-1})\, bel(\mathbf{x}_{t-1})\, d\mathbf{x}_{t-1}}_{\text{PREDICT (move)}} \qquad \underbrace{bel(\mathbf{x}_t) = \eta\, p(\mathbf{z}_t \mid \mathbf{x}_t)\, \overline{bel}(\mathbf{x}_t)}_{\text{CORRECT (measure)}}$$

What every symbol means

Symbol	Meaning	Units
$bel(\mathbf{x}_t)$	"belief": probability distribution of where we are at time t	—
$\overline{bel}$	predicted belief, before using the new measurement	—
$p(\mathbf{x}_t \mid \mathbf{u}_t, \mathbf{x}_{t-1})$	motion model with noise: where could I be after this control?	—
$p(\mathbf{z}_t \mid \mathbf{x}_t)$	measurement model (likelihood): how probable is this observation if I'm at $\mathbf{x}_t$?	—
η	normalizing constant so probabilities add to 1	—
$\int \ldots d\mathbf{x}_{t-1}$	add up over every place we might have started	—

10.4 The Kalman filter in one dimension

SIMPLE APPROACH

Two friends guess how tall a tree is. One is a careful measurer; the other is just eyeballing. You trust the careful one more, so your final answer is *between* their guesses but closer to the careful one. The Kalman filter does exactly that, with numbers that say *how much* closer.

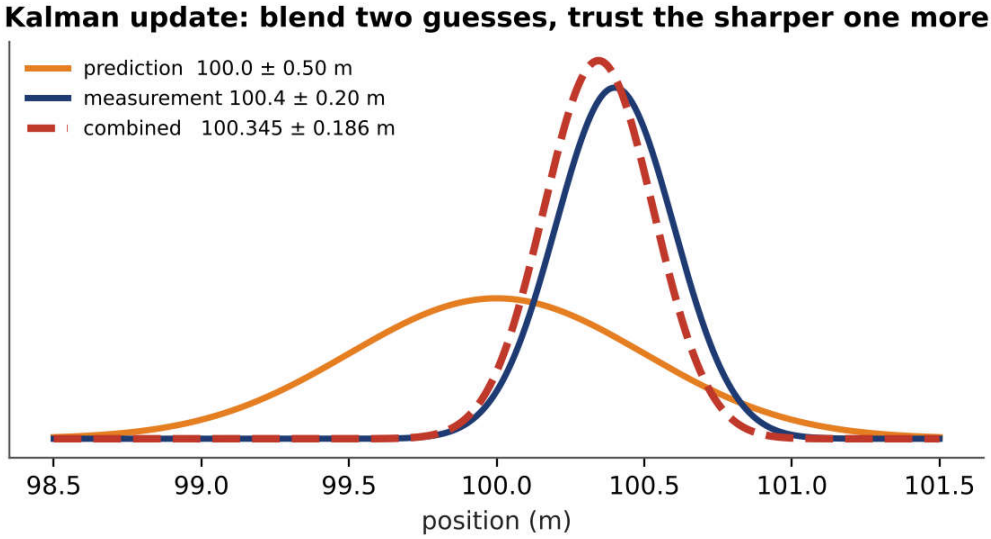


Figure 10.4: A wide prediction and a sharp measurement combine into an even sharper estimate.

KEY EQUATION — 1-D Kalman update

$$K = \frac{\bar{\sigma}^2}{\bar{\sigma}^2 + \sigma_z^2}, \qquad \mu = \bar{\mu} + K(z - \bar{\mu}), \qquad \sigma^2 = (1 - K)\,\bar{\sigma}^2$$

What every symbol means

Symbol	Meaning	Units
$\bar{\mu}, \bar{\sigma}^2$	predicted value and its variance	m, m^2
z, σ_z^2	measurement and its variance	m, m^2
K	Kalman gain, between 0 (ignore the measurement) and 1 (believe it completely)	—
$z - \bar{\mu}$	the *innovation*: how surprised we are	m
μ, σ^2	updated (posterior) value and variance — always smaller than both inputs' variances	m, m^2

WORKED EXAMPLE — 1-D Kalman update

Given: prediction $\bar{\mu} = 100.0$ m, $\bar{\sigma} = 0.50$ m; measurement $z = 100.4$ m, $\sigma_z = 0.20$ m.

Step 1 — variances

$\bar{\sigma}^2 = 0.25$ m^2; $\sigma_z^2 = 0.04$ m^2

Step 2 — gain

$K = 0.25/(0.25 + 0.04) = 0.8621$

Step 3 — update

$\mu = 100.0 + 0.8621 \times 0.4 = 100.345$ m

Step 4 — new variance

$\sigma^2 = (1 - 0.8621) \times 0.25 = 0.03448$ m^2 $\Rightarrow$ $\sigma = 0.186$ m

Result: 100.345 ± 0.186 m. Note that 0.186 m beats even the measurement's 0.20 m.

ENGINEER'S VIEW

Surveyor's equivalence. The 1-D Kalman update is the weighted mean you already know. With weights $w = 1/\sigma^2$: $\mu = (w_1\bar{\mu} + w_2 z)/(w_1 + w_2) = (4 \times 100.0 + 25 \times 100.4)/29 = 100.345$ m, and $\sigma^2 = 1/(w_1 + w_2) = 1/29 = 0.0345$ m^2. Same answer.

10.5 The extended Kalman filter (EKF)

Real motion and measurement models contain sines and cosines — they are *non-linear*. The EKF linearizes them at the current estimate using **Jacobians** (tables of partial derivatives: "if I nudge this input, how much does each output change?").

KEY EQUATION — EKF prediction

$$\bar{\mathbf{x}} = g(\mathbf{x}, \mathbf{u}), \qquad \bar{\mathbf{P}} = \mathbf{G}\,\mathbf{P}\,\mathbf{G}^{\mathsf{T}} + \mathbf{Q}, \qquad \mathbf{G} = \frac{\partial g}{\partial \mathbf{x}}$$

What every symbol means

Symbol	Meaning	Units
g	the motion model (previous section)	—
$\mathbf{P}, \bar{\mathbf{P}}$	covariance matrix of the state before and after prediction; its diagonal holds the variances	m^2, rad^2
$\mathbf{G}$	state-transition Jacobian	—
$\mathbf{Q}$	process noise: extra uncertainty added by the motion itself	m^2, rad^2

WORKED EXAMPLE — EKF prediction for our motion

Given: the motion above; $\mathbf{P} = \mathrm{diag}(0.01, 0.01, 0.001)$, $\mathbf{Q} = \mathrm{diag}(0.002, 0.002, 0.0005)$.

Step 1 — Jacobian

Differentiate x' and y' by θ: $\partial x'/\partial\theta = 2.55(\cos 9.549^\circ - 1) = -0.0353$, $\partial y'/\partial\theta = 2.55 \sin 9.549^\circ = 0.4230$:

$$\mathbf{G} = \begin{pmatrix} 1 & 0 & -0.0353 \\ 0 & 1 & 0.4230 \\ 0 & 0 & 1 \end{pmatrix}$$

Step 2 — $\mathbf{G}\mathbf{P}\mathbf{G}^{\mathsf{T}}$

$\begin{pmatrix} 0.01000 & -0.00001 & -0.00004 \\ -0.00001 & 0.01018 & 0.00042 \\ -0.00004 & 0.00042 & 0.00100 \end{pmatrix}$ (heading uncertainty leaks into y)

Step 3 — add $\mathbf{Q}$

$$\bar{\mathbf{P}} = \begin{pmatrix} 0.01200 & -0.00001 & -0.00004 \\ -0.00001 & 0.01218 & 0.00042 \\ -0.00004 & 0.00042 & 0.00150 \end{pmatrix}$$

Result: 1σ: x 11.0 cm, y 11.0 cm, heading $\sqrt{0.0015} = 0.0387$ rad $= 2.22°$. Uncertainty grew, as expected.

10.5.1 Correcting with a landmark

KEY EQUATION — Range–bearing measurement model and EKF correction

$$\hat{r} = \sqrt{\Delta x^2 + \Delta y^2}, \quad \hat{\beta} = \operatorname{atan2}(\Delta y, \Delta x) - \theta; \qquad \mathbf{S} = \mathbf{H}\bar{\mathbf{P}}\mathbf{H}^{\mathsf{T}} + \mathbf{R}_z, \quad \mathbf{K} = \bar{\mathbf{P}}\mathbf{H}^{\mathsf{T}}\mathbf{S}^{-1}$$

$$\mathbf{x} = \bar{\mathbf{x}} + \mathbf{K}(\mathbf{z} - \hat{\mathbf{z}}), \qquad \mathbf{P} = (\mathbf{I} - \mathbf{K}\mathbf{H})\bar{\mathbf{P}}, \qquad \mathbf{H} = \begin{pmatrix} -\Delta x/\hat{r} & -\Delta y/\hat{r} & 0 \\ \Delta y/\hat{r}^2 & -\Delta x/\hat{r}^2 & -1 \end{pmatrix}$$

What every symbol means

Symbol	Meaning	Units
$\Delta x, \Delta y$	landmark position minus predicted vehicle position	m
$\hat{r}, \hat{\beta}$	predicted range and bearing to the landmark	m, rad
$\mathbf{z}$	actual measured range and bearing	m, rad
$\mathbf{H}$	measurement Jacobian	—
$\mathbf{R}_z$	measurement noise covariance	m^2, rad^2
$\mathbf{S}$	innovation covariance	—
$\mathbf{K}$	Kalman gain matrix (3×2)	—

WORKED EXAMPLE — EKF correction with a pole at (25, 8)

Given: predicted pose $(10.423,\ 5.035,\ 9.549°)$ and $\bar{\mathbf{P}}$ from above. Measured $r = 14.95$ m, $\beta = 2.10°$. $\mathbf{R}_z = \operatorname{diag}(0.05^2,\ (1°)^2)$.

Step 1 — predicted measurement

$\Delta x = 14.577$, $\Delta y = 2.965$; $\hat{r} = \sqrt{221.277} = 14.875$ m; $\hat{\beta} = 11.496° - 9.549° = 1.947°$

Step 2 — innovation

$\delta r = 14.95 - 14.875 = 0.075$ m; $\delta\beta = 2.10 - 1.947 = 0.153°$

Step 3 — Jacobian

$\mathbf{H} = \begin{pmatrix} -0.9799 & -0.1993 & 0 \\ 0.0134 & -0.0659 & -1 \end{pmatrix}$

Step 4 — gain

$\mathbf{S} = \begin{pmatrix} 0.01450 & 0.00005 \\ 0.00005 & 0.00192 \end{pmatrix}$, $\mathbf{K} = \begin{pmatrix} -0.811 & 0.124 \\ -0.164 & -0.635 \\ -0.0006 & -0.798 \end{pmatrix}$

Step 5 — update state

$\mathbf{x} = (10.423, 5.035, 9.549°) + \mathbf{K}(0.075,\ 0.00267\ \text{rad}) = (10.363,\ 5.021,\ 9.424°)$

Step 6 — update covariance

1σ: x 4.9 cm, y 10.5 cm, heading 0.96°

Result: the pole pulled the pose 6 cm back and tightened x from 11.0 to 4.9 cm. y barely improved because the pole is almost straight ahead: a range measurement only constrains the direction it points.

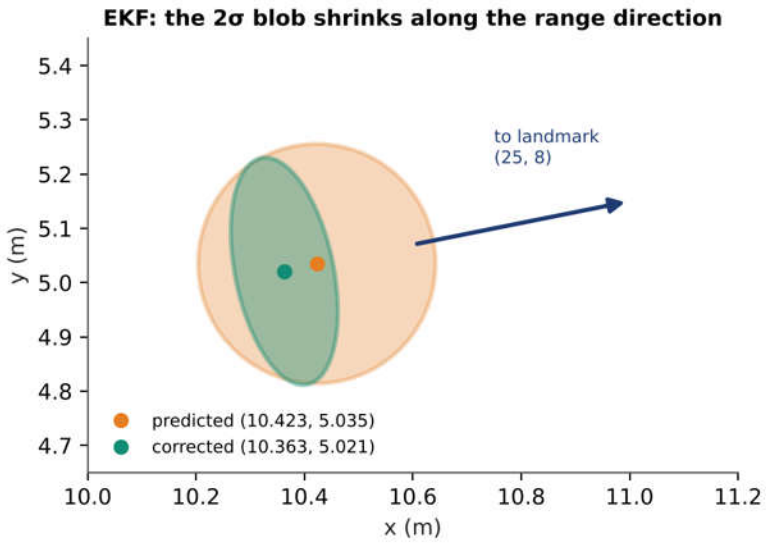


Figure 10.5: The 2σ ellipse before (orange) and after (teal) seeing the landmark.

10.6 Particle filters and EKF-SLAM



Figure 10.6: Particles can represent "I'm in corridor A *or* corridor B."

ENGINEER'S VIEW

Particle filters represent the belief by many weighted samples: move each with the noisy motion model, re-weight each by the measurement likelihood, then resample (clone heavy particles, drop light ones). They handle multi-modal situations a single Gaussian cannot. **FastSLAM** gives each particle its own small Kalman filter per landmark.

EKF-SLAM enlarges the state to hold the landmarks too: $\mathbf{x} = (x, y, \theta, m_{1x}, m_{1y}, \ldots)$. When a new landmark is seen, it is appended; when an old one is re-observed, the correlations update the pose *and every correlated landmark*. That joint estimation is the meaning of "simultaneous." Its cost grows with the square of the number of landmarks, which is why large-scale systems moved to sparse factor graphs (Module 12).

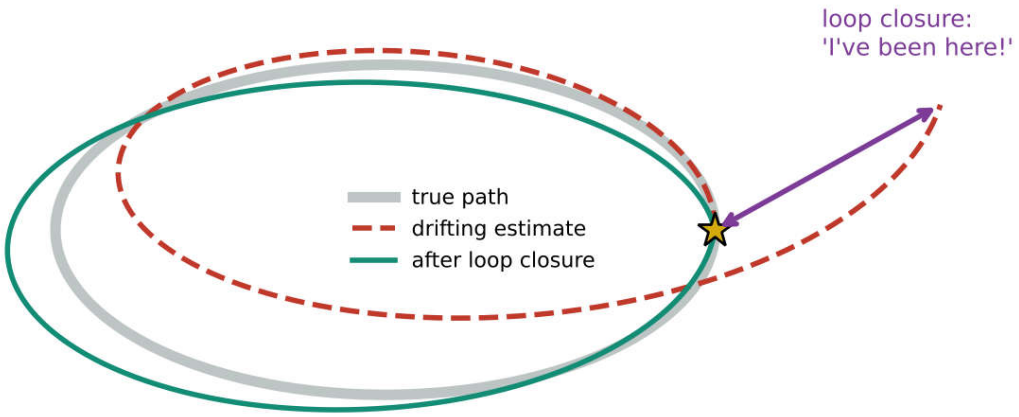


Figure 10.7: Odometry drifts like an open traverse; recognizing the start point closes the loop and distributes the misclosure.

10.7 Drift and loop closure

SURVEYOR'S FIELD TIP

Plan routes like traverses: return to the start, cross earlier paths, and add cross-ties in long corridors. Every revisit is a free closure. A route that never revisits anything is an open traverse — its end can be far off and nothing in the data will tell you.

TRY IT YOURSELF

10.1 Redo the motion-model example with $L = R = 0.40$ m.
10.2 Prediction 50.00 ± 0.30 m; measurement 50.20 ± 0.30 m. Find K, μ and σ.
10.3 (Simple) Why does seeing a landmark make the robot more sure of where it is?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Thrun, S., Burgard, W. & Fox, D. (2005), *Probabilistic Robotics*, MIT Press; Brenner, C., SLAM lecture series (Units A–G), Leibniz University Hannover.

MODULE 11

Scan Matching: Fitting Clouds Together

11.1 The puzzle-piece idea

SIMPLE APPROACH

Draw a shape on a see-through sheet. Your friend draws the same shape on another sheet, but a little shifted and turned. Lay the sheets on top of each other and slide and twist the top one until the drawings line up perfectly. How far you slid and twisted it tells you how your friend's sheet was placed! A LiDAR does this with two scans: sliding and twisting the new scan until it lines up with the old one tells it how far the car moved.

ENGINEER'S VIEW

Scan matching (registration) finds the rigid transform $(\mathbf{R}, \mathbf{t})$ that best aligns a source cloud $\{\mathbf{p}_i\}$ to a target cloud or map $\{\mathbf{q}_i\}$. Matching each new scan to the previous one gives *scan-to-scan* odometry; matching to an accumulated local map gives *scan-to-map* odometry, which drifts less. LIO-SAM matches each new keyframe to a local map built from its recent "sub-keyframes" (Module 12). The answer becomes a measurement (a factor) of how the vehicle moved.

11.2 Iterative Closest Point (ICP)

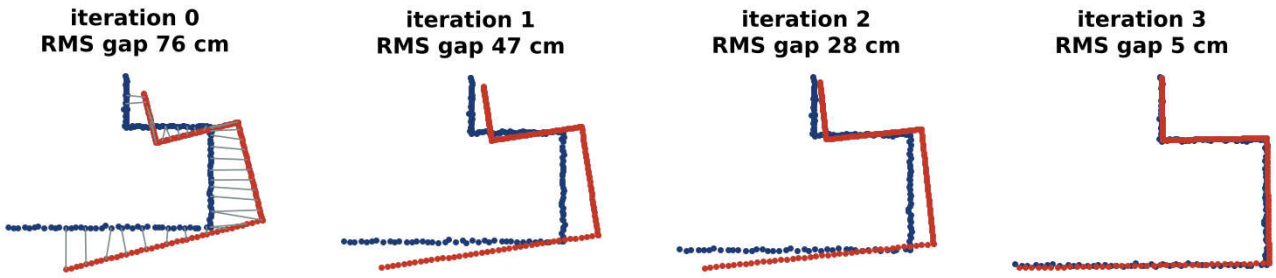


Figure 11.1: A real ICP run: pair each red point with its nearest blue point, solve the best fit, move, repeat.

KEY EQUATION — The ICP objective

$$\min_{\mathbf{R},\mathbf{t}} \; E = \sum_{i=1}^{N} \left\| \mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_{c(i)} \right\|^2$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{p}_i$	point i of the new (source) scan	m
$\mathbf{q}_{c(i)}$	its corresponding (closest) point in the target	m
$c(i)$	the correspondence: which target point is paired with $\mathbf{p}_i$	—
$\mathbf{R}, \mathbf{t}$	rotation and translation being solved for	—, m
$\|\cdot\|^2$	squared distance	m^2
E	total squared misfit (least-squares cost)	m^2

The algorithm alternates two easy problems:

1. **Associate:** for the current guess, pair every source point with its nearest target point (a k-d tree makes this fast).
2. **Solve:** with pairs fixed, find the best $(\mathbf{R}, \mathbf{t})$ in closed form (below).
3. **Apply** the transform and **repeat** until the cost stops decreasing.

ICP converges to the nearest local minimum, so it needs a good starting guess — which the IMU provides.

KEY EQUATION — Closed-form best fit (Arun/Horn/Umeyama)

$$\bar{\mathbf{p}} = \tfrac{1}{N}\textstyle\sum \mathbf{p}_i, \quad \bar{\mathbf{q}} = \tfrac{1}{N}\textstyle\sum \mathbf{q}_i, \quad \mathbf{W} = \textstyle\sum (\mathbf{p}_i - \bar{\mathbf{p}})(\mathbf{q}_i - \bar{\mathbf{q}})^{\mathsf{T}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}}, \quad \mathbf{R} = \mathbf{V}\mathbf{U}^{\mathsf{T}}, \quad \mathbf{t} = \bar{\mathbf{q}} - \mathbf{R}\bar{\mathbf{p}}$$

In 2-D this reduces to $\theta = \operatorname{atan2}\left(\textstyle\sum (p'_x q'_y - p'_y q'_x),\ \sum (p'_x q'_x + p'_y q'_y) \right)$.

What every symbol means

Symbol	Meaning	Units
$\bar{\mathbf{p}}, \bar{\mathbf{q}}$	centroids (average points) of the two sets	m
$\mathbf{p}', \mathbf{q}'$	points with their centroid subtracted ("centered")	m
$\mathbf{W}$	cross-covariance matrix of the centered sets	m^2
$\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}$	singular value decomposition of $\mathbf{W}$; if $\det(\mathbf{V}\mathbf{U}^{\mathsf{T}}) = -1$ flip the last column of $\mathbf{V}$ to avoid a mirror image	—

WORKED EXAMPLE — best fit of four matched points

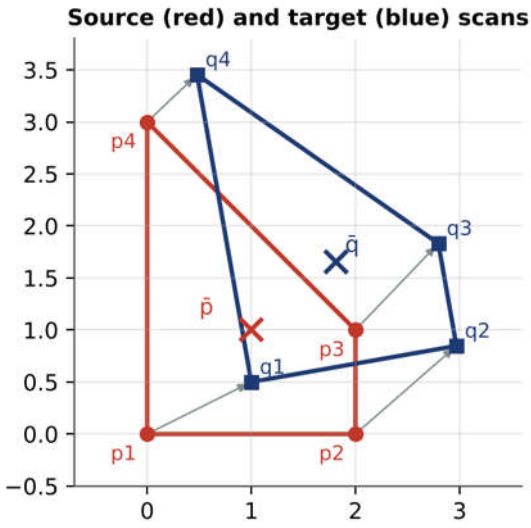
Given: source $\mathbf{p}_{1..4} = (0,0), (2,0), (2,1), (0,3)$; target $\mathbf{q}_{1..4} = (1.000, 0.500), (2.970, 0.847), (2.796, 1.832), (0.479, 3.454)$, already paired.

Step 1 — centroids

$\bar{\mathbf{p}} = (1,1)$; $\bar{\mathbf{q}} = (1.811, 1.658)$

Step 2 — center and sum

(table below)



p'_x	p'_y	q'_x	q'_y	$p'_x q'_x + p'_y q'_y$	$p'_x q'_y - p'_y q'_x$
−1	−1	−0.811	−1.158	1.970	0.347
1	−1	1.158	−0.811	1.970	0.347
1	0	0.985	0.174	0.985	0.174
−1	2	−1.332	1.796	4.924	0.868
			Sum	**9.848**	**1.736**

WORKED EXAMPLE — (continued)

Step 3 — rotation

$\theta = \text{atan2}(1.736,\ 9.848) = 10.00°$

Step 4 — translation

$\mathbf{t} = \bar{\mathbf{q}} - \mathbf{R}\bar{\mathbf{p}}$: $\mathbf{R}\bar{\mathbf{p}} = (0.9848 - 0.1736,\ 0.1736 + 0.9848) = (0.811,\ 1.158)$; $\mathbf{t} = (1.811 - 0.811,\ 1.658 - 1.158) = (1.000,\ 0.500)$

Step 5 — check with SVD

$\mathbf{W} = \begin{pmatrix} 4.287 & -1.275 \\ -3.012 & 5.562 \end{pmatrix}$ gives $\mathbf{VU}^{\mathsf{T}} = \begin{pmatrix} 0.985 & -0.174 \\ 0.174 & 0.985 \end{pmatrix} = \mathbf{R}(10°)$ ✓

Result: the second scan is the first rotated 10° and shifted (1.0, 0.5) m — exactly how the target was made. In real ICP the pairs are only guesses, so steps 1–5 repeat.

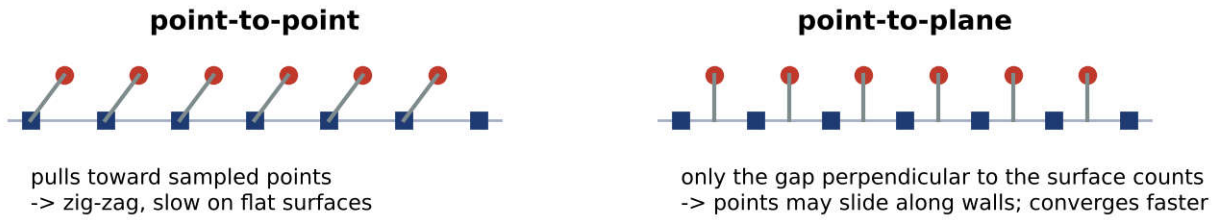


Figure 11.2: Point-to-point pulls toward sample points; point-to-plane only penalizes the gap normal to the surface.

11.3 Better cost functions: point-to-plane and NDT

KEY EQUATION — Point-to-plane ICP

$$E = \sum_i \Big(\big(\mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_i \big) \cdot \mathbf{n}_i \Big)^2$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{n}_i$	unit normal of the target surface at $\mathbf{q}_i$	—
$(\cdot) \cdot \mathbf{n}_i$	dot product: keeps only the part of the gap perpendicular to the surface	m

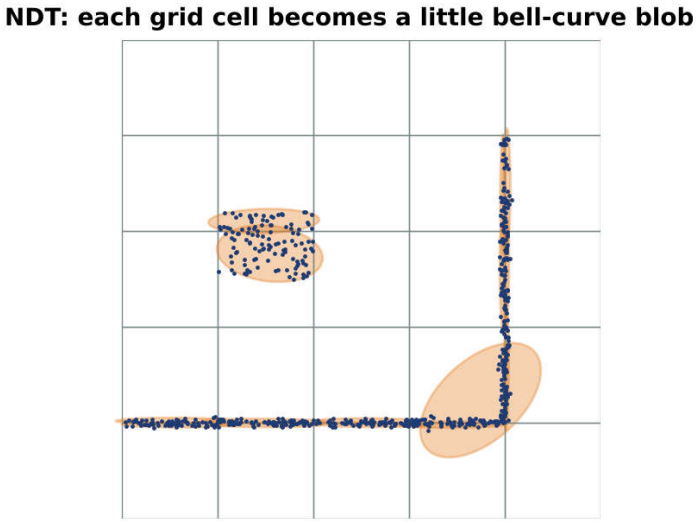


Figure 11.3: The Normal Distributions Transform summarizes each grid cell by a mean and covariance.

ENGINEER'S VIEW

NDT (Normal Distributions Transform) divides the target into voxels and fits each with a Gaussian (mean $\boldsymbol{\mu}_k$, covariance $\boldsymbol{\Sigma}_k$). The source is scored by $\sum_i \exp\big(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu}_k)^\mathsf{T} \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k)\big)$ with $\mathbf{x}_i = \mathbf{R}\mathbf{p}_i + \mathbf{t}$, and maximized. No nearest-neighbor search is needed, it is smooth, and it is widely used in automotive localization. **Generalized ICP** (GICP) blends both ideas by modeling local surface covariances on both clouds.

11.4 Feature-based matching (LOAM and LIO-SAM)

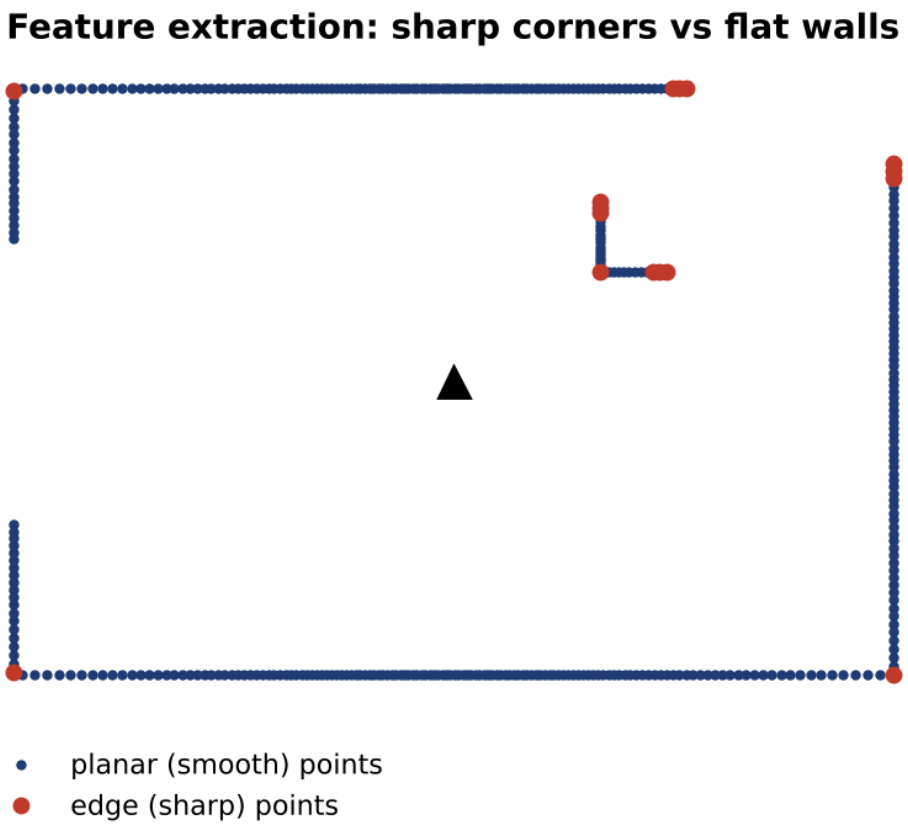


Figure 11.4: Points on sharp corners become edge features; points on smooth surfaces become planar features.

SIMPLE APPROACH

Instead of matching every single dot, look for the *special* ones: sharp corners (like the corner of a box) and flat sides (like a table top). Corners are easy to recognize and flat sides are easy to slide against. Matching only these is much faster.

KEY EQUATION — Roughness (curvature) score

$$c_i = \frac{1}{|S| \cdot \rho_i} \left| \sum_{j \in S} (\rho_j - \rho_i) \right|$$

What every symbol means

Symbol	Meaning	Units
ρ_i	range of point i along one scan line	m
S	its neighbors on the same scan line ($\lvert S \rvert$ = how many, e.g. 5 each side)	—
c_i	roughness: large ⇒ edge feature; small ⇒ planar feature	—

(Simplified range form; LOAM uses the same expression with 3-D point vectors.)

WORKED EXAMPLE — edge or plane?

Given: two neighbors each side ($|S| = 4$).

Flat wall

ranges 10.00, 10.01, 10.00, 9.99, 10.00: sum of differences $= 0 + 0.01 - 0.01 + 0 = 0 \Rightarrow c = 0$ **planar**

Corner

ranges 10.0, 10.0, 8.0, 8.0, 8.0: sum $= 2 + 2 + 0 + 0 = 4 \Rightarrow c = 4/(4 \times 8) = 0.125$ **edge**

Result: thresholds (e.g. $c > 0.1$ edge, $c < 0.1$ plane in LOAM's default) sort the points.

11.4.1 Distances to edges and planes (LIO-SAM eqs. 10–11)

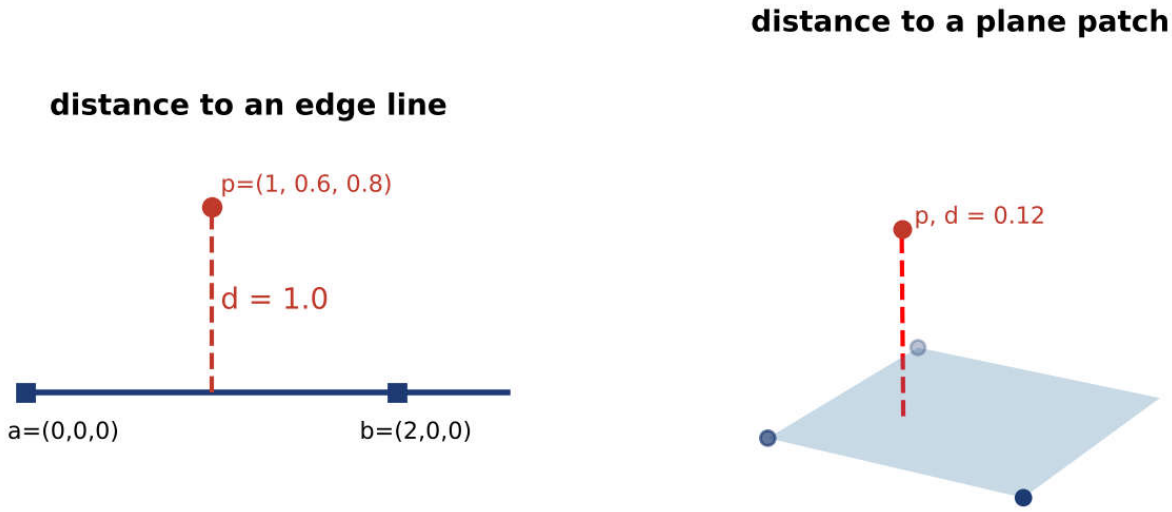


Figure 11.5: An edge feature is compared with a line through two map points; a planar feature with a plane through three.

KEY EQUATION — Point-to-line and point-to-plane distances

$$d_e = \frac{\left\|(\mathbf{p}-\mathbf{a}) \times (\mathbf{p}-\mathbf{b})\right\|}{\left\|\mathbf{a}-\mathbf{b}\right\|} \qquad d_p = \frac{\left|(\mathbf{p}-\mathbf{u}) \cdot \left((\mathbf{u}-\mathbf{v}) \times (\mathbf{u}-\mathbf{w})\right)\right|}{\left\|(\mathbf{u}-\mathbf{v}) \times (\mathbf{u}-\mathbf{w})\right\|}$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{p}$	a feature point of the new scan, transformed into the map frame with the current guess	m
$\mathbf{a}, \mathbf{b}$	two map points defining the matching edge line ($\mathbf{p}^e_{i,u}, \mathbf{p}^e_{i,v}$ in the paper)	m
$\mathbf{u}, \mathbf{v}, \mathbf{w}$	three map points defining the matching planar patch	m
$\times$	cross product (Module 2): its length is a parallelogram area; area ÷ base = height = distance	—
$(\mathbf{u}-\mathbf{v}) \times (\mathbf{u}-\mathbf{w})$	a vector normal to the plane	m^2

WORKED EXAMPLE — edge and plane distances

Edge

$\mathbf{a} = (0,0,0)$, $\mathbf{b} = (2,0,0)$, $\mathbf{p} = (1,0.6,0.8)$. $\mathbf{p}-\mathbf{a} = (1,0.6,0.8)$, $\mathbf{p}-\mathbf{b} = (-1,0.6,0.8)$; cross $= (0.6\cdot 0.8 - 0.8\cdot 0.6,\ 0.8\cdot(-1) - 1\cdot 0.8,\ 1\cdot 0.6 - 0.6\cdot(-1)) = (0,-1.6,1.2)$; length $= \sqrt{2.56+1.44} = 2.0$; divide by $\|\mathbf{a}-\mathbf{b}\| = 2$: $d_e = 1.0$ m.

Plane

$\mathbf{u} = (0,0,0)$, $\mathbf{v} = (1,0,0)$, $\mathbf{w} = (0,1,0)$, $\mathbf{p} = (0.3,0.4,0.12)$. Normal $= (\mathbf{u}-\mathbf{v}) \times (\mathbf{u}-\mathbf{w}) = (-1,0,0)\times(0,-1,0) = (0,0,1)$; $d_p = |(0.3,0.4,0.12)\cdot(0,0,1)|/1 = 0.12$ m.

Result: LIO-SAM then finds the pose that minimizes $\sum d_e + \sum d_p$ over all features using the Gauss–Newton method.

ENGINEER'S VIEW

Gauss–Newton in one paragraph. Stack all distances into a residual vector $\mathbf{d}(\mathbf{x})$ that depends on the 6-DOF pose $\mathbf{x}$. Linearize: $\mathbf{d}(\mathbf{x}+\delta\mathbf{x}) \approx \mathbf{d} + \mathbf{J}\delta\mathbf{x}$, where $\mathbf{J}$ is the Jacobian. Setting the derivative of $\|\mathbf{d}+\mathbf{J}\delta\mathbf{x}\|^2$ to zero gives the *normal equations* $\mathbf{J}^\mathsf{T}\mathbf{J}\,\delta\mathbf{x} = -\mathbf{J}^\mathsf{T}\mathbf{d}$ — exactly the normal equations of a survey least-squares adjustment. Solve, update $\mathbf{x} \leftarrow \mathbf{x}+\delta\mathbf{x}$, repeat until $\delta\mathbf{x}$ is tiny.

11.5 When matching fails: the hallway effect

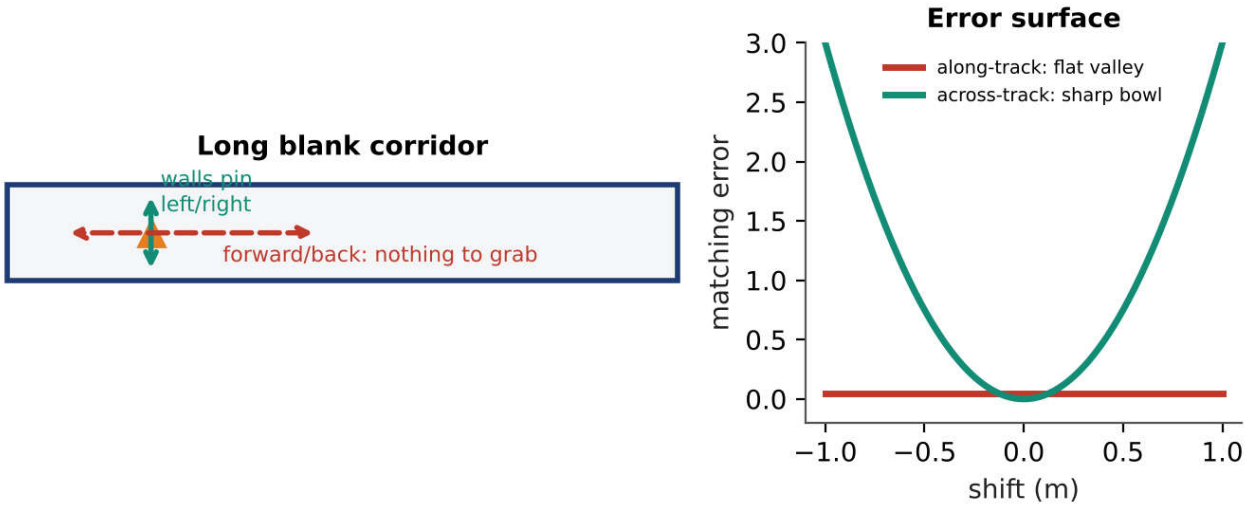


Figure 11.6: In a featureless corridor the matching error has a flat valley along the corridor — any forward shift fits equally well.

ENGINEER'S VIEW

Scan matching can only measure motion in directions where geometry changes. In a long uniform corridor, tunnel, or a highway with continuous barriers, the cost is flat along the axis, $\mathbf{J}^\mathsf{T}\mathbf{J}$ is nearly singular in that direction, and the estimate is free to stretch or shrink. Mathematically, the smallest eigenvalue of $\mathbf{J}^\mathsf{T}\mathbf{J}$ is near zero; some systems monitor this "degeneracy" and down-weight LiDAR along that axis. Surveyors recognize this as a traverse with poor geometry in one direction.

SURVEYOR'S FIELD TIP

In corridors: add control or targets along the length, walk/drive both directions, include side rooms or cross-ties to create unique geometry, place temporary objects (cones, boxes) if needed, and always check along-track distances against a taped or total-station measurement.

TRY IT YOURSELF

11.1 Source points $(0,0),(1,0)$ match target points $(2,1),(2,2)$. Find θ and $\mathbf{t}$ by the closed form.
11.2 Distance from $\mathbf{p} = (3,4,5)$ to the line through $(0,0,0)$ and $(0,0,10)$?
11.3 (Simple) Why is it hard to tell how far you have walked in a long, plain hallway with your eyes half-closed?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Besl & McKay (1992), *IEEE PAMI* 14(2); Arun, Huang & Blostein (1987), *IEEE PAMI* 9(5); Segal *et al.* (2009), Generalized-ICP; Biber & Straßer (2003), NDT; Zhang & Singh (2017), LOAM, *Autonomous Robots* 41(2); Shan *et al.* (2020), eqs. (10)–(11).

MODULE 12

Factor Graphs and LIO-SAM

12.1 From a filter to a graph

SIMPLE APPROACH

Imagine a row of people connected by rubber bands. Each rubber band says "you two should be this far apart." One person is also tied to a post that says "you belong here." If one band pulls a little wrong, everyone wiggles until the pulling is shared out as fairly as possible. A **factor graph** is that picture: the people are the car's positions over time, and the rubber bands are measurements.

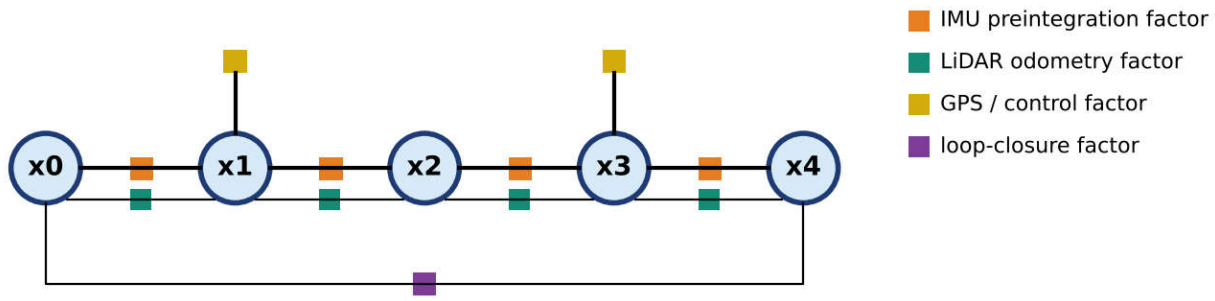


Figure 12.1: A factor graph: circles are unknown states (poses); small squares are measurements (factors) that tie them together.

ENGINEER'S VIEW

A filter (EKF) processes data once and forgets the past. A **smoother** keeps the whole trajectory as unknowns and re-solves when new information arrives — so a loop closure at minute 20 can correct the pose at minute 3. The data structure is a **factor graph**: *variable nodes* $\mathbf{x}_i$ (poses, velocities, IMU biases) and *factor nodes*, each encoding one measurement as a probability that depends on a few variables. Assuming Gaussian noise, finding the most probable trajectory (maximum a posteriori, MAP) is exactly a non-linear weighted least-squares problem — the same mathematics as a survey network adjustment.

KEY EQUATION — MAP estimation becomes least squares

$$\mathbf{X}^* = \arg\max_{\mathbf{X}} \prod_k \phi_k(\mathbf{X}_k) = \arg\min_{\mathbf{X}} \sum_k \left\| \mathbf{h}_k(\mathbf{X}_k) - \mathbf{z}_k \right\|^2_{\boldsymbol{\Sigma}_k}$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{X}$	all unknowns together (every pose, velocity, bias)	—
$\mathbf{X}^*$	the best (most probable) solution	—
ϕ_k	factor k: a Gaussian probability $\propto \exp(-\frac{1}{2}\|\cdot\|^2_{\boldsymbol{\Sigma}})$	—
$\mathbf{X}_k$	only the few variables factor k touches	—
$\mathbf{h}_k$	measurement model: what factor k *should* measure given $\mathbf{X}_k$	—
$\mathbf{z}_k$	what it actually measured	—
$\|\mathbf{e}\|^2_{\boldsymbol{\Sigma}} = \mathbf{e}^\mathsf{T}\boldsymbol{\Sigma}^{-1}\mathbf{e}$	squared Mahalanobis distance: residuals divided by their uncertainty (weights $= \boldsymbol{\Sigma}^{-1}$)	—

Taking $-\log$ of a product of Gaussians turns "maximize the product" into "minimize the sum of squares."

12.2 A factor graph you can solve by hand

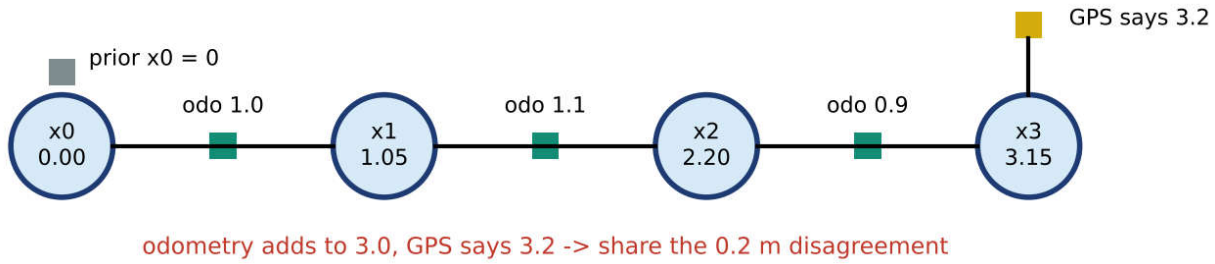


Figure 12.2: Four positions along a straight road, three odometry factors, one GPS factor and a prior.

WORKED EXAMPLE — 1-D pose graph with a GPS misclosure

Given: $x_0 = 0$ (fixed prior). LiDAR odometry: $x_1 - x_0 = 1.0$, $x_2 - x_1 = 1.1$, $x_3 - x_2 = 0.9$ m. GPS: $x_3 = 3.2$ m. All four measurements have the same $\sigma = 0.10$ m (equal weights).

Step 1 — write the residual equations $\mathbf{A}\mathbf{x} = \mathbf{b}$

$$\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1.0 \\ 1.1 \\ 0.9 \\ 3.2 \end{pmatrix}$$

Step 2 — normal equations $\mathbf{A}^\mathsf{T}\mathbf{A}\,\mathbf{x} = \mathbf{A}^\mathsf{T}\mathbf{b}$

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1.0 - 1.1 \\ 1.1 - 0.9 \\ 0.9 + 3.2 \end{pmatrix} = \begin{pmatrix} -0.1 \\ 0.2 \\ 4.1 \end{pmatrix}$$

Step 3 — solve

From row 1: $x_2 = 2x_1 + 0.1$. Row 2: $-x_1 + 2(2x_1 + 0.1) - x_3 = 0.2 \Rightarrow x_3 = 3x_1$. Row 3: $-(2x_1 + 0.1) + 6x_1 = 4.1 \Rightarrow x_1 = 1.05$. Then $x_2 = 2.20$, $x_3 = 3.15$.

Step 4 — residuals

odometry: $1.05 - 1.0 = +0.05$, $1.15 - 1.1 = +0.05$, $0.95 - 0.9 = +0.05$; GPS: $3.15 - 3.2 = -0.05$

Result: $x_1 = 1.05$, $x_2 = 2.20$, $x_3 = 3.15$ m. The 0.20 m misclosure is shared equally by the four equally weighted measurements — the compass/least-squares adjustment of a traverse, done by a factor graph.

ENGINEER'S VIEW

Notice $\mathbf{A}^\mathsf{T}\mathbf{A}$ is *sparse* (mostly zeros): each pose only links to its neighbors. Real graphs have tens of thousands of nodes (LIO-SAM's Amsterdam run: 23 304 nodes, 49 617 factors) but stay sparse, so they solve quickly. **iSAM2** (incremental smoothing and mapping with the Bayes tree) goes further: when a new node arrives it re-solves only the part of the graph that changed, which is how LIO-SAM runs in real time. GTSAM is the open-source library implementing it.

12.3 LIO-SAM, piece by piece

LIO-SAM (Shan, Englot, Meyers, Wang, Ratti & Rus, IROS 2020) is a *tightly-coupled* lidar–inertial odometry and mapping system built on a factor graph. "Tightly coupled" means the IMU and LiDAR are optimized *together* (the IMU biases are estimated from the LiDAR results), rather than the IMU merely supplying a guess.

KEY EQUATION — The LIO-SAM state (paper eq. 1)

$$\mathbf{x} = \left[\mathbf{R}^\mathsf{T},\ \mathbf{p}^\mathsf{T},\ \mathbf{v}^\mathsf{T},\ \mathbf{b}^\mathsf{T}\right]^\mathsf{T}, \qquad \mathbf{T} = [\mathbf{R} \mid \mathbf{p}] \in SE(3)$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{R} \in SO(3)$	attitude (body to world)	—
$\mathbf{p} \in \mathbb{R}^3$	position	m
$\mathbf{v}$	velocity	m/s
$\mathbf{b}$	IMU biases (gyro and accelerometer)	rad/s, m/s^2
$\mathbf{T}$	pose as a rigid transform (Module 6)	—

12.3.1 The four factors

Factor	What it measures	Surveying analog
IMU preintegration	relative motion $\Delta\mathbf{v}_{ij}, \Delta\mathbf{p}_{ij}, \Delta\mathbf{R}_{ij}$ between consecutive states (Module 7)	a measured distance and angle between traverse stations
LiDAR odometry	relative transform $\Delta\mathbf{T}_{i,i+1} = \mathbf{T}_i^{-1}\mathbf{T}_{i+1}$ from scan-to-map matching (Module 11)	a second, independent set of traverse observations
GPS	absolute position of a state (converted to local Cartesian)	tying a station to a control point
Loop closure	relative transform between a new state and an old nearby one	closing the traverse on itself

12.3.2 Keyframes and the sliding window

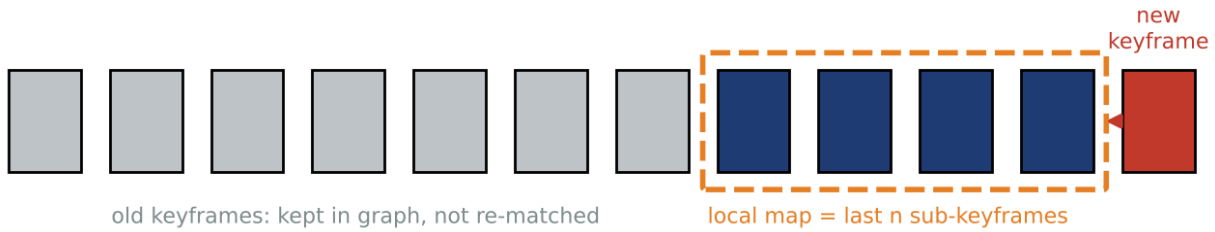


Figure 12.3: Only keyframes enter the graph; a new keyframe is matched to a local map of the last n sub-keyframes.

ENGINEER'S VIEW

Adding every LiDAR frame (10 per second) would make the graph enormous. LIO-SAM adds a new state $\mathbf{x}_{i+1}$ (a **keyframe**) only when the pose changes by more than **1 m** or **10°**; frames in between are discarded. To create a LiDAR odometry factor:

1. extract edge and planar features from the new de-skewed scan (roughness, Module 11);
2. transform the last $n = 25$ keyframes ("sub-keyframes") into the world frame and merge them into a local voxel map $\mathbf{M}_i = \{\mathbf{M}_i^e, \mathbf{M}_i^p\}$, down-sampled at 0.2 m (edges) and 0.4 m (planes);
3. using the IMU-predicted pose $\tilde{\mathbf{T}}_{i+1}$ as the starting guess, find edge/plane correspondences and minimize $\sum d_e + \sum d_p$ by Gauss–Newton;
4. the relative transform to the previous state becomes the factor.

Matching to a small local map, rather than a global one as in LOAM, is the main reason LIO-SAM is fast.

12.3.3 GPS and loop-closure factors

ENGINEER'S VIEW

GPS factor. GPS is converted to a local Cartesian frame; if not hardware-synchronized it is linearly interpolated to the keyframe time. Because lidar–inertial drift grows slowly, a GPS factor is added only when the estimated position covariance is *larger* than the GPS covariance — a sensible "only use control when you need it" rule.

Loop-closure factor. When a new state is added, LIO-SAM searches for earlier states within **15 m**. For a candidate $\mathbf{x}_3$, the new keyframe is matched to the $2m+1$ keyframes around it ($m = 12$), giving $\Delta\mathbf{T}_{3,i+1}$ as a factor. The authors note loop closures are especially valuable for correcting *altitude*, where GPS was poor (errors approaching 100 m without loops in their tests).

12.3.4 What it achieved

End-to-end translation error (m; LIO-SAM in ft in parentheses) — LIO-SAM paper, Table II

Dataset	LOAM	LIOM	LIO-odom	LIO-GPS	LIO-SAM
Campus (1.4 km / 0.9 mi)	192.43	fail	9.44	6.87	**0.12** (0.39 ft)
Park (2.9 km / 1.8 mi)	121.74	34.60	36.36	2.93	**0.04** (0.13 ft)
Amsterdam (19 km / 11.8 mi)	fail	fail	fail	1.21	**0.17** (0.56 ft)

LIO-odom uses only IMU + LiDAR factors; *LIO-GPS* adds GPS; full *LIO-SAM* adds loop closures. Each extra factor type removes another layer of drift. LIO-SAM processed scans in 42–101 ms and ran up to 13× faster than real time.

IMPROVED FROM THE SOURCE MATERIAL

These are research-grade results judged against the start/end point or GPS, on a robot, not survey-grade checkpoint accuracy. A 4–17 cm loop misclosure over kilometers is excellent odometry, but a deliverable still needs calibration, control and *independent checkpoints* (Module 13) before any accuracy can be stated. Also, the paper's eq. (12) writes $\Delta\mathbf{T} = \mathbf{T}_i^{\mathsf{T}}\mathbf{T}_{i+1}$; the correct general form is $\mathbf{T}_i^{-1}\mathbf{T}_{i+1}$.

SURVEYOR'S FIELD TIP

You do not need to write an optimizer, but you should be able to answer: Which factors were active in this dataset? Where were the loop closures? Where was GNSS used? Where did the corridor degrade LiDAR odometry? Good processing software lets you view the trajectory colored by its estimated standard deviation — inspect it before trusting any part of the cloud.

12.4 Building the trajectory: poses, scans, IMU, GNSS and images

SIMPLE APPROACH

Think of a trip recorded as a flip-book. Each page is a **pose**: where the car was and which way it was pointing at one instant. The IMU writes a new page hundreds of times a second, so the flip-book is smooth — but it slowly drifts. The LiDAR compares each new scan with the last ones and corrects the drift, like recognizing the same buildings. GNSS now and then says "this

page is exactly here on Earth." The camera takes pictures along the way, and each picture is glued to the page for the moment it was taken. Flip through all the pages and you have the **trajectory**: the path, with direction, at every instant.

The words

Pose	position (X, Y, Z) *and* orientation (roll, pitch, yaw) at one instant (Module 6)
Trajectory	the continuous sequence of poses over time — the "path with direction" (often delivered as a time-stamped file, e.g. an SBET or a pose table)
Keyframe pose	a pose that is a node in the factor graph (LIO-SAM: every 1 m or 10°)
Interpolated pose	a pose computed *between* keyframes for any time you need: each laser pulse, each image
Image pose	the camera's pose at exposure time (photogrammetrists call it exterior orientation): trajectory pose + camera lever arm and boresight

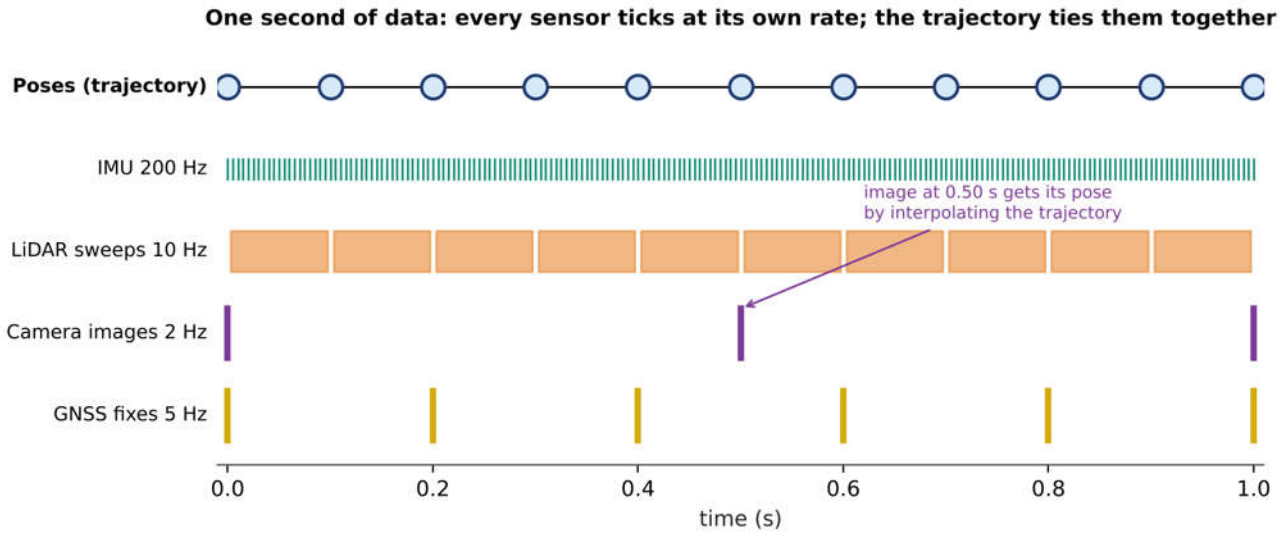


Figure 12.4: One second of data. The IMU ticks 200 times, the LiDAR completes 10 sweeps, GNSS gives 5 fixes and the camera takes 2 images. The trajectory (top row) is the common thread that gives every one of them a pose.

What each sensor contributes

Sensor	Rate	Contributes to the trajectory	Cannot do
IMU	100–1000 Hz	smooth, high-rate motion between everything else; roll and pitch from gravity; de-skews every LiDAR sweep; bridges gaps (preintegration factors)	stay accurate on its own for more than seconds; give absolute position
LiDAR scans	10–20 Hz	relative motion from scan matching (odometry factors); heading and drift control wherever there is geometry; loop closures when places are revisited; the map itself	work in featureless places (corridors, open fields); give absolute coordinates
GNSS	1–20 Hz	absolute position in the datum (GNSS factors); stops long-term drift; RTK makes it centimeter-level; dual antenna adds heading	work in tunnels, under canopy, indoors; supply attitude at high rate
Camera images	1–10 Hz	in LIO-SAM: nothing — images *receive* poses. (Visual–inertial systems also use image features as extra odometry.)	—
Control	a few points	ties the whole trajectory to surveyed coordinates; checkpoints prove it	—

WORKED EXAMPLE — giving a camera image its pose

Given: trajectory poses at $t_1 = 10.00$ s: $E = 100.00$, $N = 200.00$ m, heading $30.0°$; at $t_2 = 10.10$ s: $E = 101.50$, $N = 200.10$ m, heading $31.0°$. A photo is exposed at $t = 10.04$ s.

Step 1 — fraction of the way

$f = (10.04 - 10.00)/(10.10 - 10.00) = 0.4$

Step 2 — position

$E = 100.00 + 0.4(1.50) = 100.60$ m; $N = 200.00 + 0.4(0.10) = 200.04$ m

Step 3 — heading

$30.0 + 0.4(1.0) = 30.4°$ (real software interpolates the full rotation, e.g. quaternion "slerp")

Step 4 — camera offset

apply the camera's lever arm and boresight (Module 9) to move from the IMU to the lens

Result: the image pose is $(100.60,\ 200.04)$, heading $30.4°$. The same interpolation gives every one of the millions of laser pulses its pose.

WORKED EXAMPLE — why the camera needs calibration and timing too

Boresight error $0.1°$, point 20 m away

$20 \times 0.001745 = 0.035$ m — colors land 3.5 cm off their points

Timing error 5 ms at 10 m/s

$10 \times 0.005 = 0.05$ m — colors smear 5 cm along the road

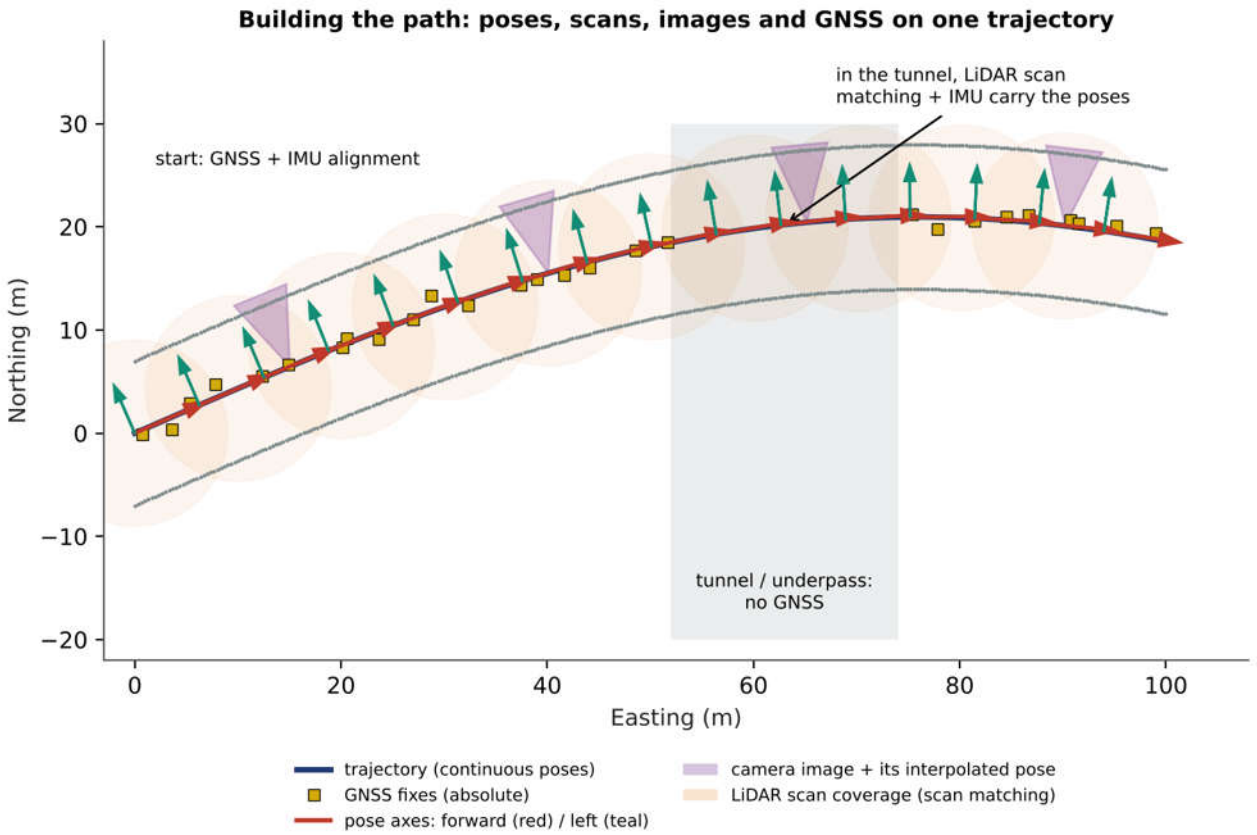


Figure 12.5: The path as built: blue trajectory with pose axes, orange LiDAR scan coverage used for matching, gold GNSS fixes (absent in the tunnel), and purple camera images, each placed with its own interpolated pose.

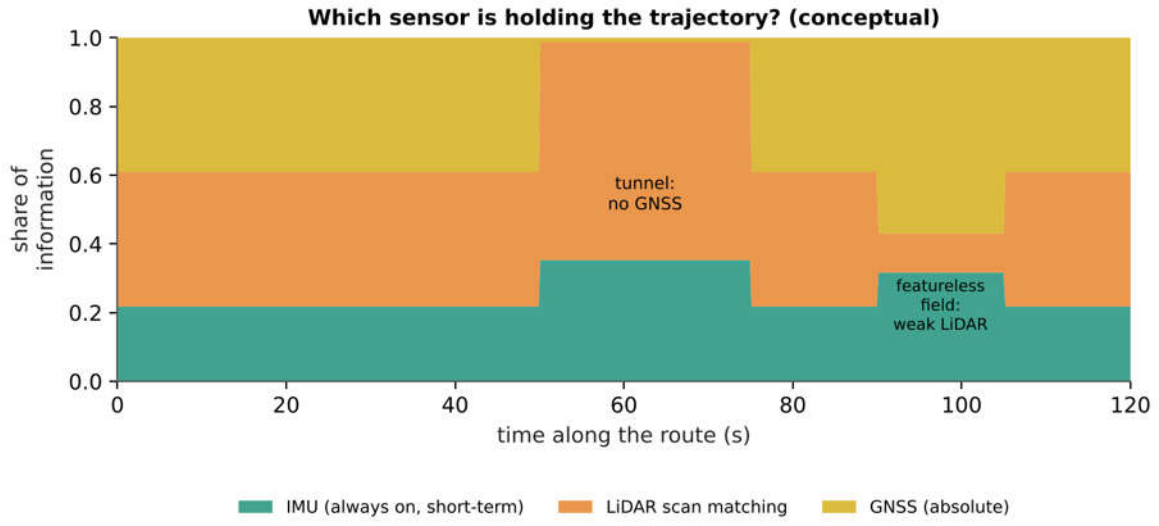


Figure 12.6: Conceptually, the factor graph leans on whichever sensors have information at each moment: LiDAR and IMU carry the tunnel; GNSS and IMU carry the featureless field.

Result: colorized point clouds and photo-measured features inherit every trajectory error *plus* the camera's own calibration and timing errors. Calibrate and synchronize the camera like the LiDAR (Modules 8–9).

SURVEYOR'S FIELD TIP

Deliver and archive the trajectory. Keep the final trajectory (time, position, attitude and their standard deviations), the image exterior orientations, the calibration file and the processing log with the project. When a client later asks "where did this point come from?", the trajectory is the answer — and it is the first thing a reviewer will inspect for drift, gaps and GNSS outages.

TRY IT YOURSELF

12.1 Repeat the 1-D example with GPS saying $x_3 = 3.0$. What happens and why?
12.2 Repeat with GPS $x_3 = 3.2$ but GPS $\sigma = 0.05$ m (weight $4\times$ the others). Solve the weighted normal equations.
12.3 (Simple) In the rubber-band picture, what happens if one band is much stiffer than the rest?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Shan *et al.* (2020), LIO-SAM (Tables I–IV, keyframe and loop-closure parameters); Kaess, M. *et al.* (2012), iSAM2, *Int. J. Robotics Research* 31(2); Dellaert & Kaess (2017), Factor graphs for robot perception.

MODULE 13

Control, Error Budgets and Accuracy Standards

13.1 Relative versus absolute

SIMPLE APPROACH

You can draw a perfect map of your bedroom — every toy in the right place compared to every other toy — but if you don't know where your house is on the street, nobody else can use it to find your room. SLAM makes the bedroom map. **Control points** — spots measured very carefully by surveyors — pin that map onto the real world. **Checkpoints** are extra careful spots that we keep secret from the computer, then use to test whether the map is truly right.

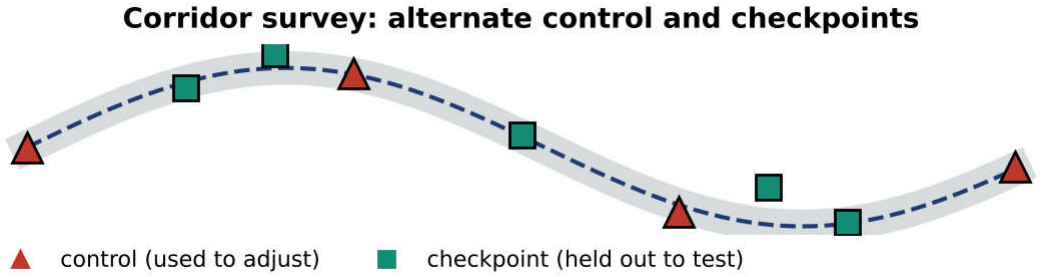


Figure 13.1: Along a corridor, alternate points used to constrain the solution (control) with points held back purely to test it (checkpoints).

ENGINEER'S VIEW

SLAM and lidar–inertial odometry are *relative*: internally consistent but floating in their own frame and slowly drifting. GNSS and surveyed **control** fix the datum (horizontal and vertical) and constrain drift. Accuracy claims, however, must come from **independent checkpoints** — surveyed points *not* used in any adjustment. Using the same points for both makes the residuals look artificially small, exactly as reporting the residuals of a constrained adjustment at its own fixed stations would. Document which points were used for what.

SURVEYOR'S FIELD TIP

- Use targets the scanner can see and the software can measure: painted chevrons or checkerboards on pavement for vertical and horizontal checks; high-intensity paint shows clearly in intensity.
- Survey control and checkpoints with a method at least **two times** more accurate than the product you are testing (ASPRS Edition 2), and include their own uncertainty in the final accuracy (below).

- Distribute checkpoints along the whole project and across the cross-section — not only at the ends near GNSS-friendly areas.
- Confirm the vertical datum and geoid model match project requirements before comparing anything.

13.2 Computing accuracy from checkpoints

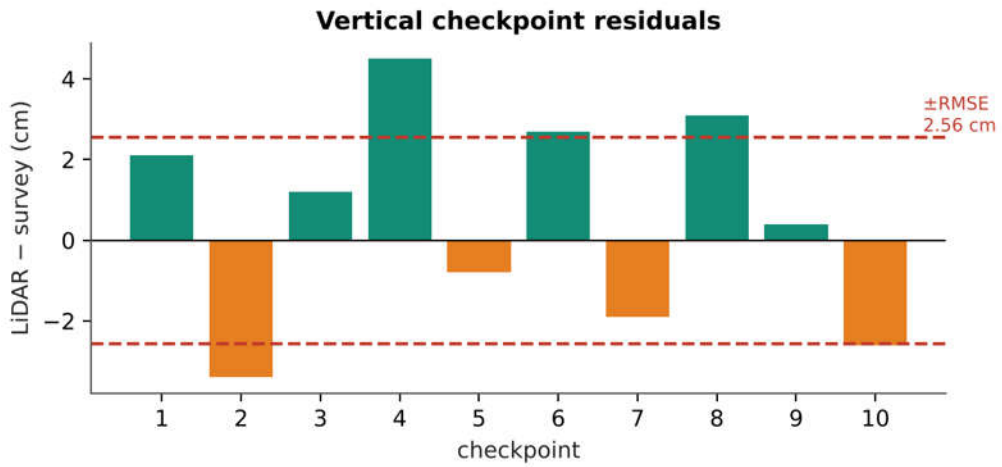


Figure 13.2: Vertical residuals (LiDAR surface minus surveyed elevation) at ten checkpoints.

KEY EQUATION — Accuracy statistics

$$\mathrm{RMSE}_z = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\Delta z_i^2}, \qquad \mathrm{RMSE}_H = \sqrt{\mathrm{RMSE}_x^2 + \mathrm{RMSE}_y^2}$$

$$\mathrm{RMSE}_{\mathrm{final}} = \sqrt{\mathrm{RMSE}_{\mathrm{data}}^2 + \mathrm{RMSE}_{\mathrm{survey}}^2}$$

What every symbol means

Symbol	Meaning	Units
Δz_i	LiDAR elevation minus checkpoint elevation at point i	m
n	number of checkpoints (ASPRS Ed. 2: at least 30 for a formal product test)	—
RMSE_x, RMSE_y	RMSE of easting and northing differences	m
RMSE_H	horizontal (radial) RMSE	m
$\mathrm{RMSE}_{\mathrm{survey}}$	uncertainty of the checkpoint survey itself	m
$\mathrm{RMSE}_{\mathrm{final}}$	accuracy to report, including checkpoint error (Ed. 2 requirement)	m

WORKED EXAMPLE — vertical accuracy from ten checkpoints

Given (cm): $\Delta z = +2.1,\ -3.4,\ +1.2,\ +4.5,\ -0.8,\ +2.7,\ -1.9,\ +3.1,\ +0.4,\ -2.6$. Checkpoints surveyed with RMSE 1.0 cm.

Step 1 — mean (bias)
sum $= 5.3$; $\bar{e} = 5.3/10 = +0.53$ cm (no meaningful bias)

Step 2 — squares
4.41, 11.56, 1.44, 20.25, 0.64, 7.29, 3.61, 9.61, 0.16, 6.76; sum $= 65.73$

Step 3 — RMSE
$\sqrt{65.73/10} = \sqrt{6.573} = 2.56$ cm

Step 4 — std. dev.
$s = 2.64$ cm (computed as in Module 2)

Step 5 — include checkpoint error
$\sqrt{2.56^2 + 1.0^2} = \sqrt{7.57} = 2.75$ cm

Result: report non-vegetated vertical accuracy $\text{RMSE}_z = 2.8$ cm (Edition 2 style). Ten points are fine for a quick field check, but a formal ASPRS test needs at least 30.

WORKED EXAMPLE — horizontal accuracy

Given: $\text{RMSE}_x = 2.25$ cm and $\text{RMSE}_y = 2.16$ cm from five well-defined points.

Combine
$\text{RMSE}_H = \sqrt{2.25^2 + 2.16^2} = \sqrt{5.06 + 4.67} = 3.12$ cm

Result: 3.1 cm horizontal (radial) RMSE.

13.3 The accuracy standards you will report against

ENGINEER'S VIEW

ASPRS Positional Accuracy Standards for Digital Geospatial Data, Edition 2 (2023; Version 2, 2024) is the reference most US contracts cite. Key changes from Edition 1 that affect mobile LiDAR:

- accuracy is stated as **RMSE only** — the old "95% confidence" figures ($1.96 \times \text{RMSE}_z$, $1.7308 \times \text{RMSE}_r$) are no longer reported;
- minimum **30 checkpoints** for product assessment (up to 120 for large projects);
- ground control and checkpoints must be at least **2**$\times$ more accurate than the product;
- checkpoint survey error must be **included** in the reported accuracy;
- vegetated vertical accuracy (VVA) is reported "as found"; acceptance rests on non-vegetated accuracy (NVA).

Florida. Surveys and maps prepared by Florida Professional Surveyors and Mappers must meet the Standards of Practice in **Chapter 5J-17, Florida Administrative Code**. In practice that means the responsible surveyor states the methods used, the horizontal and vertical datums, and the accuracy achieved and how it was verified, and signs for data captured by remote sensing just as for conventional measurements. FDOT publishes separate guidance for terrestrial mobile LiDAR on its projects. Rules are amended periodically — always work from the current rule text and the client's specification, not from a course book.

IMPROVED FROM THE SOURCE MATERIAL

The seminar's error-budget slide used "5 cm horizontal at 95% confidence." Under ASPRS Edition 2 the same requirement would be written as an RMSE. If a specification still quotes a 95% value, convert carefully: for vertical, $\text{RMSE}_z = \text{Acc}_{95}/1.96$; for horizontal, $\text{RMSE}_H = \text{Acc}_{95}/1.7308$ (valid when $\text{RMSE}_x \approx \text{RMSE}_y$), so 5 cm at 95% horizontal $\approx$ 2.9 cm RMSE_H — a much tighter target than "5 cm" sounds, and one the budget below (4.2 cm, 1σ per component) would *not* meet. Units and confidence levels must be settled before the budget is written.

13.4 ASPRS lidar data standards in detail

Two ASPRS documents govern almost every LiDAR deliverable in the United States: the **Positional Accuracy Standards for Digital Geospatial Data** (Edition 2, Version 2, 2024), which say how accurate the data are and how that is proven, and the **LAS Specification** (version 1.4), which says how the points are stored and labeled. Client specifications — FDOT, USGS and others — build on both.

SIMPLE APPROACH

One rulebook says *how close to the truth* the dots must be and how to check them with secret checkpoints. The other rulebook says *how to write the dots down* in a file, including a label on every dot: ground, building, tree, water, wire, noise… so any software can read it.

13.4.1 Accuracy classes (Edition 2)

ENGINEER'S VIEW

Edition 2 defines accuracy classes directly by RMSE, in any size the project needs: a "5 cm vertical accuracy class" means $\text{RMSE}_V \leq 5$ cm; a "3 cm horizontal accuracy class" means $\text{RMSE}_H \leq 3$ cm.

- **NVA** (non-vegetated vertical accuracy), tested on open hard or bare surfaces, decides **acceptance**.
- **VVA** (vegetated vertical accuracy) is tested and **reported as found**, with no pass/fail threshold unless the client and producer agree otherwise.
- **Checkpoints:** at least 30 for NVA (up to 120 for large projects) and at least 30 for VVA, well distributed; independent of any control used for calibration or boresighting; at least **2× more accurate** than the product.
- **Control** for calibration should likewise be twice as accurate as the product (e.g. $\text{RMSE}_V(\text{control}) \leq \frac{1}{2}\,\text{RMSE}_V(\text{product})$).
- **Product accuracy includes checkpoint error:** $\text{RMSE}_{\text{product}} = \sqrt{\text{RMSE}_{\text{fit}}^2 + \text{RMSE}_{\text{survey}}^2}$ (Module 13 worked example).
- **Horizontal accuracy of lidar** is usually not tested point by point (points do not land on well-defined corners) but estimated from GNSS and IMU error and range/altitude, or tested on well-defined features when density allows. For mobile LiDAR, painted chevrons and building corners make horizontal testing practical.

13.4.2 Data internal precision (relative accuracy)

Vertical class X	NVA $RMSE_V$	Within-swath smooth-surface max diff.	Swath-to-swath $RMSD_z$	Swath-to-swath max diff.
General rule	$\leq X$	$\leq 0.6X$	$\leq 0.8X$	$\leq 1.6X$
5 cm (0.16 ft) class	$\leq$ 5 cm / 0.16 ft	$\leq$ 3 cm / 0.10 ft	$\leq$ 4 cm / 0.13 ft	$\leq$ 8 cm / 0.26 ft
10 cm (0.33 ft) class	$\leq$ 10 cm / 0.33 ft	$\leq$ 6 cm / 0.20 ft	$\leq$ 8 cm / 0.26 ft	$\leq$ 16 cm / 0.52 ft

Source: ASPRS Positional Accuracy Standards, Edition 2, Version 2 (2024), Table 7.2, which gives the lidar data internal precision limits as $0.60X$, $0.80X$ and $1.60X$ of the vertical accuracy class X (verified October 2026).

These are *precision* checks, not accuracy: **within-swath** tests how smooth a flat surface (a parking lot) is in a single pass — range noise; **swath-to-swath** tests how well overlapping passes agree — trajectory and calibration (the cloud-to-cloud check of Module 18). A project can pass NVA at the checkpoints and still fail swath-to-swath agreement between them.

13.4.3 Reporting statements

ENGINEER'S VIEW

Edition 2 prescribes the wording:

- **Tested (30 or more checkpoints):** "This data set was tested to meet ASPRS Positional Accuracy Standards for Digital Geospatial Data, Edition 2, Version 2 (2024) for a __ (cm) $RMSE_V$ Vertical Accuracy Class. NVA accuracy was found to be $RMSE_V$ = __ cm." VVA is reported the same way, as found.
- **Fewer than 30 checkpoints:** the statement must say so: "…this test was performed using ONLY __ checkpoints."
- **Produced to meet (no independent test):** "This data set was produced to meet ASPRS Positional Accuracy Standards…" — a declaration by the producer only. Calibration control may not be reused as checkpoints.

13.4.4 The LAS 1.4 file standard

ENGINEER'S VIEW

- **Format:** LAS 1.4 (compressed LAZ is the common delivery form). Point data record formats 6–10 are the modern ones: 6 = base (x, y, z, intensity, return number, classification, scan angle, GPS time…); 7 adds RGB; 8 adds RGB + NIR; 9 and 10 add waveform.
- **Time:** *Adjusted Standard GPS Time* (GPS seconds minus 10^9), so every point can be traced back to the trajectory.
- **Coordinate system:** stored in the header as OGC WKT, including the vertical datum; it must match the report.
- **Classification codes** (ASPRS standard):

1	Unclassified	7	Low point (noise)	14	Wire — conductor
2	Ground	9	Water	15	Transmission tower
3	Low vegetation	10	Rail	17	Bridge deck
4	Medium vegetation	11	Road surface	18	High noise
5	High vegetation	13	Wire — guard (shield)	19	Overhead structure
6	Building	20	Ignored ground	21/22	Snow / temporal excl.

13.4.5 USGS quality levels (for context)

Clients often cite the **USGS Lidar Base Specification** for airborne elevation data: QL0 ($\geq$8 points/m^2, NVA RMSE$_z$ $\leq$ 5 cm), QL1 ($\geq$8 pts/m^2, $\leq$ 10 cm), QL2 ($\geq$2 pts/m^2, $\leq$ 10 cm). Mobile LiDAR easily exceeds these densities; its specifications are usually written project by project, which is why a signed client accuracy agreement (Appendix E) matters.

13.5 Error budgets

Before a job, list every error source you expect and combine them into a total — an *error budget* — to confirm the agreed accuracy class is achievable. All of the course's error budgets, from a single traverse-style example to the OS0, GX5/GQ7, RIEGL and capstone systems, are kept together in **Module 14** (Section 14.10).

TRY IT YOURSELF

13.1 Checkpoint residuals (cm): $3, -1, 2, -2, 4$. Compute mean, RMSE, and RMSE including 1.5 cm checkpoint error.
13.2 A budget has terms 1, 2, 2 and 4 cm. What is the total? Which term would you attack first?
13.3 (Simple) Why must we keep checkpoints a secret from the computer?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

ASPRS (2024), *Positional Accuracy Standards for Digital Geospatial Data*, Edition 2, Version 2 (checkpoints, NVA/VVA, Table 7.2, reporting statements); ASPRS *LAS Specification* 1.4; USGS *Lidar Base Specification* (quality levels); LIDAR Magazine overviews of Edition 2.

MODULE 14

Error Propagation and Error Budgets

Module 2 introduced the root-sum-square rule for independent errors. This module shows *where* each term comes from, how to handle errors that are *not* independent, how errors spread through the full georeferencing equation, and how to check the math with a simulation. It then collects **every error budget in the course** in one place (Sections 14.10 and following): a general job budget and the budgets for the OS0 backpack, the GX5/GQ7 IMUs, the RIEGL airborne system and the capstone corridor.

14.1 Three kinds of error

SIMPLE APPROACH

Imagine throwing darts. If your darts land all over the board, that is **random error** — your hand shakes a little differently every time. If all your darts land in a tight group, but always 5 cm to the left of the bullseye, that is **systematic error** — maybe the board is crooked. If one dart hits the wall, that is a **blunder**. You fix each one in a different way: practice to shrink random error, straighten the board to remove systematic error, and throw out the dart that hit the wall.

Type	What it looks like	LiDAR examples	How to handle it
Random	scatter around the true value; different every time	range noise, angle-encoder jitter, IMU white noise	average it, weight it, propagate it with σ
Systematic	same offset (or same pattern) every time	range bias, boresight residual, wrong lever arm, wrong geoid, clock offset	calibrate it, model it, or cancel it by differencing; averaging does *not* help
Blunder	a single large, impossible value	wrong antenna height, mixed pixel, multipath jump, wrong datum	detect it (residual tests, checkpoints), then remove it

ENGINEER'S VIEW

Precision describes random error (spread). **Accuracy** describes total closeness to the truth (spread *and* bias). A sensor datasheet usually quotes both separately — the Ouster OS0 in Module 15 lists "range precision" (1σ of 100 repeated shots) and "range accuracy" (error of their mean). Propagation formulas work on standard deviations and covariances; a bias must be carried separately or converted to an equivalent standard deviation with a stated assumption.

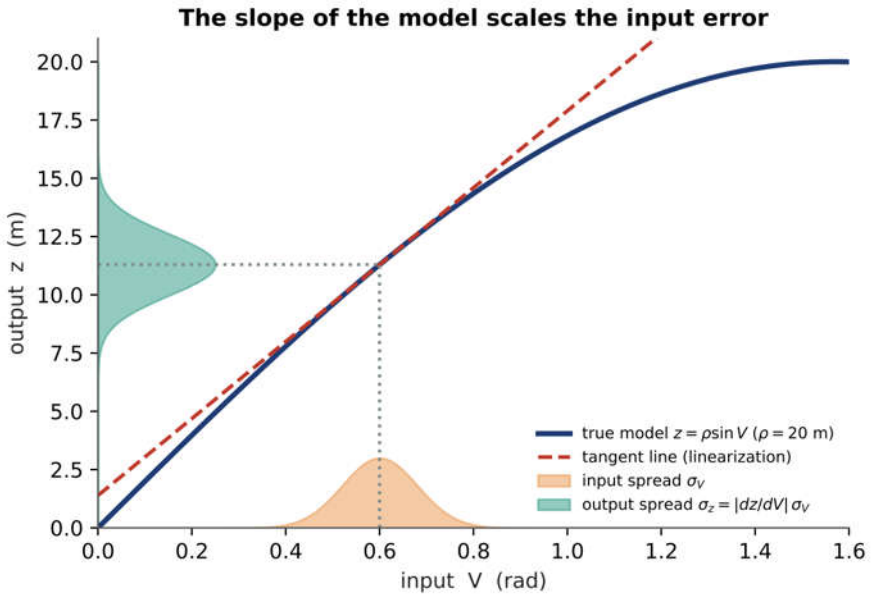


Figure 14.1: Near the operating point the model is almost a straight line. Its slope converts input spread into output spread.

14.2 The law of propagation of variances (one output)

KEY EQUATION — Propagation for one output and independent inputs

$$y = f(x_1, x_2, \ldots, x_n) \qquad \Longrightarrow \qquad \sigma_y^2 = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2$$

What every symbol means

Symbol	Meaning	Units
y	the computed result (a coordinate, a height, a distance…)	varies
f	the formula that computes it	—
x_i	the measured inputs (range, angles, pose…)	varies
$\partial f / \partial x_i$	partial derivative: how much y changes when only x_i changes by one unit (the "sensitivity" or "slope")	units of y per unit of x_i
σ_{x_i}	standard deviation of input i	units of x_i
σ_y	standard deviation of the result	units of y

ENGINEER'S VIEW

Where it comes from. Expand f in a first-order Taylor series about the measured values: $y \approx f(\bar{\mathbf{x}}) + \sum_i \frac{\partial f}{\partial x_i} \delta x_i$. The error in y is a weighted sum of the input errors, $\delta y = \sum a_i \delta x_i$ with $a_i = \partial f / \partial x_i$. For a weighted sum of independent zero-mean errors, $E[\delta y^2] = \sum a_i^2 E[\delta x_i^2]$ because all cross terms $E[\delta x_i \delta x_j]$ vanish. That is the formula. It is exact for linear models and an excellent approximation when the errors are small compared with the curvature of f — true for almost everything in surveying.

Two special cases you already know. If $y = x_1 + x_2$, both slopes are 1 and $\sigma_y = \sqrt{\sigma_1^2 + \sigma_2^2}$ (root-sum-square). If $y = kx$, then $\sigma_y = |k|\sigma_x$ — e.g. $\rho = c\,\Delta t/2$ gives $\sigma_\rho = (c/2)\sigma_t$ (Module 3).

WORKED EXAMPLE — height of a ground point seen by a scanner

Given: scanner 2.00 m above a flat road; a road point 20.00 m away horizontally. Range precision $\sigma_\rho = 0.35$ cm (dark asphalt at 20 m); vertical-angle accuracy $\sigma_V = 0.01°$. Model: $z = \rho \sin V$.

Step 1 — geometry

$\rho = \sqrt{20^2 + 2^2} = 20.100$ m; $V = -\arctan(2/20) = -5.711°$; $\sin V = -0.09950$, $\cos V = 0.99504$

Step 2 — sensitivities

$\partial z/\partial \rho = \sin V = -0.0995$; $\partial z/\partial V = \rho \cos V = 20.000$ m/rad

Step 3 — convert angle to radians

$\sigma_V = 0.01 \times \pi/180 = 1.745 \times 10^{-4}$ rad

Step 4 — each contribution

range: $0.0995 \times 0.35 = 0.035$ cm; angle: $20.000 \times 1.745 \times 10^{-4} = 0.00349$ m $= 0.349$ cm

Step 5 — combine

$\sigma_z = \sqrt{0.035^2 + 0.349^2} = 0.351$ cm

Result: 3.5 mm. Notice that on a grazing road surface the *angle* error, not the range error, controls the height — range noise points almost horizontally and barely moves z. On a wall straight ahead, the opposite is true.

14.3 When errors are correlated

KEY EQUATION — Propagation with covariance

$$\sigma_y^2 = \sum_i a_i^2 \sigma_i^2 + 2 \sum_{i<j} a_i a_j \sigma_{ij}, \qquad \sigma_{ij} = r_{ij}\,\sigma_i \sigma_j$$

What every symbol means

Symbol	Meaning	Units
a_i	sensitivity $\partial f/\partial x_i$	—
σ_{ij}	covariance of inputs i and j (how much they move together)	product of units
r_{ij}	correlation coefficient, from -1 (always opposite) through 0 (independent) to $+1$ (always together)	—

WORKED EXAMPLE — curb height from one sensor — correlation helps

Given: top of curb z_1 and gutter z_2 measured in the same sweep at about 10 m. Each has random error $\sigma = 0.25$ cm and the same range bias of 1.25 cm (fully correlated, $r = 1$ for the bias part). Curb height $h = z_1 - z_2$, so $a_1 = +1$, $a_2 = -1$.

Step 1 — random part

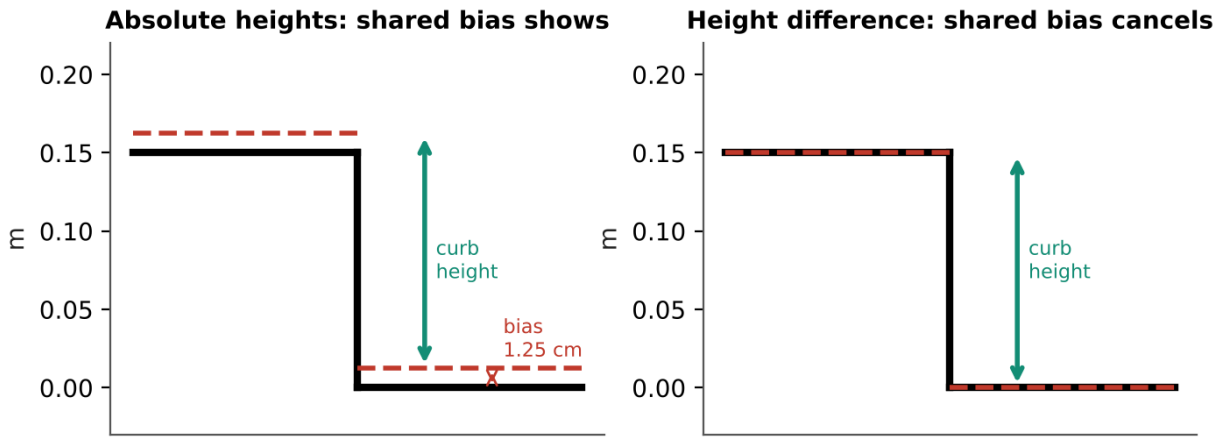


Figure 14.2: A bias shared by both points shifts absolute heights but disappears from a height *difference*.

$\sigma_h^2 = (1)^2(0.25)^2 + (-1)^2(0.25)^2 + 0 = 0.125 \text{ cm}^2 \Rightarrow 0.35 \text{ cm}$

Step 2 — bias part

$\sigma_h^2 = (1.25)^2 + (1.25)^2 + 2(1)(-1)(1)(1.25)(1.25) = 0$

Step 3 — total

$\sigma_h = 0.35 \text{ cm}$

Result: the curb height is good to 3.5 mm even though each *absolute* height is only good to about 1.3 cm. Relative measurements from the same sensor and moment are much better than absolute ones — which is why cross-sections, curb reveals and pavement cracks can be mapped more precisely than the datasheet accuracy suggests.

ENGINEER'S VIEW

Correlation can also *hurt*. If you add two quantities that share an error ($a_1 = a_2 = +1$, $r = +1$) the errors add linearly instead of in quadrature: two 1.25 cm shared biases give 2.5 cm, not 1.77 cm. Typical correlated errors in mobile LiDAR: a trajectory error that shifts every point in a stretch of road together; a boresight error that rotates every point from one sensor; a geoid error that moves a whole project. Never treat these as independent between neighboring points.

14.4 Many outputs at once: the covariance matrix

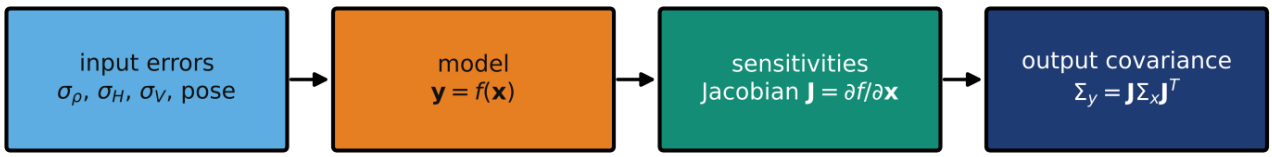


Figure 14.3: The general propagation recipe.

KEY EQUATION — General law of propagation (matrix form)

$$\mathbf{y} = f(\mathbf{x}) \qquad \Longrightarrow \qquad \boldsymbol{\Sigma}_y = \mathbf{J}\,\boldsymbol{\Sigma}_x\,\mathbf{J}^{\mathsf{T}}, \qquad J_{ki} = \frac{\partial y_k}{\partial x_i}$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{x}$	vector of all inputs (e.g. ρ, H, V)	mixed
$\mathbf{y}$	vector of all outputs (e.g. x, y, z)	mixed
$\boldsymbol{\Sigma}_x$	input covariance matrix: variances on the diagonal, covariances off it	—
$\mathbf{J}$	Jacobian: row k = how output k responds to each input	—
$\boldsymbol{\Sigma}_y$	output covariance: variances of x, y, z and how they are correlated	m^2

This single line is the same equation used for the EKF prediction ($\mathbf{GPG}^{\mathsf{T}}$, Module 10) and for adjustment results. Everything in this module is an application of it.

WORKED EXAMPLE — the error ellipsoid of one laser point

Given: OS0-class sensor 2 m above the road, a dark (10%) target at $\rho = 30.00$ m, $H = 30°$, $V = -3.823°$. $\sigma_\rho = 1.0$ cm (datasheet precision at 30 m), $\sigma_H = \sigma_V = 0.01° = 1.745 \times 10^{-4}$ rad. Point: $(25.923,\ 14.967,\ -2.000)$ m.

Step 1 — Jacobian of the polar-to-Cartesian equations (Module 5)

$$\mathbf{J} = \begin{pmatrix} \cos V \cos H & -\rho \cos V \sin H & -\rho \sin V \cos H \\ \cos V \sin H & \rho \cos V \cos H & -\rho \sin V \sin H \\ \sin V & 0 & \rho \cos V \end{pmatrix} = \begin{pmatrix} 0.8641 & -14.967 & 1.732 \\ 0.4989 & 25.923 & 1.000 \\ -0.0667 & 0 & 29.933 \end{pmatrix}$$

Step 2 — input covariance (mm, rad)

$\boldsymbol{\Sigma}_x = \mathrm{diag}(10^2,\ (1.745 \times 10^{-4})^2,\ (1.745 \times 10^{-4})^2)$

Step 3 — one entry by hand

$\Sigma_{xx} = (0.8641 \cdot 10)^2 + (14.967 \cdot 0.1745)^2 + (1.732 \cdot 0.1745)^2 = 74.67 + 6.82 + 0.09 = 81.58\ \mathrm{mm}^2$
(the 0.1745 is 1.745×10^{-4} rad $\times 1000$ mm/m)

Step 4 — full result

$$\boldsymbol{\Sigma}_y = \begin{pmatrix} 81.58 & 31.34 & -4.18 \\ 31.34 & 45.39 & -2.41 \\ -4.18 & -2.41 & 27.74 \end{pmatrix} \mathrm{mm}^2 \Rightarrow \sigma_x = 9.0,\ \sigma_y = 6.7,\ \sigma_z = 5.3\ \mathrm{mm}$$

Step 5 — principal axes (eigenvalues)

$\sqrt{\lambda} = 10.0$ mm *along the beam*, 5.2 mm and 5.2 mm across it ($= \rho\sigma_H$ and $\rho\sigma_V$)

Result: the point's uncertainty is a cigar pointing back at the scanner. The off-diagonal 31.34 mm^2 is not a mistake: because the beam runs diagonally (30°), x and y errors are correlated.

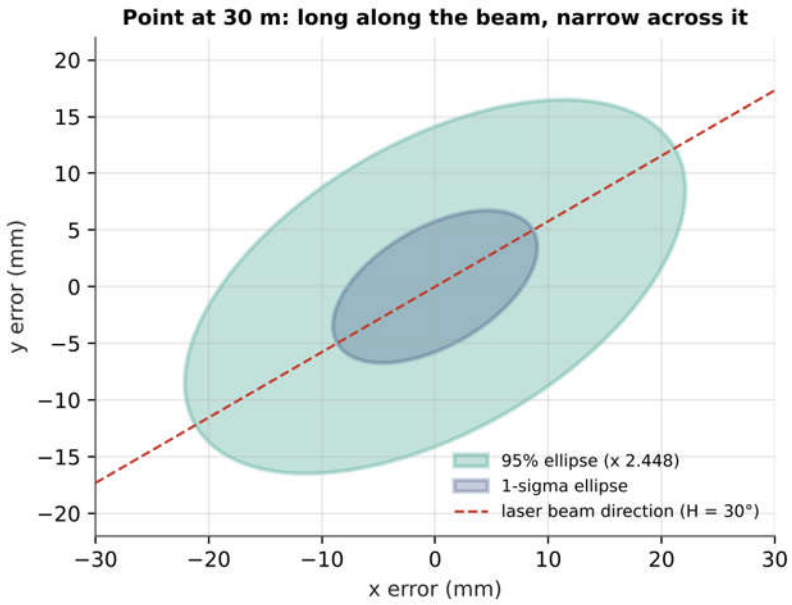


Figure 14.4: Horizontal error ellipse of the worked example. The 95% ellipse is 2.448 times the 1σ ellipse.

KEY EQUATION — From σ to a confidence region

95% regions: 1-D: $\pm 1.960\,\sigma$ 2-D: $2.448 \times$ semi-axes 3-D: $2.796 \times$ semi-axes

What every symbol means

Symbol	Meaning	Units
1.960, 2.448, 2.796	square roots of the 95th percentile of the chi-square distribution with 1, 2 and 3 degrees of freedom	—
semi-axes	square roots of the eigenvalues of the covariance matrix	m

For the worked example: 95% horizontal ellipse semi-axes $2.448 \times 10.0 = 24.4$ mm along the beam and $2.448 \times 5.2 = 12.8$ mm across it. Notice that the 95% factor grows with the number of dimensions — a point is less likely to be inside a 1.96σ box in 3-D than on a 1.96σ line.

14.5 Propagating through the full georeferencing equation

Now combine everything from Modules 5–9. The direct georeferencing equation (Module 9) has fifteen uncertain inputs:

$$\mathbf{r}_W = \mathbf{p}_W + \mathbf{R}_B^W(\phi,\theta,\psi)\left(\mathbf{R}_S^B(\delta\phi,\delta\theta,\delta\psi)\,\mathbf{r}_S(\rho,H,V) + \mathbf{l}^B\right)$$

three positions, three attitude angles, three boresight angles, three lever-arm components, and the range and two angles of the laser. The Jacobian is 3×15; it is computed by software (numerically or analytically), but every column has a simple meaning:

Input group	Sensitivity (first-order)	Grows with range?
trajectory position	1 (moves the point one-for-one)	no
attitude (roll, pitch, yaw)	ρ (small-angle rule)	yes, linearly
boresight	ρ	yes, linearly
lever arm	1 (rotated by attitude)	no
range	1 along the beam	only via precision vs. range
angle encoders	ρ across the beam	yes, linearly

WORKED EXAMPLE — full error budget of a backpack point at 20 m

Given (1σ): trajectory position 1.5 cm E, 1.5 cm N, 2.0 cm U (SLAM + control); roll and pitch 0.02°, yaw 0.05°; boresight 0.01° per axis; lever arm 2 mm per axis; OS0 range accuracy 1.25 cm combined with precision 0.40 cm; angle encoders 0.01°. The point is on a wall 20 m to the side, at sensor height.

Step 1 — position
horizontal $\sqrt{1.5^2 + 1.5^2} = 2.12$ cm; vertical 2.00 cm

Step 2 — attitude
yaw swings the point sideways: $20 \times 0.05°$ in rad $= 20 \times 8.73 \times 10^{-4} = 1.75$ cm (horizontal); roll tips it up/down: $20 \times 3.49 \times 10^{-4} = 0.70$ cm (vertical)

Step 3 — boresight
$20 \times 1.745 \times 10^{-4} = 0.35$ cm horizontal (yaw part) and 0.35 cm vertical (roll part)

Step 4 — lever arm
0.28 cm horizontal, 0.20 cm vertical (fixed size)

Step 5 — range
$\sqrt{1.25^2 + 0.40^2} = 1.31$ cm, all horizontal (the beam is level)

Step 6 — angle encoders
$20 \times 1.745 \times 10^{-4} = 0.35$ cm horizontal and 0.35 cm vertical

Step 7 — combine (RSS)
horizontal $\sqrt{2.12^2 + 1.75^2 + 0.35^2 + 0.28^2 + 1.31^2 + 0.35^2} = \sqrt{9.60} = 3.10$ cm; vertical $\sqrt{2.00^2 + 0.70^2 + 0.35^2 + 0.20^2 + 0.35^2} = \sqrt{4.78} = 2.18$ cm

Result: 3.1 cm horizontal and 2.2 cm vertical (1σ). The trajectory and the heading are the two biggest terms — the laser itself is third.

SURVEYOR'S FIELD TIP

Use the curve to set your **maximum useful range**. With this system, a 3 cm horizontal spec is met only out to about 18 m; beyond that, either filter points by range in processing or improve the heading (more loop closures, better IMU, control along the route).

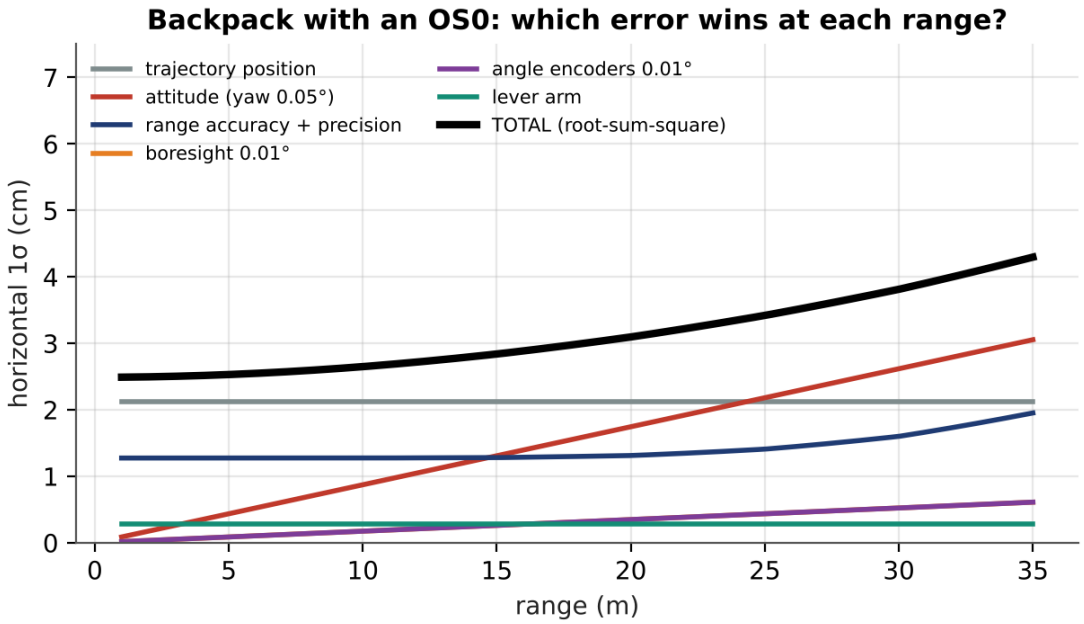


Figure 14.5: Contributions versus range for the same backpack. Fixed terms dominate close in; angle terms take over far out.

14.6 Checking with Monte Carlo simulation

SIMPLE APPROACH

Instead of doing hard math, you can just *pretend* thousands of times. Roll dice to make up small random errors for every input, compute the point, and write it down. Do it 100,000 times on a computer and look at how spread out the answers are. If the spread matches the formula, you know the formula was right.

ENGINEER'S VIEW

Monte Carlo propagation: (1) draw each input from its distribution (e.g. $\rho_k = \bar{\rho} + \sigma_\rho \varepsilon_k$, $\varepsilon_k \sim N(0,1)$); (2) evaluate the full non-linear model; (3) repeat N times; (4) compute the sample covariance of the outputs. It needs no derivatives, handles strong non-linearity and non-Gaussian inputs (for example a uniform quantization error), and is the standard cross-check for an analytic error budget. Its own precision improves as $1/\sqrt{N}$: 100 000 samples estimate σ to about 0.2%.

WORKED EXAMPLE — Monte Carlo check of the 20 m budget

Run

100 000 draws of all fifteen inputs through the full equation (rotation matrices, no approximations)

Result

horizontal $\sigma = 3.09$ cm (linear: 3.10); vertical $\sigma = 2.18$ cm (linear: 2.18)

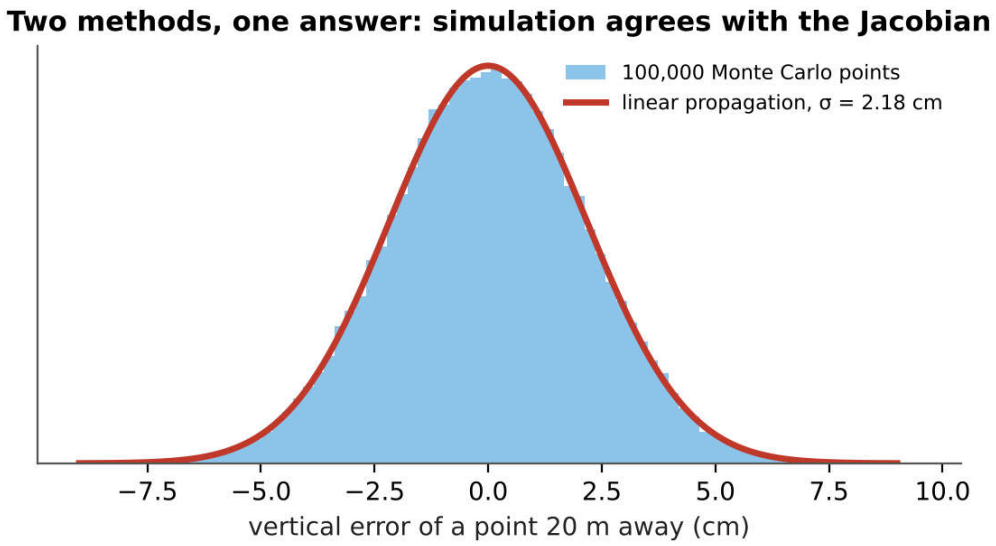


Figure 14.6: Vertical error of the 20 m point: 100 000 simulated georeferencings (bars) versus the linear-propagation Gaussian (line).

Result: agreement to better than 1%. For angles of a few hundredths of a degree the linear method is essentially exact.

14.7 Averaging, and the floor set by bias

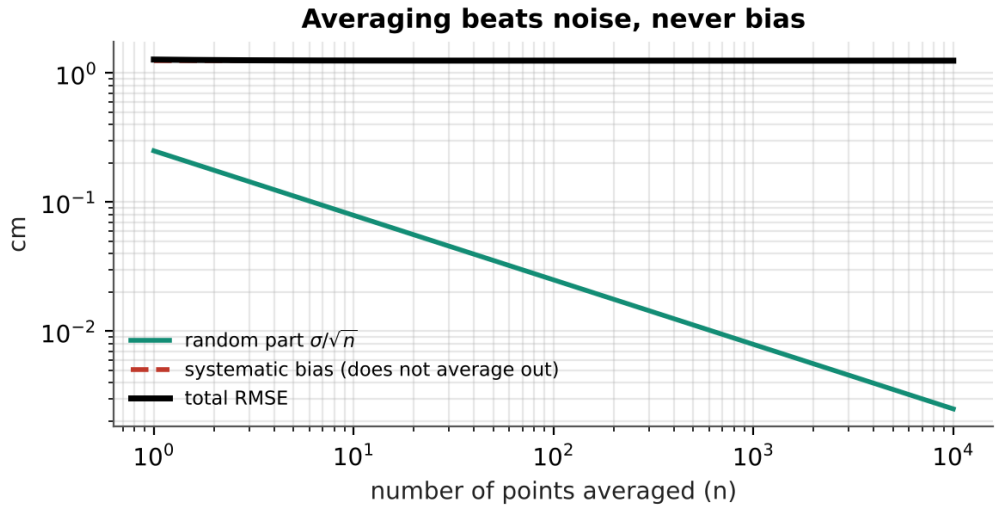


Figure 14.7: Averaging n points shrinks random error as $1/\sqrt{n}$ but cannot go below the systematic bias.

KEY EQUATION — Accuracy of an average

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \qquad \mathrm{RMSE}_{\bar{x}} = \sqrt{b^2 + \frac{\sigma^2}{n}}$$

What every symbol means

Symbol	Meaning	Units
n	number of independent measurements averaged	—
σ	random error of one measurement	m
$\sigma_{\bar{x}}$	random error of the average	m
b	systematic bias common to all measurements	m

WORKED EXAMPLE — fitting a plane to a target

Given: 400 OS0 returns on a flat 90% target at 10 m: $\sigma = 0.25$ cm each, bias $b = 1.25$ cm.

Random part

$0.25/\sqrt{400} = 0.25/20 = 0.0125$ cm

Total

$\sqrt{1.25^2 + 0.0125^2} = 1.2501$ cm

Result: averaging made the noise 20 times smaller and changed the accuracy by almost nothing. Only calibration against known distances removes the 1.25 cm. Also note that the 400 points are only independent if they came from different pulses; points from overlapping beams (Module 15) are partly correlated and average less well.

14.8 Errors that grow with time

ENGINEER'S VIEW

White noise on a gyro integrates into a **random walk** in heading: the error grows with the *square root* of time, $\sigma_\psi(t) = \mathrm{ARW}\sqrt{t}$, where ARW (angle random walk, $^\circ/\sqrt{\mathrm{h}}$) is on the IMU datasheet. A constant bias grows *linearly* ($b_g t$, Module 7). So over short times noise dominates and over long times bias dominates.

Example (assumed MEMS values): ARW $= 0.2^\circ/\sqrt{\mathrm{h}}$. After 60 s $= 1/60$ h: $\sigma_\psi = 0.2 \times \sqrt{1/60} = 0.026^\circ$. A 2°/h bias over the same minute adds $2/60 = 0.033^\circ$. The two are comparable at one minute; at ten minutes the bias term is four times larger. This is why SLAM loop closures and control every few minutes keep heading — the biggest term in the budget above — under control.

14.9 Errors in derived products

WORKED EXAMPLE — stockpile volume

Given: pile footprint 2 000 m^2; surface built from many points; a vertical bias of 1.25 cm across the whole pile; random height noise averages out over thousands of points.

Systematic part

$\delta V = A\,b = 2\,000 \times 0.0125 = 25\ \text{m}^3$

Random part

with 2 000 one-meter cells each averaging ~100 points of $\sigma = 0.4$ cm: per cell 0.04 cm; total $\sigma_V = 1\,\text{m}^2 \times 0.0004\ \text{m} \times \sqrt{2\,000} = 0.018\ \text{m}^3$

Result: 25 m^3 of possible error comes from the bias and essentially none from noise. For volumes, **vertical bias is everything**: check it on hard surfaces next to the pile.

14.10 Error budgets: from sources to a total

This section and the next hold every error budget in the course. The case-study modules (15–17) and the capstone (Module 21) point back here.

SIMPLE APPROACH

If you pack a backpack, every item takes up some room, and the bag can only hold so much. An error budget is a list of every little thing that can make a dot land in the wrong place, and how much "room" each one takes, so we know the whole bag still closes.

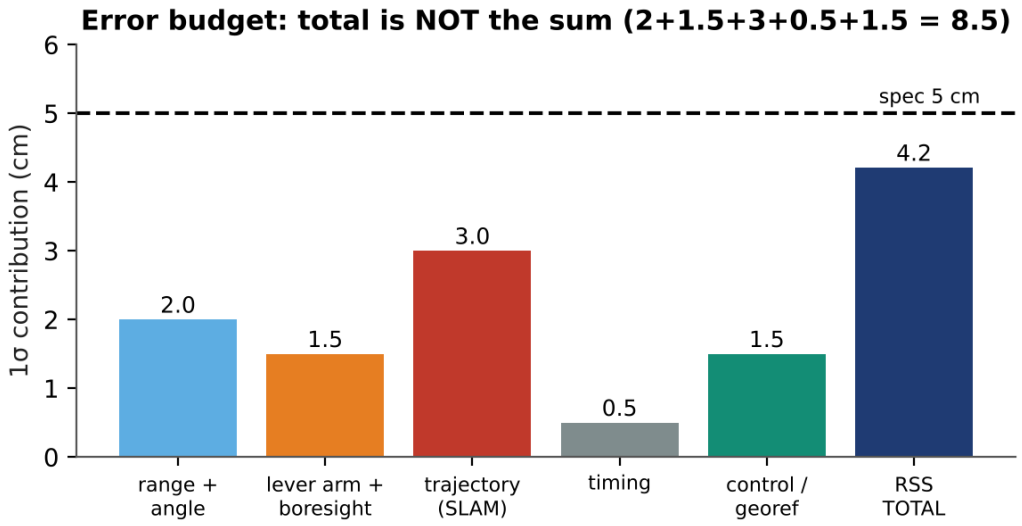


Figure 14.8: Independent 1σ contributions combine by root-sum-square, not by simple addition.

WORKED EXAMPLE — RSS budget for a 5 cm (0.16 ft, 1σ) job

Given (1σ, cm): range + angle 2.0; lever arm + boresight 1.5; trajectory 3.0; timing 0.5; control/-georeferencing 1.5.

Step 1 — squares

$4.00 + 2.25 + 9.00 + 0.25 + 2.25 = 17.75$

Step 2 — root

$\sqrt{17.75} = 4.21$ cm (0.138 ft) < 5 cm ✓

Step 3 — what if

a long featureless stretch raises the trajectory term to 5.0 cm: $\sqrt{4 + 2.25 + 25 + 0.25 + 2.25} = \sqrt{33.75} = 5.81$ cm (0.191 ft) **fails**

Result: the trajectory term dominates. To fix it: add control in the weak stretch, slow down, add a return pass for loop closures — not buy a better laser.

Error source (computed in Modules 3–9)	Size
Range noise, good unit, hard target, near	1–3 cm (0.03–0.10 ft)
Range noise, long range / dark target	5–15 cm (0.16–0.49 ft)
Beam footprint at 50 m (164 ft), 0.3 mrad	2.0 cm (0.07 ft) spot
1 ms timing at 20 m/s (45 mph)	2.0 cm (0.07 ft) along-track
0.05° boresight residual at 80 m (262 ft)	7.0 cm (0.23 ft) lateral
Tactical gyro bias 1°/h, 60 s unaided at 15 m/s	13 cm (0.43 ft) lateral
Tactical accel bias 0.5 mg, 30 s unaided	2.2 m (7.2 ft)
Consumer MEMS IMU	10–100× worse

14.10.1 Other error sources to plan around

- **GNSS multipath** near buildings and under bridges: biased positions that look precise. Watch trajectory standard deviations and check against control.
- **Dynamic occlusions**: traffic hides the curb and creates moving "ghost" objects; collect at low-traffic times or make two passes.
- **Hallway effect**: weak along-track constraint (Module 11).
- **Thermal drift**: IMU biases and mount geometry change as the system warms; warm up and do an initial calibration loop.
- **Systematic errors** (wrong geoid, datum, lever-arm typo) do not shrink with RSS or with more data — they shift everything. Only independent checks catch them.

14.11 Error budgets for the case-study systems

The same recipe — list the terms, convert each to a ground distance, root-sum-square — applied to the four systems in this course. Specifications come from the case-study modules cited in each subsection.

14.11.1 Ouster OS0 backpack (Module 15)

The backpack point at 20 m worked earlier in this module (Section 14.5) uses the OS0's range terms. Repeating it at other ranges:

WORKED EXAMPLE — what accuracy can an OS0 backpack deliver?

Given: the backpack budget of Section 14.5 (trajectory 1.5/1.5/2.0 cm; roll/pitch 0.02°, yaw 0.05°; boresight 0.01°; lever arm 2 mm) with the OS0 sensor terms of Module 15 (range accuracy 1.25 cm, precision by range, encoders 0.01°).

5 m (16 ft)

horizontal 2.5 cm (0.08 ft), vertical 2.0 cm (0.07 ft)

10 m (33 ft)

horizontal 2.7 cm (0.09 ft), vertical 2.1 cm (0.07 ft)

20 m (66 ft)

horizontal 3.1 cm (0.10 ft), vertical 2.2 cm (0.07 ft)

35 m (115 ft)

horizontal 4.3 cm (0.14 ft), vertical 2.5 cm (0.08 ft)

Result: with good SLAM and control, an OS0 system supports roughly 2–3 cm (1σ) mapping within 15–20 m. The sensor's own range bias (1.25 cm) is a floor no amount of processing removes without calibration; the trajectory, not the sensor, is the biggest term.

14.11.2 IMU attitude: GX5-25, GX5-45 and GQ7 (Module 16)

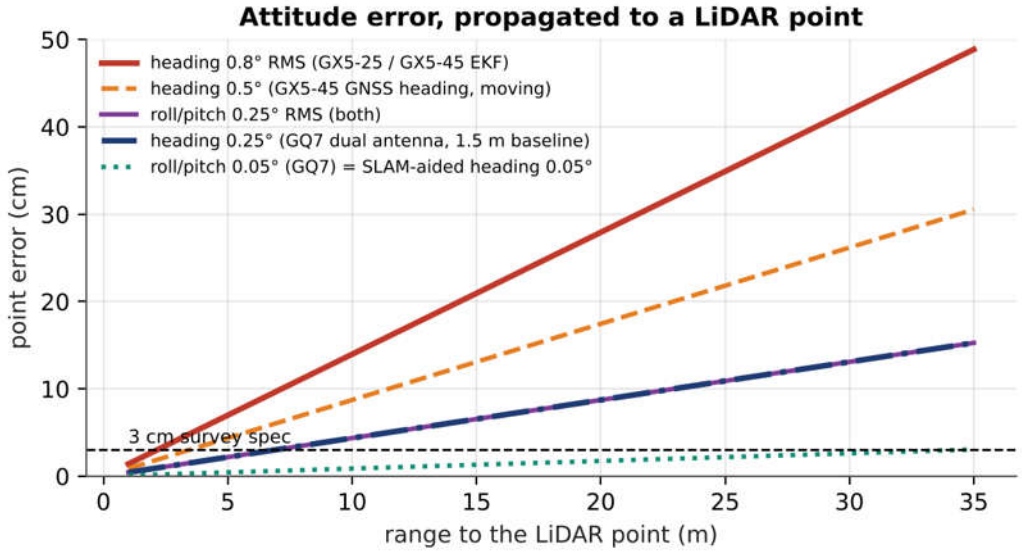


Figure 14.9: Sideways and vertical point error caused by each sensor's attitude accuracy, versus range. The GQ7 heading line coincides with the GX5 roll/pitch line, and the GQ7 roll/pitch line with SLAM-aided heading.

Attitude error source	5 m / 16 ft	10 m / 33 ft	20 m / 66 ft	35 m / 115 ft
Heading 0.8° RMS (GX5 EKF)	7.0 cm / 0.23 ft	14.0 cm / 0.46 ft	27.9 cm / 0.92 ft	48.9 cm / 1.60 ft
Heading 0.5° (GX5-45, GNSS, moving)	4.4 cm / 0.14 ft	8.7 cm / 0.29 ft	17.5 cm / 0.57 ft	30.5 cm / 1.00 ft
Heading 0.25° (GQ7, 1.5 m baseline)	2.2 cm / 0.07 ft	4.4 cm / 0.14 ft	8.7 cm / 0.29 ft	15.3 cm / 0.50 ft
Roll/pitch 0.25° RMS (GX5)	2.2 cm / 0.07 ft	4.4 cm / 0.14 ft	8.7 cm / 0.29 ft	15.3 cm / 0.50 ft
Roll/pitch 0.05° (GQ7)	0.4 cm / 0.01 ft	0.9 cm / 0.03 ft	1.7 cm / 0.06 ft	3.1 cm / 0.10 ft
SLAM-aided heading 0.05°	0.4 cm / 0.01 ft	0.9 cm / 0.03 ft	1.7 cm / 0.06 ft	3.1 cm / 0.10 ft

WORKED EXAMPLE — georeferencing an OS0 point with the GX5-45 alone

Given: a point 20 m to the side; GX5-45 position ±2 m H, ±5 m V; heading 0.8°; roll/pitch 0.25°; OS0 range error 1.31 cm.

Horizontal

$\sqrt{2.00^2 + (20 \times 0.01396)^2 + 0.0131^2} = \sqrt{4 + 0.078 + 0.0002} = 2.02$ m

Vertical

$\sqrt{5.00^2 + (20 \times 0.00436)^2} = 5.00$ m

Relative (inside one scan)

position error cancels between neighboring points; heading still shears the scan by 28 cm at 20 m

Result: 2 m horizontal and 5 m vertical. Without SLAM and control, a GX5-45 + OS0 rig is a mapping toy; with them, it can be a survey tool.

WORKED EXAMPLE — the same point with the GQ7 and RTK

Given: point 20 m to the side; GQ7 RTK 2 cm H and V; heading 0.25°; roll/pitch 0.05°; OS0 range error 1.31 cm.

Horizontal

$\sqrt{2.0^2 + (20 \times 4.36 \times 10^{-3} \times 100)^2 + 1.31^2} = \sqrt{4 + 76.2 + 1.7} = 9.0$ cm

Vertical

$\sqrt{2.0^2 + (20 \times 8.73 \times 10^{-4} \times 100)^2} = \sqrt{4 + 3.0} = 2.7$ cm

Result: vertical already meets a 3 cm spec; horizontal is held back almost entirely by heading (8.7 of the 9.0 cm). Inside SLAM, LiDAR matching tightens heading toward 0.05°, which brings the horizontal to about 3 cm (Module 14). Position from RTK plus heading from LiDAR is the winning combination.

14.11.3 RIEGL VUX-240[24] at 500 ft (Module 17)

Geometry (Case A nadir, Case B 37.5° edge) and specifications are in Module 17.

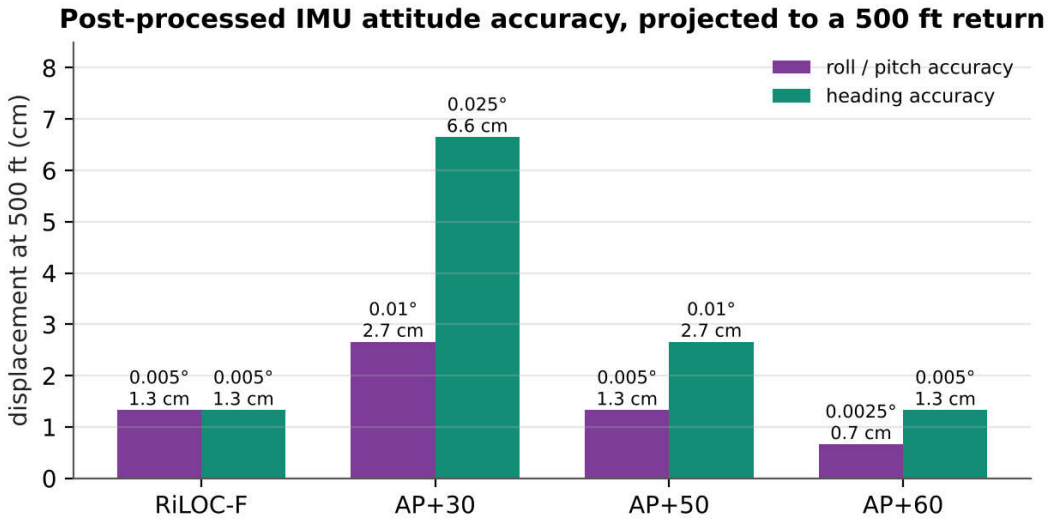


Figure 14.10: Each IMU option's post-processed attitude accuracy projected to the ground at 500 ft (small-angle rule, Module 2).

KEY EQUATION — Attitude error to ground error at 500 ft

$$\delta = \rho\,\delta\alpha \ \text{(rad)}, \qquad \rho = 152.4 \text{ m}$$

$$0.0025^\circ \rightarrow 0.7 \text{ cm}, \quad 0.005^\circ \rightarrow 1.3 \text{ cm}, \quad 0.010^\circ \rightarrow 2.7 \text{ cm}, \quad 0.025^\circ \rightarrow 6.6 \text{ cm}$$

What every symbol means

Symbol	Meaning	Units
ρ	range to the point, 500 ft = 152.4 m	m
$\delta\alpha$	attitude (roll, pitch or heading) or boresight error	rad
δ	displacement of the point on the ground	m

ENGINEER'S VIEW

Where each angle pushes the point (beam at scan angle θ from nadir, across track):

- **Roll** rotates the scan plane: horizontal $\rho\cos\theta\,\delta\phi$, vertical $\rho\sin\theta\,\delta\phi$. At nadir roll is purely horizontal; at the swath edge it starts to tilt heights.
- **Pitch** moves the point along track by $\rho\cos\theta\,\delta\theta$.
- **Heading** rotates the swath about the vertical: along-track $\rho\sin\theta\,\delta\psi$ — zero at nadir, largest at the edges.
- **Range** error lies along the beam: vertical $\delta\rho\cos\theta$, horizontal $\delta\rho\sin\theta$.
- **GNSS position** moves every point one-for-one; vertical GNSS is typically about 1.5× horizontal (assumed here).
- **Boresight** residuals act like roll, pitch and heading errors (assumed 0.005° each after calibration); **lever arm** adds its own size (assumed 5 mm).
- **Timing:** 1PPS synchronization at the microsecond level makes timing negligible (10 μs at 60 kn = 0.3 mm); a 1 ms error would add 3.1 cm along track.

RMS budget at 500 ft

WORKED EXAMPLE — RiLOC-F, Case A — nadir return from 500 ft AGL

Assumptions: GNSS 2.5 cm H, 3.75 cm V (mid-range of 2–3 cm, vertical 1.5×); roll = pitch = heading = 0.005°; boresight 0.005° per axis; lever arm 5 mm; laser range 25 mm = 2.5 cm (RIEGL 20 mm accuracy and 15 mm precision combined).

Horizontal

GNSS 2.5; roll and pitch $\sqrt{1.33^2 + 1.33^2} = 1.88$; heading 0 (nadir); boresight 1.88; lever arm 0.71; encoder 0.08

$\sqrt{2.5^2 + 1.88^2 + 1.88^2 + 0.71^2 + 0.08^2} = \sqrt{13.84} =$ **3.7 cm (0.12 ft)**

Vertical

GNSS 3.75; range 2.5; lever arm 0.5; attitude ≈ 0 at nadir

$\sqrt{3.75^2 + 2.5^2 + 0.5^2} = \sqrt{20.6} =$ **4.5 cm (0.15 ft)**

Result: both under 5 cm, but the vertical has little margin. The largest single term is GNSS, then the laser range.

WORKED EXAMPLE — RiLOC-F, Case B — 500 ft slant return at the 37.5° edge

Horizontal

GNSS 2.5; roll/pitch 1.49; heading $152.4 \times \sin 37.5° \times 8.73 \times 10^{-5} = 0.81$; range $2.5 \sin 37.5° = 1.52$; boresight 1.70; lever $0.71 \Rightarrow$ **3.9 cm (0.13 ft)**

Vertical

GNSS 3.75; roll $152.4 \sin 37.5° \times 8.73 \times 10^{-5} = 0.81$; range $2.5 \cos 37.5° = 1.98$; boresight 0.81; lever $0.5 \Rightarrow$ **4.4 cm (0.15 ft)**

Result: at the swath edge, attitude errors begin to tilt heights and heading starts to matter; the totals barely change because GNSS still dominates.

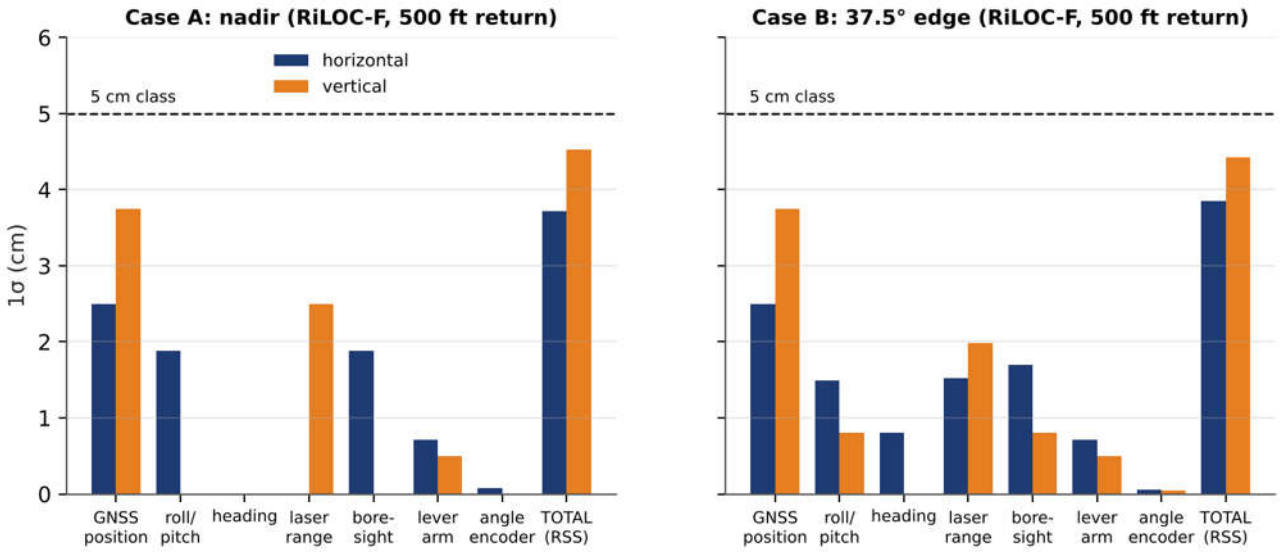


Figure 14.11: Error contributions (1σ) for the RiLOC-F at 500 ft. The dashed line is the 5 cm class.

Option	H nadir	H edge	V nadir	V edge
RiLOC-F	3.7 cm (0.12 ft)	3.9 cm (0.13 ft)	4.5 cm (0.15 ft)	4.4 cm (0.15 ft)
AP+30	5.5 cm (0.18 ft)	6.6 cm (0.22 ft)	5.8 cm (0.19 ft)	5.9 cm (0.19 ft)
AP+50	4.5 cm (0.15 ft)	4.8 cm (0.16 ft)	5.8 cm (0.19 ft)	5.8 cm (0.19 ft)
AP+60	4.1 cm (0.14 ft)	4.4 cm (0.14 ft)	5.8 cm (0.19 ft)	5.7 cm (0.19 ft)

Direct georeferencing only, before ground control or strip adjustment. Applanix GNSS taken as 3.5 cm H / 5.25 cm V (mid-range of 2–5 cm); assumptions as in the worked examples.

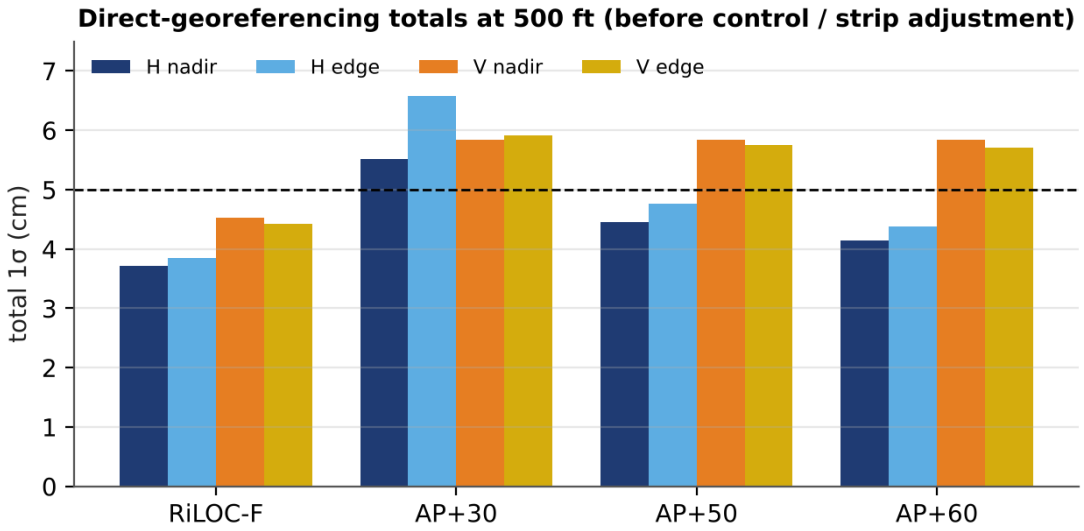


Figure 14.12: Totals for each IMU/GNSS option. Better attitude helps horizontal; vertical is limited by GNSS and laser range.

From direct georeferencing to survey grade

WORKED EXAMPLE — what ground control and strip adjustment buy you

Given: after a strip (boresight/trajectory) adjustment tied to ground control, assume the remaining GNSS-related error drops to 1.5 cm H and 1.5 cm V; checkpoint survey RMSE 1.0 cm.

RiLOC-F, nadir, horizontal

$\sqrt{1.5^2 + 1.88^2 + 1.88^2 + 0.71^2} = 3.1$ cm

RiLOC-F, nadir, vertical

$\sqrt{1.5^2 + 2.5^2 + 0.5^2} = 3.0$ cm; with checkpoint error $\sqrt{3.0^2 + 1.0^2} = 3.1$ cm

AP+30, edge, horizontal

$\sqrt{1.5^2 + 2.98^2 + 4.05^2 + 1.52^2 + 1.70^2 + 0.71^2} = 5.8$ cm — still over 5 cm, driven by the 0.025° heading

Result: with control and adjustment, the RiLOC-F (or AP+50/60) system supports a **5 cm horizontal and 5 cm vertical class with margin**, and approaches a 3 cm vertical class. The AP+30 option needs a narrower swath (heading error grows toward the edges) to meet 5 cm horizontally.

14.11.4 Capstone corridor (Module 21)

WORKED EXAMPLE — capstone vertical error budget, 3 cm (0.10 ft) specification

1σ terms (cm): range noise on pavement 0.8; boresight 1.0; lever arm 0.3; trajectory 1.8 (open sky); control 1.0; timing 0.1.

Open sky

$\sqrt{0.64 + 1.00 + 0.09 + 3.24 + 1.00 + 0.01} = \sqrt{5.98} = 2.45$ cm (0.080 ft) ✓

Underpass, trajectory 2.5
$\sqrt{8.99} = 3.00$ cm (0.098 ft) — right at the limit ×

Underpass with mid-point control, trajectory 1.5
$\sqrt{4.99} = 2.23$ cm (0.073 ft) ✓

Result: the extra control target in the underpass is what makes the job pass.

14.12 Rules of thumb

1. Square, add, root — but only for independent errors.
2. Angle errors become length errors in proportion to range ($\rho\,\delta\alpha$).
3. Biases do not average out; they cancel only in differences from the same source.
4. The error of a point is an ellipsoid, longest along the beam when range noise dominates.
5. 95%: multiply by 1.96 (1-D), 2.448 (2-D), 2.796 (3-D).
6. Verify a budget with Monte Carlo, and verify the Monte Carlo with checkpoints.

TRY IT YOURSELF

14.1 A point is 15 m straight ahead on a wall at sensor height. $\sigma_\rho = 0.3$ cm, $\sigma_H = 0.01°$. What are the along-beam and across-beam errors?
14.2 Two independent errors of 2 cm and 3 cm are *fully* correlated ($r = +1$) and added. What is the total?
14.3 How many points must be averaged to bring 0.5 cm of noise below 0.05 cm?
14.4 A 1-D error has $\sigma = 1.5$ cm. What is the 95% value?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Ghilani (2017), propagation of variances; JCGM 100:2008 (GUM) and JCGM 101:2008 (Monte Carlo propagation); Glennie (2007), kinematic LiDAR error analysis; Ouster OS0, MicroStrain 3DM-GX5/GQ7 and RIEGL VUX-240[24] datasheets for every sensor term; budget assumptions (trajectory, boresight, lever arm, GNSS for the RIEGL) are stated where used.

MODULE 15

Ouster OS0 Case Study: Errors and Resolution

This module reads a real datasheet line by line — the **Ouster OS0 Rev8** short-range imaging lidar (datasheet rev. 08/03/2026, model OS0-080) — and turns every specification into numbers a surveyor can use. Datasheet values are quoted as published; where a value had to be read from a chart, or where we make an assumption, we say so. Always check the latest datasheet for your unit.

SIMPLE APPROACH

A datasheet is like the label on a box of crayons: it tells you how many colors, how sharp the tips are, and how long they last. But to know whether you can draw a tiny ant or only a big elephant, you have to do a little math with those numbers. That is what this module does for the OS0 laser scanner.

15.1 The specifications that matter for mapping

Specification	OS0 Rev8 value (datasheet)
Range, 80% target / 10% target	75 m (246 ft) / 35 m (115 ft) (100 klx sunlight, >90% detection, 1024 @ 10 Hz)
Minimum range	0.0 m (0.3 m / 1.0 ft optional, 0.5 m / 1.6 ft default)
Channels (vertical resolution)	32, 64 or 128
Field of view	90° vertical (+45° to −45°), 360° horizontal
Horizontal resolution	512, 1024, 2048 or 4096 columns per rotation
Rotation rate	5–40 Hz (modes: 512 and 1024 @ 10–40 Hz; 2048 @ 10/20 Hz; 4096 @ 5/10 Hz)
Angular sampling accuracy	±0.01° vertical and horizontal
Range resolution	0.1 cm / 0.003 ft (0.8 cm / 0.026 ft in Low Data Rate profile)
Range precision (1σ)	±0.25 cm (0.008 ft) min to ±1.5 cm (0.049 ft) max (beyond 1 m / 3.3 ft, 1024 @ 10 Hz)
Range accuracy	±1.25 cm (0.041 ft) Lambertian; ±2.5 cm (0.082 ft) retroreflective
Returns	up to 2
Laser	865 nm, Class 1; beam 5 mm (0.2 in) at exit; divergence 0.35° (FWHM)
Points per second (max)	10 485 760
Timestamp resolution / latency	< 1 μs / < 10 ms
Time sync	PTP, gPTP, external PPS: < 1 ms error; internal clock: < 20 ppm
Built-in IMU	6-axis (InvenSense IAM-20685HP), 640 / 1280 / 2560 Hz (default)
Data rate, 128 ch @ 1024×10 Hz	43–295 Mbps

15.2 Angular resolution and point spacing

KEY EQUATION — Angular steps of a spinning sensor

$$\Delta\theta_H = \frac{360^\circ}{N_{\text{col}}}, \qquad \Delta\theta_V \approx \frac{\text{vertical FOV}}{N_{\text{ch}} - 1}, \qquad s = \rho\,\Delta\theta\ (\text{rad})$$

What every symbol means

Symbol	Meaning	Units
N_{col}	columns (firings per channel) per rotation: 512–4096	—
N_{ch}	number of channels: 32, 64, 128	—
$\Delta\theta_H, \Delta\theta_V$	horizontal and vertical angular step between samples	deg
s	spacing between neighboring points on a surface facing the sensor	m

(The OS0's actual beam elevation angles can be queried from the sensor's intrinsic calibration and need not be exactly evenly spaced; the formula gives the average.)

Mode	Step	5 m / 16 ft	10 m / 33 ft	20 m / 66 ft	35 m / 115 ft
512 columns	0.703°	6.1 cm / 0.20 ft	12.3 cm / 0.40 ft	24.5 cm / 0.80 ft	43.0 cm / 1.41 ft
1024 columns	0.352°	3.1 cm / 0.10 ft	6.1 cm / 0.20 ft	12.3 cm / 0.40 ft	21.5 cm / 0.71 ft
2048 columns	0.176°	1.5 cm / 0.05 ft	3.1 cm / 0.10 ft	6.1 cm / 0.20 ft	10.7 cm / 0.35 ft
4096 columns	0.088°	0.8 cm / 0.03 ft	1.5 cm / 0.05 ft	3.1 cm / 0.10 ft	5.4 cm / 0.18 ft
Vertical, 128 ch	0.709°	6.2 cm / 0.20 ft	12.4 cm / 0.41 ft	24.7 cm / 0.81 ft	43.3 cm / 1.42 ft
Vertical, 64 ch	1.43°	12.5 cm / 0.41 ft	24.9 cm / 0.82 ft	49.9 cm / 1.64 ft	87.3 cm / 2.86 ft
Vertical, 32 ch	2.90°	25.3 cm / 0.83 ft	50.7 cm / 1.66 ft	101 cm / 3.31 ft	177 cm / 5.81 ft
Beam footprint (FWHM)	0.35°	3.6 cm / 0.12 ft	6.6 cm / 0.22 ft	12.7 cm / 0.42 ft	21.9 cm / 0.72 ft

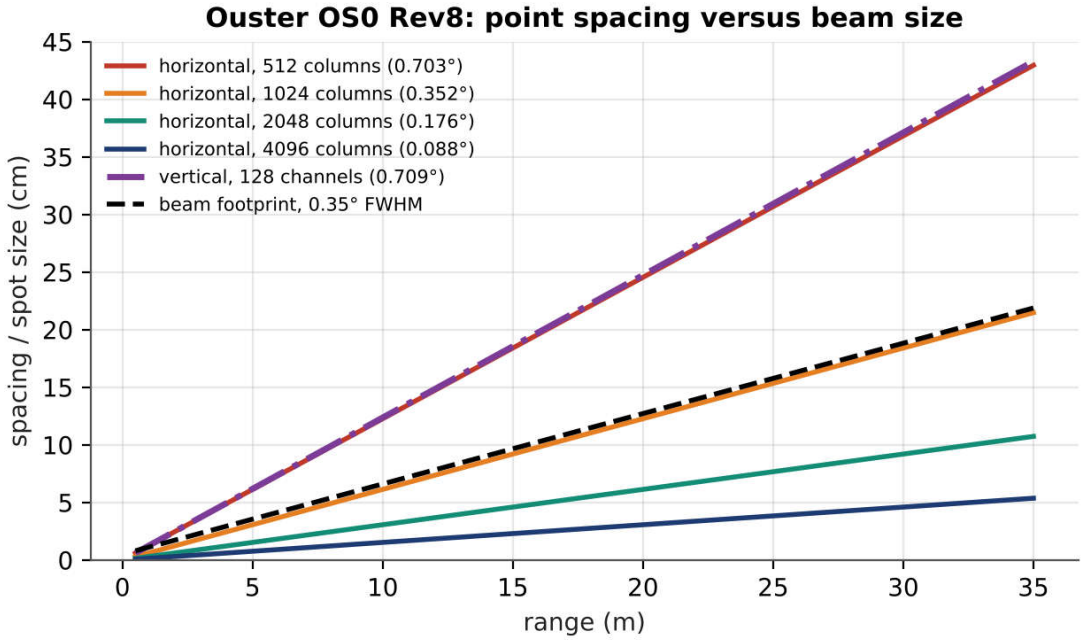


Figure 15.1: Point spacing for each OS0 mode, compared with the size of the laser spot.

WORKED EXAMPLE — spacing on a building face

Given: OS0-128 in 1024 @ 10 Hz mode; building face 20 m away, square to the beam.

Horizontal step

$360/1024 = 0.3516^\circ = 0.006136$ rad

Horizontal spacing

$20 \times 0.006136 = 0.123$ m

Vertical step

$90/127 = 0.7087^\circ = 0.01237$ rad $\Rightarrow 20 \times 0.01237 = 0.247$ m

Result: 12 cm × 25 cm grid in a single sweep. Motion (a walking operator sweeps the scan lines across the wall) fills the vertical gaps over time; that is why mobile clouds are denser than one sweep suggests.

15.3 Beam footprint: the real resolution limit

KEY EQUATION — Footprint from FWHM divergence

$$D(\rho) \approx d_0 + \rho\,\gamma_{\text{FWHM}}, \qquad d_0 = 5 \text{ mm}, \quad \gamma_{\text{FWHM}} = 0.35^\circ = 6.11 \times 10^{-3} \text{ rad}$$

What every symbol means

Symbol	Meaning	Units
γ_{FWHM}	full-width-at-half-maximum divergence: the angle across which the beam's power is at least half its peak	rad
D	spot diameter at range ρ (to half power; some light falls outside it)	m

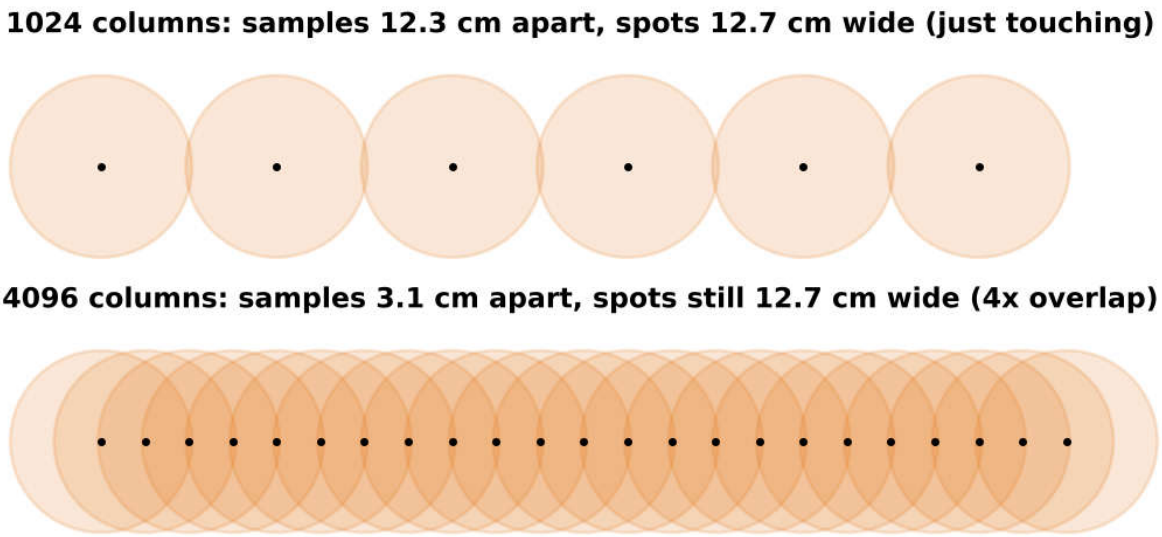


Figure 15.2: At 20 m, the 1024 mode places spots edge to edge; the 4096 mode overlaps each spot with its neighbors.

WORKED EXAMPLE — does 4096 mode make the map sharper?

Spot at 20 m

$0.005 + 20 \times 0.00611 = 0.127$ m

1024-mode spacing

0.123 m — about one spot per spot-width (contiguous coverage)

4096-mode spacing

0.031 m — four samples inside every spot

Result: the 4096 mode gives *more* points but not proportionally *sharper* detail: each sample still averages a 12.7 cm spot. Edges (poles, curb lips) are located only to about ±half a spot, ±6 cm at 20 m. Fine features are best captured *close* to the sensor, where the spot is small.

IMPROVED FROM THE SOURCE MATERIAL

The OS0's 0.35° divergence is about 12 times wider than the 0.3 mrad (0.017°) beam assumed in Module 4 for a survey-grade mapping head. That difference — not the point count — is the main reason short-range robotic lidars and survey scanners produce such different-looking edges. Read the divergence line of any datasheet before reading the point rate.

15.4 Range resolution versus precision versus accuracy

SIMPLE APPROACH

Think of a ruler. **Resolution** is how small the marks are (millimeters). **Precision** is how much your answer wiggles if you measure the same thing ten times. **Accuracy** is whether the ruler itself is the right length. A ruler can have tiny marks and still be stretched!

15.4.1 Resolution (quantization)

KEY EQUATION — Quantization error

$$\sigma_q = \frac{q}{\sqrt{12}}$$

What every symbol means

Symbol	Meaning	Units
q	size of the smallest reported step (the "resolution")	m
σ_q	standard deviation of rounding to that step (a uniform error between $-q/2$ and $+q/2$)	m
$\sqrt{12}$	comes from the variance of a uniform distribution of width q: $q^2/12$	—

WORKED EXAMPLE — does the data profile matter?

Standard profile, $q = 0.1$ cm

$\sigma_q = 0.1/3.464 = 0.029$ cm — negligible against 0.25 cm precision

Low Data Rate profile, $q = 0.8$ cm

$\sigma_q = 0.8/3.464 = 0.23$ cm — *as large as the best precision*

Combined at short range

$\sqrt{0.25^2 + 0.23^2} = 0.34$ cm instead of 0.25 cm

Result: for survey work never use the Low Data Rate profile: it adds a 2.3 mm rounding error to every range, a 36% increase in noise on bright targets.

15.4.2 Precision and accuracy versus range and reflectivity

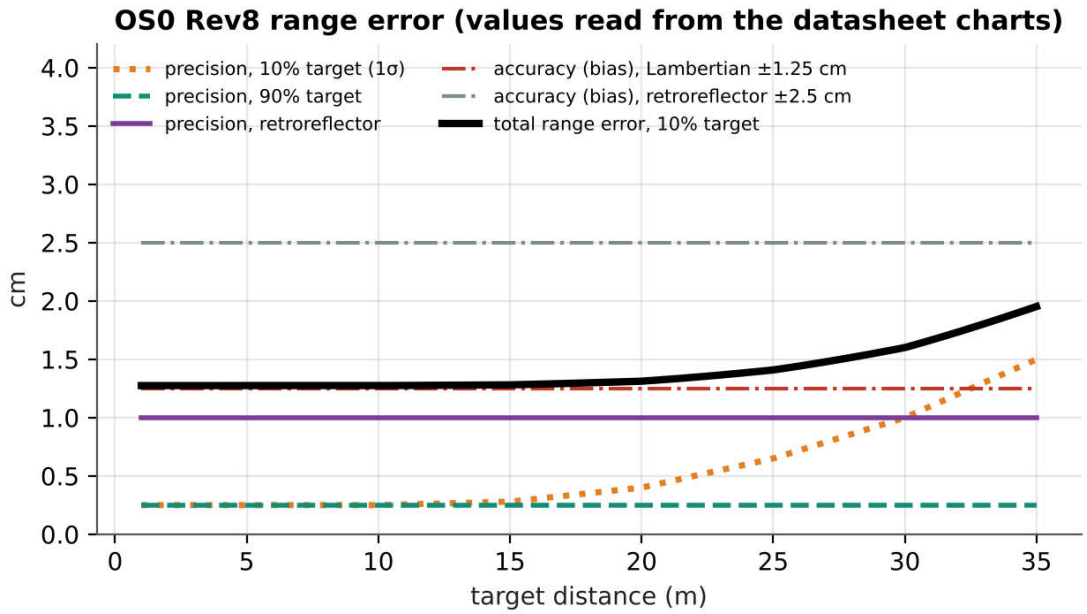


Figure 15.3: Range precision (1σ) and accuracy for the OS0 Rev8. Curves for the 10% target are read from the datasheet chart and are approximate.

Range (m)	5	10	15	20	25	30	35
Range (ft)	16	33	49	66	82	98	115
Precision, 10% target (cm, read from chart)	0.25	0.25	0.28	0.40	0.65	1.00	1.50
Precision, 90% target (cm)	0.25	0.25	0.25	0.25	0.25	0.25	0.25
Precision, retroreflector (cm)	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Accuracy, Lambertian (cm)	1.25	1.25	1.25	1.25	1.25	1.25	1.25
Precision, 10% target (ft)	0.008	0.008	0.009	0.013	0.021	0.033	0.049
Accuracy, Lambertian (ft)	0.041	0.041	0.041	0.041	0.041	0.041	0.041

KEY EQUATION — Total range error from a datasheet

$$\sigma_{\rho,\text{total}} \approx \sqrt{b^2 + \sigma_p^2 + \sigma_q^2}$$

What every symbol means

Symbol	Meaning	Units
b	range accuracy (bias) from the datasheet, treated here as a 1σ systematic error (a conservative assumption; "±" values are not always 1σ)	m
σ_p	range precision at that range and reflectivity	m
σ_q	quantization (usually negligible)	m

WORKED EXAMPLE — total range error on dark asphalt

At 10 m (10% target)
$\sqrt{1.25^2 + 0.25^2 + 0.03^2} = 1.27$ cm

At 20 m
$\sqrt{1.25^2 + 0.40^2} = 1.31$ cm

At 30 m
$\sqrt{1.25^2 + 1.00^2} = 1.60$ cm

At 35 m (maximum range for 10%)
$\sqrt{1.25^2 + 1.50^2} = 1.95$ cm

Result: out to about 25 m the bias dominates and the total barely changes; beyond that, noise on dark surfaces takes over. Because bias is common to nearby points, relative measurements (curb height, cracks) are much better than these totals (Module 14).

SURVEYOR'S FIELD TIP

Retroreflective targets are a trap. The datasheet gives a 2.5 cm accuracy and 1.0 cm precision on retroreflectors — worse than on ordinary surfaces — because the very strong return distorts the pulse timing. Do not use retro-tape targets as checkpoints for absolute height unless the bias has been calibrated; use matte painted targets (Lambertian) and use retro targets for horizontal identification only.

15.5 Angular accuracy

WORKED EXAMPLE — the $\pm 0.01^\circ$ encoder spec

In radians
$0.01^\circ = 1.745 \times 10^{-4}$ rad

At 10 / 20 / 35 m
0.17 / 0.35 / 0.61 cm across the beam (each axis)

Result: smaller than the range error at every range the OS0 can reach — the encoders are not the weak link. (We treat $\pm 0.01^\circ$ as 1σ; if it is a maximum bound, the 1σ is smaller.)

15.6 Maximum range and detection

ENGINEER'S VIEW

The range figures (75 m at 80% reflectance, 35 m at 10%) are for >90% *detection probability* in 100 klx sunlight — they say when points *exist*, not when they are *accurate*. Near the limit the 10% precision is already 1.5 cm and the footprint is 22 cm. Typical asphalt is 5–20% reflective, so on a sunny day expect road returns to thin out beyond roughly 30–35 m. The false-positive rate of 1 in 10 000 means about 131 spurious points per second at 1.31 million points per second: outlier removal (Module 18) is mandatory. With a minimum range setting of 0.0 m, returns from the operator's own body or the vehicle will appear; mask them by azimuth or range.

15.7 Timing, synchronization and skew

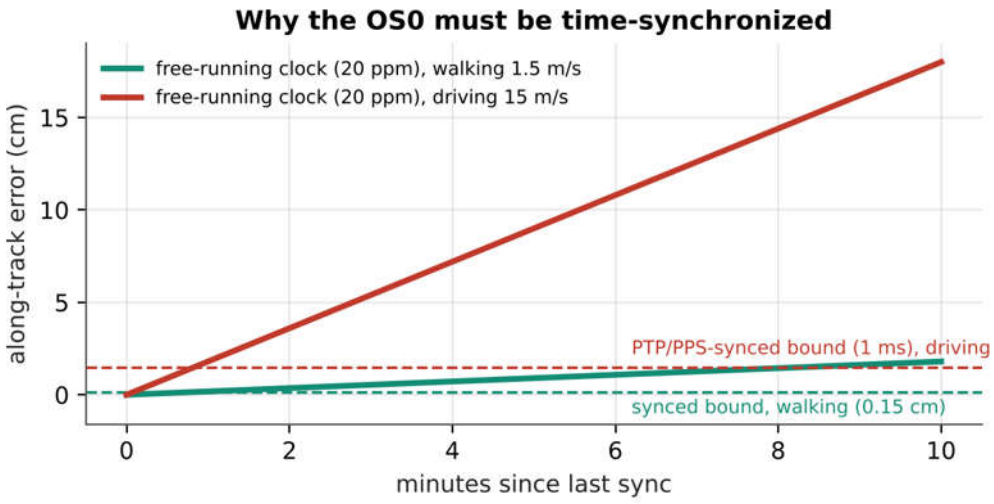


Figure 15.4: Along-track error from timing for a free-running OS0 clock versus a PTP/PPS-synchronized one.

WORKED EXAMPLE — why the OS0 must be synchronized

Given: synced accuracy < 1 ms (PTP or PPS); internal clock drift < 20 ppm.

Synced, walking 1.5 m/s

$1.5 \times 0.001 = 0.15$ cm

Synced, driving 15 m/s

$15 \times 0.001 = 1.5$ cm

Synced, operator turning 90 °/s, point at 20 m

$90^\circ/\text{s} \times 0.001\ \text{s} = 0.09^\circ = 1.57 \times 10^{-3}$ rad; $20 \times 1.57 \times 10^{-3} = 3.1$ cm

Free-running 10 minutes

$600 \times 20 \times 10^{-6} = 0.012$ s; at 15 m/s $\Rightarrow$ 18 cm

Result: the < 1 ms spec is fine for walking, marginal for driving, and poor for fast turns at range. For survey work, synchronize with PPS/PTP to the best your timing source can deliver, and verify with the turn-signature test of Module 8.

WORKED EXAMPLE — scan skew inside one rotation

Given: 1024 @ 10 Hz.

Time per rotation

$1/10 = 0.1$ s; time per column $= 0.1/1024 = 97.7$ μs

Walking at 1.4 m/s

the sensor moves $1.4 \times 0.1 = 0.14$ m between the first and last column

Turning 60 °/s

the head rotates 6° in one sweep; a point at 20 m would smear $20 \times 0.105 = 2.1$ m if not de-skewed

Result: per-column timestamps (< 1 μs resolution) plus IMU motion make de-skewing possible (Module 8). Raising the rotation rate to 20 Hz halves the skew but halves the columns available at each rate setting.

15.8 Built-in IMU and data volume

ENGINEER'S VIEW

The OS0's internal 6-axis MEMS IMU (InvenSense IAM-20685HP) is sampled up to 2 560 Hz ($\Delta t = 0.39$ ms) and time-stamped on the same clock as the lidar — exactly what tightly-coupled SLAM such as LIO-SAM needs for de-skewing and preintegration (Modules 7, 12). Its rigid offset to the lidar is reported through the queryable "IMU pose correction matrix," a factory lever arm and boresight. Its bias and noise are those of an automotive/industrial MEMS part (read its own datasheet for values); expect it to need LiDAR and control aiding continuously, not to coast. Survey backpacks often add an external tactical IMU and GNSS.

Data volume. At 128 channels, 1024 @ 10 Hz: $128 \times 1024 \times 10 = 1\,310\,720$ points/s; maximum data rate 295 Mbps $= 36.9$ MB/s $\approx$ **133 GB per hour**. The quoted maximum 10 485 760 points/s equals $128 \times 2048 \times 20 \times 2$ — consistent with the 2048 @ 20 Hz mode with both returns enabled (our interpretation).

15.9 Putting it together

In a complete backpack system the OS0's own range error is only one term. The full budget — trajectory, attitude, boresight, lever arm, range and encoders — is worked in Module 14 (Section 14.11.1): about **2.5 cm horizontal / 2.0 cm vertical at 5 m, 3.1 / 2.2 cm at 20 m and 4.3 / 2.5 cm at 35 m** (0.08/0.07, 0.10/0.07, 0.14/0.08 ft, 1σ).

15.10 Choosing a mode: a worked decision

WORKED EXAMPLE — interior as-built of a mechanical room

Requirement: 5 cm point spacing on walls up to 15 m; 2 cm (1σ) accuracy; walking survey.

Needed step

$0.05/15 = 0.00333$ rad $= 0.19° \Rightarrow$ 2048 mode (0.176°) or finer

Rotation rate

2048 mode allows 10 or 20 Hz; choose 10 Hz (more points per rotation, still low skew at walking speed)

Footprint check

at 15 m: $0.005 + 15 \times 0.00611 = 0.097$ m — edges will be soft at ± 5 cm; capture pipe and valve details from closer than 5 m (spot 3.6 cm)

Range error

indoor bright walls: $\sqrt{1.25^2 + 0.25^2} = 1.27$ cm; budget with indoor SLAM trajectory (assume 1.0 cm) and 0.03° heading: at 15 m $\sqrt{1.0^2 + 0.79^2 + 1.27^2 + 0.26^2 \cdot 2} = 1.8$ cm ✓

Data

2048 @ 10 Hz $\approx$ 2.6 M points/s, about 2$\times$ the 1024 data volume: plan $\sim$270 GB per hour

Result: 2048 @ 10 Hz, 128 channels, standard (not low-data-rate) profile, PTP sync, scan within 15 m and close up for fine detail. Control: at least three matte targets per room plus checkpoints.

TRY IT YOURSELF

15.1 In 512 mode, what is the horizontal spacing at 8 m?
15.2 What is the OS0 spot diameter at 3 m?
15.3 Compute the total range error for a 10% target at 25 m.
15.4 An unsynchronized OS0 runs for 30 minutes on a vehicle at 10 m/s. Estimate the worst-case along-track error from clock drift alone.

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Ouster, *OS0 Short-Range High-Resolution Imaging Lidar, Rev8 datasheet* (REV 08/03/2026); precision values for 10% targets read from its precision chart.

MODULE 16

Choosing an IMU: AHRS, GNSS/INS and Tactical GNSS/INS

Two small MicroStrain sensors are popular on backpack and robot LiDAR rigs: the **3DM-GX5-25**, an attitude and heading reference system (AHRS), and the **3DM-GX5-45**, a GNSS-aided inertial navigation system (GNSS/INS). They share the same MEMS inertial core, but one knows only which way it is pointing while the other also knows where it is. In this module we drive both around a city block — the *Block Loop Challenge* — and watch their errors propagate into a traverse and into LiDAR points. Then we step up to a third, higher-grade sensor, the **3DM-GQ7**: a dual-antenna, RTK-capable GNSS/INS with a 1.5°/h gyro, and ask how much of the problem money can solve.

SIMPLE APPROACH

Imagine two friends walking around the block with their eyes closed. The first friend has a compass and counts steps. The second friend has the same compass and step-counter but also gets a phone message every second saying "you are about here, give or take two meters." Who gets home closest? And who can tell you exactly which way the mailbox is? That is the whole story of this module.

16.1 Meet the two sensors

Specification (manufacturer datasheets)	3DM-GX5-25 AHRS	3DM-GX5-45 GNSS/INS
What it outputs	attitude (roll, pitch, heading), raw IMU	attitude, *position*, *velocity*, raw IMU, GNSS
Heading reference	magnetometer	GNSS (when moving) + magnetometer
Roll/pitch accuracy (EKF)	$\pm 0.25^\circ$ RMS	$\pm 0.25^\circ$ RMS
Heading accuracy (EKF)	$\pm 0.8^\circ$ RMS	$\pm 0.8^\circ$ RMS (GNSS heading 0.5°)
Position accuracy	none	± 2 m (6.6 ft) RMS horizontal, ± 5 m (16.4 ft) RMS vertical
Velocity accuracy	none	± 0.1 m/s (0.33 ft/s) RMS
GNSS receiver	—	72-channel, L1-band GPS/QZSS, GLONASS, BeiDou, Galileo, SBAS; 1–4 Hz
RTK corrections	—	not listed (single-frequency L1; SBAS only)
Time pulse	—	30 ns RMS
Gyro bias instability	$8^\circ/\text{h}$	$8^\circ/\text{h}$
Gyro noise density	$0.005^\circ/\text{s}/\sqrt{\text{Hz}}$	$0.005^\circ/\text{s}/\sqrt{\text{Hz}}$
Gyro initial bias error	$\pm 0.04^\circ/\text{s}$	—
Accel bias instability	± 0.04 mg	± 0.04 mg
Accel noise density	25 $\mu g/\sqrt{\text{Hz}}$	20 $\mu g/\sqrt{\text{Hz}}$
Accel initial bias error	± 0.002 g (2 mg)	—
Output rate	IMU to 1 kHz; EKF to 500 Hz	IMU to 1 kHz; EKF to 500 Hz
Size, weight	$36\times36.6\times11$ mm, 16.5 g	$44.2\times36.6\times11.1$ mm, 20 g

Values from the MicroStrain 3DM-GX5-25 datasheet (8400-0093) and the 3DM-GX5 GNSS/INS datasheet; "—" means not stated in the document we used. Check the current datasheet and manual for your unit's firmware.

16.2 Turning datasheet noise into random walks

KEY EQUATION — Noise density to angle and velocity random walk

$$\text{ARW}\ [^\circ/\sqrt{\text{h}}] = N_g\ [^\circ/\text{s}/\sqrt{\text{Hz}}] \times 60, \qquad \text{VRW}\ [\text{m/s}/\sqrt{\text{s}}] = N_a\ [g/\sqrt{\text{Hz}}] \times 9.81$$

$$\sigma_\psi(t) = \text{ARW}\sqrt{t}, \qquad \sigma_v(t) = \text{VRW}\sqrt{t}$$

What every symbol means

Symbol	Meaning	Units
N_g	gyro noise density from the datasheet	$^\circ/\text{s}/\sqrt{\text{Hz}}$
N_a	accelerometer noise density	$g/\sqrt{\text{Hz}}$
ARW	angle random walk: heading wander from white noise	$^\circ/\sqrt{\text{h}}$
VRW	velocity random walk	$\text{m/s}/\sqrt{\text{s}}$
60	$\sqrt{3600}$: converts $\sqrt{\text{s}}$ to $\sqrt{\text{h}}$	—

WORKED EXAMPLE — the GX5's random walks

ARW

$0.005 \times 60 = 0.30^\circ/\sqrt{\text{h}}$

Heading wander after one 70 s lap

$0.30 \times \sqrt{70/3600} = 0.30 \times 0.139 = 0.042^\circ$

VRW

$25 \times 10^{-6} \times 9.81 = 2.45 \times 10^{-4}\ \text{m/s}/\sqrt{\text{s}}$

Velocity wander after 70 s

$2.45 \times 10^{-4} \times \sqrt{70} = 0.0021\ \text{m/s}$

Result: the white noise is small. The danger is in the *biases*, the *tilt* and the *magnetometer*, as the Block Loop will show.

16.3 Pros and cons

	3DM-GX5-25 AHRS	3DM-GX5-45 GNSS/INS
Pros	tiny (16.5 g), low power (∼0.5 W), low cost; good roll/pitch from gravity; high-rate IMU data for SLAM de-skew; works indoors and underground	absolute position (meter level) with no setup; heading from GNSS motion, not only magnetism; precise time pulse (30 ns) usable as a sync source; velocity output; bounded error over any distance
Cons	heading depends on the magnetometer — disturbed by vehicles, rebar, steel bridges, motors; no position at all; free-inertial position drifts to tens of meters in a minute	L1-band receiver with no RTK in its datasheet: ±2 m H, ±5 m V — 100× too coarse for survey, and it cannot use network RTK such as FDOT's FPRN; needs open sky; heading weak at standstill; same MEMS core coasts poorly in tunnels
Best role	IMU inside LiDAR SLAM (indoor, handheld)	IMU + coarse georeference for SLAM outdoors; GNSS factor and timing for a factor graph

16.4 The Block Loop Challenge

We run a closed traverse around one city block, starting and ending at control point A, exactly like a four-sided compass-and-tape traverse.

Leg	From–To	Distance	Azimuth	$\Delta E = L\sin\alpha$ (m)	$\Delta N = L\cos\alpha$ (m)
1	A–B	200 m (656.17 ft)	090°	+200.000	0.000
2	B–C	150 m (492.13 ft)	180°	0.000	−150.000
3	C–D	200 m (656.17 ft)	270°	−200.000	0.000
4	D–A	150 m (492.13 ft)	000°	0.000	+150.000
		700 m (2,296.59 ft)		0.000	0.000

Control coordinates: A (1000, 1000), B (1200, 1000), C (1200, 850), D (1000, 850). The vehicle drives at 10 m/s, so one lap takes 70 s. A steel bridge carries leg B–C over a creek, and an underpass covers part of leg C–D.

Run 1 — AHRS heading + odometer, systematic errors only

WORKED EXAMPLE — a traverse that closes perfectly and is still wrong

Given: AHRS heading reads 0.8° too high on every leg (uncorrected magnetic declination/hard-iron error, equal to the RMS spec); odometer reads 0.2% long.

Leg 1

$L = 200 \times 1.002 = 200.400$ m, $\alpha = 90.8°$: $\Delta E = 200.400 \sin 90.8° = +200.380$, $\Delta N = 200.400 \cos 90.8° = -2.798 \Rightarrow$ B′ = (1200.380, 997.202)

Legs 2–4

computed the same way; every leg is turned by +0.8° and stretched by 0.2%

Closure

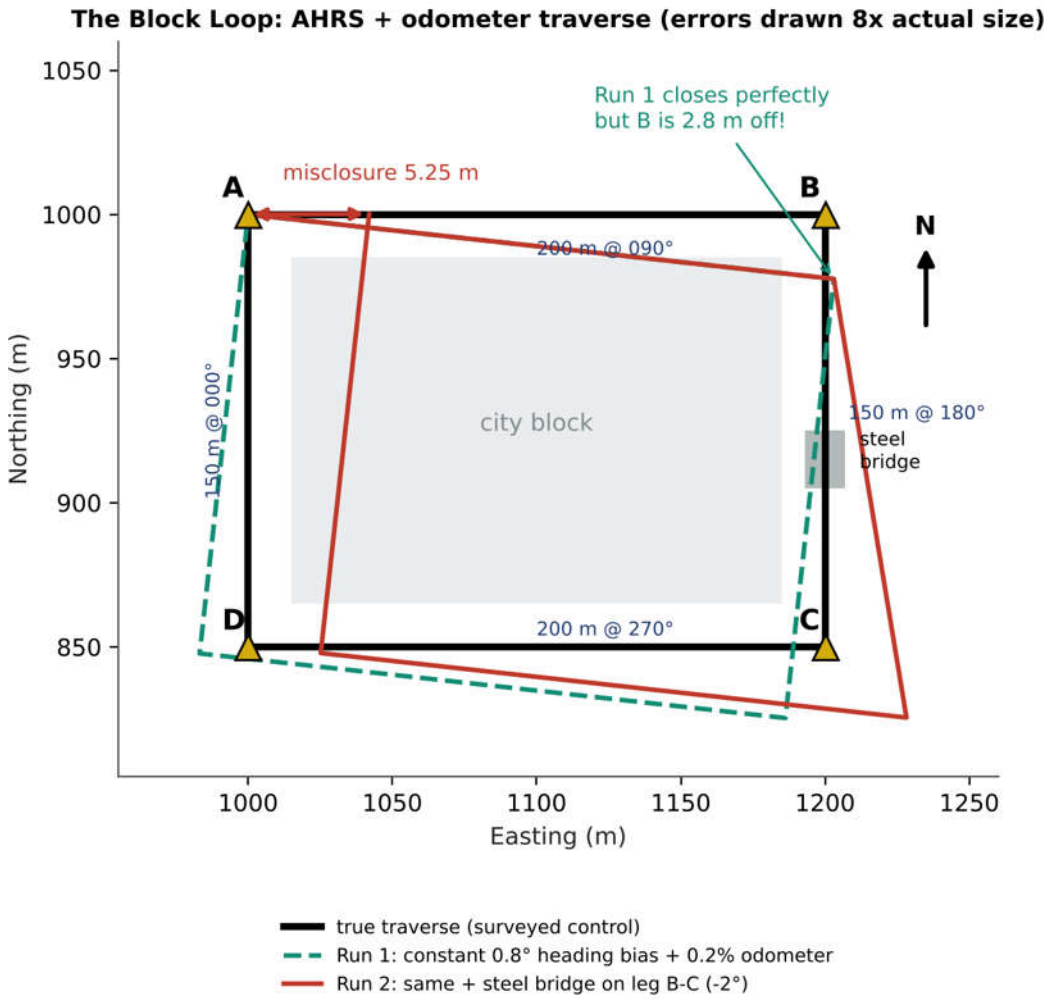


Figure 16.1: Sketch of the traverse with the AHRS runs. Errors are drawn 8 times larger than computed so they can be seen.

$\sum \Delta E = 0.000$, $\sum \Delta N = 0.000$ — **perfect closure**

But

B′ is $\sqrt{0.380^2 + 2.798^2} = 2.82$ m from the true B; C′ is 3.53 m from C

Result: a closed loop hides any error that rotates or scales the whole figure. The misclosure is zero because a rotated, stretched rectangle still closes. Only a *second* control point (or an azimuth mark) reveals it — the same reason a surveyor checks into a second known point instead of trusting a loop.

BIG IDEA

Closure tests catch errors that *change* around the loop. They are blind to errors that are the *same* on every leg.

Run 2 — add the steel bridge

WORKED EXAMPLE — a local magnetic disturbance

Given: Run 1, plus the steel bridge deflects the magnetometer by $-2.0°$ on leg B–C (so that leg's total heading error is $0.8 - 2.0 = -1.2°$).

Recompute

B′ = (1200.380, 997.202); C′ = (1203.528, 846.935); D′ = (1003.148, 849.733); A′ = (1005.246, 1000.018)

Misclosure

$\sqrt{5.246^2 + 0.018^2} = 5.25$ m over 700 m $\Rightarrow$ precision ratio 1:133

Compass-rule adjustment

distribute the misclosure in proportion to cumulative distance (200, 350, 550, 700 m): corrections -1.50, -2.62, -4.12 m in Easting at B, C, D

Adjusted errors vs. truth

B 3.02 m, C 3.20 m, D 1.01 m

Result: the adjustment makes the traverse *look* closed, but B and C are still about 3 m wrong because the constant 0.8° bias was never seen. A 1:133 traverse would fail any survey standard by a factor of 50 or more.

Run 3 — random heading errors (Monte Carlo)

WORKED EXAMPLE — what does a typical lap look like?

Given: on each leg an independent heading error drawn from $N(0,\ 0.8°)$ and an odometer scale error from $N(0,\ 0.2\%)$; 20 000 simulated laps.

Result

mean misclosure 4.3 m; 95% of laps below 8.6 m

Hand check (Module 14 propagation)

a heading error $\delta\alpha$ on a leg of length L moves the end point by $L\,\delta\alpha$ sideways. With $\delta\alpha = 0.8° = 0.01396$ rad: legs of 200 m give 2.79 m, legs of 150 m give 2.09 m. RSS of the four: $\sqrt{2(2.79^2) + 2(2.09^2)} = 4.93$ m (RMS radial misclosure) — consistent with the simulation.

Result: an AHRS-plus-odometer traverse is a meter-level tool. Heading error, multiplied by distance, is the whole story.

Run 4 — pure inertial navigation with the GX5-25

WORKED EXAMPLE — could the IMU alone drive the loop?

Given: the GX5 IMU integrated with no odometer, no magnetometer, no GNSS, for one 70 s lap at 10 m/s.

Tilt error

$0.25°$ roll/pitch error leaks $9.81 \sin 0.25° = 0.0428$ m/s^2 of gravity: $\frac{1}{2}(0.0428)(70^2) = 105$ m

Uncalibrated accel bias

2 mg $= 0.0196$ m/s^2: $\frac{1}{2}(0.0196)(70^2) = 48$ m

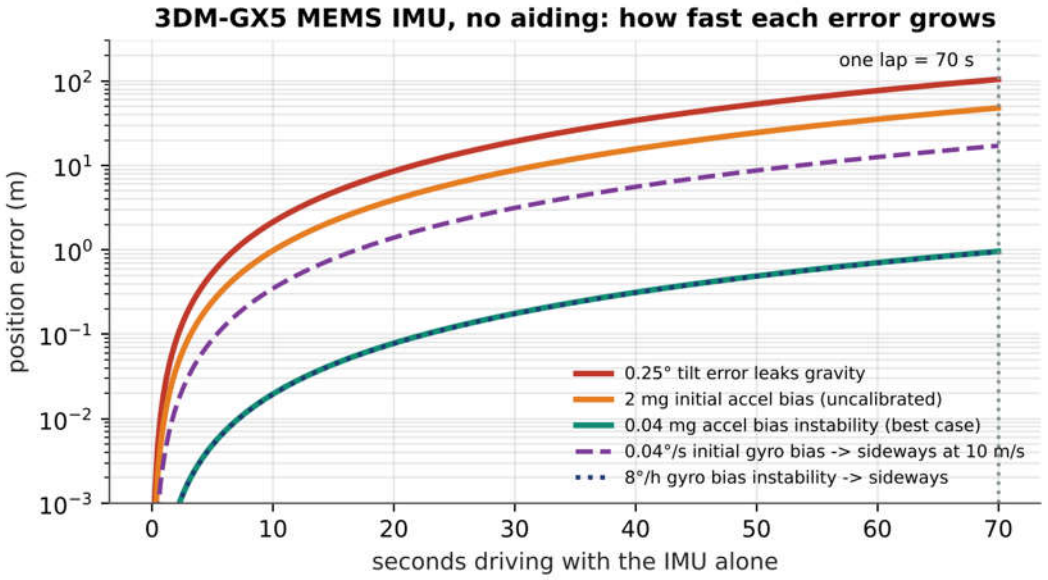


Figure 16.2: Each IMU error source, integrated with no aiding at all, over one 70 s lap.

Uncalibrated gyro bias

$0.04^\circ/\text{s} \Rightarrow$ heading off by 2.8° after the lap; sideways error $\frac{1}{2} \times 10 \times 6.98 \times 10^{-4} \times 70^2 = 17$ m

Best case, biases estimated at start

accel bias instability 0.04 mg: $\frac{1}{2}(3.92 \times 10^{-4})(70^2) = 0.96$ m; gyro 8°/h: 0.95 m

Result: tens to hundreds of meters in one minute. A MEMS IMU is a superb *short-term* motion sensor (milliseconds to seconds) and a hopeless *navigator*. This is not a defect of the GX5 — it is the nature of every MEMS IMU (Module 7).

Run 5 — the 3DM-GX5-45 GNSS/INS

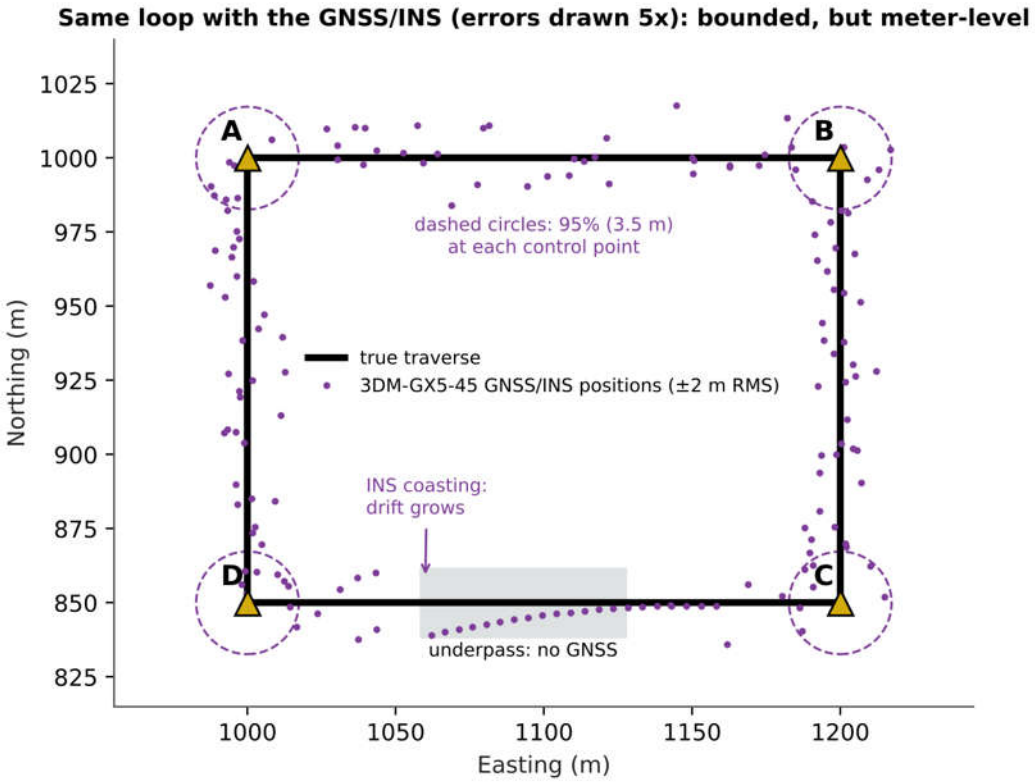


Figure 16.3: The GNSS/INS keeps every position within a few meters; in the underpass it coasts on the IMU and drift begins to grow. Errors drawn 5 times actual size.

WORKED EXAMPLE — bounded, but meter-level

Given: the GX5-45 EKF with ±2 m RMS horizontal and ±5 m RMS vertical; a 20 s underpass on leg C–D.

Each control point

95% horizontal circle radius $= 2.448 \times (2/\sqrt{2}) = 3.46$ m (Module 14: split 2 m radial RMS into 1.41 m per axis, scale by 2.448)

Misclosure

start and end are independent fixes: $\sigma = \sqrt{2} \times 2 = 2.8$ m RMS — *no growth with distance*

Underpass coast (20 s)

if the filter has the tilt to 0.05° and the accel bias to 0.04 mg: $\frac{1}{2}(9.81 \sin 0.05^\circ + 3.9 \times 10^{-4})(20^2) = 1.8$ m; at the full 0.25° tilt spec it would be 8.6 m

Time pulse

30 ns RMS at 10 m/s is 0.3 μm — timing is no longer an error source

Result: the GNSS/INS never gets lost, but its errors are meters, not centimeters. It answers "which street am I on?", not "where is this curb?"

16.5 Stepping up: the 3DM-GQ7 dual-antenna RTK GNSS/INS

SIMPLE APPROACH

Now give the second friend a much better step-counter, a pair of ears far apart that hear exactly which way the street runs, and phone messages that say "you are here, give or take two centimeters" instead of two meters. That friend can walk the block and point to the mailbox almost perfectly — as long as the phone has a signal.

Specification	3DM-GX5-45	3DM-GQ7
Gyro bias instability	8°/h	**1.5°/h**
Angle random walk	0.30°/$\sqrt{\text{h}}$ (from noise density)	**0.15°/$\sqrt{\text{h}}$**
Accel bias instability	40 μg	**5** μg
Accel noise density	20 $\mu g/\sqrt{\text{Hz}}$	20 $\mu g/\sqrt{\text{Hz}}$
Roll / pitch	0.25° RMS	**0.05°**
Heading	0.8° RMS (magnetometer/GNSS)	**0.25° RMS, dual antenna, 1.5 m (4.9 ft) baseline**
GNSS	one L1-band receiver	**two multiband receivers** (L1/L2, E1/E5b, B1/B2), 184 channels
Position	2 m (6.6 ft) H, 5 m (16.4 ft) V	1.25 m (4.1 ft) H, 2 m (6.6 ft) V standalone; **2 cm (0.07 ft) with RTK** (1 cm + 1 ppm)
Velocity	0.1 m/s (0.33 ft/s)	0.05 m/s (0.16 ft/s)
Output rate	up to 1 kHz (IMU) / 500 Hz (EKF)	up to 1 kHz (IMU and EKF)
Size, weight, power	44×37×11 mm, 20 g	76×69×13 mm, **78 g**, 2.0 W typ.

GQ7 values from the MicroStrain 3DM-GQ7 GNSS/INS datasheet. Marketing calls the GQ7 IMU "tactical grade"; by the bias ranges in Module 7 its 1.5°/h gyro sits at the boundary between high-end industrial and entry tactical.

Why two antennas give heading

KEY EQUATION — Dual-antenna heading accuracy

$$\psi = \text{atan2}(\Delta E,\ \Delta N) \ \text{ of the antenna-to-antenna vector,} \qquad \sigma_\psi \approx \frac{\sigma_{\text{rel}}}{b}$$

What every symbol means

Symbol	Meaning	Units
b	baseline: distance between the two GNSS antennas	m
σ_{rel}	precision of the relative (carrier-phase) position between the antennas, across the baseline	m
σ_ψ	heading standard deviation (small-angle rule, Module 2)	rad

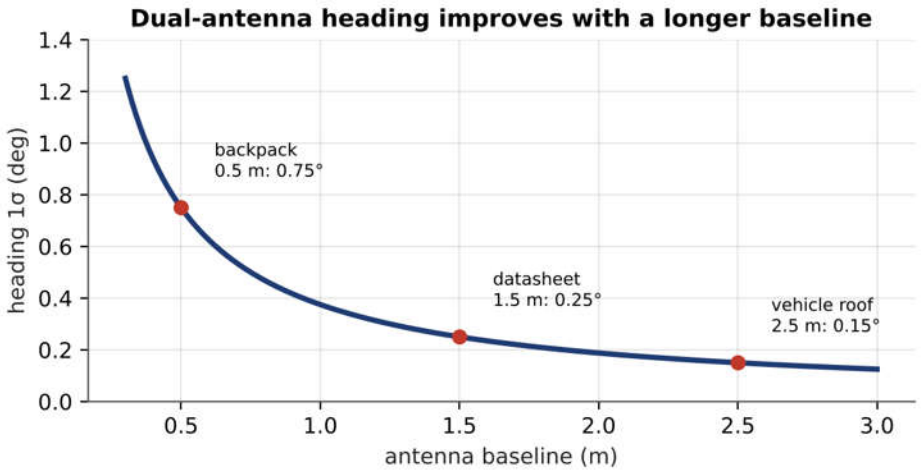


Figure 16.4: The same relative-position precision gives better heading over a longer baseline.

WORKED EXAMPLE — heading on a backpack versus a truck

Back out σ_{rel} from the datasheet

$0.25^\circ = 4.36 \times 10^{-3}$ rad; $\sigma_{\text{rel}} = 1.5 \times 4.36 \times 10^{-3} = 6.5$ mm

Backpack, 0.5 m baseline

$\sigma_\psi = 0.0065/0.5 = 0.013$ rad $= 0.75^\circ$

Vehicle roof, 2.5 m baseline

$0.0065/2.5 = 0.0026$ rad $= 0.15^\circ$

Result: on a short backpack baseline the GQ7's GNSS heading is no better than the GX5's magnetometer. Dual-antenna heading shines on vehicles. Unlike a magnetometer it is immune to steel and works at standstill, but it needs open sky at both antennas.

Real-time corrections in Florida: the FDOT FPRN

SIMPLE APPROACH

RTK is like having a friend standing on a spot whose location is known exactly. Your friend radios you, "The satellites are fooling me by this much right now." You subtract the same mistake from your own reading and suddenly you know where you are to within a couple of centimeters. In Florida, the state runs over a hundred of these "friends" for free.

ENGINEER'S VIEW

The **Florida Permanent Reference Network (FPRN)**, operated by FDOT, is a statewide network of 100+ continuously operating, dual-frequency geodetic GNSS reference stations (CORS). It is currently **free**; each rover needs its own account (register online; activation in 24–48 hours). Corrections are streamed over the internet by **NTRIP**:

- **Solutions:** single baseline (nearest station), VRS (virtual reference station), MAX and iMAX (network solutions), and FKP.
- **Formats:** RTCM 2.3, **RTCM 3.1**, CMR+, and RTCM 3.3 MSM4 (format availability varies

by solution type).

- **Datum:** NAD 83(2011), epoch 2010.0, on the GRS 80 ellipsoid — RTK heights are *ellipsoid* heights until a geoid model (e.g. GEOID18) converts them to NAVD 88.
- **Typical RTK accuracy:** about 1–2 cm horizontal and 2–3 cm vertical with good sky, geometry and low multipath (FDOT).

Connecting the GQ7. The GQ7 accepts RTCM 3.1 corrections. MicroStrain's own 3DM-RTK cellular modem connects to its SensorCloud base network; to use FPRN instead, an **NTRIP client** (a field computer, data logger or cellular router) logs into the FPRN caster with the rover's account, selects a mountpoint (VRS or iMAX recommended), and forwards the RTCM stream to the GQ7's correction input. Many NTRIP mountpoints, including VRS, expect the rover to send its approximate position (an NMEA GGA message) back to the caster.
Why not the GX5-45? Its datasheet lists a single-frequency L1 receiver with SBAS and does not list RTK or RTCM input, so it cannot produce a centimeter RTK fix from FPRN corrections. For FDOT-network RTK in this course's sensor lineup, use the GQ7 (or an external RTK receiver).

WORKED EXAMPLE — single baseline versus VRS

Given: GQ7 RTK spec 1 cm + 1 ppm. The nearest FPRN station is 22 km away.

Single baseline

$1\text{ cm} + 1 \times 10^{-6} \times 22\,000\text{ m} = 1\text{ cm} + 2.2\text{ cm} = 3.2\text{ cm}$ horizontal — exceeds the 2 cm budget used in Run 6

VRS / iMAX

the network models the errors and creates a virtual station a few km away, e.g. 3 km: $1 + 0.3 = 1.3$ cm

Result: in rural Florida, choose a network (VRS or iMAX) mountpoint rather than the nearest single station, or the ppm term will dominate.

SURVEYOR'S FIELD TIP

Standards of practice for network RTK. Before and after each session, *check in on a known control point* and record the difference; log the mountpoint, solution type (VRS/iMAX/single), fix status, PDOP and the number of satellites; record the network datum and epoch (NAD 83(2011) epoch 2010.0) and the geoid model used; and keep independent checkpoints out of the adjustment (Module 13). This is the documentation a reviewer will expect when the accuracy statement cites RTK positioning under Florida's Chapter 5J-17 and FDOT practice.

Run 6 — the Block Loop with the GQ7 and RTK

WORKED EXAMPLE — centimeters instead of meters

Given: RTK corrections from an FPRN VRS mountpoint (virtual station about 2 km away); RTK 1 cm + 1 ppm; dual-antenna heading on a 1.5 m roof baseline; the same 20 s underpass.

Each control point

$1\text{ cm} + 1\text{ ppm} \times 2\,000\text{ m} = 1.2\text{ cm}$, within the 2 cm figure; 95% circle $\approx 2.448 \times (2/\sqrt{2}) = 3.5$ cm

Misclosure
independent fixes at start and end: $\sqrt{2} \times 2 = 2.8$ cm RMS

Traverse precision
2.8 cm in 700 m $\approx$ 1:25 000 — a respectable conventional traverse

Underpass coast
see below: 0.2–1.0 m after 20 s

Result: with RTK, the Block Loop closes like a good total-station traverse in open sky. The underpass is still the weak link.

How coasting errors grow

KEY EQUATION — Three-term coast error during a GNSS outage

$$\delta x(t) \approx \sqrt{(\sigma_v t)^2 + \left(\tfrac{1}{2} b_a t^2\right)^2 + \left(\tfrac{1}{6} g\, b_g t^3\right)^2}$$

What every symbol means

Symbol	Meaning	Units
σ_v	velocity error at the moment GNSS is lost — grows the error *linearly*	m/s
b_a	accelerometer bias (instability) — grows with t^2	m/s^2
b_g	gyro bias: tilts the platform a little more every second, leaking gravity — grows with t^3	rad/s
g	gravity, 9.81	m/s^2
t	time since GNSS was lost	s

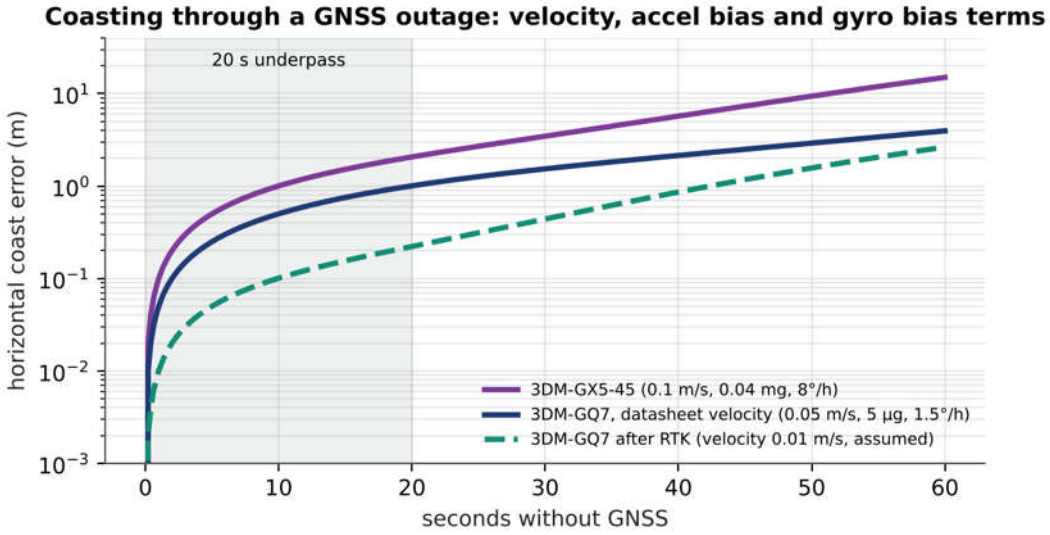


Figure 16.5: Coast error versus outage length. Velocity dominates first; the gyro's cubic term dominates later.

WORKED EXAMPLE — 20 s and 60 s outages

GX5-45, 20 s

$\sigma_v t = 0.1 \times 20 = 2.0$; $\frac{1}{2}(3.9 \times 10^{-4})(400) = 0.08$; $\frac{1}{6}(9.81)(3.88 \times 10^{-5})(8000) = 0.51 \Rightarrow 2.07$ m

GQ7 (datasheet velocity 0.05 m/s), 20 s

1.00; 0.01; $\frac{1}{6}(9.81)(7.27 \times 10^{-6})(8000) = 0.095 \Rightarrow 1.00$ m

GQ7 after RTK (velocity 0.01 m/s, assumed), 20 s

0.20; 0.01; 0.095 $\Rightarrow$ 0.22 m

60 s outage

GX5-45: 15.0 m; GQ7 (RTK velocity): $\sqrt{0.6^2 + 0.09^2 + 2.57^2} = 2.64$ m

Result: the better gyro wins big on long outages (the t^3 term is 5× smaller), and RTK-quality velocity wins on short ones. Even so, no MEMS INS holds centimeters through a minute-long tunnel: that still takes LiDAR SLAM, control in the tunnel, or a navigation-grade INS.

GQ7 pros	centimeter RTK position; dual-antenna heading that ignores magnetism and works at standstill; 5–6× better gyro and accelerometer biases than the GX5; multiband GNSS for faster, more robust RTK fixes; up to 1 kHz output for de-skewing
GQ7 cons	needs RTK corrections (base station or network) and open sky at *two* antennas; heading quality depends on the baseline you can physically mount; heavier, larger and more power-hungry; higher cost; still a MEMS INS — long outages need other aiding

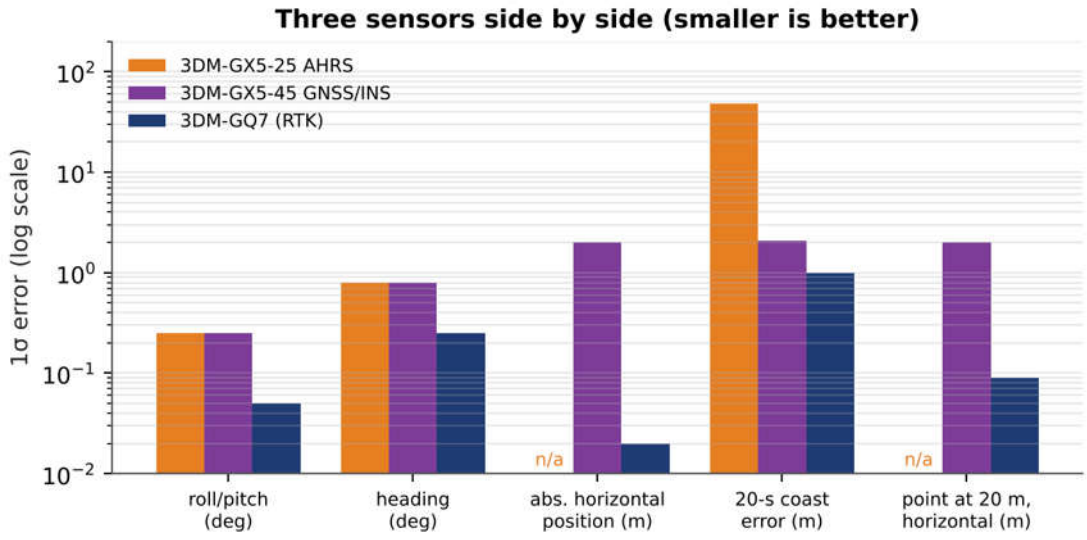


Figure 16.6: The three sensors side by side. For the AHRS, the "20-s coast" bar is free-inertial drift with an uncalibrated accelerometer bias; it has no absolute position.

16.6 What these attitude errors do to LiDAR points

The attitude specs above are propagated into LiDAR point errors in Module 14 (Section 14.11.2). In short, at 20 m a 0.8° heading moves a point 27.9 cm (0.92 ft); the GX5-45 alone georeferences an OS0 point to about 2.0 m H / 5.0 m V (6.6 / 16.4 ft), while the GQ7 with RTK reaches about 9.0 cm H / 2.7 cm V (0.30 / 0.09 ft) — limited by heading until SLAM tightens it.

16.7 The right job for each sensor: inside SLAM

ENGINEER'S VIEW

Inside a tightly-coupled SLAM system such as LIO-SAM (Module 12), the EKF attitude spec hardly matters. What matters is the **raw IMU** over the short gap between keyframes, because that is what preintegration uses:

$$\sigma_\theta(\Delta t) = N_g\sqrt{\Delta t}, \qquad \sigma_v(\Delta t) = N_a\, g\,\sqrt{\Delta t}$$

For a 0.1 s keyframe gap: $\sigma_\theta = 0.005 \times \sqrt{0.1} = 0.0016°$ and $\sigma_v = 25 \times 10^{-6} \times 9.81 \times \sqrt{0.1} = 7.8 \times 10^{-5}$ m/s. The 8°/h bias contributes $8/3600 \times 0.1 = 0.0002°$. These are tiny: the LiDAR scan matching then pins heading to a few hundredths of a degree (the 0.05° of Module 14), and the biases are estimated online as graph variables. So:

- **Indoors, underground, under canopy:** the GX5-25 is enough. Its magnetometer heading should be *down-weighted or ignored*; LiDAR supplies heading. Georeference with surveyed control targets.
- **Outdoors, open sky:** the GX5-45 adds a GNSS factor that removes long-term drift and ties the map to the datum at the meter level, plus a precise time pulse for synchronizing the LiDAR. Survey-grade absolute accuracy still requires RTK/PPK GNSS or ground control (Module 13).
- **Outdoors with RTK (in Florida, from FPRN):** the GQ7 supplies centimeter GNSS factors and steel-proof heading, turning the factor graph's absolute accuracy from meters into centimeters in open sky. It is the only one of the three that can georeference a survey-grade map without ground control — but you should still check it with independent checkpoints (Module 13).
- **None of the three** replaces a navigation-grade or high-end tactical INS on a highway mobile-mapping vehicle, where GNSS outages of tens of seconds must be bridged with centimeter accuracy (Module 7).

SURVEYOR'S FIELD TIP

Magnetometer hygiene. Perform the hard- and soft-iron calibration with the sensor mounted on the final rig (the rig's own steel and motors are the biggest disturbance), keep the unit as far as practical from motors and batteries, and treat magnetic heading as suspect near vehicles, rebar, steel bridges and buried utilities — Run 2's 2° bridge error is a conservative example.

TRY IT YOURSELF

16.1 A GX5-25 heading has a constant +0.5° bias around a closed 1 km square loop. What is the misclosure? How far off is the far corner (about 707 m away diagonally)?
16.2 Convert a gyro noise density of $0.01°/s/\sqrt{Hz}$ to ARW in $°/\sqrt{h}$.
16.3 What roll/pitch error keeps vertical LiDAR error below 3 cm at 25 m?
16.4 (Simple) Why can a loop close perfectly and still be wrong?
16.5 With the GQ7's 6.5 mm relative precision, what antenna baseline gives 0.1° heading?
16.6 Estimate the GQ7 coast error after a 30 s outage with RTK-quality velocity (0.01 m/s).

Answers: worked solutions are in Appendix A, “Answers to Try It Yourself,” starting on page 158.

SOURCES FOR THIS MODULE

MicroStrain/HBK *3DM-GX5-25 datasheet* (8400-0093); *3DM-GX5 GNSS/INS datasheet*; *3DM-GQ7 GNSS/INS datasheet*; *3DM-RTK* sheet; FDOT *FPRN Frequently Asked Questions*. The Block Loop traverse and coast-error values are computed in this course from those specifications.

MODULE 17

Survey-Grade Case Study: RIEGL VUX-240[24]

This module is a project-style case study with a survey-grade airborne scanner, the **RIEGL VUX-240[24]** (datasheet dated 2026-08-17), flown on a crewed aircraft, helicopter or large UAS. Every laser and IMU error is pushed to the ground for a **500 ft (152.4 m) return** — conveniently, almost exactly the 150 m range at which RIEGL specifies its accuracy and precision — and the results are compared with an ASPRS accuracy class.

SIMPLE APPROACH

This scanner is like a super-fast, super-careful version of the ones we met before. It fires up to 2.4 million laser flashes a second from an airplane. To know exactly where each dot lands 500 feet below, we have to add up all the little mistakes: the laser's own wobble, the tiny tilt and turn errors of the motion sensor, and where the GPS thinks the airplane is. The surprise: with a scanner this good, the *airplane's position* is the biggest mistake, not the laser.

17.1 The scanner in one table

Specification (RIEGL datasheet)	**VUX-240[24]**
Measuring principle	time of flight; echo digitization; online waveform processing; multiple-target and multiple-time-around (MTA) processing
Pulse repetition rate (PRR)	300, 600, 1200, 1800, 2400 kHz (up to 2.0 M meas./s at 2400 kHz, 75° FOV)
Max. range, ≥20% targets	1200 m (300 kHz) … 460 m (2400 kHz)
Max. operating altitude AGL, ≥20%	2900 ft (300 kHz) … 1100 ft (2400 kHz)
Targets per pulse	up to 30 (300 kHz) … 5 (2400 kHz)
Minimum range	5 m
Accuracy / precision	**20 mm / 15 mm** (one sigma @ 150 m, RIEGL test conditions)
Beam divergence; footprint	0.35 mrad ($1/e^2$); 35 mm @ 100 m, 175 mm @ 500 m
Scanner	rotating polygon mirror, parallel scan lines, FOV ±37.5°, 40–600 lines/s
Angular step / angle resolution	0.002°–0.18° / 0.001°
Timing	GNSS 1PPS input; camera trigger and exposure interfaces (up to 4 cameras)
Laser class	**Class 3R** (IEC 60825-1:2014); NOHD 0 m and ENOHD 0 m when flying faster than 1 kn; stationary: NOHD 0.3 m, ENOHD 3.5 m
Size, weight, power	292×162×185 mm; approx. 4.3 kg; typ. 77 W

IMU/GNSS (post-processed)	RiLOC-F	Applanix AP+30	AP+50	AP+60
Roll, pitch	0.005°	0.010°	0.005°	0.0025°*
Heading	0.005°	0.025°	0.010°	0.005°
IMU sampling	up to 700 Hz	200 Hz	200 Hz	200 Hz
Position (typ.)	0.02–0.03 m	0.02–0.05 m	0.02–0.05 m	0.02–0.05 m
in feet	0.07–0.10 ft	0.07–0.16 ft	0.07–0.16 ft	0.07–0.16 ft

*May require a local gravity model to achieve full accuracy (RIEGL datasheet).

17.2 The project

Client need	corridor topography for a highway widening; hard-surface vertical $\text{RMSE}_V \leq 5$ cm and horizontal $\text{RMSE}_H \leq 5$ cm (ASPRS Edition 2 classes), checkpoints ≤ 2.5 cm
Platform	helicopter or large UAS, 500 ft AGL, 60 kn (30.9 m/s); UAS alternative 10 m/s
Scanner settings	2400 kHz PRR, 75° FOV
Trajectory	IMU/GNSS post-processed (PPK) against a nearby CORS or base; RiLOC-F as the baseline option
Question	which laser and IMU errors matter, how big are they at 500 ft, and does the system meet the class?

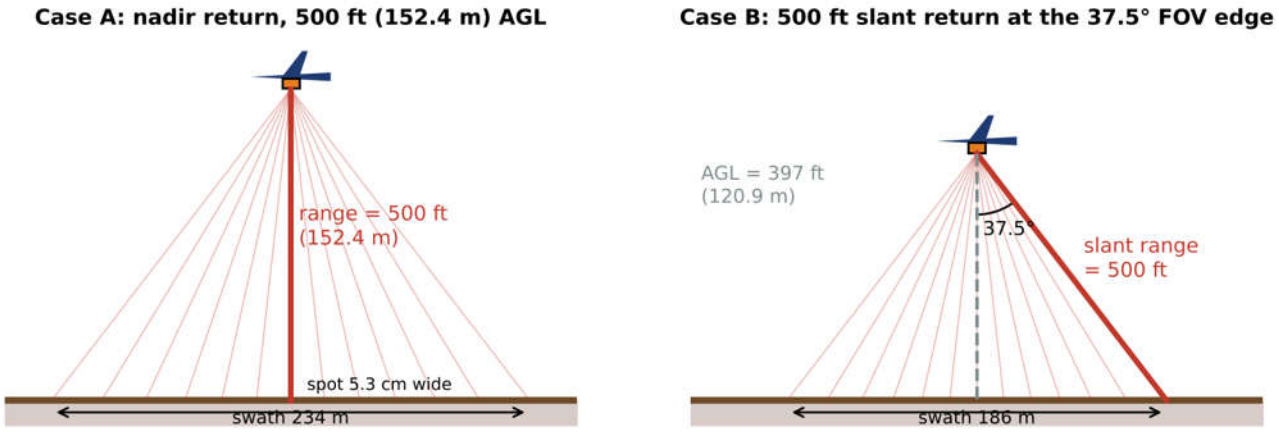


Figure 17.1: The two returns we analyze: Case A straight down from 500 ft AGL; Case B a 500 ft slant return at the edge of the 75° field of view (aircraft at 397 ft AGL).

17.3 Laser errors to consider

RIEGL states accuracy and precision in **millimeters**: 20 mm and 15 mm (one sigma at 150 m). The rest of this module works in **centimeters** so that laser, IMU and GNSS terms can be added on one scale, and gives feet for U.S. practice. The conversions are simply 10 mm = 1 cm and 1 ft = 304.8 mm, so 20 mm = 2.0 cm = 0.066 ft and 15 mm = 1.5 cm = 0.049 ft. Because RIEGL specifies the values at 150 m and our return is at 152.4 m (500 ft), they apply essentially unchanged.

Laser error source	Why it matters	At 500 ft (152.4 m)
Range accuracy (bias)	systematic offset along the beam; removed only by calibration	20 mm = 2.0 cm (0.066 ft)
Range precision (noise)	random scatter; averages down over surfaces	15 mm = 1.5 cm (0.049 ft)
Combined range error	$\sqrt{20^2 + 15^2} = \sqrt{625}$	**25 mm = 2.5 cm (0.082 ft)**
Beam footprint	edge location and mixed returns on curbs, rails, wires	53 mm (5.3 cm) spot $\Rightarrow \pm 27$ mm edge uncertainty
Angle encoder	0.001° resolution $\Rightarrow$ quantization $0.001°/\sqrt{12}$	0.8 mm = 0.08 cm — negligible
Polygon mirror facets	each facet needs its own angular calibration; a bad facet gives repeating stripes	checked in strip comparisons
Range walk / incidence	strong or grazing echoes shift timing slightly; waveform processing reduces it	sub-cm to cm; check on targets
MTA ambiguity	at high PRR several pulses are in the air; wrong zone $\Rightarrow$ gross blunder	see below
Atmosphere	refractive index changes range by ~300 ppm if uncorrected	46 mm = 4.6 cm at 500 ft if ignored!

WORKED EXAMPLE — multiple-time-around (MTA) at 2400 kHz

Unambiguous range

$c/(2\,f_{\mathrm{PRR}}) = 2.997 \times 10^8/(2 \times 2.4 \times 10^6) = 62.4$ m (Module 3)

Which zone?

$152.4/62.4 = 2.44 \Rightarrow$ zone 3: two newer pulses leave before this echo returns

Result: the processor must assign every echo to the correct zone. A wrong assignment moves a point by a whole multiple of 62.4 m — obvious "ghost layers" that are blunders, not noise. At 300 kHz the zone is 500 m deep and the problem disappears at this altitude.

WORKED EXAMPLE — atmospheric correction

Refractive index of air

$n \approx 1.0003$, i.e. 300 ppm

At 500 ft

$152.4 \times 300 \times 10^{-6} = 0.046$ m

Result: 4.6 cm — nearly twice the laser's whole error budget. Processing software applies the correction automatically from temperature and pressure; confirm it is on and the values are sensible (Module 3).

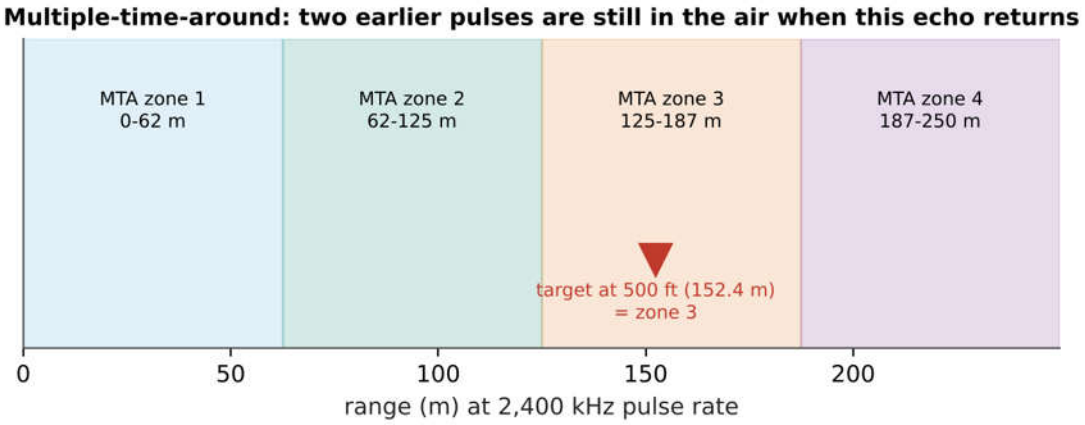


Figure 17.2: At 2.4 MHz the 500 ft echo belongs to the third pulse back.

17.4 IMU/GNSS errors and the 500 ft error budget

The post-processed IMU/GNSS options (table above) are carried through a full error budget for a 500 ft return in Module 14 (Section 14.11.3). Summary (1σ, before ground control): RiLOC-F 3.7–3.9 cm H and 4.4–4.5 cm V (0.12–0.13 ft H, 0.15 ft V); AP+30 5.5–6.6 cm H and 5.8–5.9 cm V (0.18–0.22 ft H, 0.19 ft V). After ground control and strip adjustment the RiLOC-F reaches about 3.1 cm H and 3.1 cm V (0.10 ft), supporting a 5 cm class with margin. GNSS position — not the laser — is the largest term.

17.5 Point density at 500 ft

WORKED EXAMPLE — point density at 500 ft

Swath

$2 \times 152.4 \times \tan 37.5^\circ = 234$ m

At 60 kn (30.9 m/s), 2.0 M meas./s

$2.0 \times 10^6 / (234 \times 30.9) = 277$ points/m^2

UAS at 10 m/s

$2.0 \times 10^6 / (234 \times 10) = 855$ points/m^2

Result: consistent with RIEGL's example of ~126 pts/m^2 at 1,100 ft and 60 kn (density scales with 1/altitude). At 277 pts/m^2 points are about 6 cm apart — similar to the 5.3 cm footprint, so features are sampled about once per spot.

17.6 Survey-grade checklist for this system

Laser	range calibration current; atmospheric correction on; MTA zones resolved (no ghost layers); facet calibration checked; minimum range 5 m respected
IMU/GNSS	post-processed (PPK) trajectory with short baselines; IMU option matched to the class (heading drives swath-edge horizontal error); gravity model if using AP+60
Calibration	lever arm measured to mm; boresight and facet calibration flown over a calibration site with opposite and cross strips
Timing	1PPS connected and logged; camera exposure events time-stamped
Control	ground control for strip adjustment; independent checkpoints ($\geq$30), 2$\times$ more accurate than the product (Module 13)
Safety	Class 3R: safe from the air ($>$1 kn), but on the ground the NOHD is 0.3 m and ENOHD 3.5 m — never look into the aperture or test-fire toward people
Reporting	ASPRS Edition 2 statement; swath-to-swath and within-swath precision results; client accuracy agreement (Appendix E)

BIG IDEA

At 500 ft, a survey-grade laser is no longer the weak link: the RIEGL's own range error (2.5 cm) is comparable to the GNSS error, and the IMU's attitude error is 1–7 cm depending on the option. Survey-grade results come from the *whole system* — trajectory, calibration, control and checkpoints — not from the laser alone.

TRY IT YOURSELF

17.1 What ground displacement does a 0.010° heading error cause at the swath edge for a 500 ft slant return?
17.2 At 1200 kHz PRR, which MTA zone does a 500 ft echo fall in?
17.3 If GNSS vertical error could be cut to 1.0 cm, what is the RiLOC-F nadir vertical total (range 2.5 cm, lever 0.5 cm)?
17.4 (Simple) Why is the RIEGL safe to fly over people but not safe to stare into on the ground?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

RIEGL, *VUX-240[24] datasheet* (2026-08-17), including RiLOC-F and Applanix AP+30/50/60 IMU/GNSS options; Glennie (2007) for airborne error geometry; Module 3 for MTA ranging.

MODULE 18

Processing, QA/QC and Deliverables

18.1 From raw data to a clean cloud

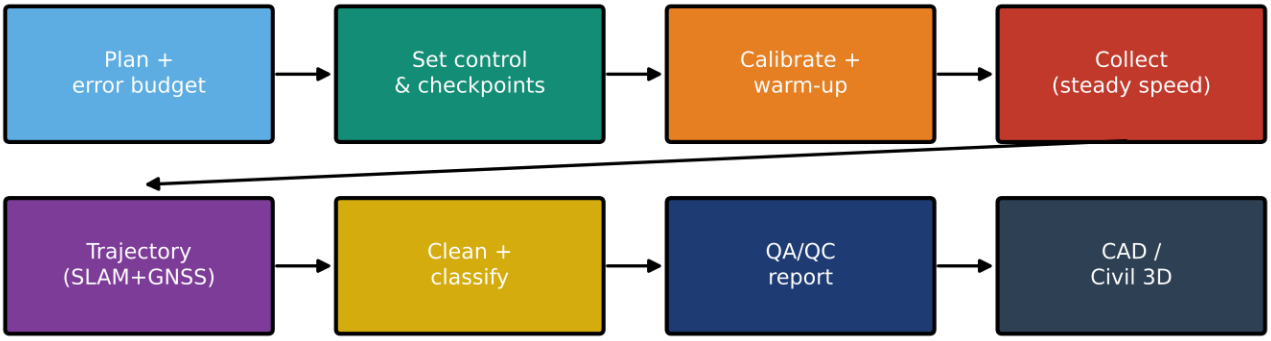


Figure 18.1: Field-to-office workflow.

Step	What happens
1 Ingest	copy raw LiDAR, IMU, GNSS, camera and field logs; verify time bases and the calibration file
2 Trajectory	GNSS/INS post-processing (PPK) and/or SLAM with loop closures; inspect the trajectory's standard deviations
3 Georeference	apply lever arm, boresight, de-skew; generate the point cloud (Module 9)
4 Clean	remove noise, outliers and moving objects
5 Adjust	introduce control; strip-to-strip alignment if needed
6 QA/QC	checkpoints, cloud-to-cloud comparisons, accuracy report (Module 13)
7 Classify & extract	ground, buildings, vegetation, poles, line work, surfaces
8 Deliver & archive	LAS/LAZ, surfaces, CAD, report, metadata; keep raw data

18.2 Removing outliers

SIMPLE APPROACH

Imagine a crowd of people standing close together in a park — except one person way out by the fence. You can tell that person is "alone" because their nearest neighbors are far away. Outlier removal looks for dots whose nearest neighbors are far away, and throws them out: they are usually dust, rain, birds, or reflections.

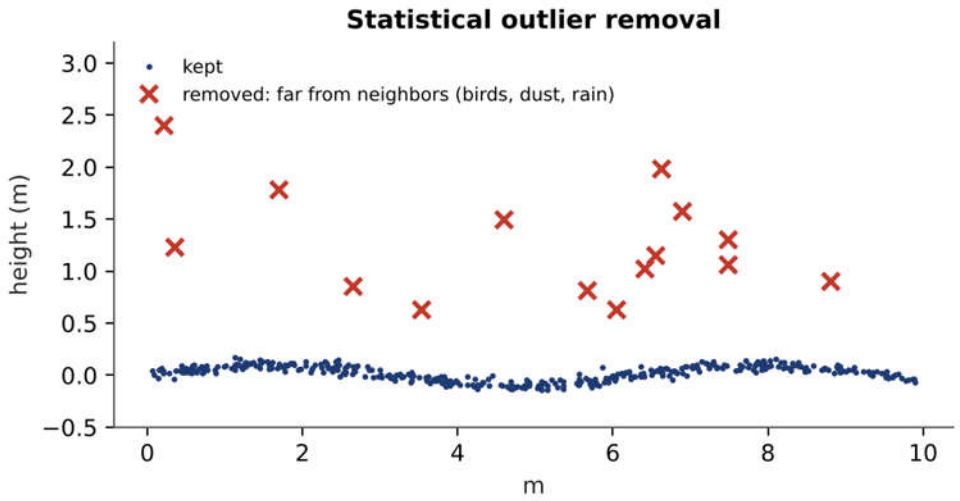


Figure 18.2: Statistical outlier removal keeps points whose neighbors are close.

KEY EQUATION — Statistical outlier removal (SOR)

$$\bar{d}_i = \frac{1}{k}\sum_{j=1}^{k} \|\mathbf{p}_i - \mathbf{p}_j\|, \qquad \text{remove } i \text{ if } \bar{d}_i > \mu_{\bar{d}} + \alpha\,\sigma_{\bar{d}}$$

What every symbol means

Symbol	Meaning	Units
$\bar{d}_i$	average distance from point i to its k nearest neighbors	m
k	neighbors examined (e.g. 8–50)	—
$\mu_{\bar{d}}, \sigma_{\bar{d}}$	mean and standard deviation of all the $\bar{d}_i$ values	m
α	how many standard deviations count as "too far" (e.g. 1–3)	—

WORKED EXAMPLE — SOR on eight points

Given: mean-neighbor distances (m): $0.05, 0.06, 0.04, 0.05, 0.07, 0.05, 0.06, 0.42$; $\alpha = 1$.

Step 1 — mean

$\mu = 0.80/8 = 0.100$ m

Step 2 — standard deviation

$\sigma = 0.121$ m

Step 3 — threshold

$0.100 + 1 \times 0.121 = 0.221$ m

Result: only the 0.42 m point is removed. Notice the outlier inflated both μ and σ; robust versions use the median instead.

18.3 Moving objects

Cars, people and bicycles appear as smeared streaks or as "ghost" copies when two passes see them in different places. Dynamic-object removal compares overlapping passes (or a voxel map over time): points that appear in a space later seen *through* (the laser passed through it and hit something

behind) are flagged as transient. Critical work still gets a visual edit. Record what was removed.

18.4 Thinning: voxel down-sampling

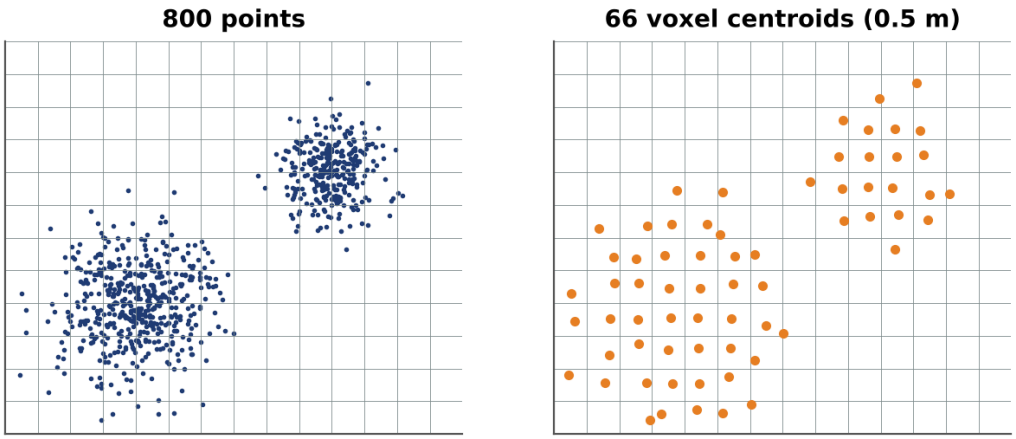


Figure 18.3: Each 0.5 m cube keeps one representative point (the centroid of its points).

KEY EQUATION — Voxel index

$$(i, j, k) = \left(\left\lfloor \frac{x}{s} \right\rfloor, \left\lfloor \frac{y}{s} \right\rfloor, \left\lfloor \frac{z}{s} \right\rfloor \right)$$

What every symbol means

Symbol	Meaning	Units
s	voxel (cube) size, e.g. 0.2 m (LIO-SAM edge map), 0.4 m (plane map)	m
$\lfloor \cdot \rfloor$	floor: round down to the whole number below	—
(i, j, k)	which cube the point falls in; all points with the same index are averaged	—

Example: point $(3.27,\ 1.81,\ 0.44)$ with $s = 0.5$ m falls in voxel $(6,\ 3,\ 0)$. Thinning makes matching fast and evens out density near the vehicle, but deliverable clouds should keep full resolution where detail matters.

WARNING — voxel size trades accuracy for speed

Making your voxel (the 3-D box) **too large reduces accuracy** while it improves performance. Every point inside a box is replaced by one average point, so a large box:

- rounds off edges, curb lips, pipe walls and other fine features — a 0.5 m voxel cannot show a 15 cm curb reveal;
- blends points from different surfaces (a wall and the ground beside it) into one point that lies on neither;
- thins the features that scan matching relies on, which can weaken SLAM and let the trajectory drift (Modules 11–12).

A small voxel keeps detail but leaves more points, so processing, matching and file sizes all

cost more. **There is no single correct size: you have to balance the cost with trial and error.** Start with a voxel well below the smallest feature you must map (for example, one-third of it), process a test section, compare the result against checkpoints and the full-resolution cloud, then enlarge the voxel only as far as accuracy still meets the specification. Record the voxel size you used in the processing report.

18.5 Classification and the bare-earth surface

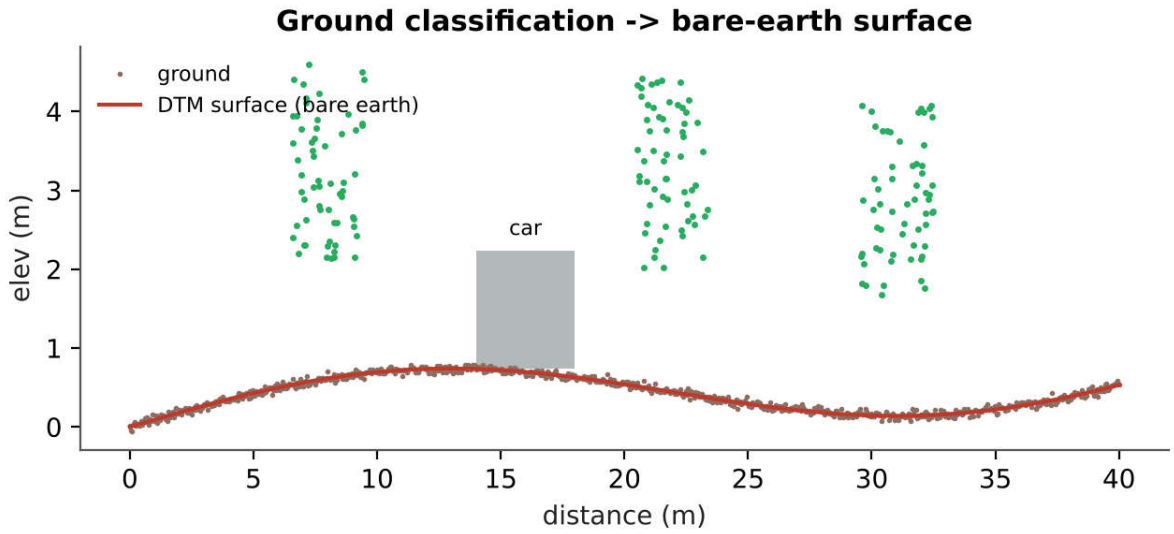


Figure 18.4: Classifying ground points and building a bare-earth surface (DTM).

ENGINEER'S VIEW

Ground filters (progressive morphological filters, cloth simulation, TIN densification) separate ground from objects. ASPRS LAS classification codes used in deliverables include: 1 unclassified, 2 ground, 3/4/5 low/medium/high vegetation, 6 building, 7 low noise, 9 water, 17 bridge deck, 18 high noise. From ground points a TIN or grid DTM is built; breaklines (curb lines, edges of pavement, ditch bottoms) are digitized from the cloud and enforced in the surface so it follows sharp grade changes.

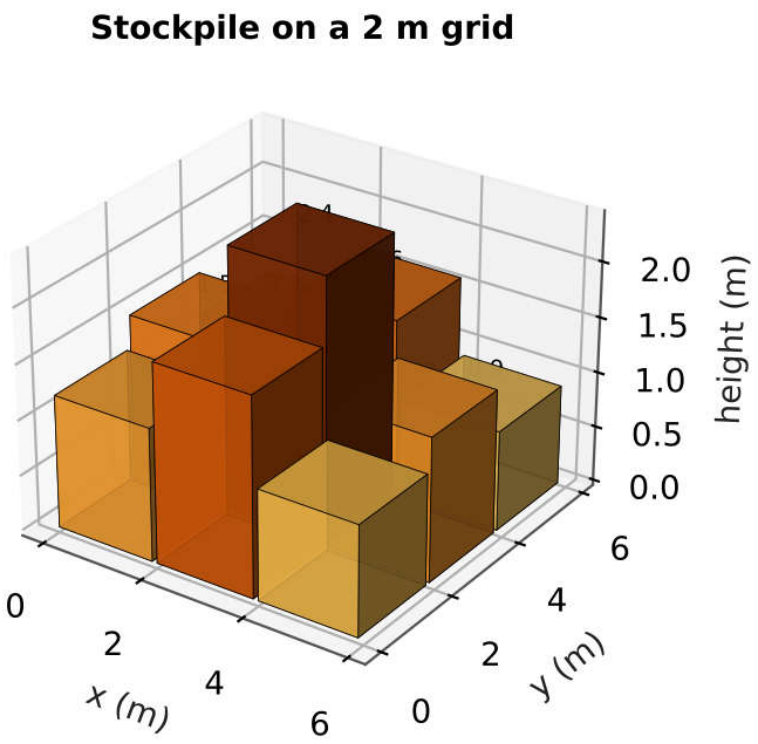


Figure 18.5: A stockpile gridded at 2 m: each column is height × cell area.

18.6 Volumes

KEY EQUATION — Grid volume

$$V = \sum_{\text{cells}} A_{\text{cell}} \left(z_{\text{top}} - z_{\text{base}} \right)$$

What every symbol means

Symbol	Meaning	Units
A_{cell}	area of one grid cell	m^2
$z_{\text{top}} - z_{\text{base}}$	height of the surface above the base surface in that cell	m
V	volume	m^3

WORKED EXAMPLE — stockpile volume

Given: 3×3 grid of 2 m cells ($A = 4$ m^2); heights above base (m): 1.2, 1.5, 1.1; 1.8, 2.4, 1.6; 1.0, 1.3, 0.9.

Step 1 — sum heights

$3.8 + 5.8 + 3.2 = 12.8$ m

Step 2 — times area

$12.8 \times 4 = 51.2$ m^3

Result: 51.2 m^3 (about 67 cubic yards). Finer grids or TIN-to-TIN methods (as in Civil 3D's volume surfaces) give better answers on steep edges.

18.7 Cloud-to-cloud QA

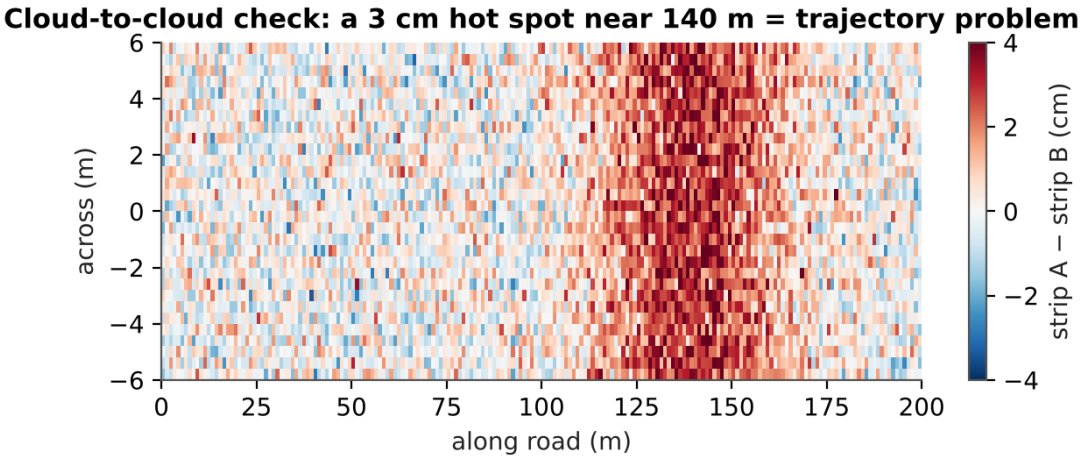


Figure 18.6: Differencing two overlapping passes reveals trajectory problems that checkpoints between them could miss.

ENGINEER'S VIEW

Compute, for each point of pass A, the distance to the nearest point (or local plane) of pass B (a C2C or M3C2 comparison). On hard surfaces, the result should be centered on zero with a spread close to the range noise. Systematic patterns tell you what went wrong: a constant offset suggests datum/GNSS; a tilt across the road suggests roll boresight; a localized blob suggests a trajectory problem (GNSS outage, corridor); a smooth ramp along the road suggests drift.

18.8 Deliverables and CAD

- **Point clouds**: LAS 1.4 / LAZ (compressed), classified, with intensity and GPS time; coordinate reference system and vertical datum/geoid in the header and metadata.
- **Surfaces**: TIN/DTM with breaklines for Civil 3D; contours if required.
- **Planimetrics**: edge of pavement, curb, striping, signs, poles, manholes, utility as-builts — extracted as 2-D/3-D line work.
- **Cross-sections and volumes**.
- **Report**: equipment, calibration, trajectory method, control used, checkpoint results, accuracy statement, datum, limitations.

SURVEYOR'S FIELD TIP

Civil 3D handles point clouds through ReCap (RCP/RCS): index the LAS/LAZ in ReCap, attach to the drawing, create a surface from the ground class, then add breaklines. Very large clouds

should be tiled and thinned for the surface while the full cloud is kept for reference. Match deliverable size to what the client's hardware and software can actually open.

18.9 A peek inside the data stream (ROS topics)

Topic	Message type	Rate	Survey meaning
`/imu/data`	`sensor_msgs/Imu`	100–400 Hz	angular velocity (rad/s) and linear acceleration (m/s^2, gravity included)
`/points`	`sensor_msgs/PointCloud2`	10 Hz	x, y, z, intensity, ring, and per-point *time* (for deskew)
`/gnss/fix`	`sensor_msgs/NavSatFix`	1–20 Hz	latitude, longitude, height, covariance
`/tf_static`	`tf2_msgs/TFMessage`	once	lever arm and boresight

A sample IMU epoch at rest might read angular velocity $(0.0021, -0.0154, 0.0842)$ rad/s and acceleration $(0.12, -0.03, 9.78)$ m/s^2: the 9.78 is gravity, and the 0.12 m/s^2 forward reading at rest would suggest a 0.7° pitch tilt (Module 7's "gravity leak") or a bias.

TRY IT YOURSELF

18.1 Which voxel (0.25 m) contains the point $(10.13, -2.40, 1.99)$?
18.2 A 4×4 grid of 1 m cells has an average height of 0.75 m. What is the volume?
18.3 Two passes differ by a tilt: +3 cm on the left edge, −3 cm on the right edge of a 10 m road. Which calibration angle would you suspect?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Rusu, R. B. *et al.* (2008), statistical outlier removal (PCL); Lague, D. *et al.* (2013), M3C2 cloud comparison, *ISPRS J.* 82; ASPRS *LAS Specification* 1.4 (classes); Autodesk ReCap/Civil 3D documentation.

MODULE 19

Florida Standards of Practice, FDOT and the New Datums

This module ties mobile LiDAR to the rules a Florida Professional Surveyor and Mapper (PSM) works under: the **Standards of Practice in Chapter 5J-17, Florida Administrative Code (F.A.C.)**, the **FDOT Surveying and Mapping Handbook** requirements for Terrestrial Mobile LiDAR (TML), and the **modernization of the National Spatial Reference System (NSRS)**. Rule text is paraphrased from the versions cited in the sources box; always work from the current official text.

REGULATORY REVIEW REQUIRED BEFORE TEACHING

This module was compiled from the rule and agency texts listed at the end of the module (retrieved October 2026). It is **not legal advice** and has **not yet been reviewed by a second Florida PSM**. Before the seminar, a reviewing PSM should check every citation against the current official text and sign below.

Reviewer: _____, PSM No. _____ Date: _____ Rule versions checked: _____

SIMPLE APPROACH

Every game has a rulebook. Surveyors in Florida have one too, written by the state. It says what every map must show, how accurate it must be and how you prove it, and how you sign your name to it. When the map is made with a laser scanner instead of a total station, the rulebook still applies — the scanner is just a new kind of measuring tape.

19.1 Who sets the rules

Chapter 472, Florida Statutes	the law that defines and licenses the practice of surveying and mapping
Board of Professional Surveyors and Mappers	adopts the rules; administratively housed in the Florida Department of Agriculture and Consumer Services
Chapter 5J-17, F.A.C.	the rules, including the Standards of Practice (5J-17.050 to 5J-17.053) and electronic signing and sealing (5J-17.062)
FDOT Surveying and Mapping Handbook	contract requirements for surveys on FDOT projects, including Terrestrial Mobile LiDAR (Sections 35–40)
NGS / Federal Register	the national datums (NSRS) that every Florida coordinate is tied to

19.2 LiDAR is inside the Standards of Practice: definitions (5J-17.050)

5J-17.050(10)	**Remote sensing**: acquiring knowledge about physical features by analyzing data obtained from instrumentation — a mobile LiDAR survey is remote sensing.
5J-17.050(11)	**Survey** includes control surveys and photogrammetric surveys, "whether it is measured by direct or remote sensing methods."
5J-17.050(7)	**Map of survey**: a graphical *or digital* depiction — a point cloud or digital surface delivered as a survey product is a map.
5J-17.050(8)	**Measurement records**: sufficient data to reconstruct the measurements — for LiDAR, raw sensor data, trajectory and calibration.
5J-17.050(12)	**Survey and map report**: the map and report identify the responsible surveyor and mapper.
5J-17.050(13)	**Work product**: includes work made under the supervision of a PSM in responsible charge.

19.3 General requirements applied to mobile LiDAR (5J-17.051)

Rule	Requirement (paraphrased)	What it means for a mobile LiDAR deliverable
.051(2)	accuracy depends on survey type and intended use; *the accuracy of data shall be independently verified*	test against **independent checkpoints** never used as control (Module 13); report RMSE in ASPRS Edition 2 terms
.051(2)(a)	products with elevation data state the **vertical datum** and describe the control; document vertical accuracy and the method used	e.g. NAVD 88 via GEOID18; list benchmarks/control; include the NVA test (30+ checkpoints)
.051(2)(b)	products showing mapped features state the **horizontal datum** (and control)	e.g. NAD 83(2011) epoch 2010.0, SPCS Florida East; FPRN mountpoint if RTK used
.051(2)(c)	if an assumed datum is used, adequate control points are established and communicated	avoid local-only SLAM frames; tie to control so the survey can be reproduced
.051(2)(d)	measurement and computation records dated to the acquisition date, with enough data to support the products	archive raw LiDAR/IMU/GNSS, trajectory, calibration file, processing logs (Section 20.6)
.051(3)(a)–(b)	results communicated by a map, or a report with an attached map; type and/or purpose stated	e.g. "Topographic Survey — Mobile LiDAR" with purpose (design, as-built)
.051(3)(c)	business name, certificate of authorization number, addresses; surveyor's name and license number	on the map, report and data index
.051(3)(d)	survey date = **last date of data acquisition**; revisions dated	the last collection pass date (not the processing date)
.051(3)(e)	unsigned copies state they are not valid without the signature and seal (electronic versions excepted)	see 5J-17.062 below
.051(3)(f)–(h)	abbreviations defined; bearing reference stated; north arrow and scale	applies to plotted LiDAR products and CAD sheets
.051(3)(i)	responsibility stated for all mapped features; if data from different surveyors are combined, identify who is primarily responsible	when LiDAR from another firm is merged with your field survey, say so

SURVEYOR'S FIELD TIP

5J-17.052 still governs boundaries. A LiDAR topographic survey does not satisfy boundary-survey requirements (monuments, measured dimensions, water-boundary descriptions and so on in 5J-17.052). If a LiDAR product is part of a boundary survey, the boundary items must still be established and shown under 5J-17.052.

19.4 Signing and sealing digital LiDAR deliverables (5J-17.062)

.062(1)	electronic files representing documents required to be sealed under Ch. 472, F.S. are signed, dated and sealed by the PSM **in responsible charge**
.062(2)	a computer-generated seal may be used; final hard copies carry an original signature, seal and date (or an electronic signature); a **scanned signature is not acceptable**
.062(3)	an **electronic signature** has the same effect as an original signature and seal if it is unique to the user, verifiable, under the user's sole control, and **invalidated if any data in the document change**
.062(4)	alternatively, a **signature file** lists the PSM's name and number, a description, and each file's name and **secure authentication code** (a cryptographic hash); a printed report of it is manually signed, dated and sealed

WORKED EXAMPLE — sealing a 40-tile LAZ delivery with a signature file

Step 1

Compute a cryptographic hash (e.g. SHA-256) for every delivered file: 40 LAZ tiles, the TIN, the CAD file, the trajectory file and the PDF report.

Step 2

Build the signature file: PSM name and number, project description, and one line per file — `tile_0101.laz SHA-256: 9f2c...e41a` (64 hexadecimal characters each).

Step 3

Print the signature file report; sign, date and seal it manually.

Step 4 — verification

anyone can recompute each file's hash; if it matches the sealed list, that file is the sealed file. If one point in one tile is edited, that tile's hash changes completely and it is no longer sealed.

Result: this is how a multi-gigabyte point cloud can be sealed without printing it — and why every later edit must be re-delivered with a new sealed signature file.

19.5 FDOT Terrestrial Mobile LiDAR (TML) requirements

ENGINEER'S VIEW

On FDOT work, TML surveys follow the **FDOT Surveying and Mapping Handbook** (current edition dated February 28, 2025, superseding the 3/29/2019 edition). Its TML part is organized as follows; read these sections before scoping a project:

Section	Topic
35	Introduction
36, 36.1	TML methods; types of TML surveys
37	TML project selection
38, 38.1	TML equipment; minimum TML system sensor components
39.1–39.5	mission; base station control; maintenance and boresight calibration; redundancy; monitoring
39.6, 39.6.1, 39.6.2	project control and checkpoints; Type "A" control layout; TML survey specifications
39.7, 39.8	accuracy analysis; quality assurance/quality control
40, 40.1, 40.1.1, 40.2	deliverables; all-type deliverables; LiDAR point classes; TML survey report
Appendix C	accuracy requirements

FDOT defines three survey types by accuracy (originally in its 2012 TML guidelines, adapted from Caltrans): **Type A** (high accuracy) for design engineering and as-built surveys; **Type B** (medium) for detailed asset inventory and earthwork; **Type C** (low) for planning and transportation statistics.

Type	Typical use	Accuracy requirement (enter from Handbook §39.6.2 / App. C)	Control/checkpoint layout (enter from §39.6, §39.6.1)
A	design engineering, as-built	_____	_____
B	asset inventory, earthwork	_____	_____
C	planning, statistics	_____	_____

The numeric tolerances are deliberately left blank: copy them from the Handbook edition named in your contract so the seminar never teaches an outdated number.

SURVEYOR'S FIELD TIP

District practice example (FDOT District 5 QA/QC plan). "Imagery" covers aerial photogrammetry *and* TML. Coordinate with the FDOT Central Office for a Mobile Survey Tracking System (MSTS) number; submit all raw, interim and final imagery data to the Central Office with a letter of transmittal; deliver the derived DGN/TIN files to the District; and note in the Survey and Map Report that the Central Office has received a separate certified report covering the imagery data. Check your own District's plan.

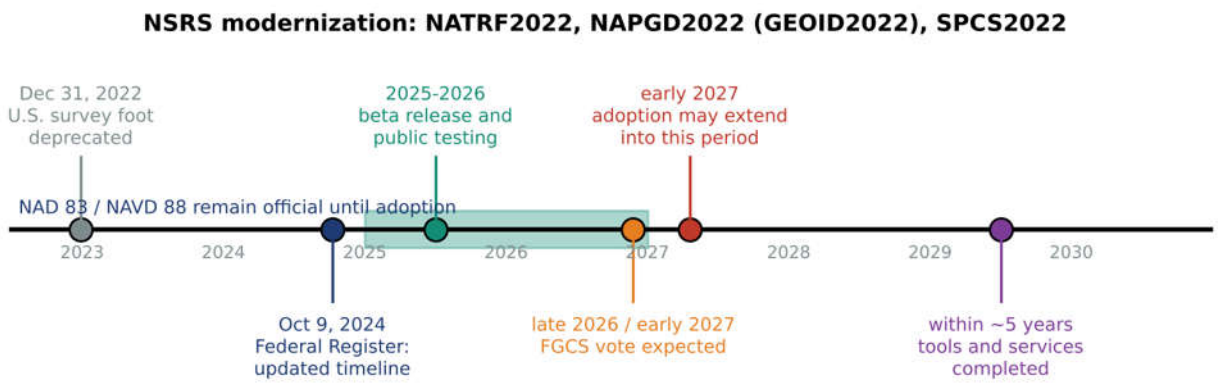


Figure 19.1: NSRS modernization timeline as published by NGS and the Federal Register. Dates after 2026 are NGS expectations, not fixed.

19.6 The new national datums (NSRS modernization)

ENGINEER'S VIEW

NGS is replacing NAD 83 and NAVD 88 with a modernized NSRS:

- **NATRF2022** (North American Terrestrial Reference Frame of 2022), with PATRF2022, CATRF2022 and MATRF2022 for other plates — replacing the NAD 83 realizations.
- **NAPGD2022** (North American-Pacific Geopotential Datum of 2022), including **GEOID2022** — replacing NAVD 88 and the hybrid geoids such as GEOID18.
- **SPCS2022** — new State Plane zones, defined in meters with the **international foot** as the only foot.

Timeline. The Federal Register notice of October 9, 2024 scheduled the modernization for 2025 or 2026, with components posted for public beta testing (at least six months after the last component), a vote by the Federal Geodetic Control Subcommittee (FGCS), FGDC endorsement and a Federal Register adoption notice, followed by tools and services within five years. NGS's February 2026 update placed the FGCS vote in **late 2026 or early 2027** and said full adoption **may extend into the first part of 2027**. **Until adoption, the current NSRS (NAD 83, NAVD 88) remains official.**

Units. NIST and NOAA deprecated the **U.S. survey foot on December 31, 2022** in favor of the international foot (0.3048 m exactly). The difference is 2 parts per million — about 0.2 ft over the 100 000 ft magnitudes of Florida State Plane coordinates. Section 177.151, F.S. expresses Florida State Plane coordinates "in feet and decimals of a foot": state explicitly which foot each deliverable uses, and confirm the current statute and the client's requirement.

WORKED EXAMPLE — why the foot matters

Given: a State Plane easting of 650 000 ft.

U.S. survey foot

$1\ \text{ft} = 1200/3937\ \text{m} = 0.3048006096\ \text{m} \Rightarrow 650\,000 \times 0.3048006096 = 198\,120.396\ \text{m}$

International foot

$650\,000 \times 0.3048 = 198\,120.000\ \text{m}$

Difference
0.396 m = 1.30 ft
Result: a mislabeled foot moves every point by more than a foot — far larger than any LiDAR error in this course. Always label the foot.

SURVEYOR'S FIELD TIP

Future-proof today's LiDAR. Record datum, realization, epoch, geoid model and foot in the report, the LAS header (WKT) and the metadata; keep ellipsoid heights and the trajectory so the project can be transformed to NATRF2022/NAPGD2022 later; and expect clients to ask for re-delivery in the new datums once they are adopted.

19.7 Checklist: a Florida-compliant mobile LiDAR deliverable

- ☐ map or report with attached map; type and purpose stated (.051(3)(a)–(b))
- ☐ business, certificate of authorization, PSM name and license (.051(3)(c))
- ☐ survey date = last acquisition date; revisions dated (.051(3)(d))
- ☐ horizontal datum, realization, epoch, SPCS zone, foot; vertical datum and geoid; control described (.051(2)(a)–(b))
- ☐ accuracy independently verified with checkpoints; method documented; ASPRS statement (.051(2))
- ☐ measurement and computation records archived and dated (.051(2)(d))
- ☐ responsibility for merged data stated (.051(3)(i))
- ☐ electronic files sealed by electronic signature or sealed signature file (.062)
- ☐ FDOT projects: Handbook TML type, control/checkpoint layout, report and Central Office submittal

TRY IT YOURSELF

19.1 (Concept) Which 5J-17.051 subsection requires that the accuracy of data be independently verified, and how does a LiDAR survey satisfy it?
19.2 (Concept) What is the "survey date" of a mobile LiDAR project collected on March 3 and March 5 and processed on March 20?
19.3 (Concept) A client edits one LAZ tile after delivery. Is it still sealed under a 5J-17.062(4) signature file? Why?
19.4 Convert a 1 200 000 ft State Plane northing using both feet. How large is the difference?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Fla. Admin. Code R. 5J-17.050, 5J-17.051 (eff. 11/13/2017), 5J-17.052 and 5J-17.062 (eff. 7/15/2018), as published by Cornell LII; Ch. 472 and s. 177.151, F.S.; FDOT *Surveying and Mapping Handbook* (Feb. 28, 2025), Sections 35–40 and App. C; FDOT *Terrestrial Mobile LiDAR Surveying & Mapping Guidelines* (2012) as summarized by LiDAR News; FDOT District 5 *Survey QA/QC Management Plan*; Federal Register 2024-23347, "Updated Implementation Timeline for the Modernized NSRS" (Oct. 9, 2024); GPS World, "NGS presents the latest NSRS news at Geo Week 2026" (Mar. 3, 2026); Esri ArcNews, "Change is afoot with the US survey foot."

MODULE 20

Professional Practice for LiDAR Surveys

The technology modules explain how a LiDAR survey works. This module covers what a licensed surveyor must manage around it: responsible charge, sealing digital data, contract scope and liability, drone regulations, field safety, and records. It is professional guidance, **not legal advice**; consult counsel and the current rules for your situation.

SIMPLE APPROACH

Having a great camera does not make you a professional photographer. A professional also makes sure the right person is in charge, writes down what was promised, works safely, keeps the originals, and signs only what they have checked. The same is true for a laser scanner.

20.1 Responsible charge when the data come from a machine (and a vendor)

ENGINEER'S VIEW

Under Chapter 472, F.S. and 5J-17.050(13), a work product is made by or under the supervision of the **PSM in responsible charge**. With mobile LiDAR, parts of the work are often done by others — a scanning vendor, a processing service, a trajectory specialist. Responsible charge means the PSM:

- sets the accuracy requirements and the method (the client agreement and SOP, Appendix E);
- directs the control and checkpoint survey and keeps the checkpoints *independent* (5J-17.051(2));
- reviews calibration, trajectory quality and the accuracy test before sealing — not just the final CAD file;
- states responsibility for any merged data from others (5J-17.051(3)(i)).

Sealing a vendor's point cloud without directing or verifying the work is not responsible charge.

20.2 What gets sealed

Survey and Map Report	always signed and sealed; carries datums, accuracy statement, methods, limitations
Maps and CAD sheets	signed and sealed (electronically or on hard copy) per 5J-17.051(3)(e) and 5J-17.062
Point clouds, surfaces, trajectory	sealed through an electronic signature or a sealed signature file of hashes (5J-17.062(4), Module 19)
Data delivered "for information only"	state clearly that it is not a sealed survey product and what it may not be used for

20.3 Contract scope and liability

Scope	limits of work, features to be mapped, deliverables and formats, schedule; reference the signed client accuracy agreement (Appendix E)
Accuracy	horizontal and vertical classes (RMSE, ASPRS Edition 2), which surfaces are tested, checkpoint count and accuracy
Limitations	LiDAR does not see through solid objects, under dense vegetation, below water (near-infrared) or underground (Module 4); occluded areas and utilities need other methods; state them in the report
Intended use	e.g. design topography, not boundary; prohibit reuse for other purposes without review
Third-party data	control or base data supplied by others: identify it and its accuracy; state reliance
Datums and units	datum, realization, epoch, geoid, foot (Module 19); who pays for later transformation to NATRF2022
Changes	how added scope, recollection after construction changes, or new accuracy demands are handled
Data license	who owns raw and processed data; reuse, resale and retention terms
Insurance	professional liability coverage appropriate to the project and contract requirements

20.4 Drone LiDAR and FAA Part 107

ENGINEER'S VIEW

UAS LiDAR flown for compensation in the U.S. falls under **14 CFR Part 107** (small UAS under 55 lb), unless another authorization applies. Key operating limits as published by the FAA:

- maximum altitude **400 ft above ground level**, higher only within 400 ft of a structure; maximum speed 100 mph (87 kn);
- keep the aircraft within **visual line of sight** (a visual observer is needed with first-person view);
- no flight over people not directly participating, unless the operation qualifies under the operations-over-people rules or a waiver;
- controlled airspace (Class B, C, D, E) requires FAA authorization, usually through **LAANC**;
- a Remote Pilot Certificate is required; operations outside the rules require a **waiver** (FAA generally responds within about 90 days).

Later amendments added night operations with anti-collision lighting and recurrent training, operations over people by category, and **Remote ID** (14 CFR Part 89). Check the current FAA rules before every project.

WORKED EXAMPLE — can the RIEGL case study be flown by drone?

Given: Module 17 plans 500 ft AGL; the VUX-240[24] weighs about 4.3 kg (9.5 lb) without IMU, 4.6–6.8 kg (10–15 lb) with an IMU/GNSS option.

Weight

a large UAS carrying it can stay under 55 lb — Part 107 can apply

Altitude

500 ft AGL exceeds the 400 ft limit (no structure nearby) ⇒ needs a **waiver**, or fly at 400 ft or lower, or use a crewed helicopter/airplane

At 400 ft AGL instead

nadir range 121.9 m: attitude terms shrink by 20%; swath 187 m (614 ft); density at 10 m/s $\approx 2.0 \times 10^6/(187 \times 10) = 1\,070$ pts/m^2

Result: flying at 400 ft is both legal without an altitude waiver and slightly more accurate. Regulatory limits belong in the error budget and the flight plan from day one.

20.5 Field safety and maintenance of traffic

- **Traffic:** mobile mapping on public roads needs a maintenance-of-traffic plan consistent with the MUTCD (Part 6, temporary traffic control) and, on FDOT roads, the FDOT Standard Plans for temporary traffic control and any permit conditions. Plan speeds, shoulders, shadow vehicles and stops for control and checkpoint shots.
- **People on foot:** high-visibility apparel; never set targets or checkpoints in live lanes without protection.
- **Laser safety:** street and handheld scanners are typically Class 1; airborne scanners such as the RIEGL VUX-240[24] are **Class 3R** with a ground nominal ocular hazard distance (Module 17). Follow the manufacturer's limits and a written laser-safety procedure.
- **Railroads and restricted sites:** obtain right-of-entry and follow the owner's safety training and flagging requirements.

20.6 Records, data management and retention

Record	Why	Typical size
Raw LiDAR, IMU, GNSS, camera	measurement records that reconstruct the survey (5J-17.050(8), .051(2)(d))	OS0 at full rate: ~133 GB/hour
Base station / CORS data and logs	reprocess the trajectory; prove the datum	small
Trajectory (with standard deviations)	shows where accuracy is strong or weak	0.1–1 GB
Calibration file and calibration flight/drive	ties lever arm and boresight to the data	small
Control and checkpoint survey	proves the accuracy statement	small
Processing logs and settings (voxel, filters, classification)	reproducibility (Module 18 warning)	small
Deliverables and sealed signature file	what the client received	tens to hundreds of GB

Keep records for the period required by contract, client policy and law (public agencies may also be subject to public-records retention schedules). Use checksums (the same hashes used for sealing) to prove archives are unchanged, keep two copies in different places, and record the software versions used.

20.7 Quality, competence and disclosure

- Write and follow an SOP (Appendix E); have an independent reviewer run the accuracy test.
- Practice within competence: new sensors, platforms (UAS) or datums (NSRS 2022) deserve training and a pilot project before production work.
- Disclose limitations plainly in the report: occlusions, vegetation, water, range limits, GNSS outages, areas below the stated accuracy.

TRY IT YOURSELF

20.1 (Concept) A vendor delivers a classified point cloud; you never saw the checkpoints. What must you do before sealing it?
20.2 (Concept) List three LiDAR limitations that belong in the report's limitations section.
20.3 (Concept) Why does flying the RIEGL at 400 ft instead of 500 ft help both legally and technically?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," starting on page 158.

SOURCES FOR THIS MODULE

Ch. 472, F.S.; Fla. Admin. Code R. 5J-17.050, 5J-17.051, 5J-17.062 (Cornell LII); FAA, "Small Unmanned Aircraft Systems (UAS) Regulations (Part 107)" newsroom summary, and 14 CFR Parts 89 and 107; RIEGL VUX-240[24] datasheet (2026-08-17); Ouster OS0 Rev8 datasheet; FHWA *Manual on Uniform Traffic Control Devices*, Part 6; FDOT Standard Plans (temporary traffic control). Legal and regulatory statements are summaries for teaching and must be confirmed against current official texts.

MODULE 21

Capstone and Sample Project Dataset

This module walks through a complete mobile-LiDAR project, using a tool from every earlier module. Work through it with a calculator.

SIMPLE APPROACH

This is the "big game" at the end of the season. You use every skill you practiced: measuring with light, turning angles into dots, keeping track of the car, fixing mistakes with landmarks, and checking your work with secret checkpoints. If every step is done carefully, the final map can be trusted to build a real road.

21.1 The assignment

Project	topographic survey of a 2.0 km (6,562 ft) urban road for resurfacing design
Specification	hard-surface vertical $\text{RMSE}_z \leq 3$ cm (0.10 ft); horizontal $\text{RMSE}_H \leq 5$ cm (0.16 ft) (ASPRS Ed. 2 style)
Features	curbs, edges of pavement, striping, manholes, signs; along-track point spacing ≤ 0.25 m (0.82 ft) on pavement
Complication	a 300 m (984 ft) underpass with no GNSS
System	tilted survey-grade head, 32 channels, 20 Hz, 4 channels effectively sweep the road per rotation; tactical IMU (gyro 1°/h, accel 0.5 mg); PTP time sync; dual-frequency GNSS

21.2 Step-by-step plan

WORKED EXAMPLE — 1 — drive speed from point spacing (Module 5)

Requirement

$$\Delta_{\text{along}} = \frac{v}{f_{\text{rot}} \times N_{\text{eff}}} \leq 0.25 \text{ m}$$

Solve for v

$v \leq 0.25 \times 20 \times 4 = 20$ m/s

Choose

13 m/s (≈30 mph) for traffic and margin: $\Delta_{\text{along}} = 13/80 = 0.16$ m ✓

WORKED EXAMPLE — 2 — the underpass: how bad is an unaided IMU? (Module 7)

Outage time

$300/13 = 23.1$ s

Accelerometer drift

$\frac{1}{2} \times 0.0049 \times 23.1^2 = 1.31$ m along-track

Gyro drift

$\frac{1}{2} \times 13 \times 4.85 \times 10^{-6} \times 23.1^2 = 0.017$ m lateral

Result: lateral is fine; along-track is not. LiDAR SLAM helps but an underpass is a corridor (Module 11) with weak along-track constraint. **Plan:** chevron targets at the entrance, midpoint and exit as control, plus a return pass for loop closures.

WORKED EXAMPLE — 3 — timing and calibration checks (Modules 8–9)

Timing

PTP sync to ± 0.1 ms: $13 \times 0.0001 = 1.3$ mm — negligible

Boresight

after the morning calibration loop the residual is 0.02°; at the longest useful range (30 m): $30 \times 0.000349 = 1.0$ cm

WORKED EXAMPLE — 4 — vertical error budget (Section 14.11.4)

Open sky

2.45 cm (0.080 ft) $\leq$ 3 cm ✓

Underpass, trajectory 2.5 cm

3.00 cm (0.098 ft) — at the limit ×

Underpass with mid-point control

2.23 cm (0.073 ft) ✓

Result: the extra control target in the underpass is what makes the job pass. The full budget is in Module 14.

WORKED EXAMPLE — 5 — control and checkpoint layout (Module 13)

Checkpoints

ASPRS minimum 30 over 2.0 km: about one every 67 m, alternating sides of the road and including the underpass

Control

every ~250 m plus the three underpass targets; never reused as checkpoints

Survey accuracy

RTK/total station at $\leq$ 1.5 cm vertical, i.e. at least 2× better than the 3 cm product

21.3 Collecting and processing

1. Morning: power and storage checks, PPS/PTP lock, calibration file loaded, 5-minute static IMU alignment, short loop past known features.
2. Collect northbound and southbound at 13 m/s, log traffic and GNSS outages.
3. Office: PPK trajectory; SLAM with loop closures; inspect trajectory σ (expect a bump in the underpass); georeference; de-skew; strip-to-strip check.
4. Clean noise and traffic; classify ground; adjust to control.
5. Measure the 30 checkpoints *last*, without changing anything afterwards.

WORKED EXAMPLE — 6 — the accuracy results

Measured from 30 checkpoints: vertical $\text{RMSE}_{z,\text{data}} = 1.9$ cm (mean +0.3 cm); horizontal $\text{RMSE}_x = 2.0$, $\text{RMSE}_y = 1.7$ cm. Checkpoint survey RMSE: 0.8 cm vertical, 1.0 cm horizontal.

Vertical

$\sqrt{1.9^2 + 0.8^2} = \sqrt{3.61 + 0.64} = 2.06 \text{ cm} \leq 3 \text{ cm}$ ✓

Horizontal

$\text{RMSE}_H = \sqrt{2.0^2 + 1.7^2} = 2.62$ cm; with survey error $\sqrt{2.62^2 + 1.0^2} = 2.81 \text{ cm} \leq 5 \text{ cm}$ ✓

Cloud-to-cloud

northbound vs southbound on pavement: mean 0.2 cm, spread 0.9 cm; one 2.5 cm patch at the underpass exit flagged, re-checked against an extra shot, within tolerance

Result: the data meet specification. Write it down.

SURVEYOR'S FIELD TIP

Sample accuracy statement. "This dataset was tested for vertical accuracy at 30 independent checkpoints on hard surfaces and found to have RMSE_z = 2.1 cm, including checkpoint survey uncertainty, and RMSE_H = 2.8 cm horizontally, in accordance with the ASPRS Positional Accuracy Standards, Edition 2. Horizontal datum: [NAD83(2011) State Plane Florida East, US ft]; vertical datum: [NAVD88, GEOID18]." Fill in the datums your project actually used, and comply with the content requirements of the governing standards of practice (in Florida, Ch. 5J-17 F.A.C.).

21.4 Sample project dataset: trajectory, checkpoints and accuracy report

The following dataset is **realistic but simulated**: it was generated to match the Main Street plan above (coordinates are fictitious). It is supplied with the course as two CSV files, so participants can recompute every number:

- `sample_trajectory.csv`: time, station, position, heading, trajectory standard deviations, GNSS status;
- `sample_checkpoints.csv`: 30 checkpoints with survey and LiDAR elevations and residuals (cm and ft).

BIG IDEA

Read the trajectory before reading the point cloud. The bottom plot shows exactly where accuracy is strong (open sky, 1.4 cm / 0.05 ft) and where it is weak (underpass, up to 3.6 cm / 0.12 ft) — which is where extra checkpoints belong.

Checkpoint table

ID	Station m / ft	Side	Survey elev. m / ft	dZ cm / ft	dE cm	dN cm
CP01	33 / 108	R	6.207 / 20.365	-0.9 / -0.030	-3.0	-1.6
CP02	100 / 327	L	6.421 / 21.066	+0.1 / +0.003	-0.8	-1.6
CP03	166 / 546	R	6.627 / 21.742	+0.4 / +0.013	+3.8	+2.1
CP04	233 / 765	L	6.821 / 22.377	-0.3 / -0.010	-0.9	-1.3

ID	Station m / ft	Side	Survey elev. m / ft	dZ cm / ft	dE cm	dN cm
CP05	300 / 983	R	6.999 / 22.963	+2.8 / +0.092	-0.1	-2.3
CP06	366 / 1202	L	7.161 / 23.494	-3.6 / -0.118	+1.4	-2.7
CP07	433 / 1421	R	7.307 / 23.973	-1.0 / -0.033	+0.9	-1.1
CP08	500 / 1640	L	7.439 / 24.408	+1.7 / +0.056	-0.8	+1.5
CP09	567 / 1859	R	7.562 / 24.809	-0.5 / -0.016	-1.0	+1.5
CP10	633 / 2077	L	7.679 / 25.195	-0.4 / -0.013	+3.2	+0.5
CP11	700 / 2296	R	7.797 / 25.581	-0.6 / -0.020	-2.7	-1.5
CP12	767 / 2515	L	7.921 / 25.987	+2.4 / +0.079	-3.7	+0.9
CP13	833 / 2734	R	8.055 / 26.427	+1.8 / +0.059	+2.2	-0.7
CP14	900 / 2953	L	8.203 / 26.914	-3.0 / -0.098	+2.5	+0.6
CP15	967 / 3171	R	8.368 / 27.453	-2.5 / -0.082	-2.8	+3.0
CP16	1033 / 3390	L	8.549 / 28.047	-0.8 / -0.026	+1.6	-0.7
CP17	1100 / 3609	R	8.744 / 28.689	-0.2 / -0.007	+0.4	-2.5
CP18	1167 / 3828	L	8.952 / 29.369	+2.3 / +0.075	-0.7	+2.6
CP19	1233 / 4047	R	9.166 / 30.073	-0.1 / -0.003	-0.2	-0.6
CP20	1300 / 4265	L	9.383 / 30.783	+3.2 / +0.105	+4.0	-0.6
CP21	1367 / 4484	R	9.595 / 31.481	+4.5 / +0.148	+1.4	-2.9
CP22	1433 / 4703	L	9.800 / 32.151	-1.9 / -0.062	+1.0	+0.8
CP23	1500 / 4922	R	9.991 / 32.779	+0.5 / +0.016	+0.8	+1.3
CP24	1567 / 5141	L	10.167 / 33.357	+2.6 / +0.085	-1.8	+0.0
CP25	1634 / 5359	R	10.327 / 33.880	+0.7 / +0.023	-1.1	+1.3
CP26	1700 / 5578	L	10.470 / 34.351	+1.0 / +0.033	-0.7	-1.0
CP27	1767 / 5797	R	10.601 / 34.780	-0.9 / -0.030	-2.5	+1.7
CP28	1834 / 6016	L	10.722 / 35.178	-1.1 / -0.036	+0.9	+0.9
CP29	1900 / 6235	R	10.839 / 35.562	+0.6 / +0.020	+2.3	+2.4
CP30	1967 / 6453	L	10.958 / 35.950	+2.5 / +0.082	-0.6	-2.9
Mean (bias)				+0.31 / +0.010		
RMSE				**1.90 / 0.062**	2.00	1.71

dZ = LiDAR − survey on hard surfaces (NVA). dE, dN measured on painted chevrons and curb corners. Checkpoint survey accuracy: 0.8 cm (0.026 ft) vertical, 1.0 cm (0.033 ft) horizontal.

Computations

Vertical, data fit	sum of squares 107.99 cm^2; $\text{RMSE}_z = \sqrt{107.99/30} = 1.90$ cm (0.062 ft); mean +0.31 cm
Vertical, with checkpoint error	$\sqrt{1.90^2 + 0.8^2} = 2.06$ cm (0.068 ft)
Horizontal, data fit	$\text{RMSE}_x = 2.00$, $\text{RMSE}_y = 1.71$; $\text{RMSE}_H = \sqrt{2.00^2 + 1.71^2} = 2.63$ cm (0.086 ft)
Horizontal, with checkpoint error	$\sqrt{2.634^2 + 1.0^2} = 2.82$ cm (0.092 ft), using unrounded RMSE_H
Largest residual	4.5 cm (0.148 ft) $< 3 \times$ RMSE — no blunders

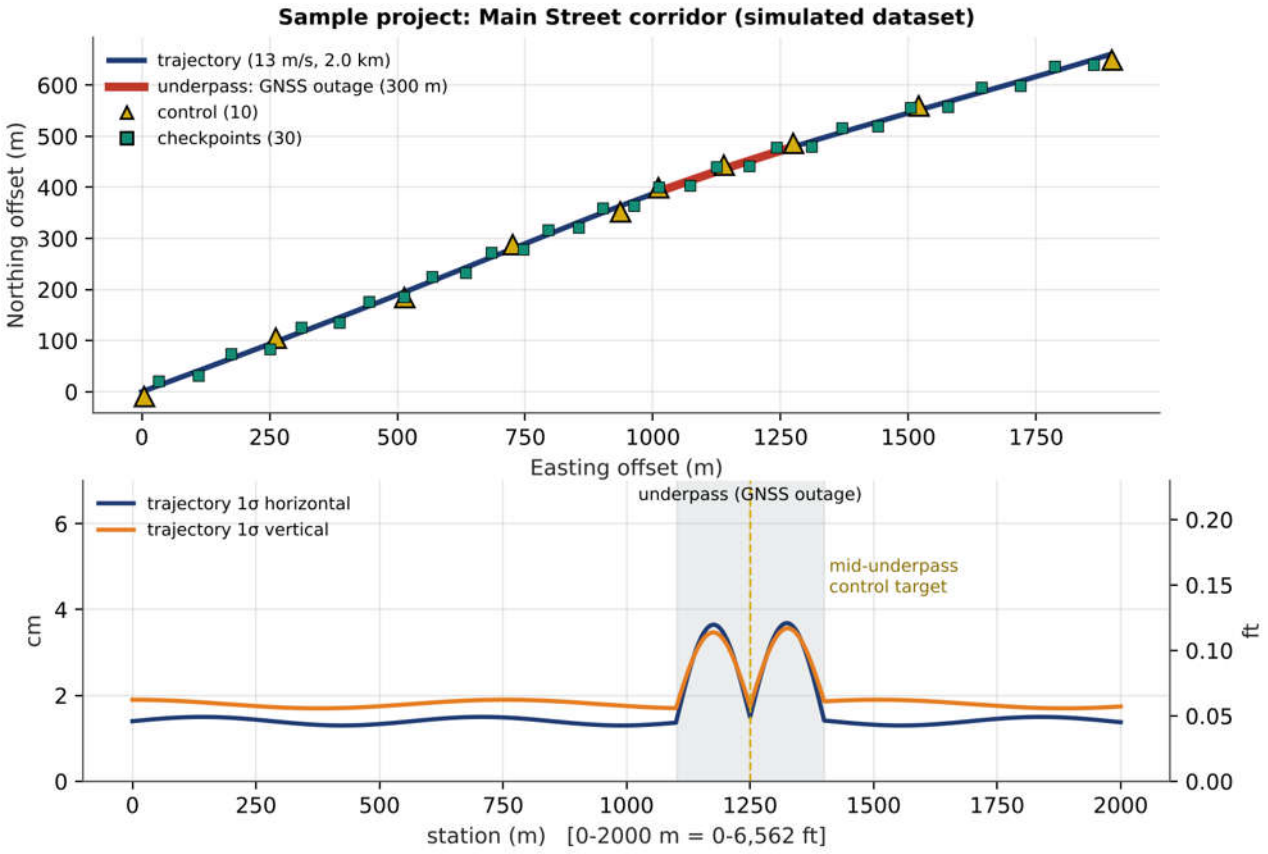


Figure 21.1: Top: the 2.0 km (6,562 ft) trajectory with 10 control points and 30 checkpoints; the red stretch is the GNSS outage under the underpass. Bottom: the trajectory's own 1σ estimates rise in the underpass and are pulled back down by the mid-underpass control target.

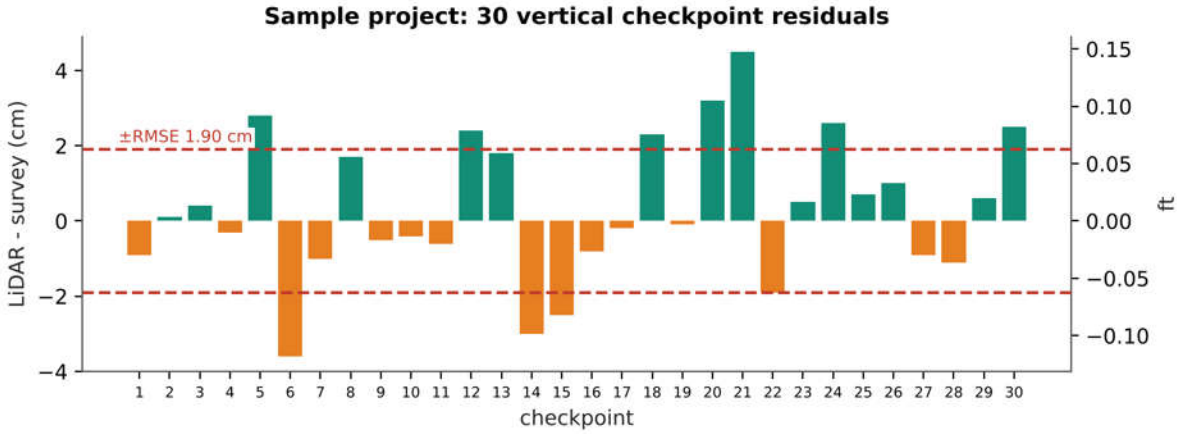


Figure 21.2: Vertical residuals at the 30 checkpoints. No residual exceeds $3\times$RMSE (5.7 cm), so none is flagged as a blunder.

Filled-in accuracy report (sample)

SURVEY AND MAP REPORT — ACCURACY SECTION (sample, simulated data)

Project	Main Street corridor topographic survey (Module 21 sample), 2.0 km / 6,562 ft
Survey type and purpose	Topographic survey by terrestrial mobile LiDAR, for roadway resurfacing design
Survey date	last date of data acquisition: _____ (5J-17.051(3)(d))
Horizontal datum	NAD 83(2011), epoch 2010.0; SPCS 83 Florida East zone; units: international foot (stated per contract)
Vertical datum	NAVD 88 via GEOID18
Control	10 control points (RTK/static GNSS and digital leveling); FPRN VRS for rover RTK; mid-underpass control target
Equipment	tilted survey-grade mobile scanner; tactical IMU; dual-frequency GNSS; PTP timing
Trajectory	PPK GNSS/INS with LiDAR SLAM and loop closures; 300 m GNSS outage bridged by SLAM and control
Accuracy agreement	3 cm $RMSE_V$ (NVA) and 5 cm $RMSE_H$ classes (Appendix E)
NVA test	30 independent hard-surface checkpoints: **$RMSE_V$ = 2.06 cm (0.068 ft)** incl. checkpoint error — **pass**
Horizontal test	30 well-defined points: **$RMSE_H$ = 2.82 cm (0.092 ft)** incl. checkpoint error — **pass**
VVA	30 vegetated checkpoints: $RMSE_V$ = 5.6 cm (0.18 ft), reported as found
Data internal precision	within-swath smooth-surface max diff 1.4 cm ($\leq$1.8); swath-to-swath $RMSD_z$ 0.9 cm ($\leq$2.4), max 3.1 cm ($\leq$4.8) — pass
Point density	median 620 points/m^2 (58 points/ft^2) on pavement within 20 m
Limitations	no data beneath vehicles parked during collection (re-shot by total station); underground utilities not mapped; points beyond 25 m excluded from design surface
Accuracy statement	"This data set was tested to meet ASPRS Positional Accuracy Standards for Digital Geospatial Data, Edition 2, Version 2 (2024) for a 3 cm $RMSE_V$ Vertical Accuracy Class and a 5 cm $RMSE_H$ Horizontal Accuracy Class. NVA was found to be $RMSE_V$ = 2.06 cm and horizontal accuracy $RMSE_H$ = 2.82 cm, including checkpoint survey uncertainty. VVA was found to be $RMSE_V$ = 5.6 cm."
Records	raw data, trajectory, calibration, control/checkpoint data and processing logs archived (5J-17.051(2)(d))
Signature and seal	PSM in responsible charge; digital deliverables sealed by signature file (5J-17.062(4))

21.5 One pulse, one more time: the master checklist

#	Operation	Module
1	Range from time of flight $\rho = c\Delta t/2$ (with atmospheric correction)	3
2	Polar to sensor x, y, z	5
3	De-skew: assign each point its own pose time	8
4	Boresight rotation $\mathbf{R}_S^B$	9
5	Add lever arm $\mathbf{l}^B$	9
6	Rotate by attitude $\mathbf{R}_B^W(t)$ (roll, pitch *and* yaw)	6, 9
7	Add trajectory position $\mathbf{p}_W(t)$	9
8	Trajectory from IMU + GNSS + scan matching + loop closures in a factor graph	7, 10–12
9	Tie to control; propagate errors; test on independent checkpoints; report RMSE	13, 14
10	Clean, classify, extract, deliver, archive	18

BIG IDEA

Measure carefully, control what you can, check what you must, and document what you did. That is how survey-grade work is produced with any instrument — including a laser that never stands still.

SOURCES FOR THIS MODULE

ASPRS Edition 2, Version 2 (2024); Fla. Admin. Code R. 5J-17.051 and 5J-17.062; Appendix E (SOP and client agreement). The sample dataset is simulated for this course; its CSV files reproduce every number in the checkpoint table and report.

APPENDIX A

Answers to Try It Yourself

1.1 $343 \times 1/2 = 171.5$ m.
1.2 IMU; time synchronization; lever arm and boresight; trajectory estimation (GNSS/INS + SLAM); ground control and checkpoints.

2.1 8.727×10^{-4} rad; $80 \times 8.727 \times 10^{-4} = 6.98$ cm.
2.2 $\sqrt{4 + 2.25 + 9} = \sqrt{15.25} = 3.91$.
2.3 $\sqrt{144 + 25} = 13$ m.
2.4 5 steps (a 3-4-5 triangle).

3.1 $2.997 \times 10^{8} \times 166.8 \times 10^{-9}/2 = 25.0$ m.
3.2 $\delta t = 2 \times 0.01/c \approx 66.7$ ps.

4.1 $0.008 + 40 \times 0.0005 = 0.028$ m $= 2.8$ cm.
4.2 $(60/20)^2 = 9$ times weaker.
4.3 The white sign: it reflects much more light (higher reflectance); black rubber absorbs it.
4.4 The tree (through gaps between leaves) and the glass storefront (the laser mostly passes through clear glass). Not the brick wall (opaque) or the pond (near-infrared light stops at the water surface).

5.1 $\rho \cos V = 39.848$; $x = 34.509$, $y = -19.924$, $z = 3.486$ m; $\sqrt{34.509^2 + 19.924^2 + 3.486^2} = 40.000$ ✓
5.2 $\rho = \sqrt{9 + 16 + 144} = 13.0$ m; $H = \text{atan2}(4, 3) = 53.13°$; $V = \text{atan2}(12, 5) = 67.38°$.
5.3 $32 \times 900 \times 20 = 576\,000$ pts/s; $10/20 = 0.5$ m per head line.

6.1 $(0, 10)$ then back to $(10, 0)$: a rotation by $-\psi$ undoes $+\psi$ ($\mathbf{R}^{\mathsf{T}} = \mathbf{R}^{-1}$).
6.2 $\psi = 90° - 135° = -45°$.
6.3 $0.02/40 = 5 \times 10^{-4}$ rad $= 0.029°$ (about 1.7 arcminutes).

7.1 $10\,mg = 0.098$ m/s^2; $\frac{1}{2} \times 0.098 \times 10^2 = 4.9$ m.
7.2 $10°/\text{h} = 4.85 \times 10^{-5}$ rad/s; $\frac{1}{2} \times 20 \times 4.85 \times 10^{-5} \times 30^2 = 0.44$ m.
7.3 $9.83 - 9.80 = 0.03$ m/s$^2 \approx 3$ mg.

8.1 $30 \times 0.0005 = 1.5$ cm.
8.2 $90°/\text{s} \times 0.002 = 0.18° = 0.00314$ rad; $\times 15 = 4.7$ cm.
8.3 The car moves while the picture is being taken, so different parts are captured at different positions — just like a LiDAR sweep.

9.1 0.765 m error in the body z direction; it does *not* depend on range (a lever-arm error is a constant offset in the body frame).
9.2 $0.02/60 = 3.3 \times 10^{-4}$ rad $= 0.019°$.
9.3 $Z = 0.974$ m, so the point height is 10.974 m (31 cm lower than with roll and pitch).

10.1 Straight line: $(10.40,\ 5.00,\ 0°)$.
10.2 $K = 0.09/0.18 = 0.5$; $\mu = 50.10$ m; $\sigma^2 = 0.045$, $\sigma = 0.212$ m.
10.3 The landmark is somewhere known, so seeing it at a measured distance and direction rules out many possible positions.

11.1 Centered: $\mathbf{p}' = (\mp 0.5, 0)$, $\mathbf{q}' = (0, \mp 0.5)$; sums 0 and $0.5 \Rightarrow \theta = 90°$; $\mathbf{t} = (2, 1.5) - \mathbf{R}(0.5, 0) = (2, 1.0)$.
11.2 The line is the z-axis: $d = \sqrt{3^2 + 4^2} = 5$.
11.3 Everything looks the same as you move, so there is nothing to tell you how far you have gone.

12.1 Odometry sums to exactly 3.0, so there is no misclosure: $x = (1.0,\ 2.1,\ 3.0)$, all residuals 0.
12.2 $\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 5 \end{pmatrix}\mathbf{x} = \begin{pmatrix} -0.1 \\ 0.2 \\ 13.7 \end{pmatrix} \Rightarrow x = (1.062,\ 2.223,\ 3.185)$; odometry residuals $+0.062$ each, GPS -0.015. The stiffer GPS gets most of its way.
12.3 The stiff band hardly stretches; the others stretch more to share the strain.

13.1 Mean 1.2 cm; RMSE $\sqrt{34/5} = 2.61$ cm; with checkpoint error $\sqrt{6.8 + 2.25} = 3.01$ cm.
13.2 $\sqrt{1 + 4 + 4 + 16} = 5$ cm; attack the 4 cm term first (it is 64% of the variance).
13.3 If the computer used them to fit the map, they would of course agree with it — the test would prove nothing.

14.1 Along the beam: 0.3 cm; across: $15 \times 1.745 \times 10^{-4} = 0.26$ cm.
14.2 Fully correlated errors add linearly: $2 + 3 = 5$ cm (not 3.6 cm).
14.3 $0.5/\sqrt{n} < 0.05 \Rightarrow n > 100$.
14.4 $1.96 \times 1.5 = 2.94$ cm.

15.1 $8 \times 0.01227 = 0.098$ m (9.8 cm).
15.2 $0.005 + 3 \times 0.00611 = 0.023$ m (2.3 cm).
15.3 $\sqrt{1.25^2 + 0.65^2} = 1.41$ cm.
15.4 $1800 \times 20 \times 10^{-6} = 0.036$ s; $\times 10$ m/s $= 0.36$ m.

16.1 Misclosure 0 (a constant bias rotates the whole

loop); far corner $707 \times 0.008727 = 6.2$ m off.
16.2 $0.01 \times 60 = 0.6^\circ/\sqrt{\text{h}}$.
16.3 $0.03/25 = 0.0012$ rad $= 0.069^\circ$.
16.4 Because an error that is the same on every leg rotates or stretches the whole figure, and a rotated or stretched closed figure still closes.
16.5 $b = 0.0065/1.745 \times 10^{-3} = 3.7$ m.
16.6 $\sqrt{0.30^2 + 0.022^2 + 0.321^2} = 0.44$ m.

17.1 $152.4 \times \sin 37.5^\circ \times 1.745 \times 10^{-4} = 0.016$ m (1.6 cm).
17.2 Zone depth $c/(2 \times 1.2 \times 10^6) = 124.9$ m; $152.4/124.9 = 1.22 \Rightarrow$ zone 2.
17.3 $\sqrt{1.0^2 + 2.5^2 + 0.5^2} = 2.7$ cm.
17.4 It is Class 3R: from a moving aircraft the beam sweeps past any eye too fast to harm it (NOHD 0 m above 1 kn), but a stationary beam has a hazard distance (0.3 m, 3.5 m with optical aids).

18.1 $(\lfloor 40.52 \rfloor, \lfloor -9.6 \rfloor, \lfloor 7.96 \rfloor) = (40, -10, 7)$.
18.2 $16 \times 1 \times 0.75 = 12$ m^3.
18.3 A roll boresight error (it tilts the road in opposite senses on opposite passes).

19.1 5J-17.051(2): the accuracy of data shall be independently verified. A LiDAR survey satisfies it by testing against independent checkpoints that were never used as control, and documenting the method and RMSE.
19.2 March 5 — the survey date is the last date of data acquisition (5J-17.051(3)(d)), not the processing date.
19.3 No. The edited tile's hash no longer matches the code listed in the sealed signature file, so that file is no longer sealed (5J-17.062(4)).
19.4 U.S. survey foot: 365 760.7315 m; international foot: 365 760.0000 m; difference 0.7315 m (2.40 ft).

20.1 Exercise responsible charge: obtain and review the calibration, trajectory quality and control; run (or direct) an independent checkpoint test; confirm datums, units and classification; state responsibility for the vendor's data before sealing.
20.2 Any three of: no data behind solid objects or under dense vegetation; no near-infrared returns below water; no underground utilities; occlusions by traffic; reduced accuracy beyond the usable range; areas affected by GNSS outages.
20.3 Part 107 limits altitude to 400 ft AGL (absent a waiver or nearby structure); lower altitude also shortens the range, so attitude and range terms shrink and point density rises.

APPENDIX B

Final Exam (Administered Separately)

The final exam and its answer keys are **not printed in this book**. They are in the separate *Instructor Packet*, which contains:

- **two exam versions** (A and B) of 50 conceptual and standards questions each — 15 true/false, 20 multiple choice on concepts, and 15 on standards and professional practice;
- the answer keys, a map of every question to the seminar's learning objectives, and administration instructions;
- the **pass mark: 70%** (35 of 50 correct).

No calculations are required and no calculator is needed. To prepare, review the figures, Simple Approach boxes, Big Ideas, Field Tips, and Modules 13, 14, 19 and 20.

APPENDIX C

Equation Summary Sheet

Quantity	Equation	Mod.
Time-of-flight range	$\rho = c\,\Delta t/2,\ \ c = c_0/n$	3
Max. unambiguous range	$\rho_{\max} = c/(2f_{\mathrm{PRF}})$	3
Phase range	$\rho = \frac{\lambda_m}{2}(N + \Delta\varphi/2\pi)$	3
Beam footprint	$D = d_0 + \rho\gamma$	4
Received power	$P_r \propto \rho_{\mathrm{refl}} \cos\theta_i / R^2$	4
Polar to Cartesian	$x = \rho\cos V\cos H,\ y = \rho\cos V\sin H,\ z = \rho\sin V$	5
Point spacing	$\Delta_{\mathrm{along}} = v/f,\ \Delta_{\mathrm{across}} = \rho\Delta\theta$	5
Small-angle error	$\delta = \rho\,\delta\alpha$ (rad)	2, 6
3-D rotation	$\mathbf{R} = \mathbf{R}_z(\psi)\mathbf{R}_y(\theta)\mathbf{R}_x(\phi)$	6
Rigid transform	$\mathbf{r}_W = \mathbf{R}\mathbf{r}_B + \mathbf{p};\ \Delta\mathbf{T}_{ij} = \mathbf{T}_i^{-1}\mathbf{T}_j$	6
IMU model	$\hat{\boldsymbol{\omega}} = \boldsymbol{\omega} + \mathbf{b}^{\omega} + \mathbf{n}^{\omega},\ \hat{\mathbf{a}} = \mathbf{R}^{BW}(\mathbf{a} - \mathbf{g}) + \mathbf{b}^{a} + \mathbf{n}^{a}$	7
Strapdown	$\mathbf{v}' = \mathbf{v} + \mathbf{g}\Delta t + \mathbf{R}\hat{\mathbf{a}}\Delta t;\ \mathbf{p}' = \mathbf{p} + \mathbf{v}\Delta t + \frac{1}{2}(\mathbf{g} + \mathbf{R}\hat{\mathbf{a}})\Delta t^2$	7
Bias growth	$\delta y = \frac{1}{2} v b_g t^2;\ \delta x = \frac{1}{2} b_a t^2$	7
Preintegration	$\Delta\mathbf{R}_{ij} = \mathbf{R}_i^{\mathsf{T}}\mathbf{R}_j, \ldots$	7
Timing error	$\delta = v\,\delta t + \rho\dot{\psi}\,\delta t$	8
Direct georeferencing	$\mathbf{r}_W = \mathbf{p}_W + \mathbf{R}_B^W(\mathbf{R}_S^B\mathbf{r}_S + \mathbf{l}^B)$	9
Motion model	$\alpha = (R - L)/w;\ x' = x + (r + \frac{w}{2})(\sin(\theta + \alpha) - \sin\theta)$	10
Kalman (1-D)	$K = \bar{\sigma}^2/(\bar{\sigma}^2 + \sigma_z^2);\ \mu = \bar{\mu} + K(z - \bar{\mu});\ \sigma^2 = (1 - K)\bar{\sigma}^2$	10
EKF	$\bar{\mathbf{P}} = \mathbf{G}\mathbf{P}\mathbf{G}^{\mathsf{T}} + \mathbf{Q};\ \mathbf{K} = \bar{\mathbf{P}}\mathbf{H}^{\mathsf{T}}(\mathbf{H}\bar{\mathbf{P}}\mathbf{H}^{\mathsf{T}} + \mathbf{R})^{-1}$	10
ICP	$\min\sum\|\mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_i\|^2;\ \mathbf{R} = \mathbf{V}\mathbf{U}^{\mathsf{T}},\ \mathbf{t} = \bar{\mathbf{q}} - \mathbf{R}\bar{\mathbf{p}}$	11
Edge / plane distance	$d_e = \|(\mathbf{p} - \mathbf{a}) \times (\mathbf{p} - \mathbf{b})\|/\|\mathbf{a} - \mathbf{b}\|;\ d_p = \|(\mathbf{p} - \mathbf{u}) \cdot \mathbf{n}\|/\|\mathbf{n}\|$	11
Factor graph	$\mathbf{X}^* = \arg\min\sum\|\mathbf{h}_k(\mathbf{X}) - \mathbf{z}_k\|^2_{\Sigma_k}$	12
Normal equations	$\mathbf{A}^{\mathsf{T}}\mathbf{W}\mathbf{A}\,\mathbf{x} = \mathbf{A}^{\mathsf{T}}\mathbf{W}\mathbf{b}$	12
RMSE	$\sqrt{\frac{1}{n}\sum e_i^2};\ \mathrm{RMSE}_H = \sqrt{\mathrm{RMSE}_x^2 + \mathrm{RMSE}_y^2}$	2, 13
Final accuracy	$\sqrt{\mathrm{RMSE}_{\mathrm{data}}^2 + \mathrm{RMSE}_{\mathrm{survey}}^2}$	13
Error budget (RSS)	$\sigma = \sqrt{\sum\sigma_j^2}$	2, 13
Outlier test	$\bar{d}_i > \mu + \alpha\sigma$	18
Grid volume	$V = \sum A(z_{\mathrm{top}} - z_{\mathrm{base}})$	18
Variance propagation	$\sigma_y^2 = \sum(\partial f/\partial x_i)^2\sigma_i^2 + 2\sum a_i a_j \sigma_{ij}$	14

Quantity	Equation	Mod.
Covariance propagation	$\mathbf{\Sigma}_y = \mathbf{J}\mathbf{\Sigma}_x\mathbf{J}^\mathsf{T}$	14
Averaging with bias	$\text{RMSE} = \sqrt{b^2 + \sigma^2/n}$	14
95% factors	1.960 (1-D), 2.448 (2-D), 2.796 (3-D)	14
Quantization	$\sigma_q = q/\sqrt{12}$	15
Angular step	$\Delta\theta = 360^\circ/N_{\text{col}}$; $s = \rho\Delta\theta$	15
Random walks	$\text{ARW} = 60N_g$; $\sigma_\psi = \text{ARW}\sqrt{t}$	16
Dual-antenna heading	$\sigma_\psi \approx \sigma_{\text{rel}}/b$	16
Coast error	$\sqrt{(\sigma_v t)^2 + (\frac{1}{2}b_a t^2)^2 + (\frac{1}{6}g b_g t^3)^2}$	16

APPENDIX D

Hands-On Labs

These five labs turn the modules into things you can do with your hands, a tape measure, a spreadsheet or a laptop. Labs 1–2 use the simple approach and need no math; Labs 3–5 are for engineers.

Lab 1 — Be a human LiDAR (Modules 3 and 5)

SIMPLE APPROACH

You need: a protractor taped flat on a chair, a tape measure, graph paper, a friend.
Do it: Sit at the chair — you are the scanner. Point at a corner of the room, a door edge, a table leg. For each, read the angle on the protractor (that is H) and have your friend measure the distance (that is ρ). Write them in a table.
Then: On graph paper, put yourself in the middle. For each object, turn your ruler to the angle and mark a dot at the distance (1 square = 50 cm). Join the dots — you have drawn a *point-cloud map* of your room!
Engineer's extension: compute each dot with $x = \rho \cos H$, $y = \rho \sin H$ and compare with the drawing.

Lab 2 — The drifting blindfold walk (Modules 7 and 10)

SIMPLE APPROACH

Do it: Put a cone at the start. Blindfolded, try to walk 10 steps forward, turn around, and walk 10 steps back to the cone. Mark where you stop. Repeat three times. You will miss the cone — that is *drift*. Now try again, but you may touch one wall halfway (a *landmark*). Do you get closer?
What you learned: counting steps is like an IMU: little mistakes add up. Touching something you know is like a LiDAR match or a loop closure: it fixes the mistakes.

Lab 3 — ICP in 30 lines of Python (Module 11)

Run this with Python 3 and NumPy. It builds an L-shaped "room," makes a sparse, moved and turned copy (a new scan), and recovers the motion with ICP. Watch the RMS gap fall each iteration — exactly Figure 11.1.

icp_lab.py

```
import numpy as np

def best_fit(P, Q):
    """Closed-form rigid fit (Module 11): returns R, t with Q ~ R P + t."""
    pb, qb = P.mean(0), Q.mean(0)
    W = (P - pb).T @ (Q - qb)
    U, S, Vt = np.linalg.svd(W)
```

```
    R = Vt.T @ U.T
    if np.linalg.det(R) < 0:          # avoid a mirror image
        Vt[-1] *= -1; R = Vt.T @ U.T
    return R, qb - R @ pb

def icp(P, Q, iters=30):
    cur = P.copy()
    for k in range(iters):
        d = ((cur[:, None, :] - Q[None, :, :])**2).sum(-1)
        match = Q[d.argmin(1)]                # 1. nearest neighbors
        R, t = best_fit(cur, match)           # 2. best rigid fit
        cur = cur @ R.T + t                   # 3. apply
        rms = np.sqrt(d.min(1).mean())
        print(f"iter {k:2d}  RMS gap = {rms:.4f} m")
        if rms < 1e-4: break                  # 4. repeat until stable
    return best_fit(P, cur)                   # total transform

def room(n):                          # an L-shaped wall plus a short wall
    s = np.linspace(0, 1, n)
    return np.vstack([np.c_[6*s, 0*s], np.c_[6+0*s, 3*s], np.c_[3.5+0*s, 3+1.5*s]])
Q = room(600)                         # dense "map"
th = np.radians(6); Rt = np.array([[np.cos(th), -np.sin(th)], [np.sin(th), np.cos(th)]])
P = (room(40) - [0.25, -0.15]) @ Rt   # sparse new scan, moved and turned
R, t = icp(P, Q)
print("recovered angle (deg):", np.degrees(np.arctan2(R[1, 0], R[0, 0])).round(3))
```

Expected output: RMS gap 0.18 m at iteration 0 falling to about 0.002 m; recovered angle 6.0°. **Experiments:** (a) increase the turn to 25° and watch ICP get stuck in a local minimum — the reason it needs an IMU guess; (b) delete the short wall from `room()` and add a shift along x only: the long walls let the scan slide — the hallway effect.

Lab 4 — Field check of boresight from two passes (Module 9)

ENGINEER'S VIEW

Procedure: pick a sharp vertical feature beside the road — a building corner or a pole. Drive past it in both directions in the same lane. In the cloud, measure the along-track offset Δ between the feature's position in the two passes, and the range ρ from the drive line to the feature.

Why it works: a yaw boresight error $\delta\psi$ shifts the feature by $\rho\,\delta\psi$ forward on one pass and backward on the other, so the passes differ by $2\rho\,\delta\psi$:

$$\delta\psi \approx \frac{\Delta}{2\rho}.$$

Example: $\Delta = 4.2$ cm along-track at $\rho = 30$ m $\Rightarrow \delta\psi = 0.042/60 = 7.0 \times 10^{-4}$ rad $= 0.040°$. The same logic applied to the *height* of a flat road edge on both sides gives the roll residual, and to the height of features ahead/behind gives pitch.

Lab 5 — Checkpoint report in a spreadsheet (Module 13)

ENGINEER'S VIEW

Set up columns: Point ID | Survey Z | LiDAR Z | dZ = LiDAR − Survey | dZ^2. Then:

- mean: `=AVERAGE(D2:D31)`
- standard deviation: `=STDEV.S(D2:D31)`
- RMSE: `=SQRT(SUMSQ(D2:D31)/COUNT(D2:D31))`
- final RMSE with checkpoint survey error in G1: `=SQRT(RMSE^2+G1^2)`

Enter the ten residuals from the Module 13 worked example and confirm 0.53 cm, 2.64 cm, 2.56 cm and 2.75 cm. Then add a deliberate blunder (+25 cm) to one point and see how strongly RMSE reacts — and why each large residual must be investigated, not just averaged.

APPENDIX E

Sample SOP and Client Accuracy Agreement

This appendix is a **template**. It shows how the ideas in this course become a written Standard Operating Procedure (SOP) and a signed agreement on horizontal and vertical accuracy. Replace every sample value with your project's requirements, and have the surveyor in responsible charge review it against the current standards and contract.

SOP number	ML-SOP-001	**Revision**	0 (sample)
Title	Mobile LiDAR Topographic Survey — Collection, Processing, QA/QC and Reporting		
Effective date	_____	**Approved by**	_____, PSM

1. Purpose

To produce mobile LiDAR point clouds and derived topographic products whose horizontal and vertical accuracy is agreed with the client *before* collection, measured independently *after* processing, and reported in the language of the ASPRS Positional Accuracy Standards.

2. Scope

Vehicle-, backpack- or cart-mounted mobile laser scanning for topographic surveys, corridor mapping, as-builts and volumes. It does not cover boundary determination or subsurface utility locating.

3. References

- ASPRS Positional Accuracy Standards for Digital Geospatial Data, Edition 2, Version 2 (2024).
- ASPRS LAS Specification, version 1.4 (current revision).
- Chapter 5J-17, Florida Administrative Code, Standards of Practice (current text).
- FDOT guidance for terrestrial mobile LiDAR and the FDOT Florida Permanent Reference Network (FPRN), where applicable.
- IEC 60825-1, Safety of laser products (Class 1 equipment only).
- Manufacturer manuals and datasheets for the scanner, IMU/INS and GNSS receivers.

4. Roles and responsibilities

Surveyor in responsible charge (PSM)	approves the SOP and accuracy agreement; signs the final report and accuracy statement
Project manager	client contact; schedule; scope changes; nonconformance decisions with the PSM
Field crew chief	equipment checks, calibration loop, collection, field log, safety
Control surveyor	control and checkpoint survey; keeps checkpoint coordinates *sealed* from processing staff
Processing analyst	trajectory, georeferencing, cleaning, classification, products
QA reviewer (independent)	accuracy testing against checkpoints, swath checks, report review

5. Client accuracy agreement

Complete and sign this agreement **before mobilization**. It fixes what will be delivered, how accurate it must be, and how accuracy will be proven.

Item	Agreed value (sample — edit for your project)
Project / location	Main Street corridor, 2.0 km, curb to right-of-way line
Horizontal datum and realization	NAD 83(2011), epoch 2010.0; State Plane Coordinate System, Florida East zone
Vertical datum and geoid	NAVD 88 via GEOID18 (state the model)
Linear units	_____ (state U.S. survey foot or international foot explicitly; NGS deprecated the U.S. survey foot after 2022 — confirm statutory and client requirements)
Horizontal accuracy class	**RMSE$_H \leq$ 5 cm** (0.16 ft) on well-defined features within 25 m (82 ft) of the trajectory
Vertical accuracy class (NVA)	**RMSE$_V \leq$ 3 cm** (0.10 ft) on hard, non-vegetated surfaces — *acceptance criterion*
Vegetated vertical accuracy (VVA)	tested and reported as found (no pass/fail)
Data internal precision	within-swath smooth-surface max difference $\leq$ 1.8 cm (0.06 ft); swath-to-swath RMSD$_z \leq$ 2.4 cm (0.08 ft), max difference $\leq$ 4.8 cm (0.16 ft) ($0.6X$ / $0.8X$ / $1.6X$ for X = 3 cm)
Checkpoints	$\geq$30 NVA checkpoints on hard surfaces; $\geq$30 well-defined horizontal checkpoints (painted chevrons, corners); distributed along the full length and both sides; never used as control
Checkpoint / control accuracy	checkpoints RMSE$_V \leq$ 1.5 cm (0.05 ft), RMSE$_H \leq$ 2.5 cm (0.08 ft) (at least 2$\times$ better than the product); method and redundancy documented
Accuracy reporting	ASPRS Edition 2 wording; checkpoint survey error included in the reported RMSE
Point density	$\geq$ 400 points/m^2 (37 points/ft^2) on pavement within 20 m (66 ft) (verified on test patches)
Usable range	points beyond 25 m (82 ft) from the trajectory excluded from accuracy-critical products
Classification (LAS codes)	1, 2, 6, 7, 11, 13–15, 17, 18 (others by agreement)
Deliverables	LAS 1.4 (PDRF 7, RGB) as LAZ tiles; ground surface (TIN) with breaklines; planimetric line work in CAD format; trajectory file; control and checkpoint report; survey report with accuracy statement; metadata
Collection conditions	dry pavement; daylight or as agreed; traffic control per permit; GNSS/RTK availability checked (e.g. FPRN VRS)
Acceptance and remedy	if NVA or horizontal RMSE exceeds the class, producer investigates, reprocesses, or recollects the affected area at no added cost; outliers beyond 3$\times$RMSE are investigated and documented, never silently removed
Client representative	Name _____ Signature _____ Date _____
Surveyor in responsible charge	Name _____ PSM No. _____ Signature _____ Date _____

SURVEYOR'S FIELD TIP

Agree on what "accuracy" means before anyone drives. Most disputes come from four gaps: (1) the class was stated without saying RMSE or 95%; (2) horizontal and vertical were not separated; (3) nobody said which surfaces are tested (hard surfaces vs. grass); (4) datum, geoid and units were assumed. The table above closes all four.

6. Procedure

6.1 Planning

a. Build an error budget for the system and route (Modules 13–14); confirm the agreed classes are achievable at the planned range and speed.

b. Plan routes for loop closures and passes in both directions; identify GNSS-poor areas (tunnels, canopy, urban canyons) and add control there.

c. Lay out control (every ∼250 m / 820 ft and in GNSS-poor areas) and checkpoints (independent, distributed, both sides).

d. Confirm RTK source (base or network such as FPRN), mountpoint, datum and epoch.

e. Obtain permits and maintenance-of-traffic plans; schedule for dry surfaces.

6.2 Equipment checks and calibration

a. Verify firmware, calibration file (lever arm, boresight) and its date; record in the field log.

b. Confirm time synchronization (PPS/PTP lock) and IMU, LiDAR and GNSS streams at nominal rates.

c. Static IMU alignment and a calibration loop over known features; review boresight residuals before production.

d. Laser safety: street and terrestrial scanners Class 1; airborne Class 3R scanners (e.g. RIEGL VUX-240[24]) operated only per the manufacturer's NOHD/ENOHD and the company laser-safety program; post crew safety briefing.

6.3 Control and checkpoint survey

a. Survey control and checkpoints with methods meeting the agreed accuracy (redundant RTK, static GNSS, or total station / digital levels).

b. Assign IDs; deliver checkpoint coordinates only to the QA reviewer (sealed from processing).

c. Check into a known monument at start and end of each session; record differences.

6.4 Data collection

a. Collect at or below the planned speed; drive both directions; avoid stops in traffic where possible.

b. Keep a field log: times, weather, GNSS outages, sensor warnings, deviations from plan.

c. Verify data on site (coverage, density, no gaps) before demobilizing.

6.5 Processing

a. Trajectory: GNSS/INS (RTK or PPK) with LiDAR SLAM and loop closures; review trajectory standard deviations and flag weak stretches.

b. Georeference with the approved calibration; de-skew; apply control.

c. Clean noise and moving objects; classify to the agreed LAS codes; document voxel size and filter settings.

d. Extract products: surface with breaklines, planimetrics, volumes as ordered.

6.6 QA/QC and acceptance testing

Test	Criterion (sample)	Pass?
NVA, ≥30 hard-surface checkpoints	$RMSE_V$ (incl. checkpoint error) $\leq$ 3 cm (0.10 ft)	☐
Horizontal, ≥30 well-defined points	$RMSE_H$ (incl. checkpoint error) $\leq$ 5 cm (0.16 ft)	☐
VVA	reported as found	☐
Within-swath precision	max difference $\leq$ 1.8 cm (0.06 ft) on flat surfaces	☐
Swath-to-swath	$RMSD_z \leq$ 2.4 cm (0.08 ft); max $\leq$ 4.8 cm (0.16 ft)	☐
Point density	$\geq$ 400 pts/m^2 on test patches	☐
Classification	spot check; no ground points on vehicles/buildings	☐
Datum / geoid / units	match the agreement and file headers	☐
Residual review	every residual >3×RMSE investigated and documented	☐

6.7 Reporting

The report states methods, equipment, calibration, trajectory solution, control used, checkpoints used (kept separate), datums, geoid, units, the test results and the accuracy statement. Sample statement:

Sample accuracy statement

"This data set was tested to meet ASPRS Positional Accuracy Standards for Digital Geospatial Data, Edition 2, Version 2 (2024) for a 3 cm $RMSE_V$ Vertical Accuracy Class and a 5 cm $RMSE_H$ Horizontal Accuracy Class. Non-Vegetated Vertical Accuracy (NVA) was found to be $RMSE_V$ = 2.1 cm using 30 independent checkpoints, including checkpoint survey uncertainty. Horizontal accuracy was found to be $RMSE_H$ = 2.8 cm using 30 well-defined checkpoints. Vegetated Vertical Accuracy (VVA) was found to be $RMSE_V$ = __ cm. Horizontal datum: NAD 83(2011) epoch 2010.0, SPCS Florida East; vertical datum: NAVD 88 (GEOID18); units: ___."

6.8 Nonconformance and corrective action

If any acceptance test fails: stop delivery of the affected area, identify the cause (trajectory, calibration, control, datum, processing), correct and retest, and record the event, cause and fix in the project file. Never delete checkpoints to make a test pass.

6.9 Records and retention

Archive raw sensor data, trajectory, calibration file, control and checkpoint data, processing logs and settings (including voxel and filter parameters), QA results, deliverables and the signed agreement, for the retention period required by law and contract.

7. Revision history

Rev.	Date	Description	Approved
0	_____	Initial issue (sample from course text)	_____

APPENDIX F

Glossary

Attitude Orientation of the platform: roll, pitch, yaw.

Bias A slowly changing offset in a sensor reading; integrates into drift.

Boresight The small rotation between the LiDAR and IMU frames.

Checkpoint A surveyed point used only to test accuracy, never to adjust.

Control point A surveyed point used to fix the datum and constrain the solution.

Covariance A matrix describing uncertainties and how they are correlated.

De-skew Correcting each point of a sweep for the motion during that sweep.

Dead reckoning Computing position by adding up measured motion from a known start.

DTM Digital terrain model: the bare-earth surface.

EKF Extended Kalman filter: a Kalman filter linearized with Jacobians.

Factor graph A graph of unknown states and the measurements (factors) that connect them, solved by least squares.

Footprint Size of the laser spot on the target.

GNSS Global Navigation Satellite Systems (GPS, GLONASS, Galileo, BeiDou).

Hallway effect Weak along-track constraint in featureless corridors.

ICP Iterative Closest Point: repeatedly pair and fit two point clouds.

IMU Inertial measurement unit: accelerometers + gyroscopes.

Innovation Measurement minus its prediction.

iSAM2 Incremental smoothing and mapping: updates only the changed part of a factor graph.

Jacobian Matrix of partial derivatives used to linearize a model.

Kalman gain Weight given to a new measurement versus the prediction.

Keyframe A selected scan whose pose becomes a node in the graph.

LAS/LAZ Standard point-cloud file format / its compressed version.

Lever arm Offset vector between two sensor origins.

LIO-SAM Tightly-coupled lidar–inertial odometry via smoothing and mapping (Shan *et al.*, 2020).

Loop closure Recognizing a revisited place and adding that constraint.

MAP Maximum a posteriori: the most probable solution given all data.

NDT Normal Distributions Transform: registration against per-voxel Gaussians.

Point cloud A set of 3-D points, often with intensity and time.

Pose Position plus orientation (6 degrees of freedom).

PPS Pulse per second: GNSS timing signal for synchronization.

Preintegration Summarizing many IMU samples into one relative-motion measurement.

RMSE Root-mean-square error: the standard accuracy measure.

RSS Root-sum-square: how independent errors combine.

SLAM Simultaneous localization and mapping.

Smoother An estimator that uses past *and* future data to refine each state.

Strapdown IMU fixed to the body; orientation tracked computationally.

Tightly coupled Sensors are fused in one estimation problem, at raw-measurement level.

Voxel A small 3-D grid cube.

APPENDIX G

References and Further Study

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BIG IDEA

Where to go next. Run LIO-SAM or a similar open-source pipeline on a public dataset; then take a commercial system to a small, well-controlled site, survey 30 checkpoints, and follow Module 21 from plan to accuracy statement. Nothing teaches like closing your own loop.